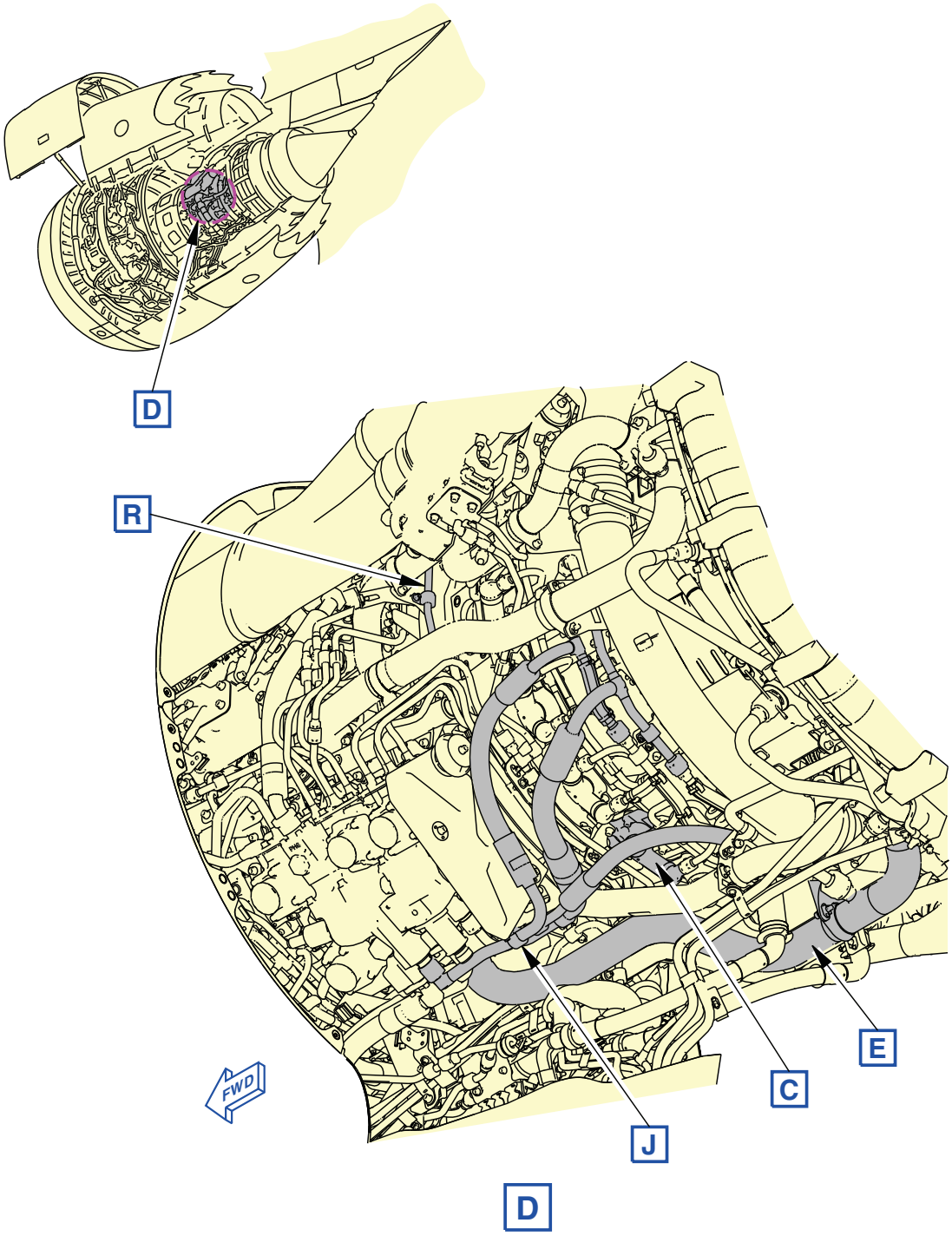
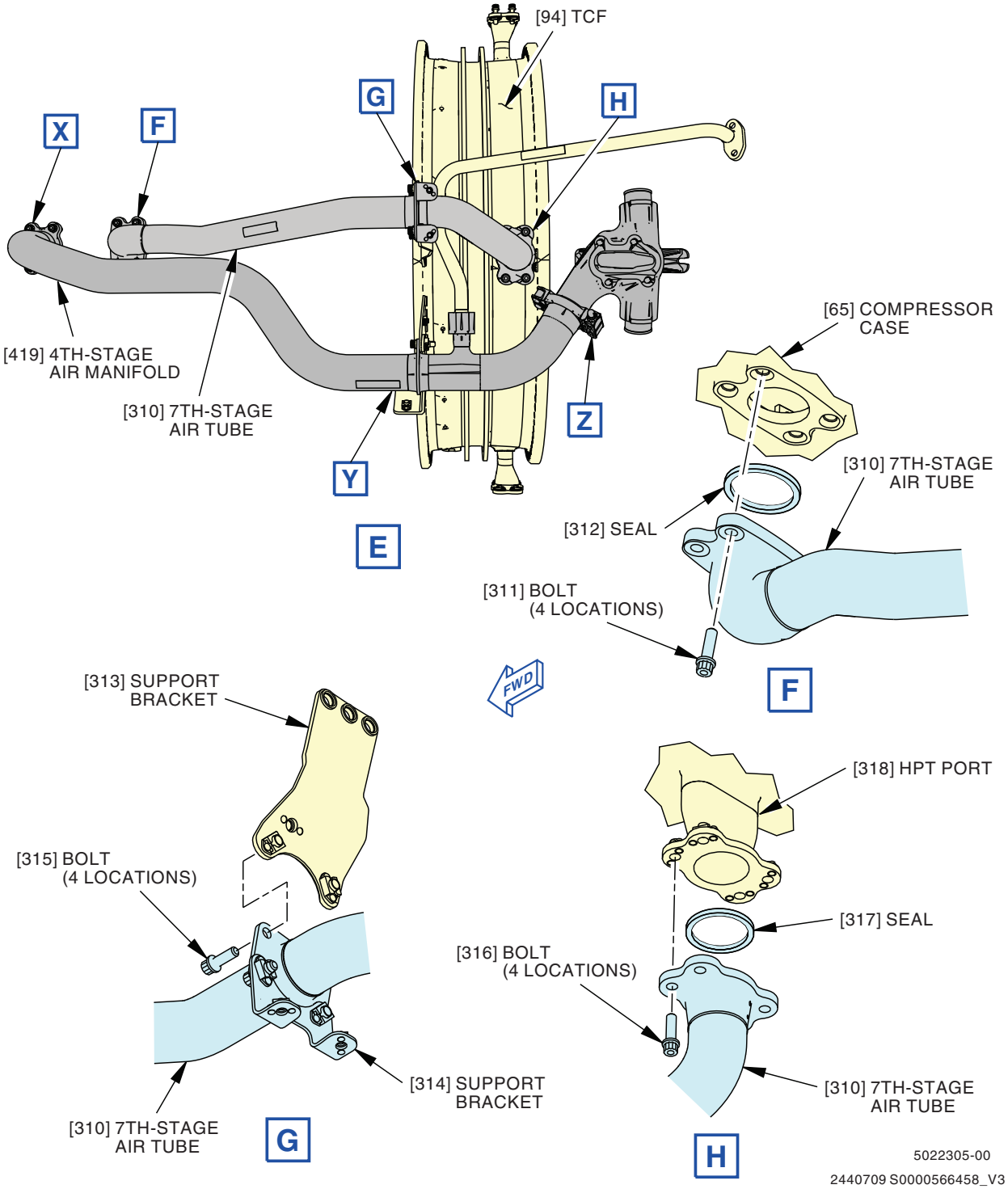


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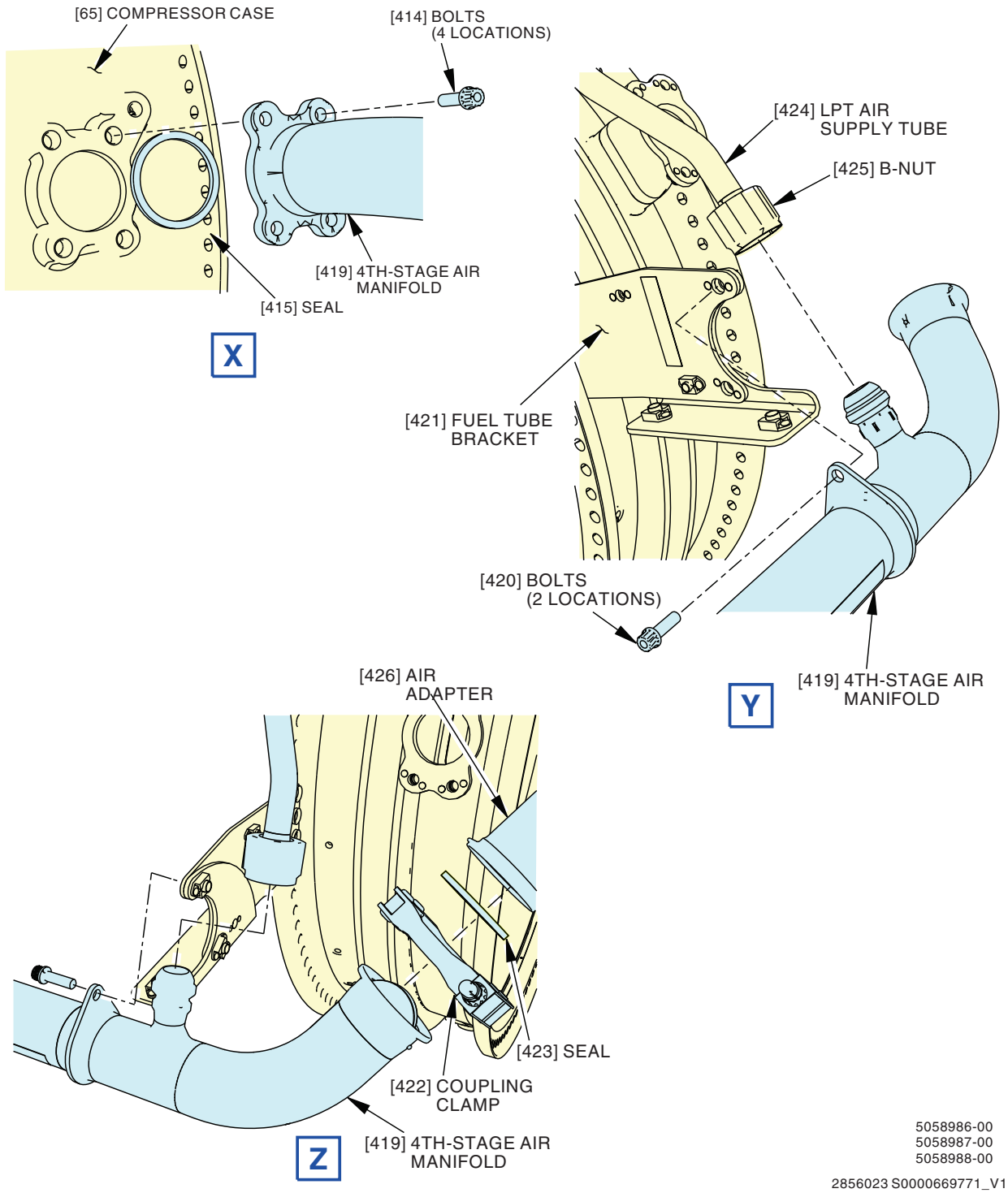
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 2 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 3 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

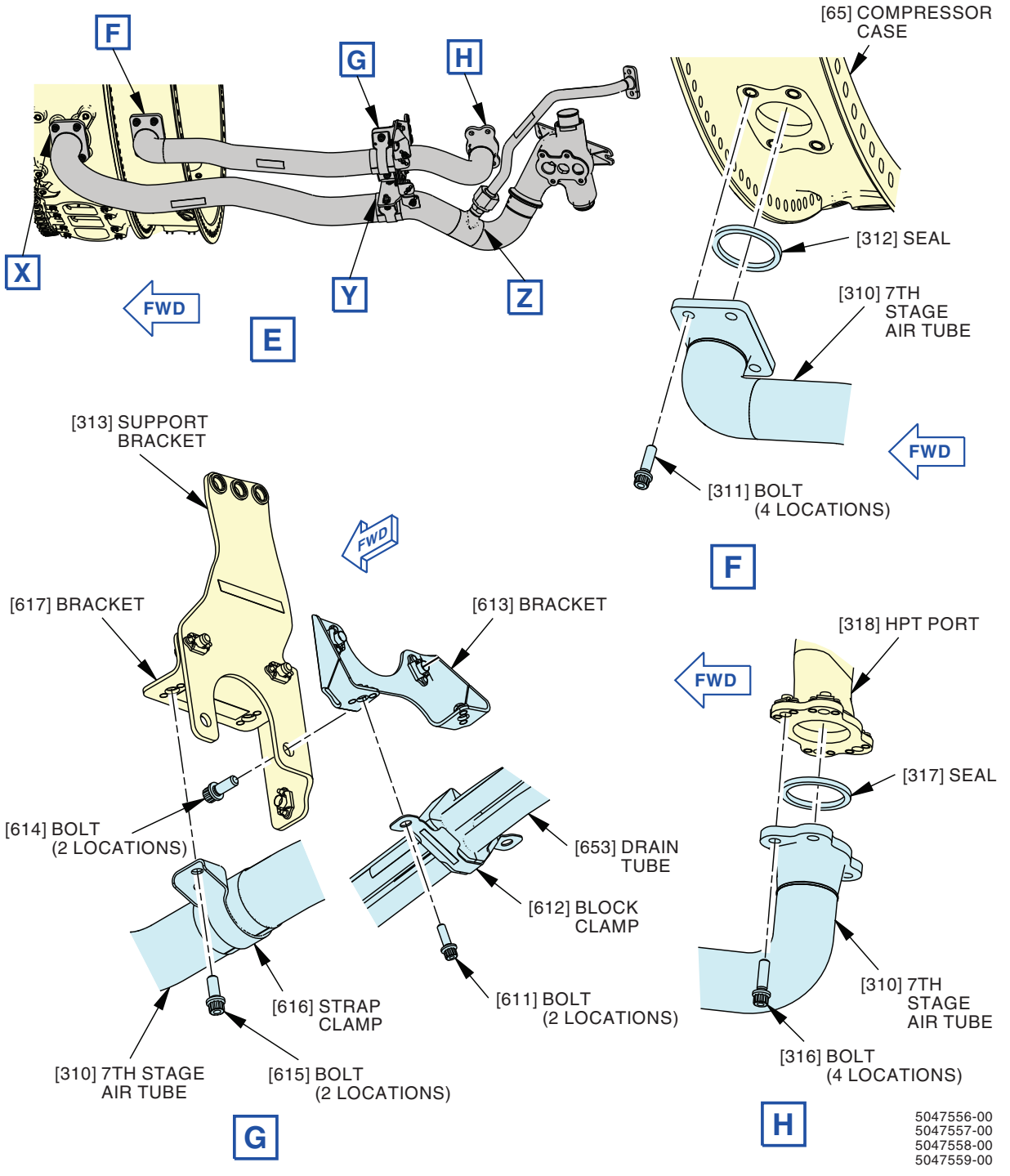


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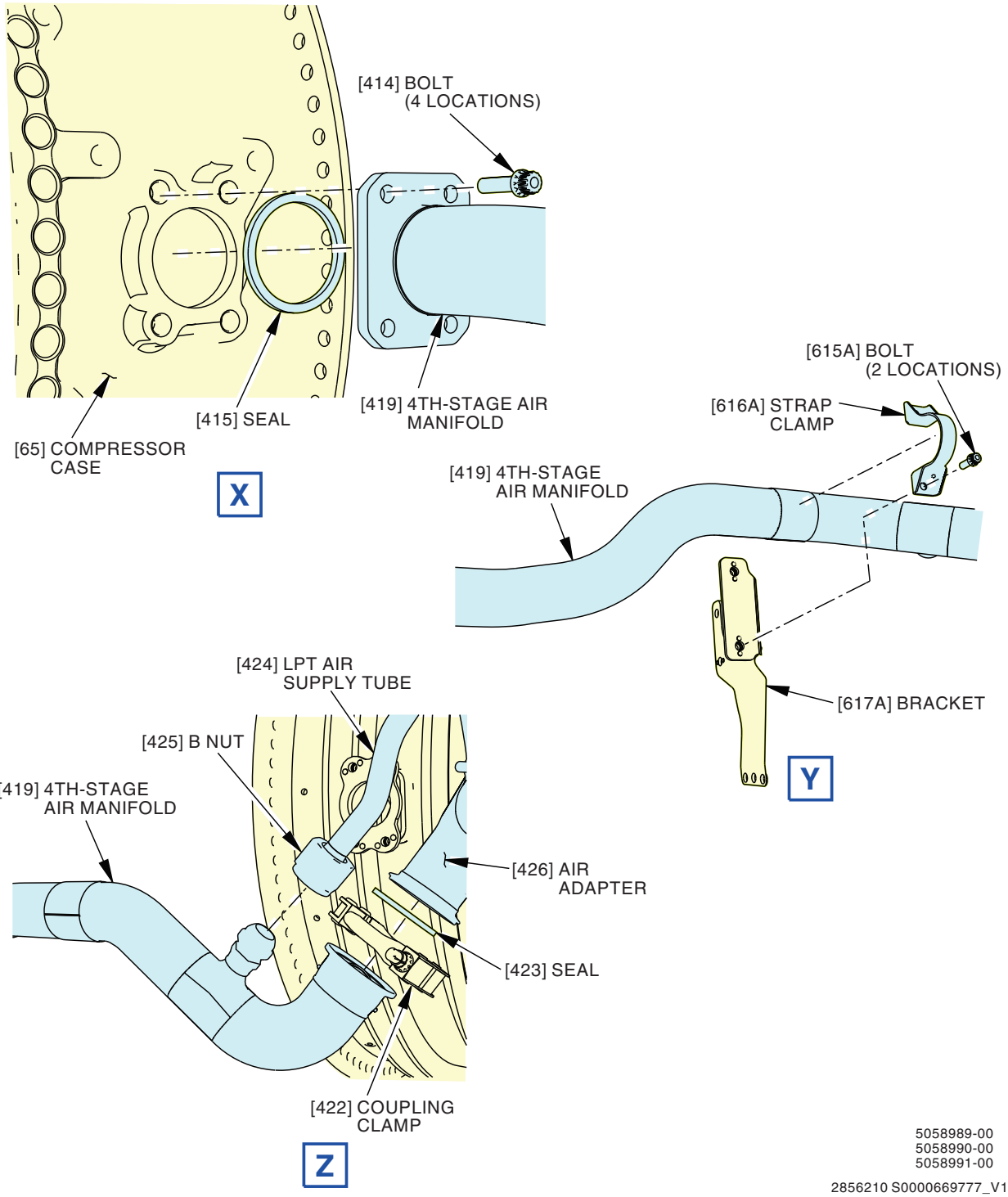
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 4 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



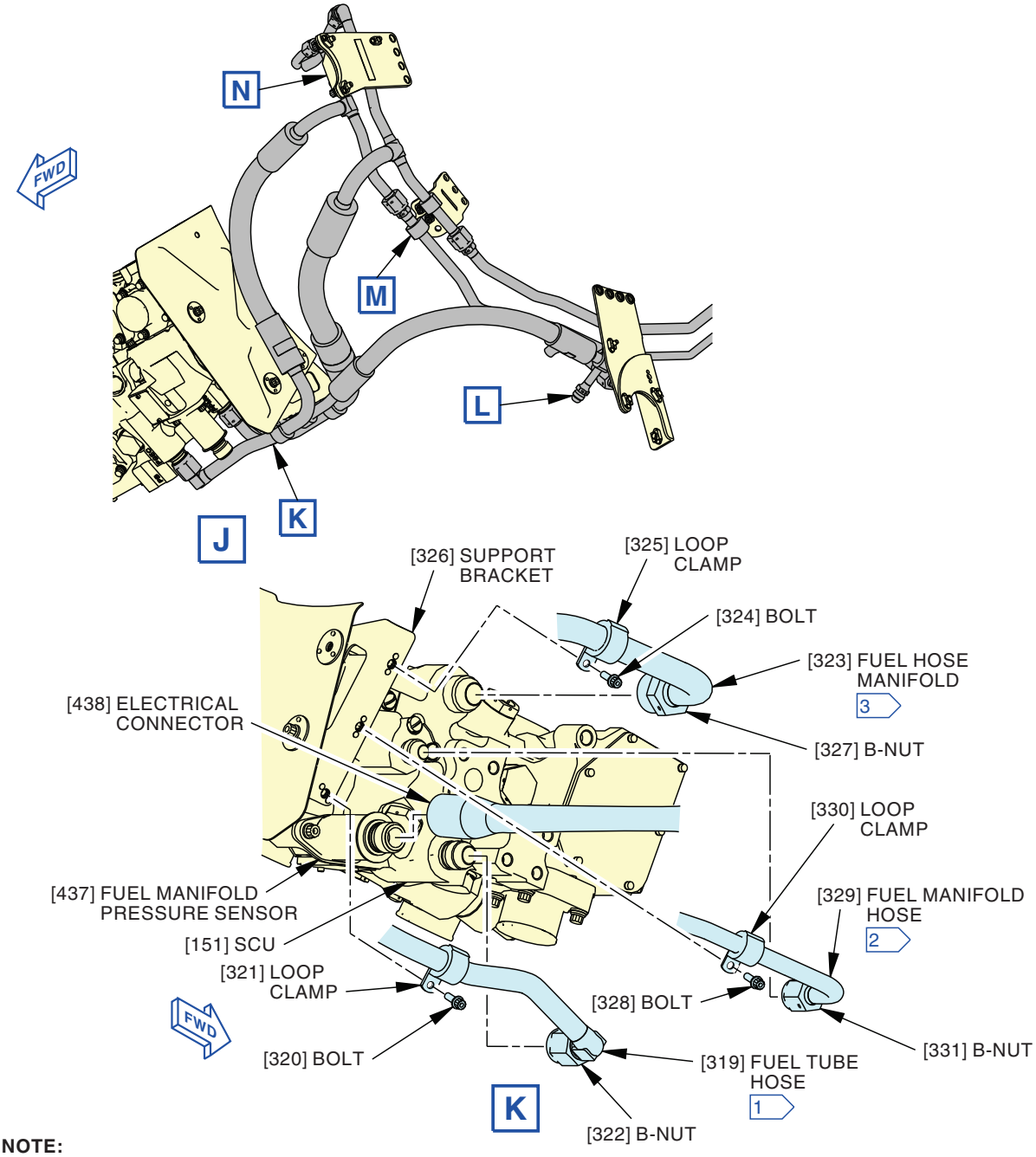
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 5 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 6 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



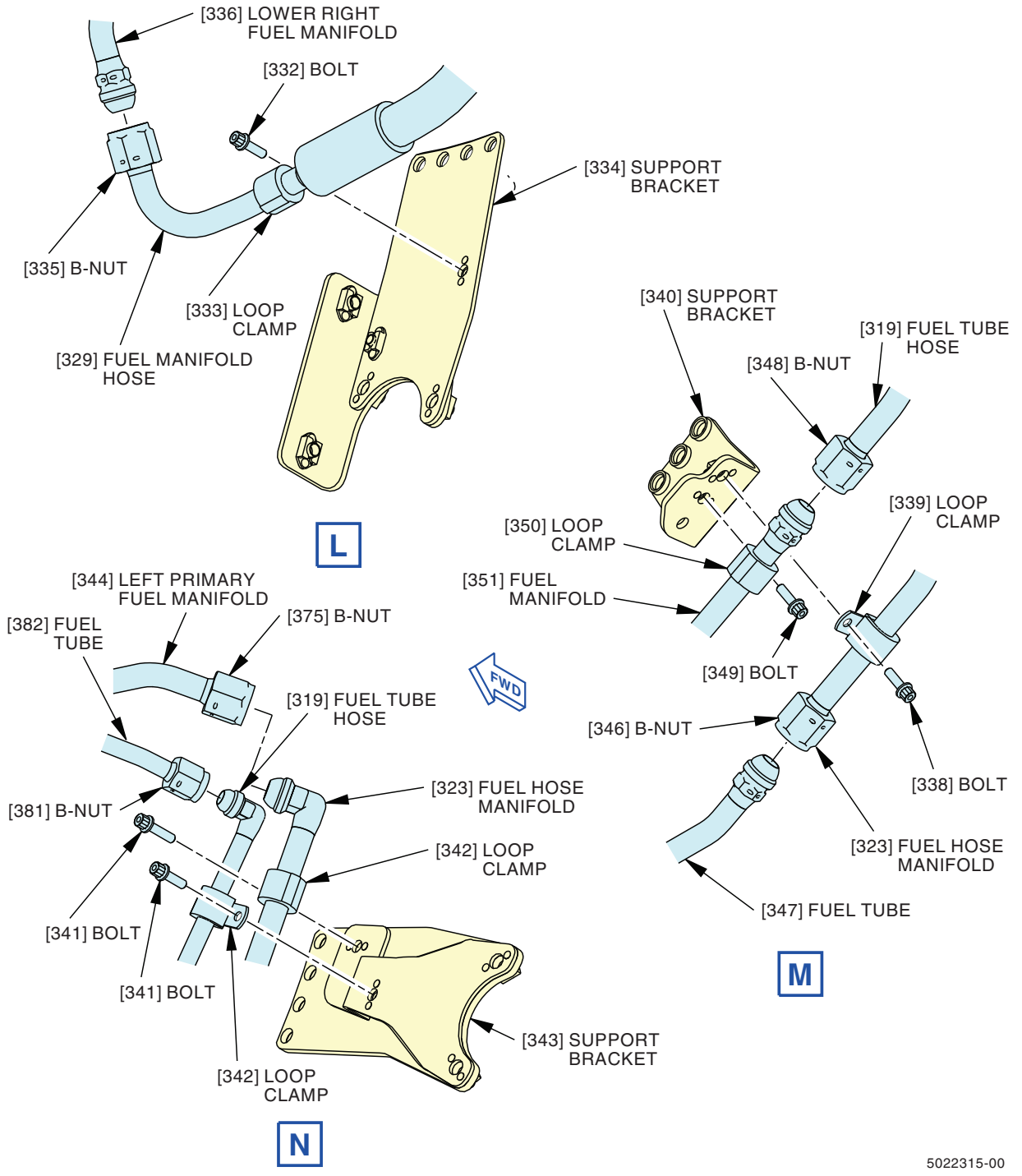
NOTE:

1. PILOT SECONDARY (PSEC) FUEL MANIFOLD
2. PILOT PRIMARY AND MAIN ENRICHED (PPME) FUEL MANIFOLD
3. PILOT PRIMARY AND MAIN NON-ENRICHED (PPMNE) FUEL MANIFOLD

5022312-00
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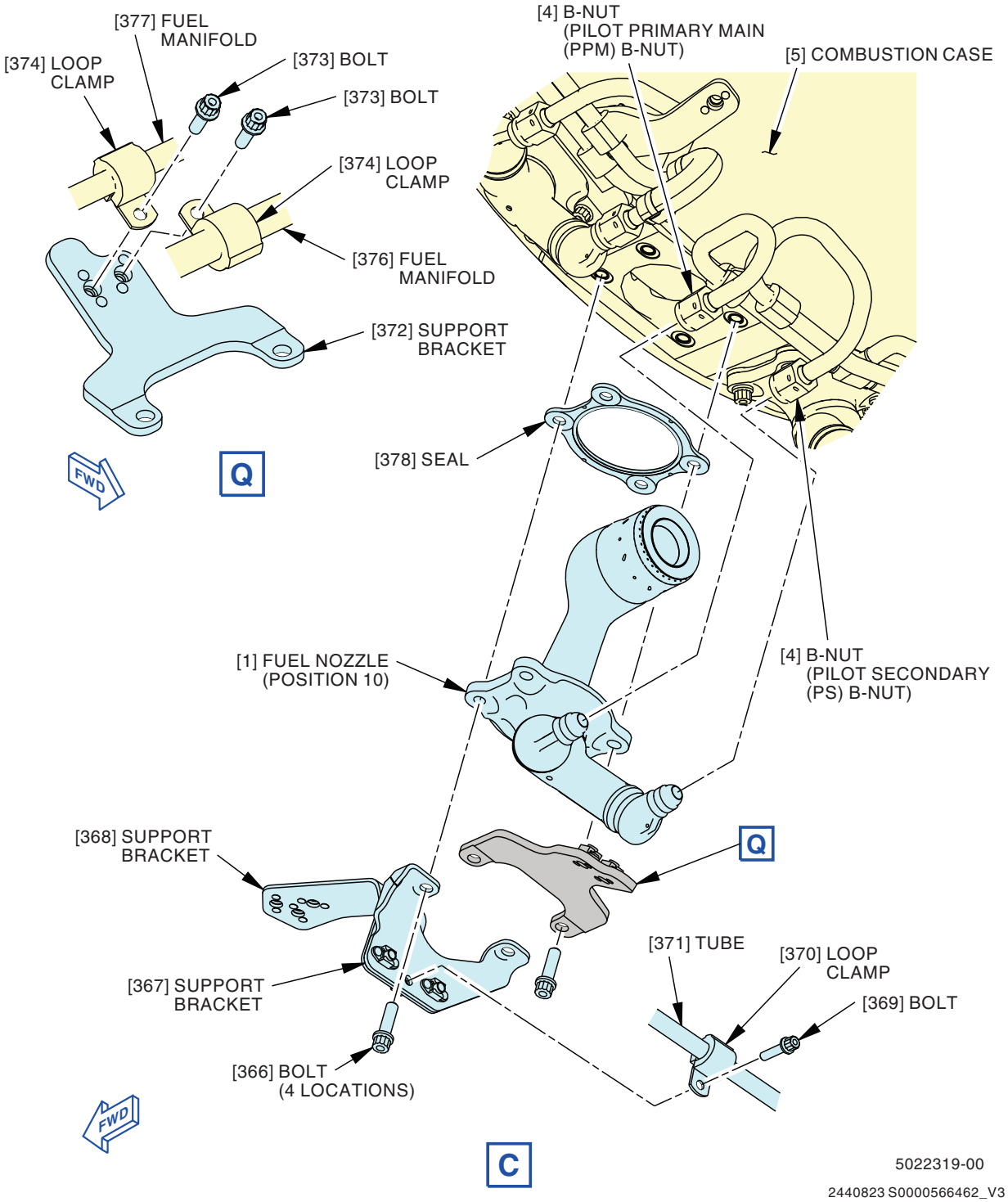
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 7 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



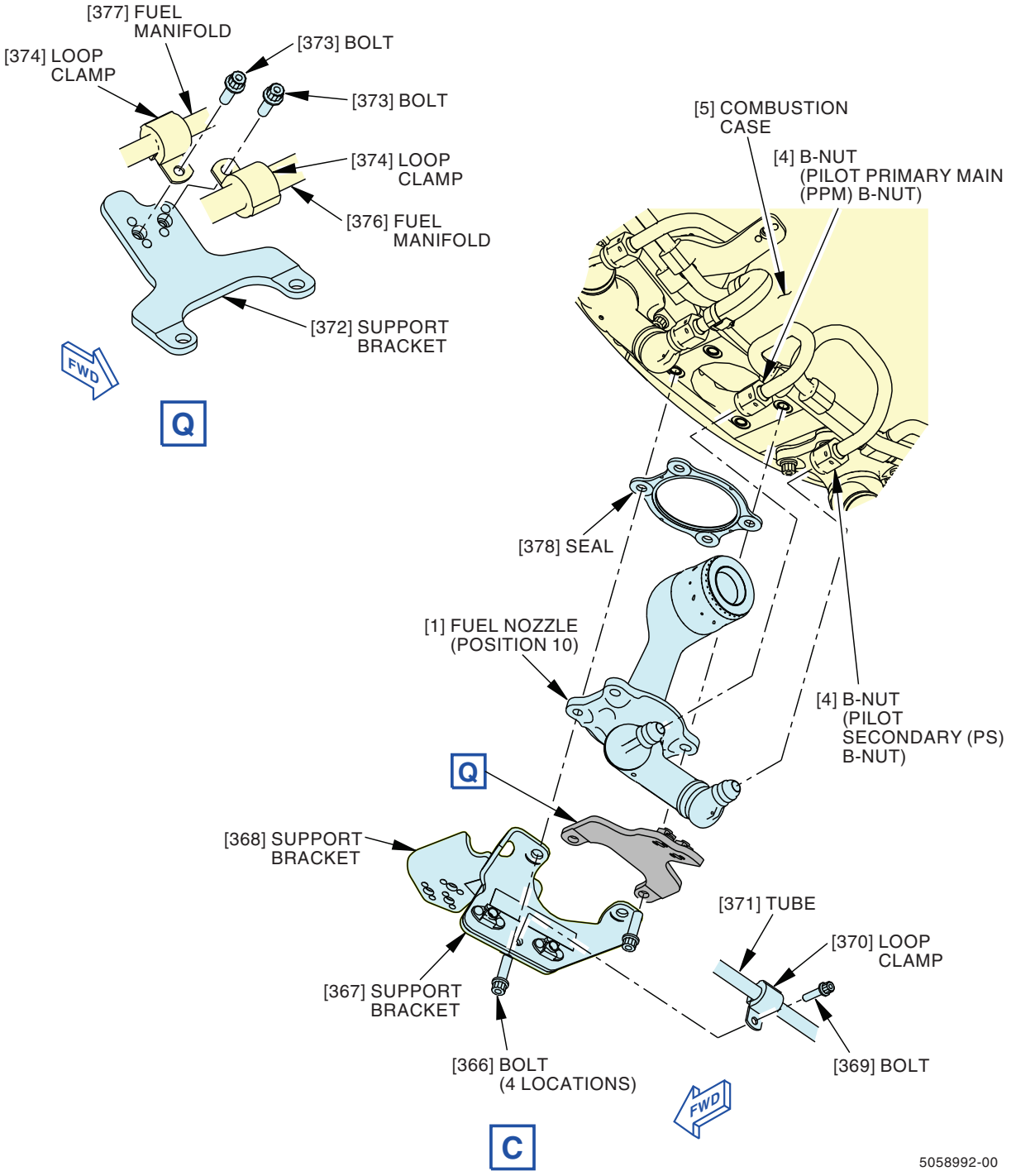
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 8 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



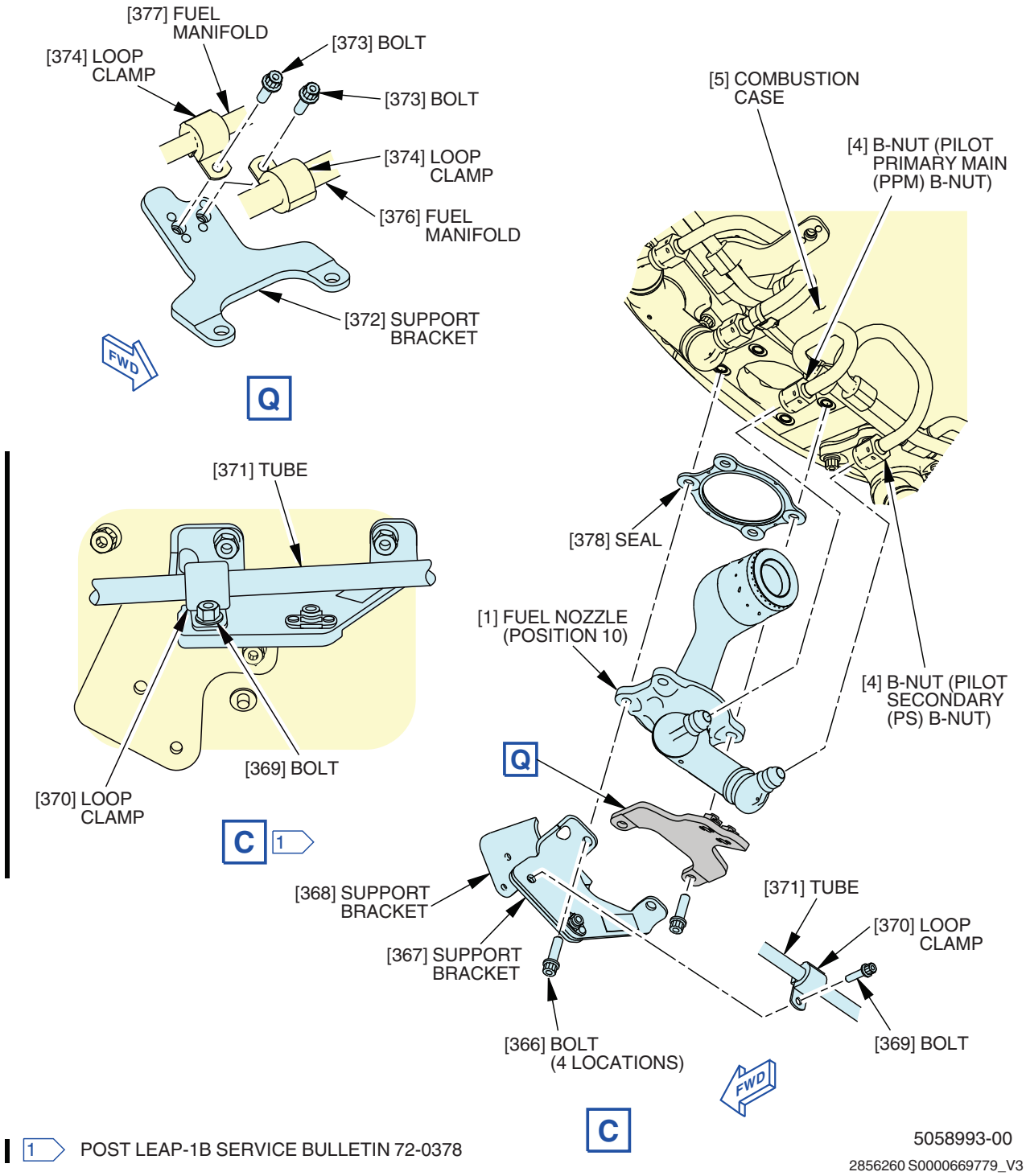
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 9 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



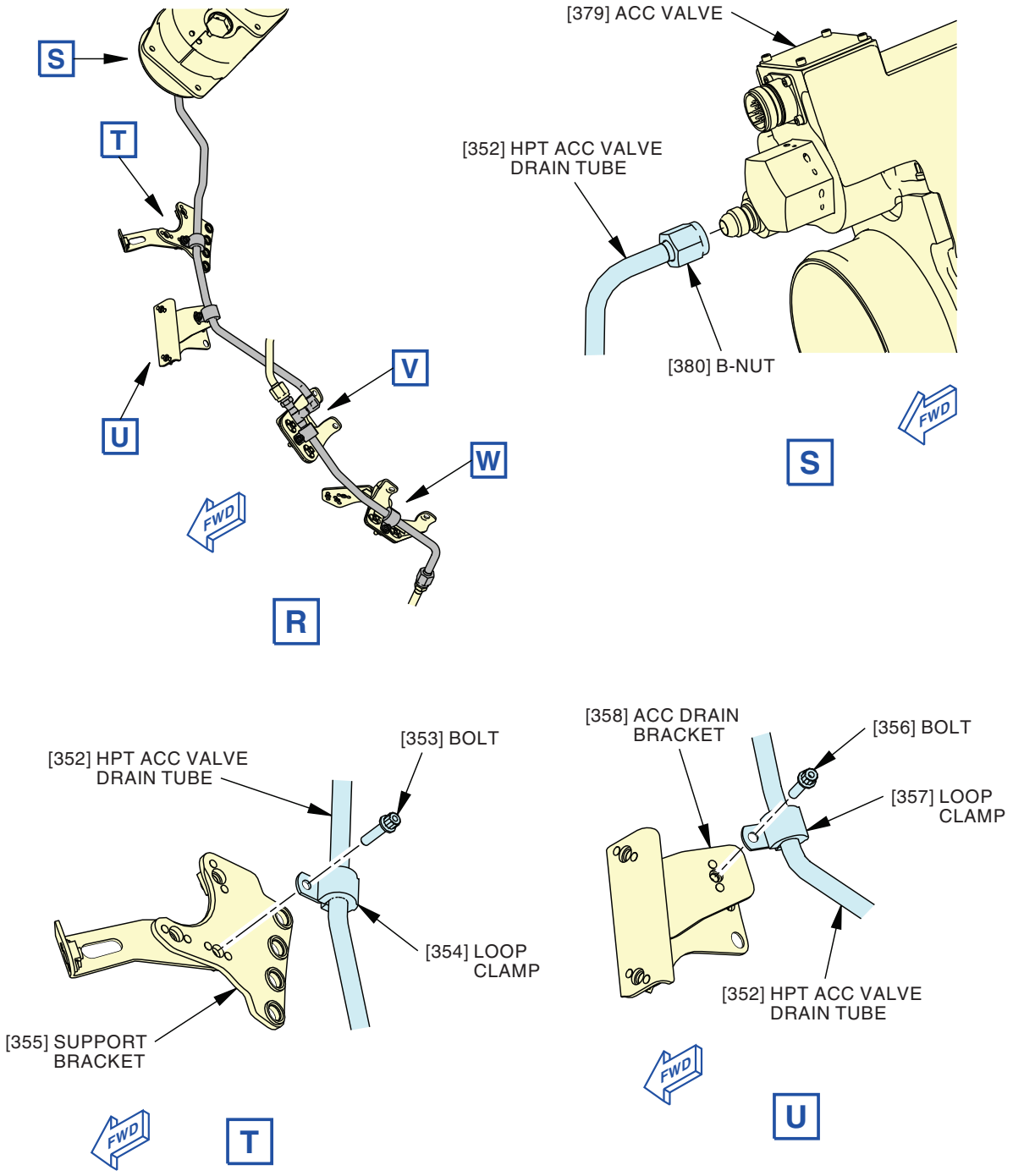
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 10 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



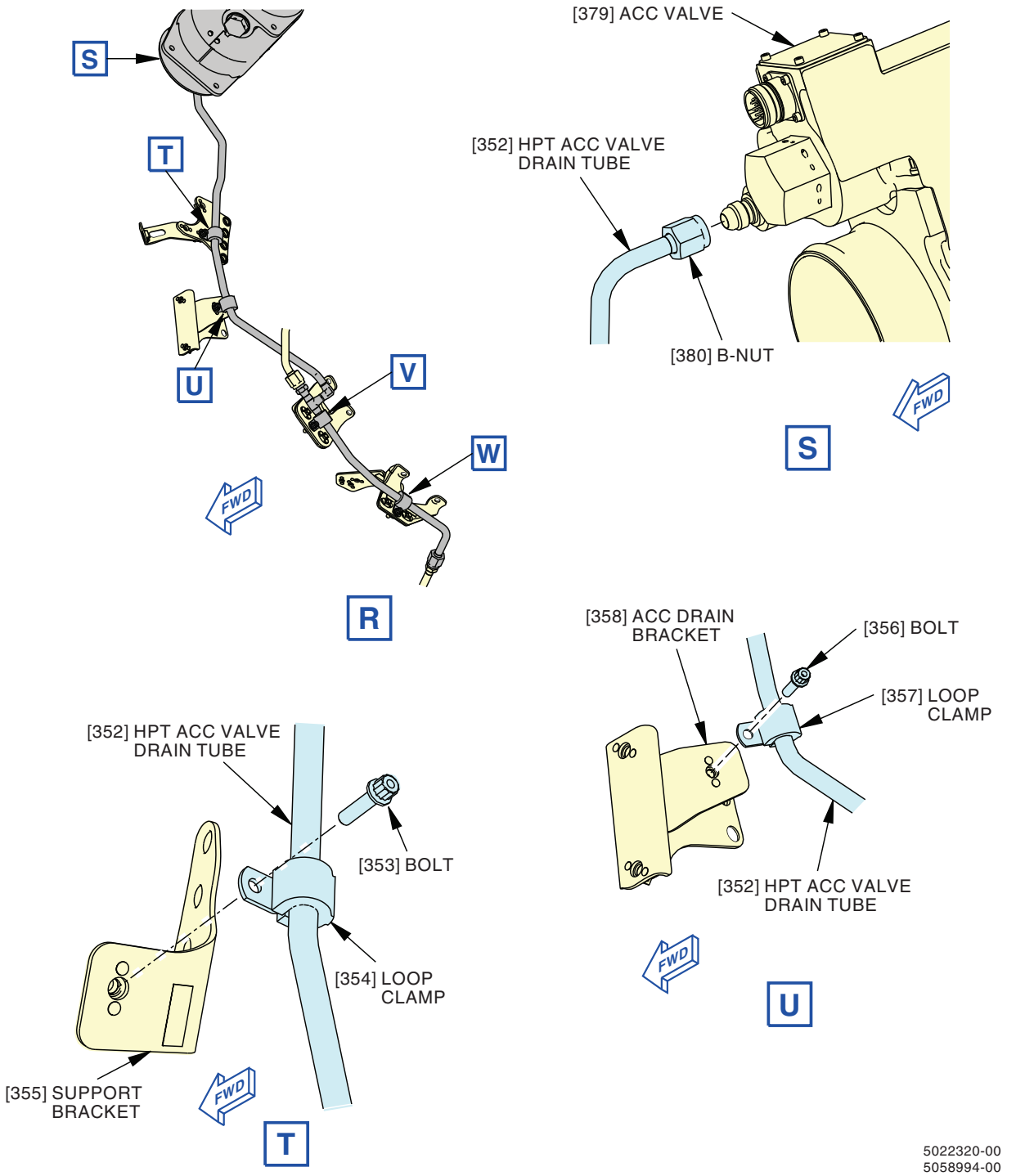
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 11 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 12 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

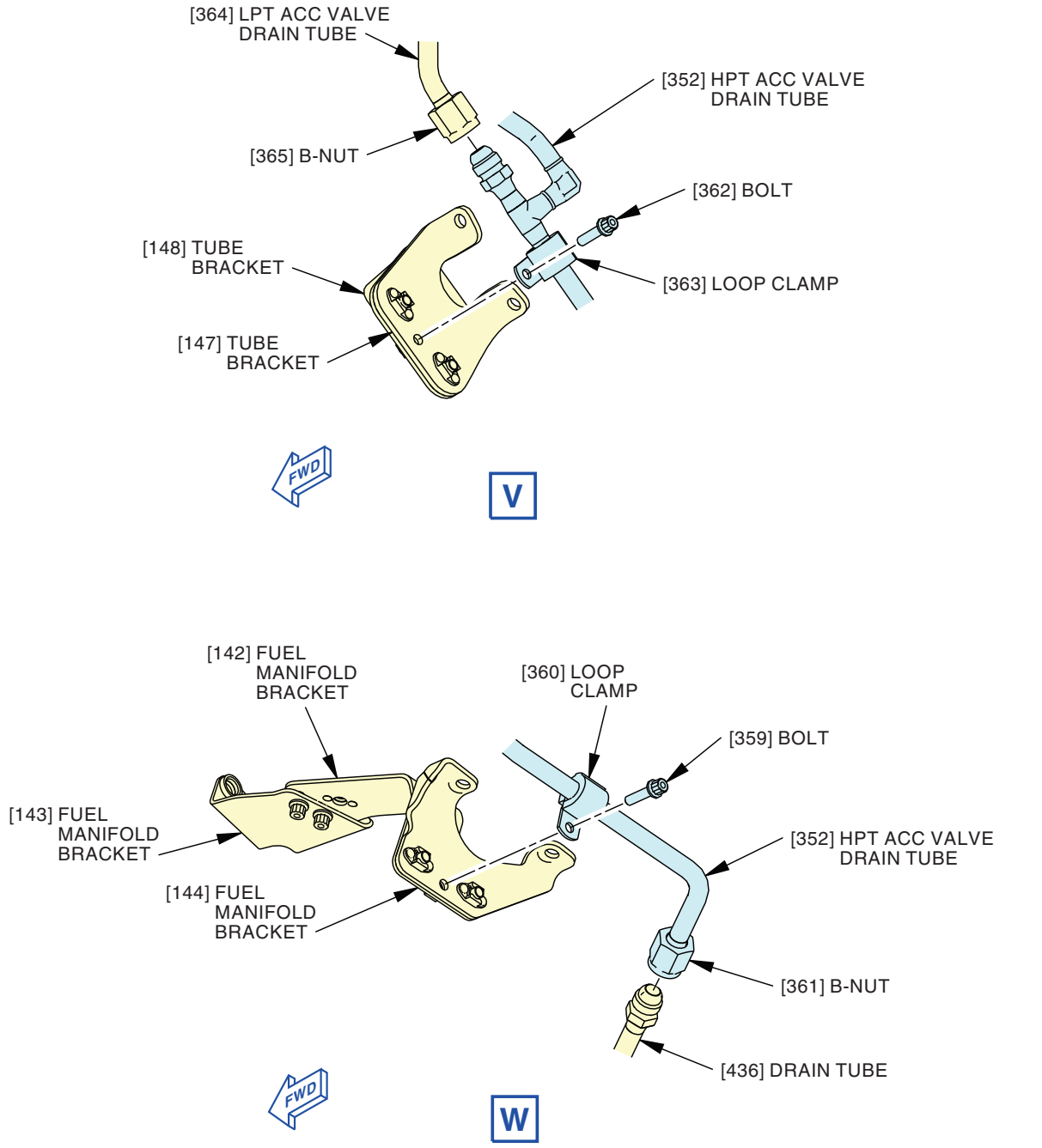


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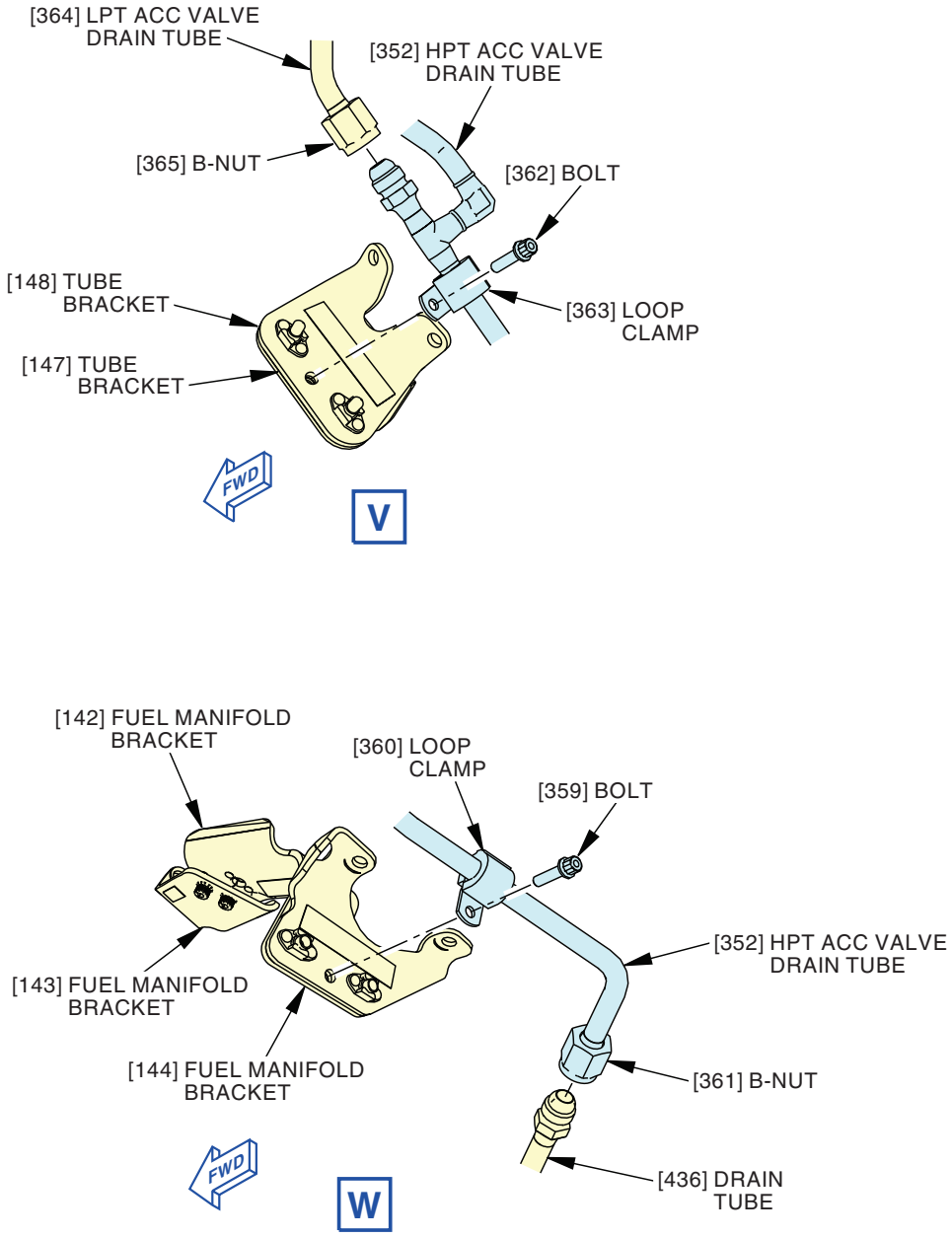
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 13 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 14 of 18)

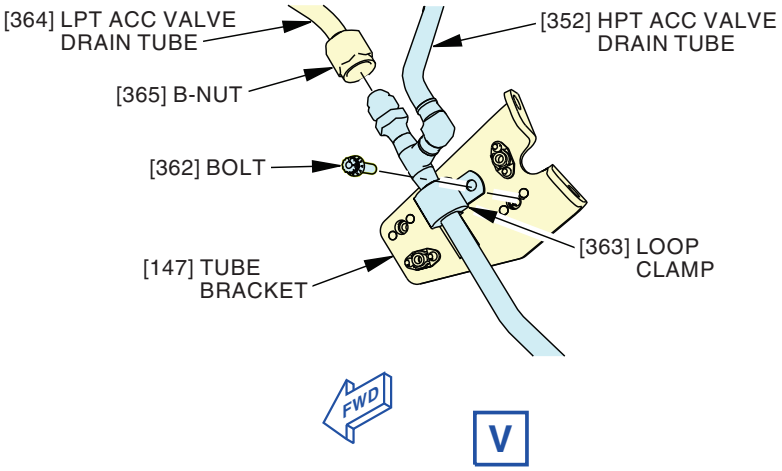
ECCN 9E991 BOEING PROPRIETARY - See title page for details

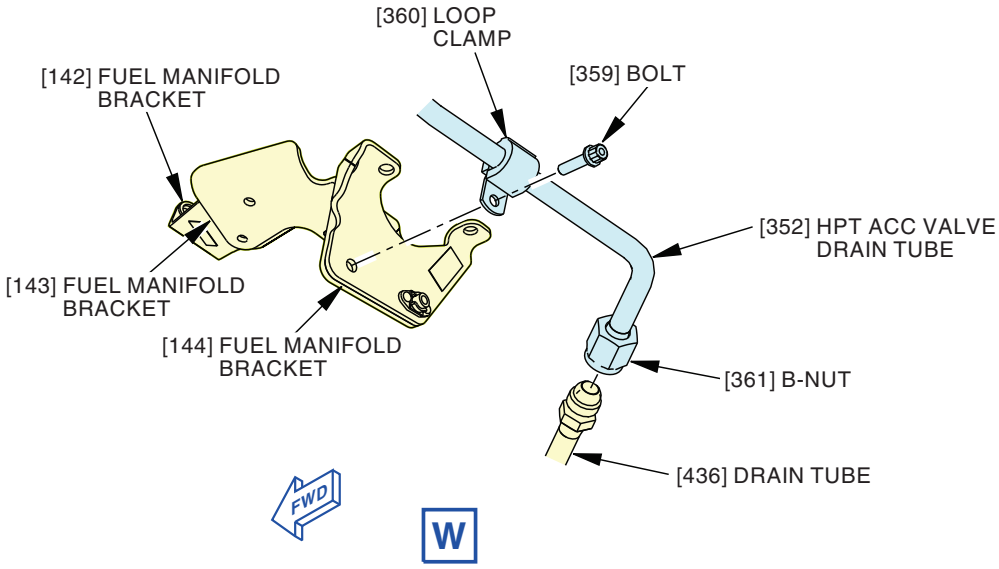


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Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 15 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

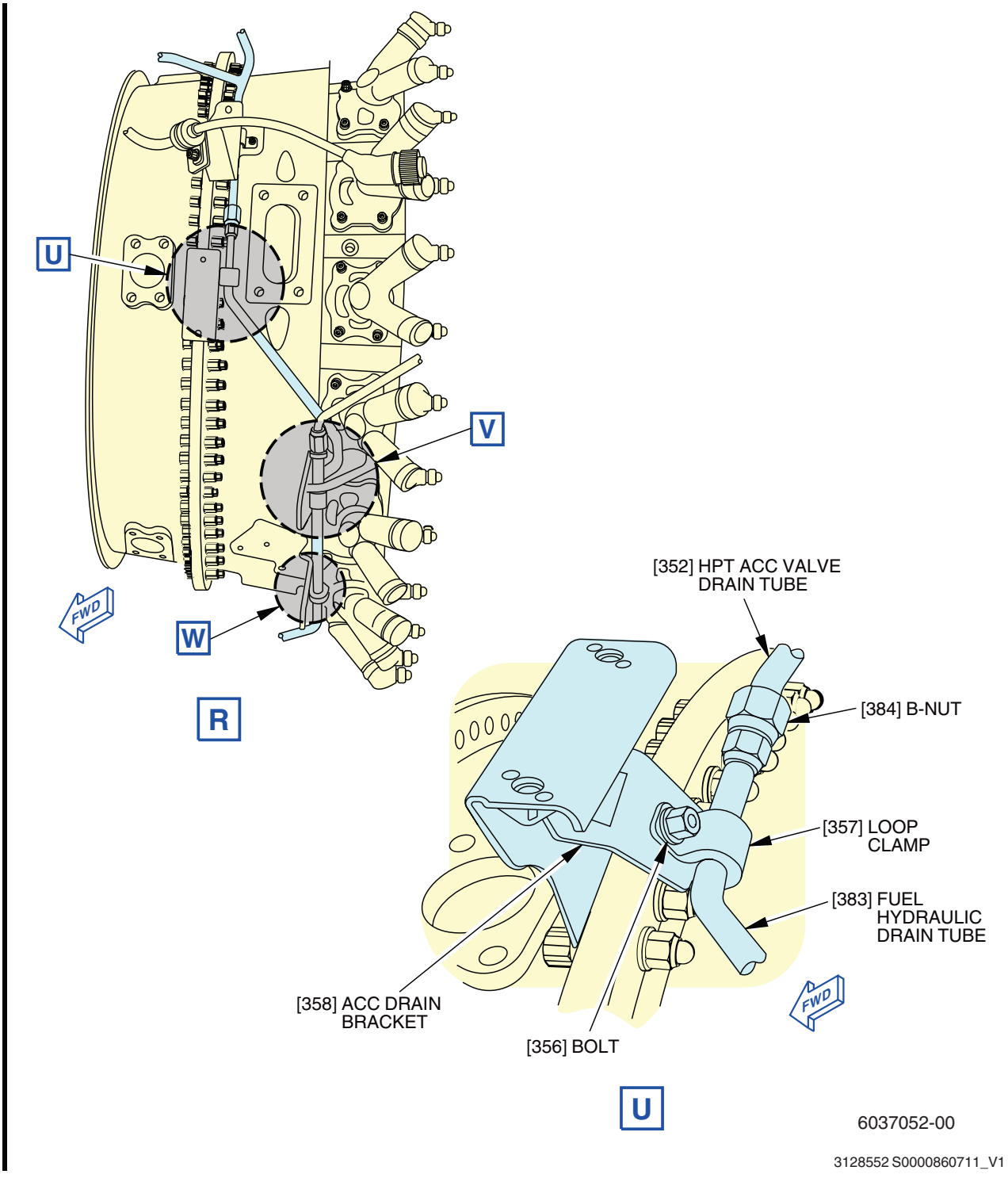




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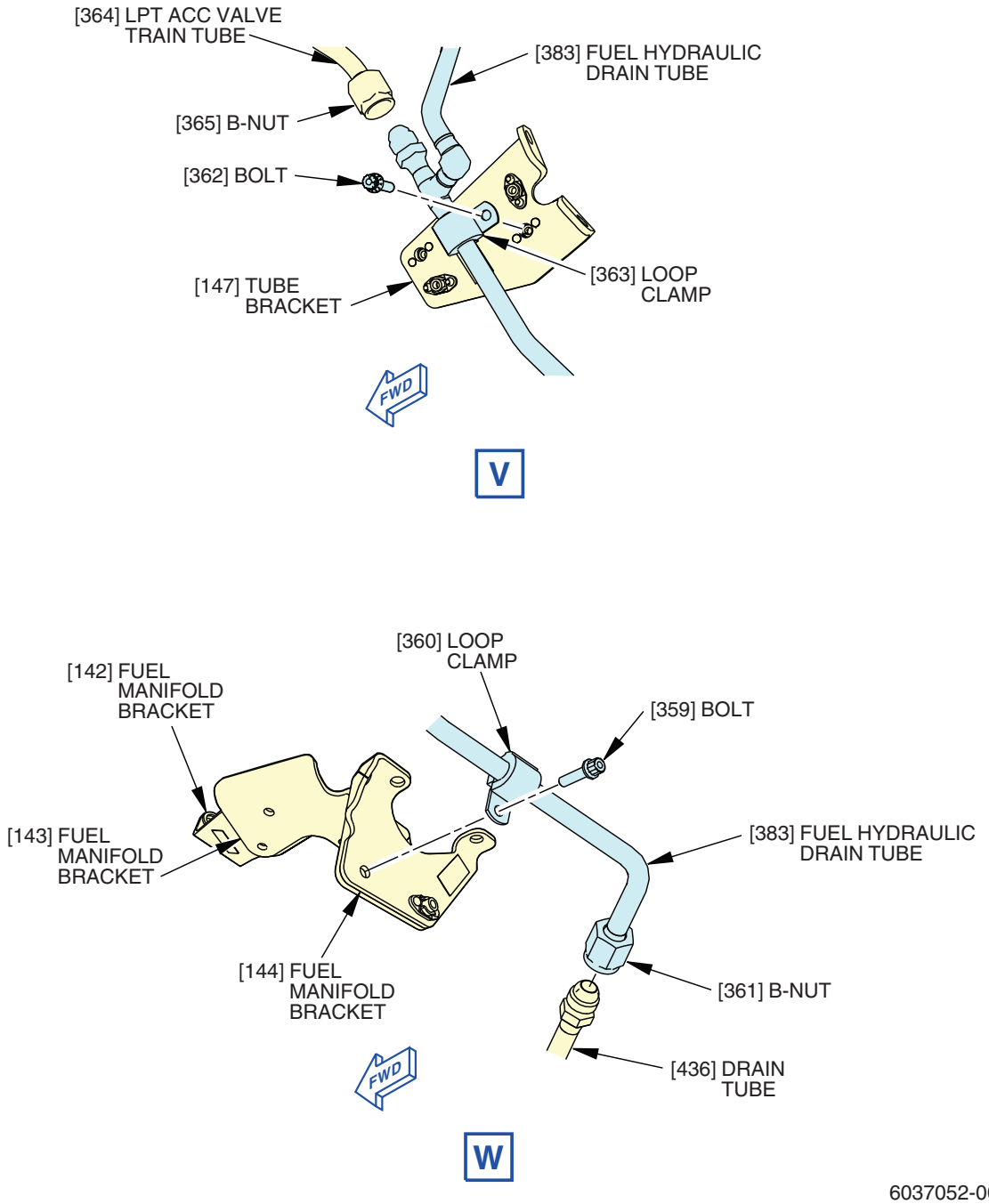
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 16 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



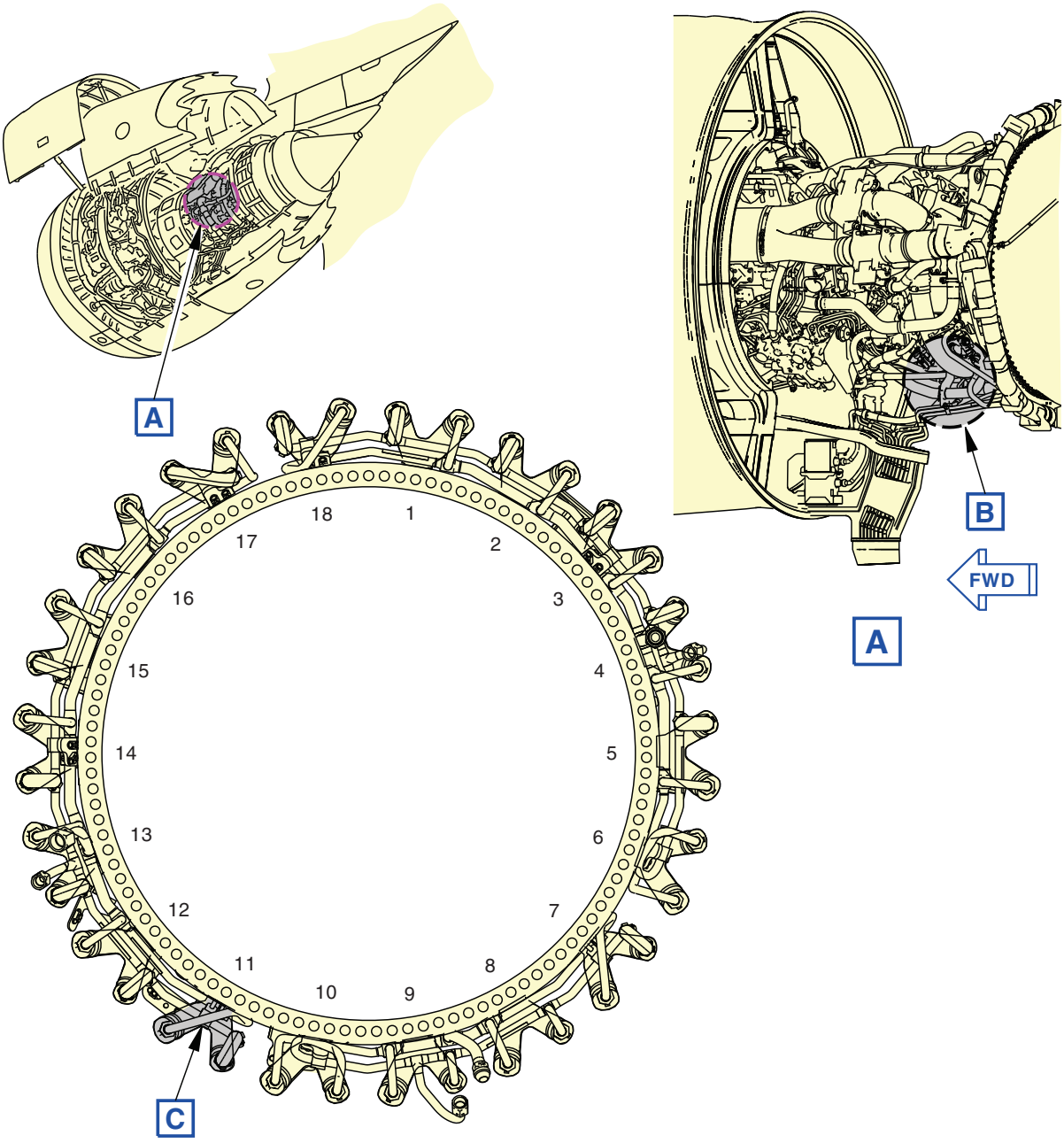
Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 17 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 18 of 18)

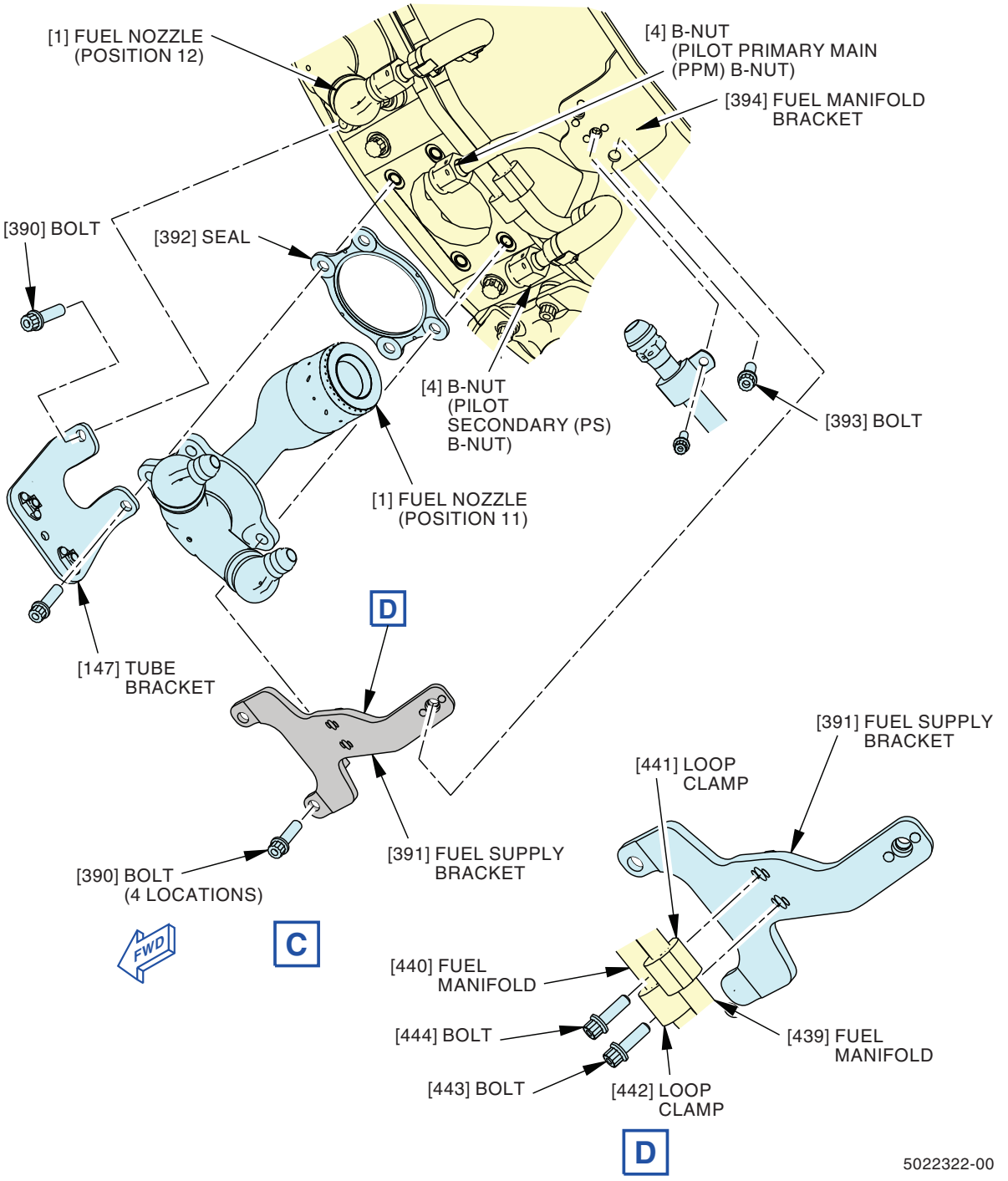
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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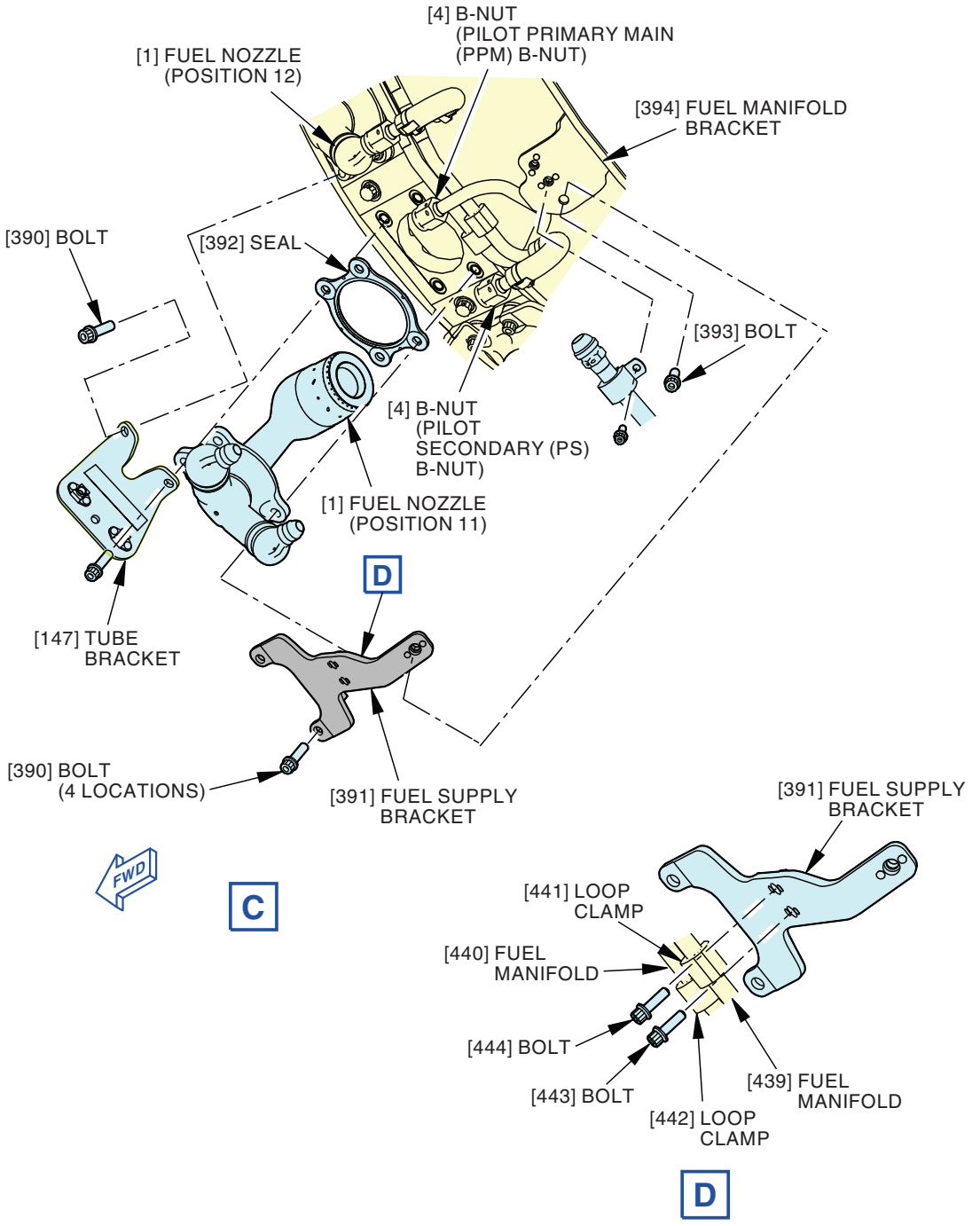
Fuel Nozzle (Position 11) Installation
Figure 412/73-11-05-990-898-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



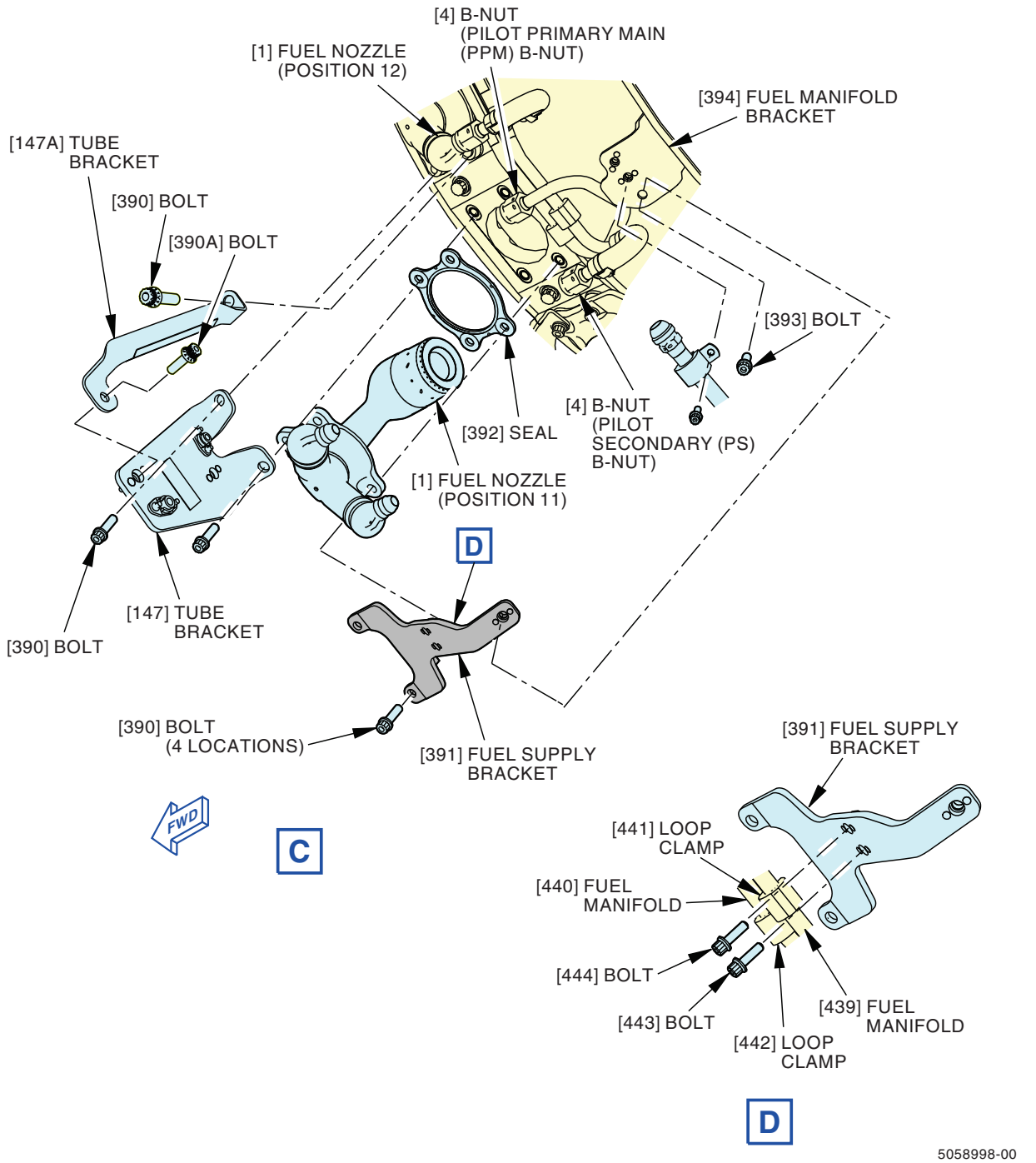
Fuel Nozzle (Position 11) Installation
Figure 412/73-11-05-990-898-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



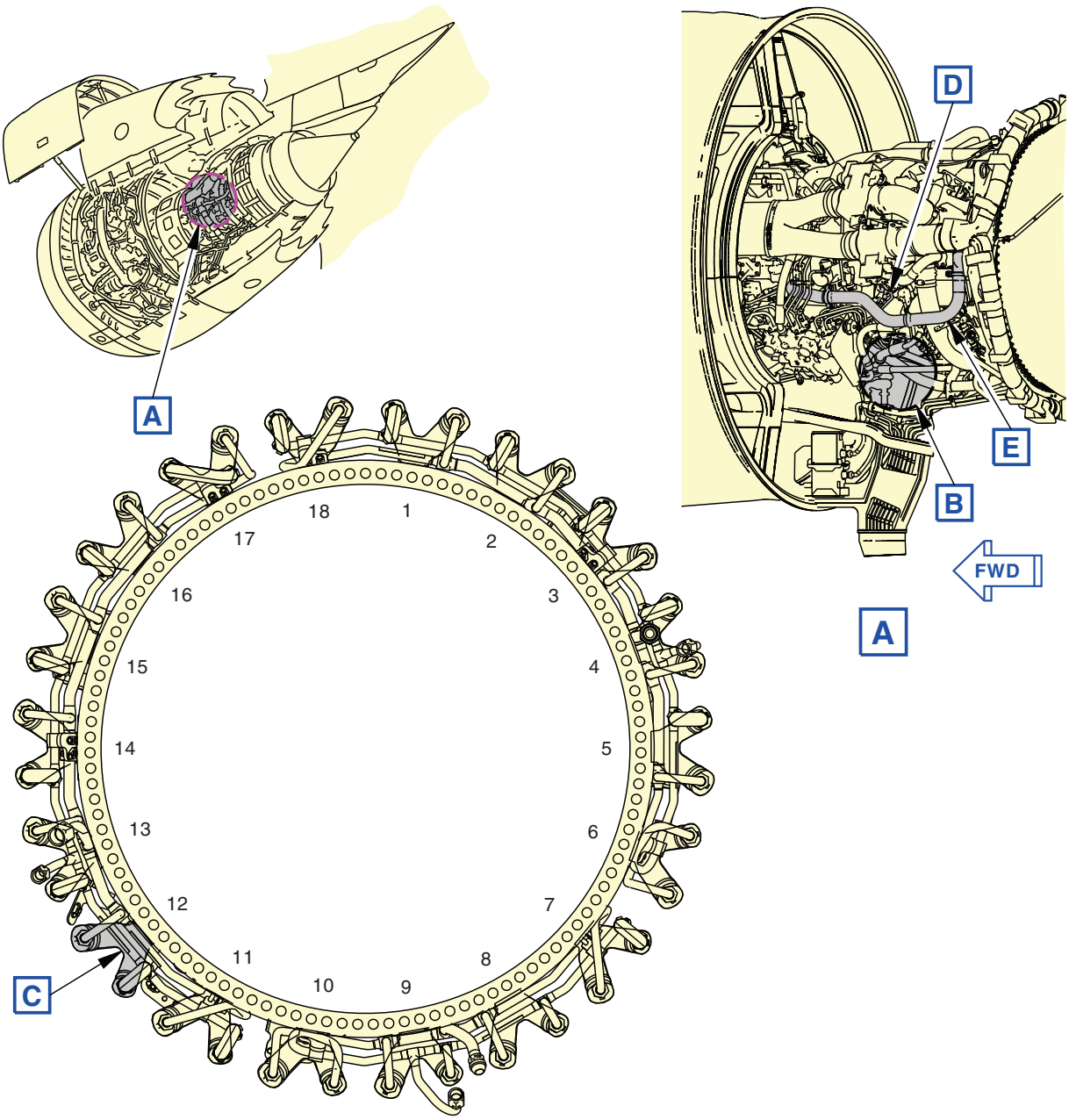
Fuel Nozzle (Position 11) Installation
Figure 412/73-11-05-990-898-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 11) Installation
Figure 412/73-11-05-990-898-G00 (Sheet 4 of 4)

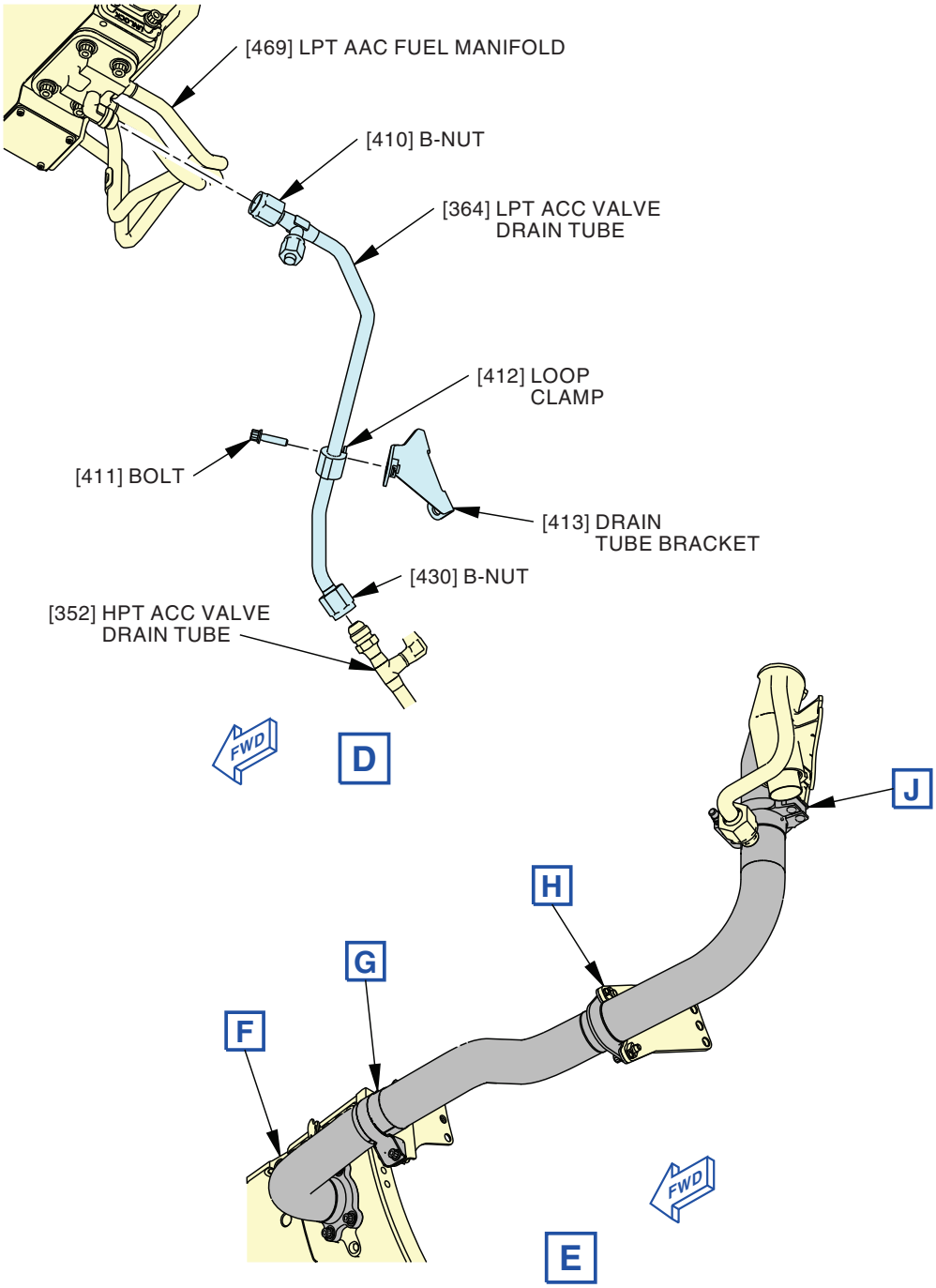
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 1 of 8)

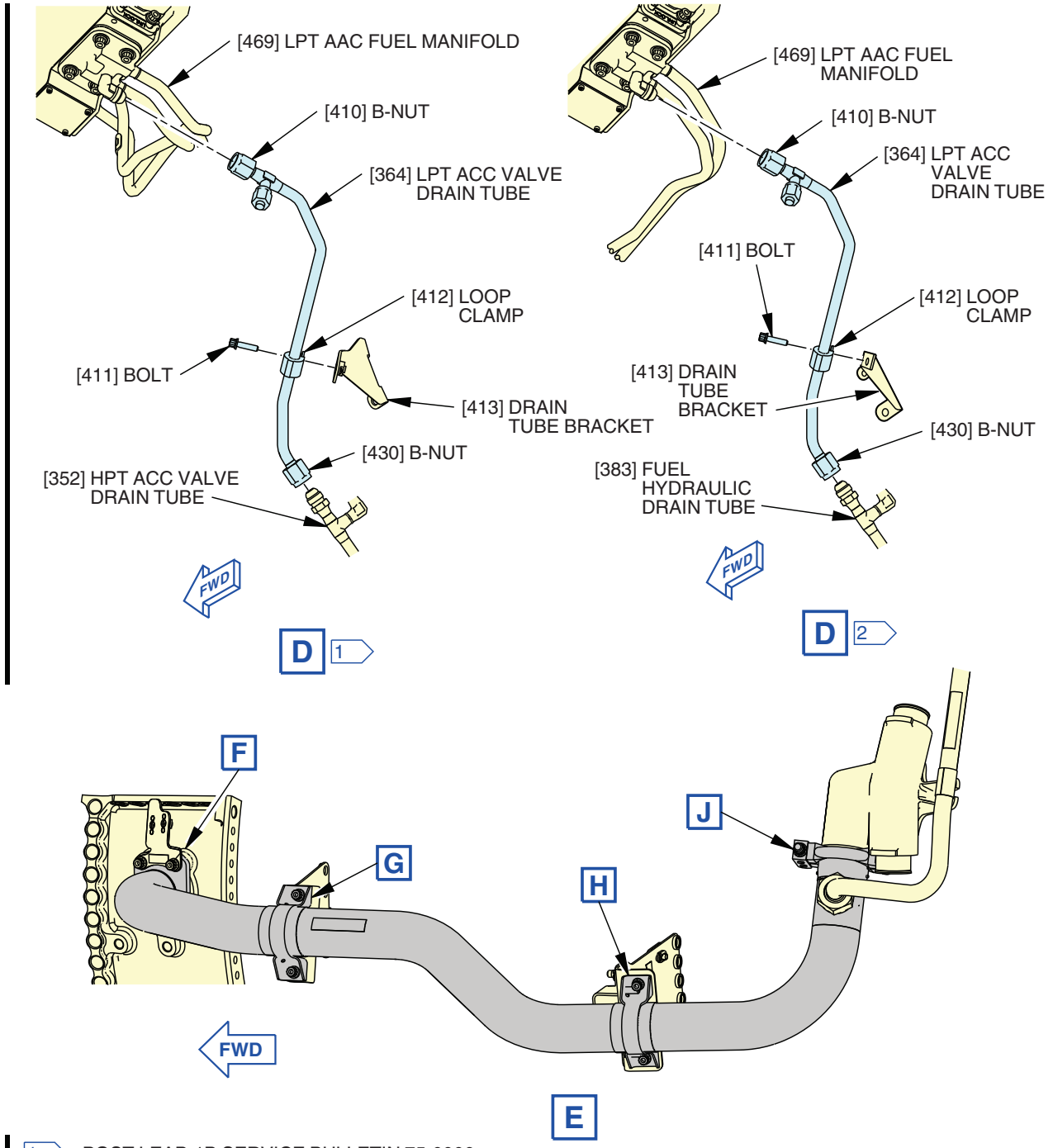
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022331-00
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Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

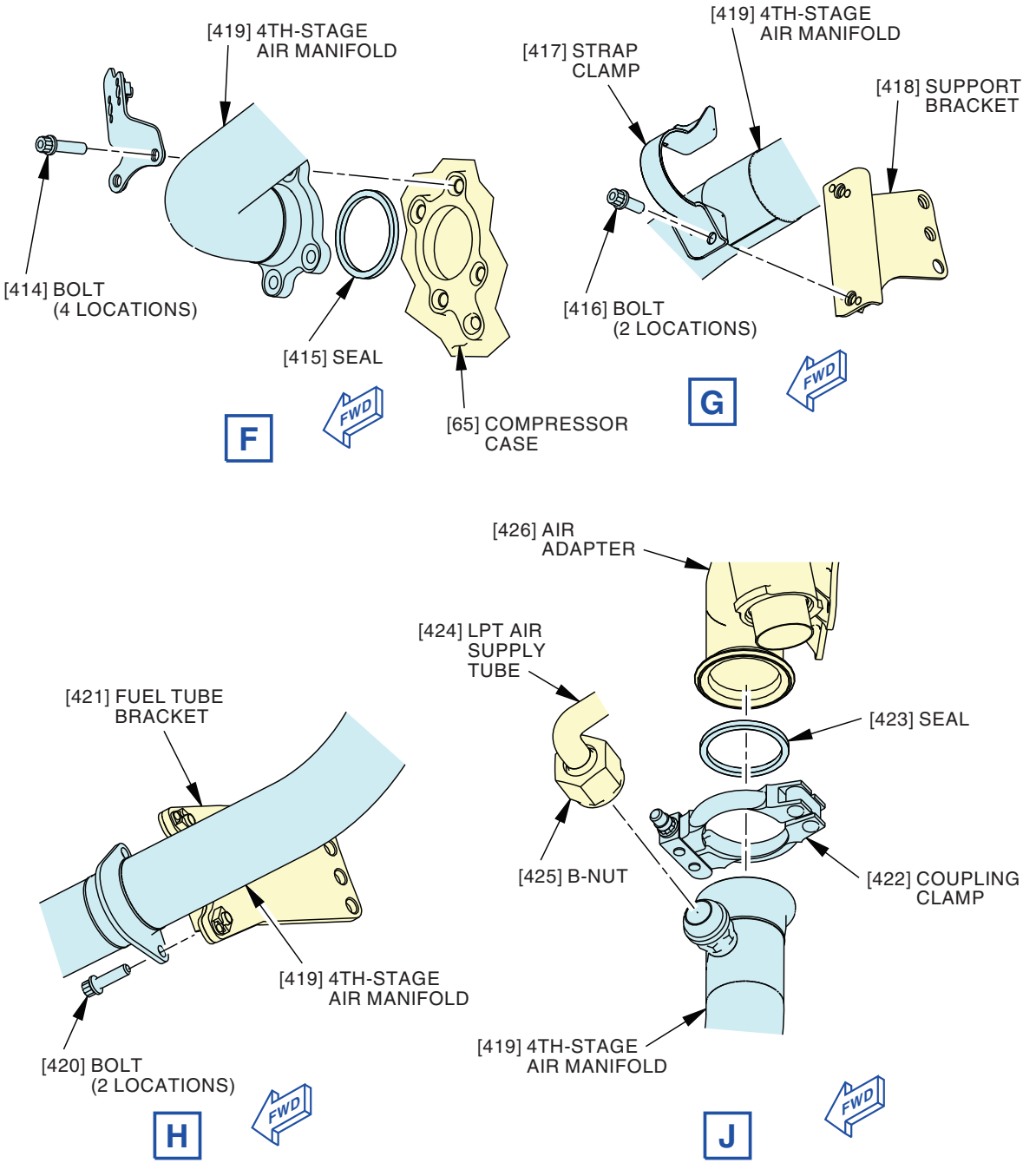


1 POST LEAP-1B SERVICE BULLETIN 75-0006
2 POST LEAP-1B SERVICE BULLETIN 72-0378

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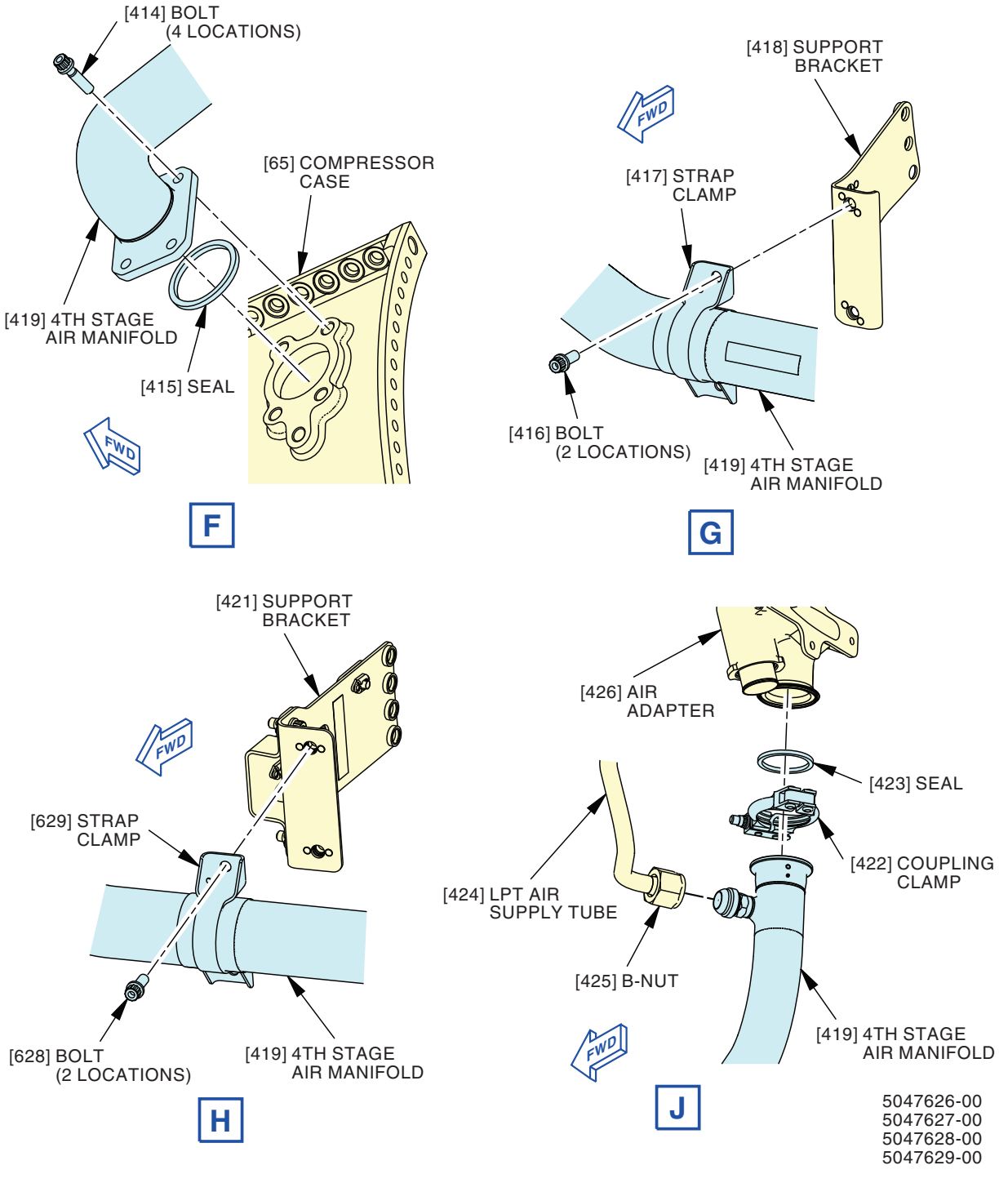
Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



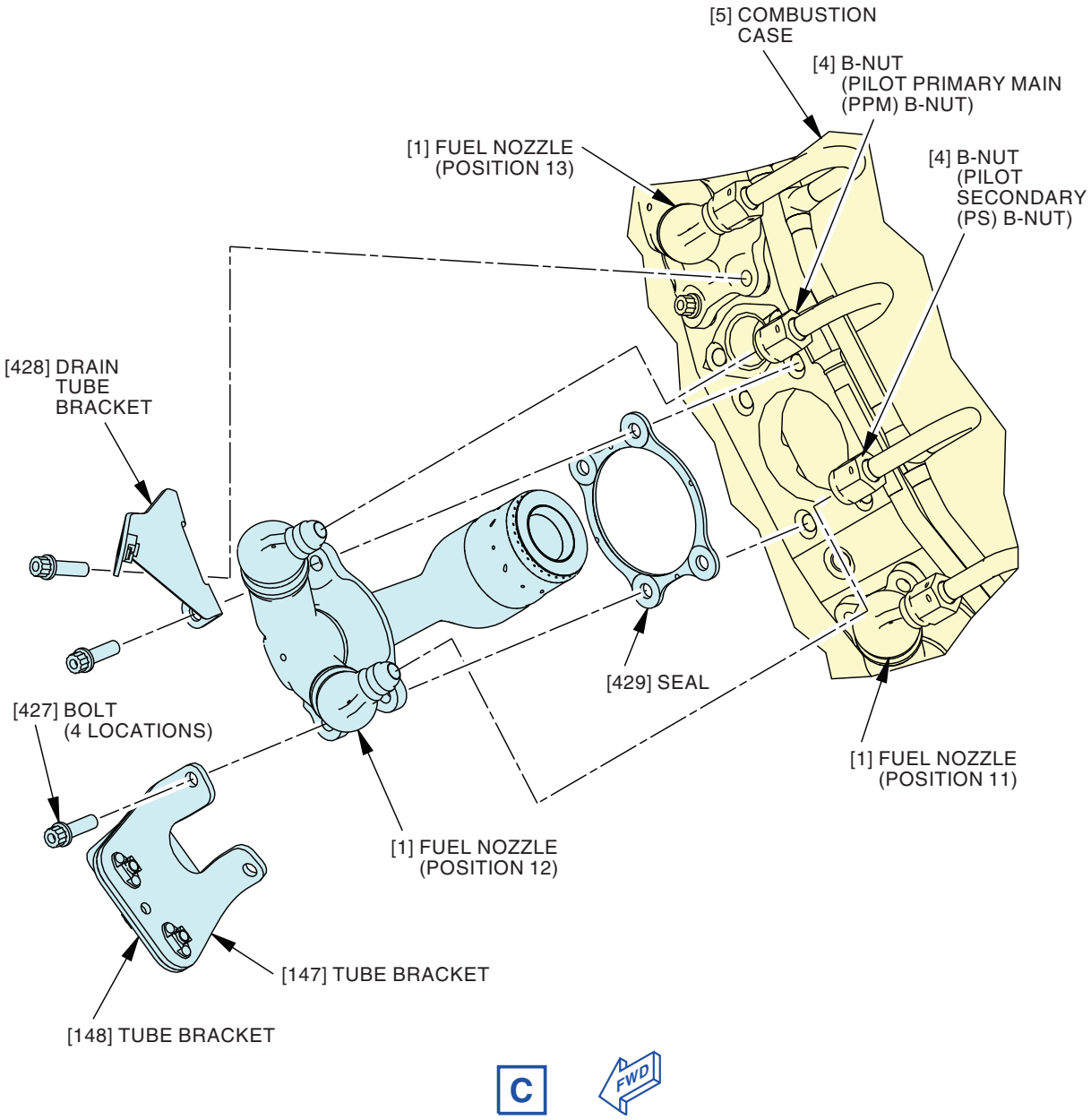
Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 5 of 8)

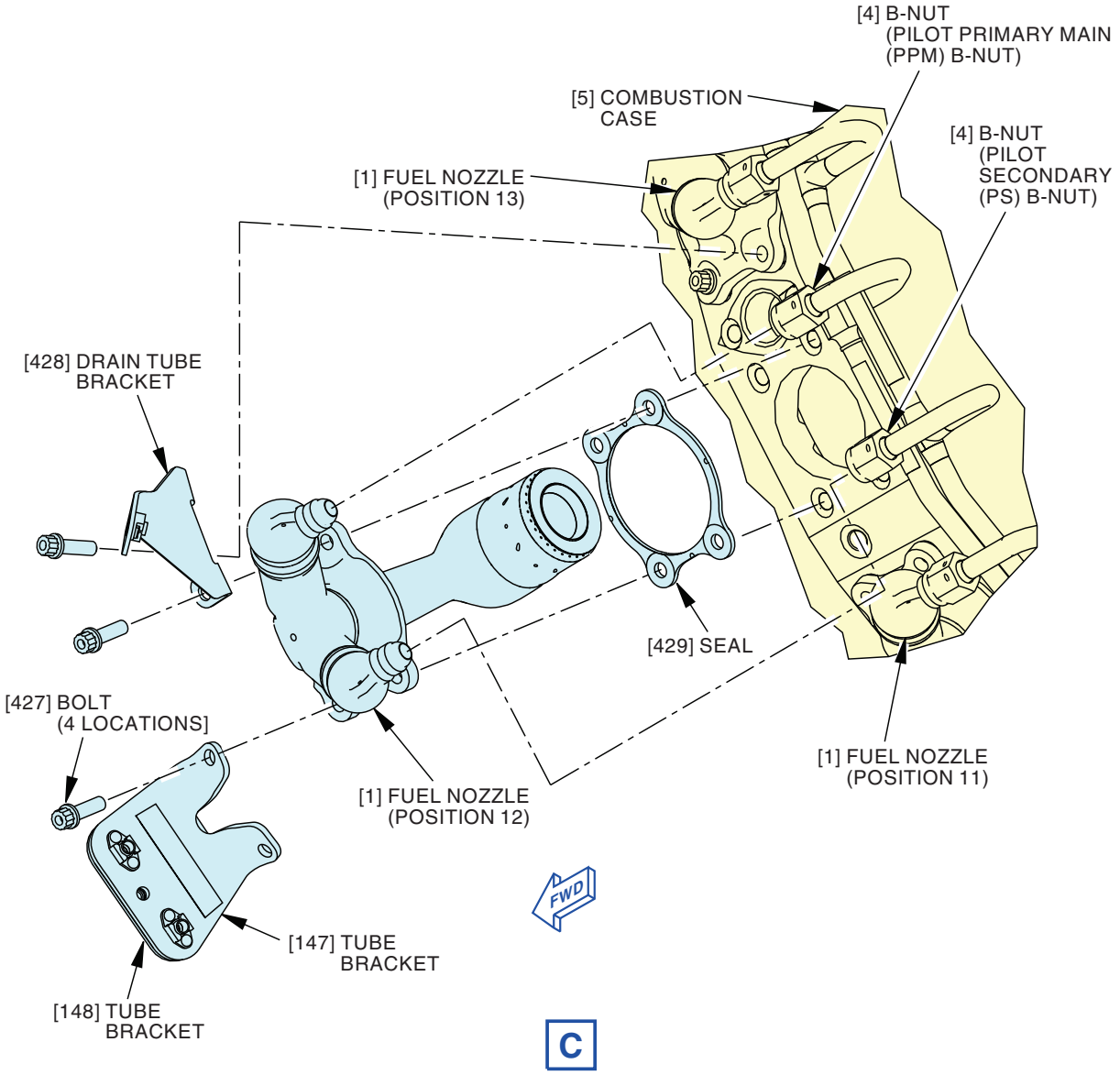
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 6 of 8)

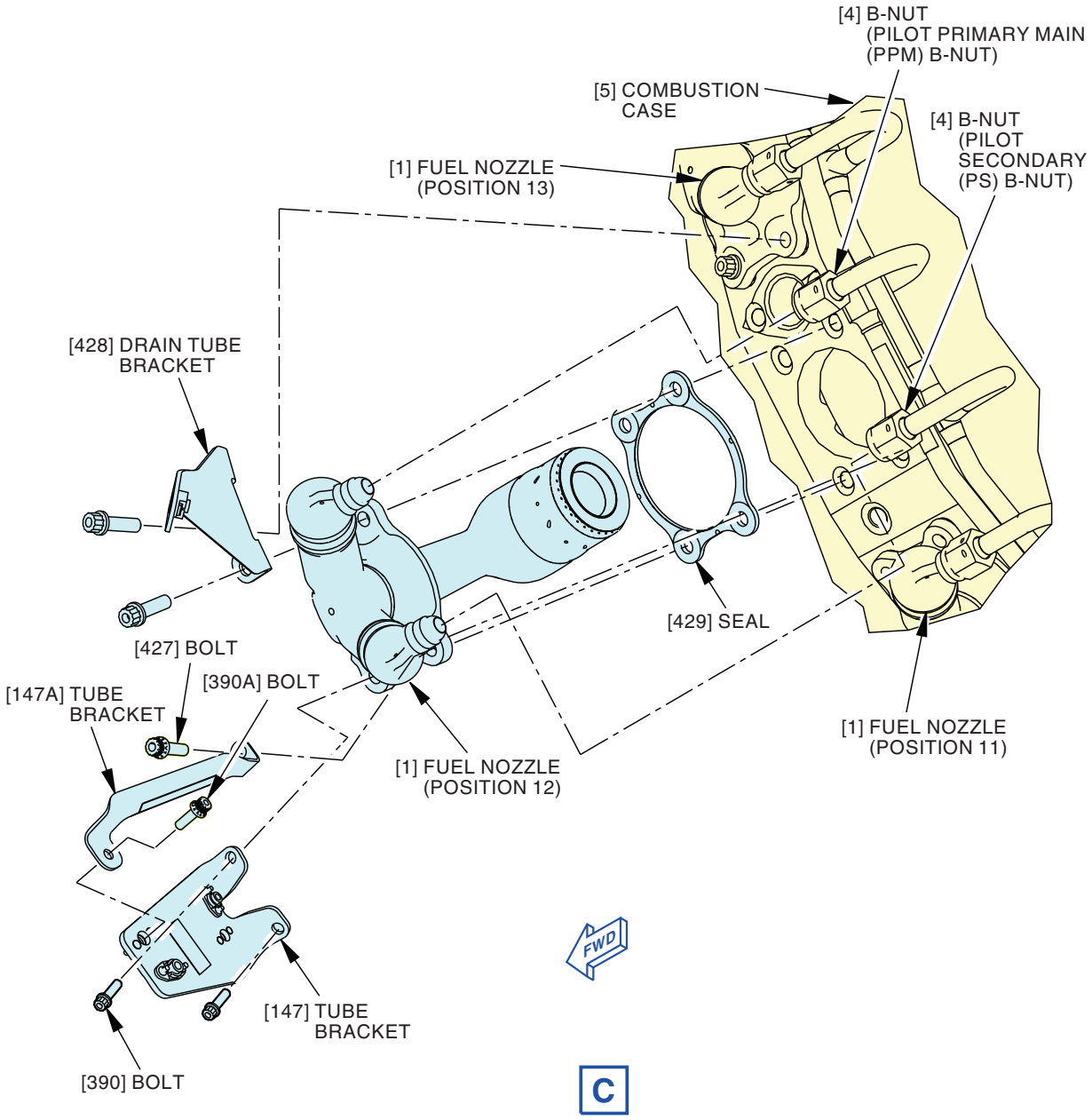
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 7 of 8)

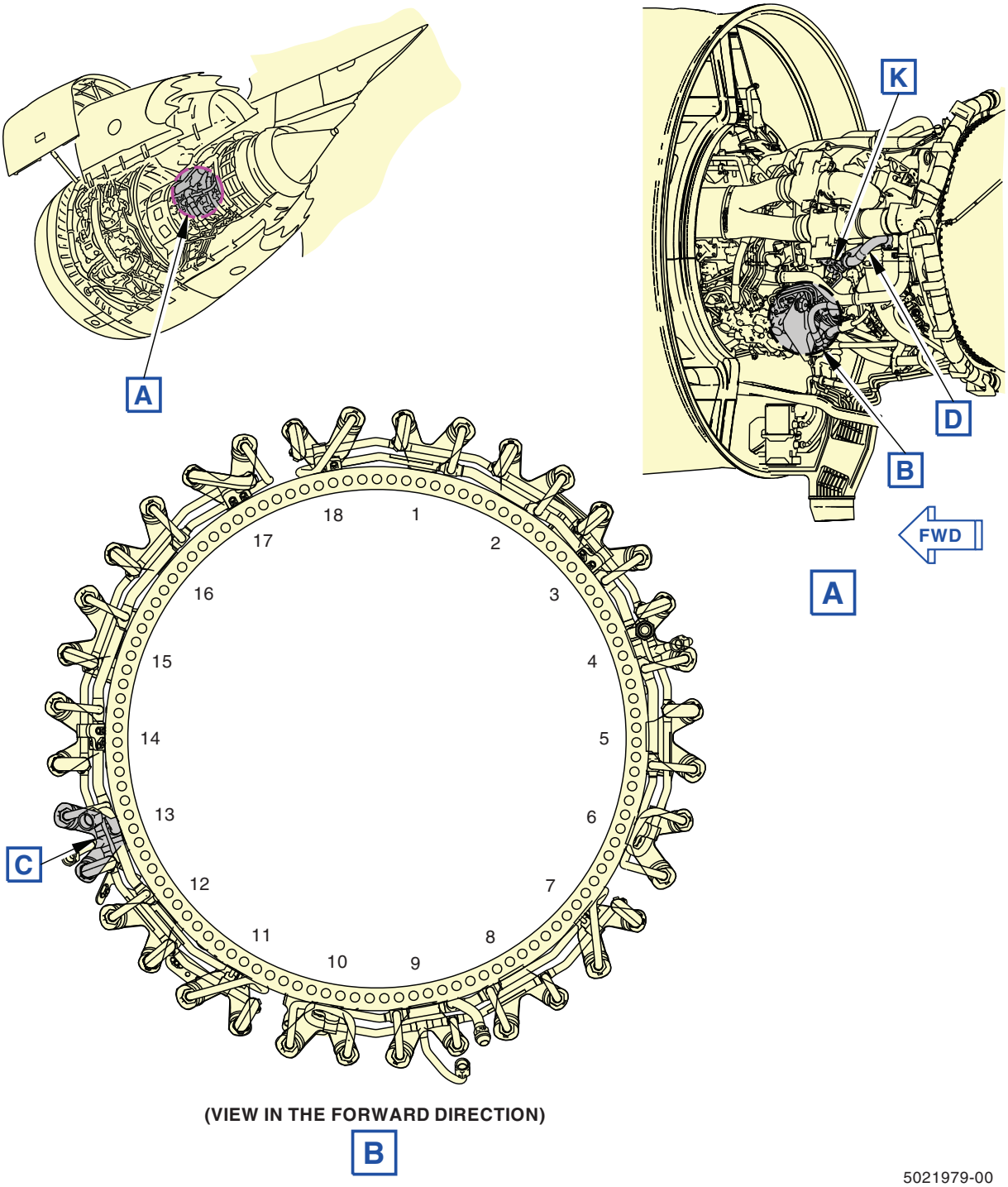
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 12) Installation
Figure 413/73-11-05-990-899-G00 (Sheet 8 of 8)

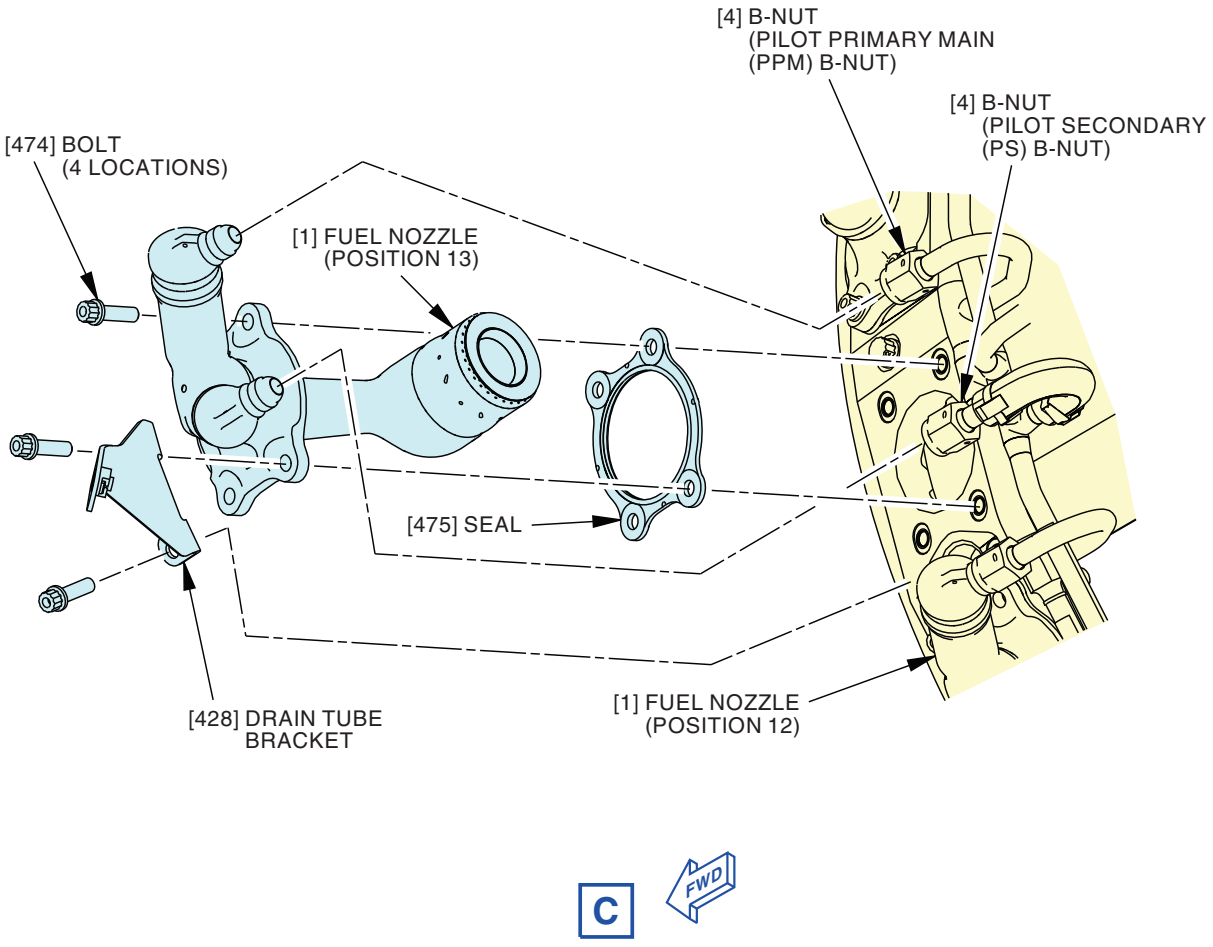
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 1 of 9)

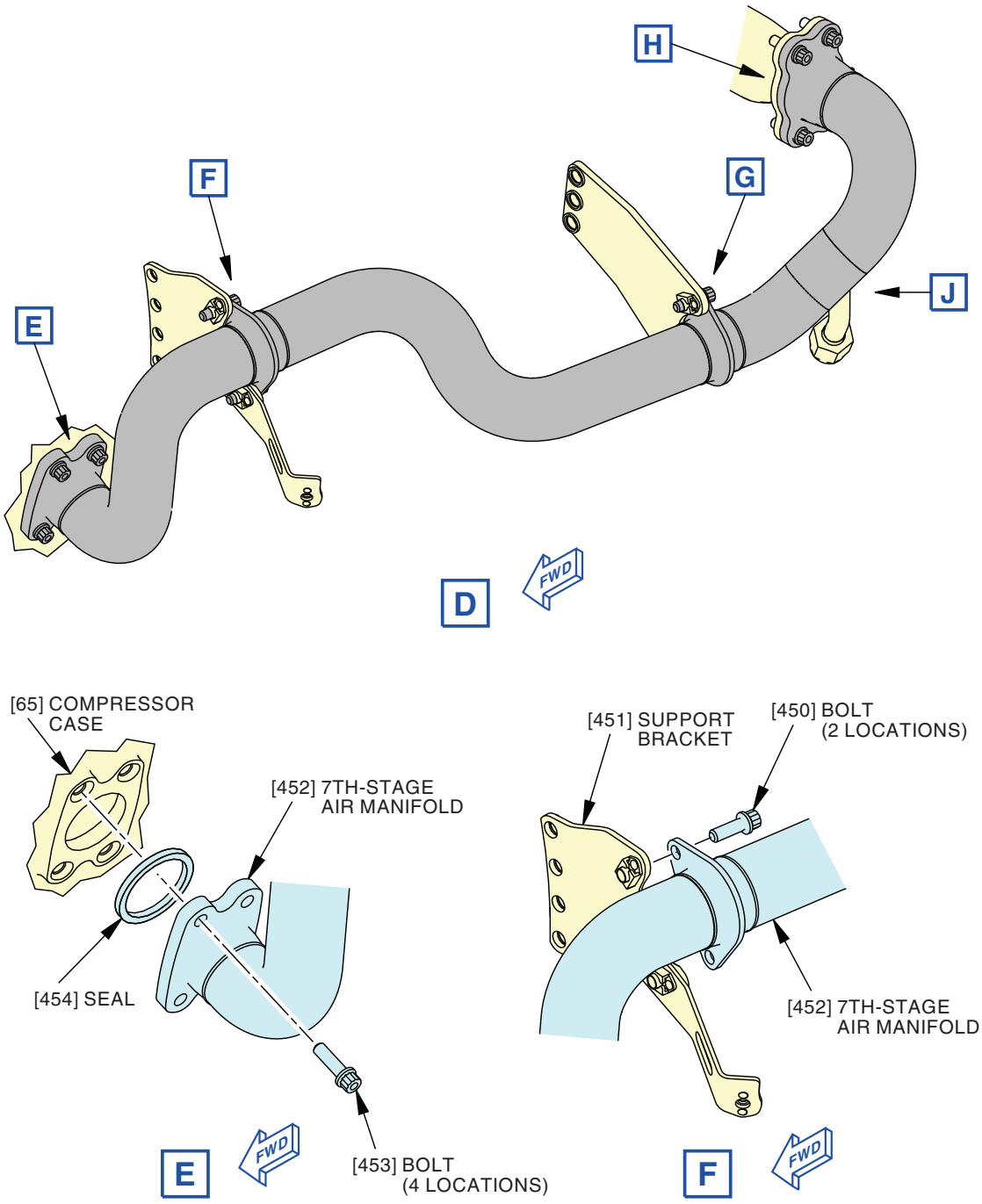
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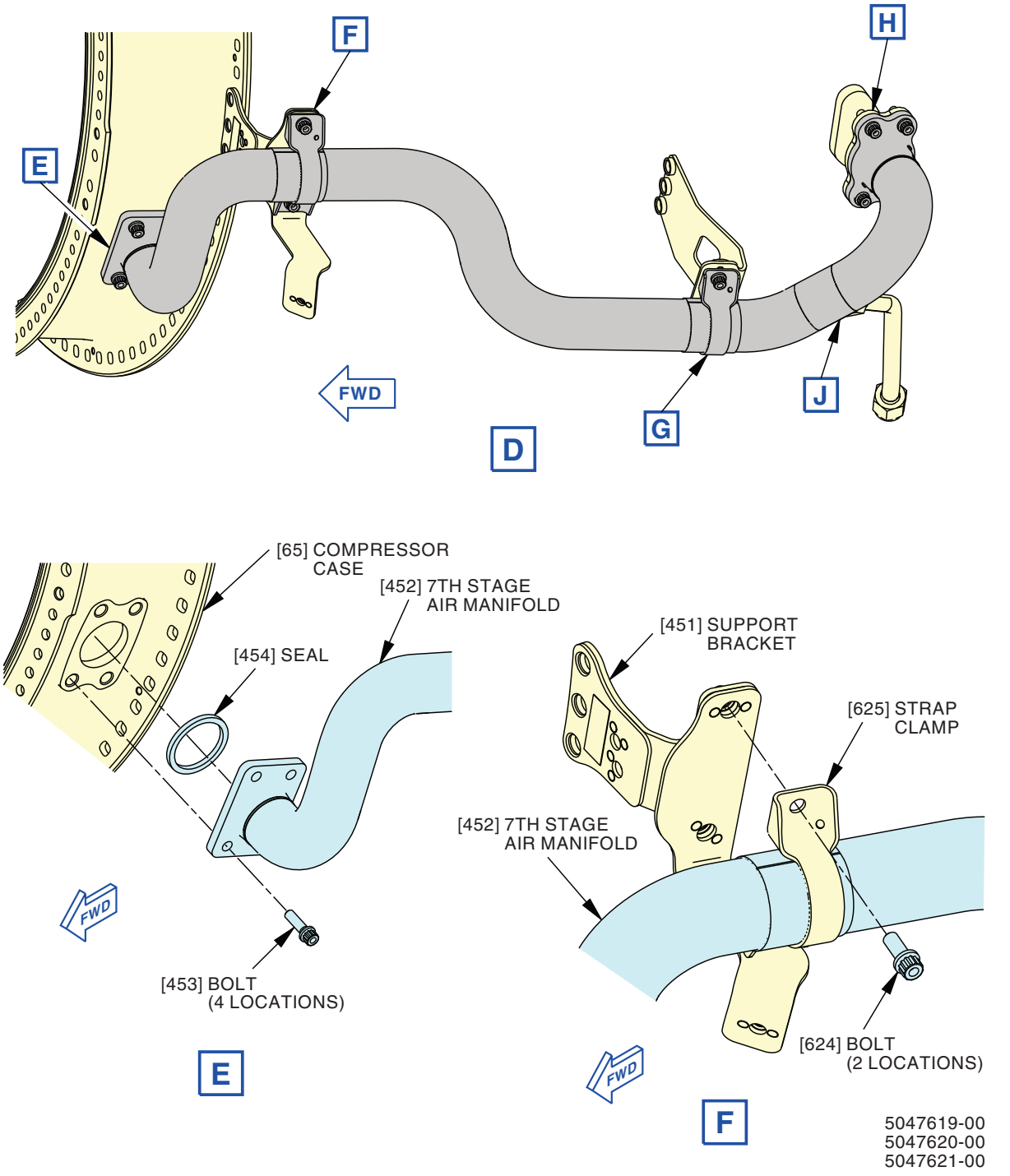
Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



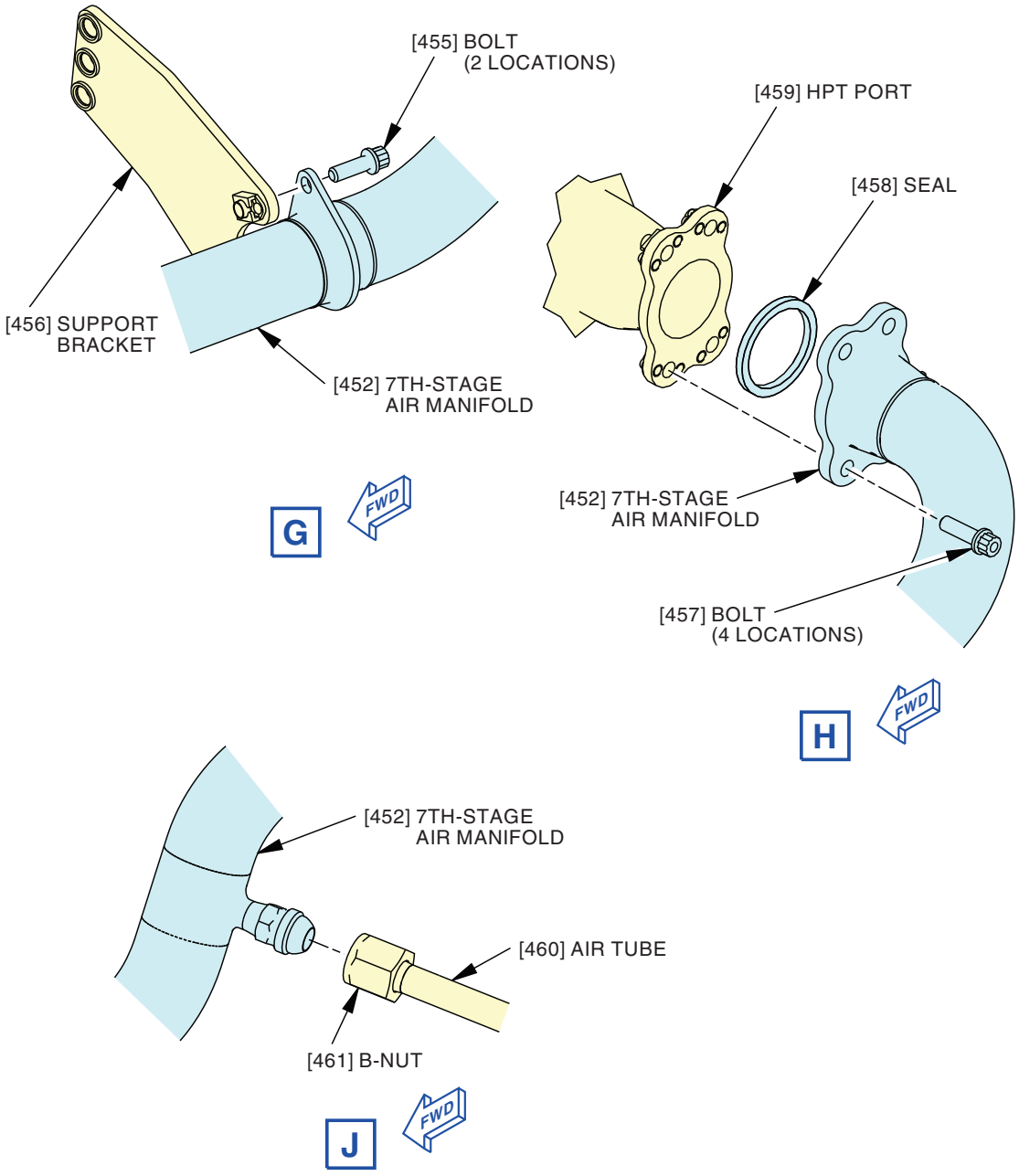
Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

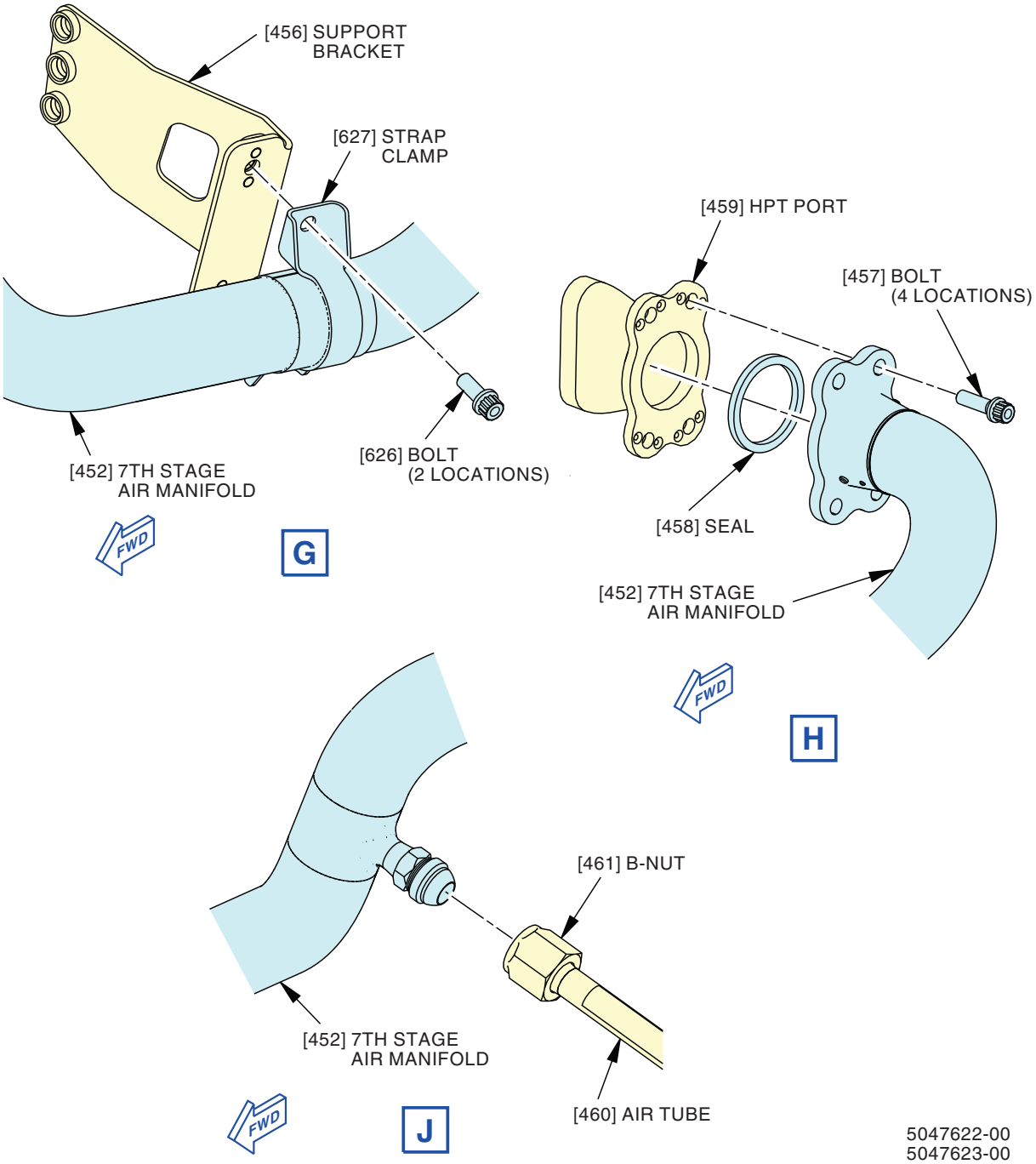


5022335-00

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Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

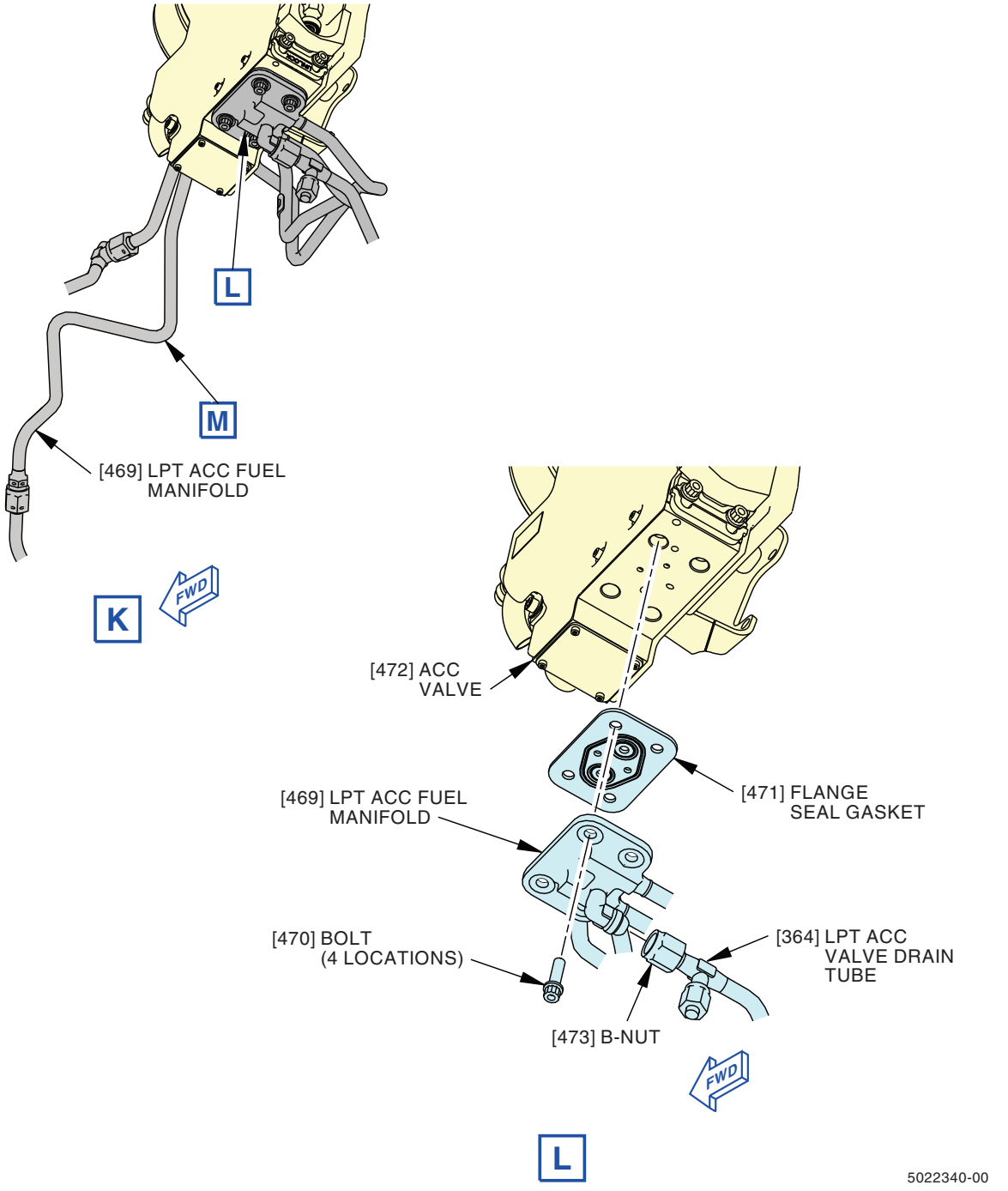


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5047624-00

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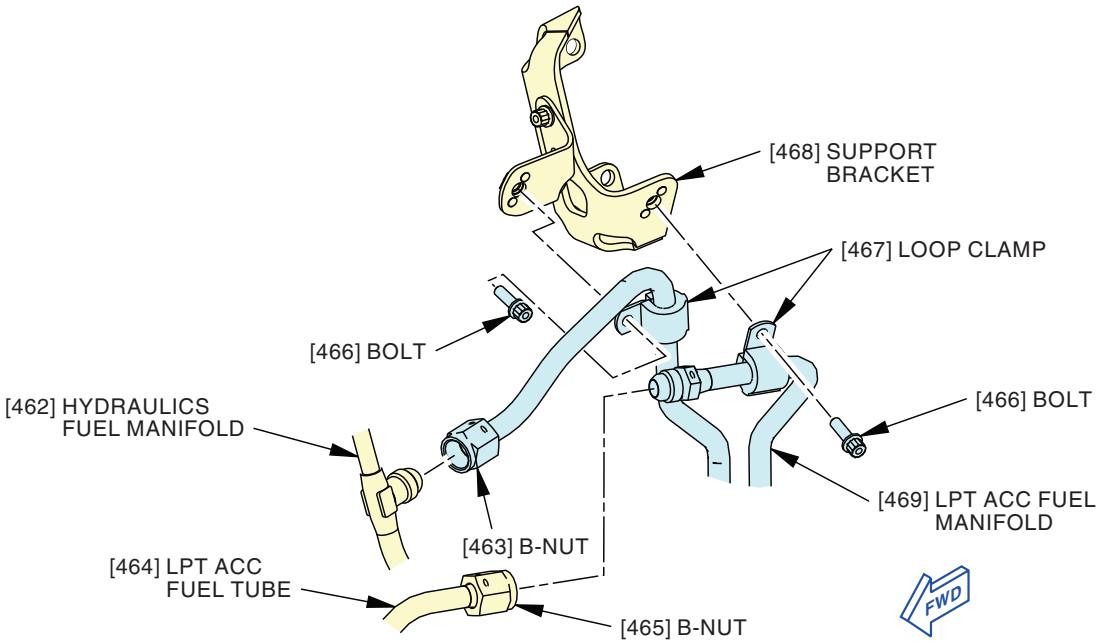
Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

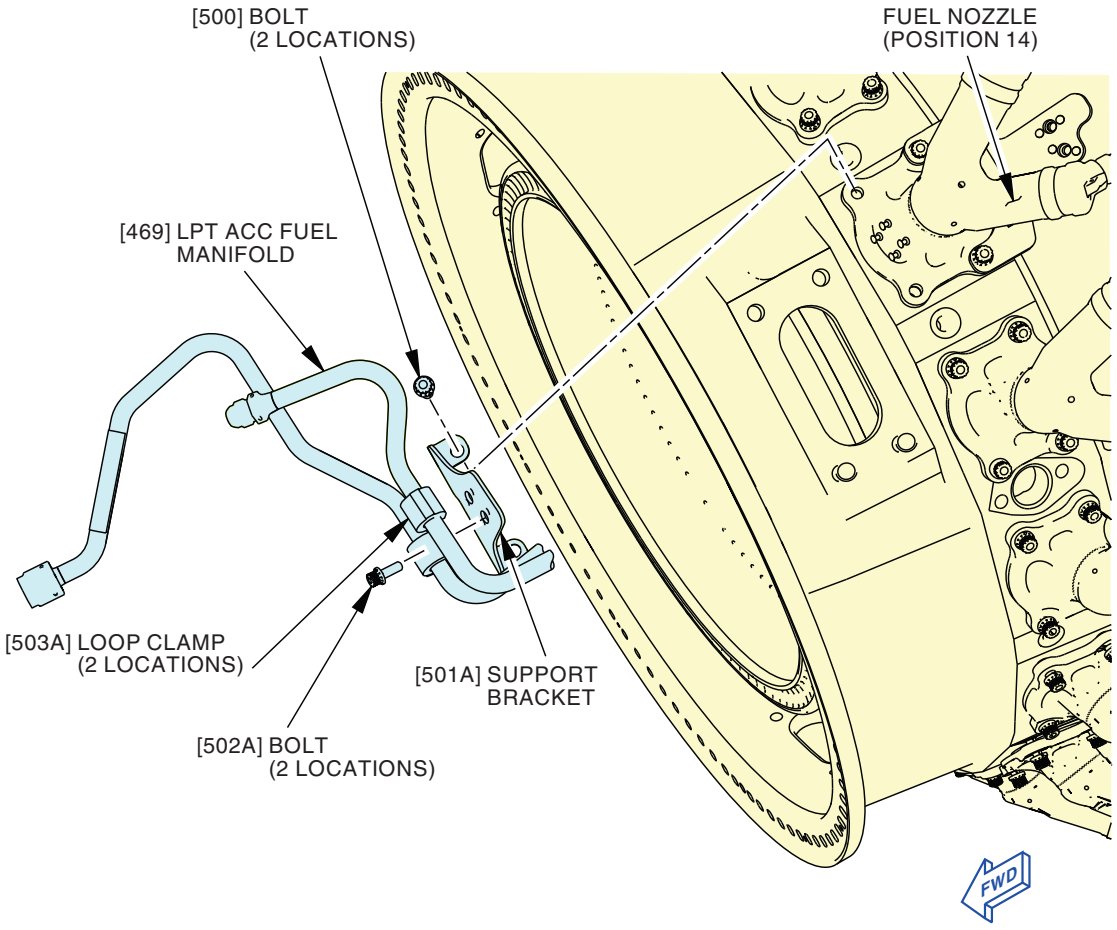


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2892644 S0000688934_V1

Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

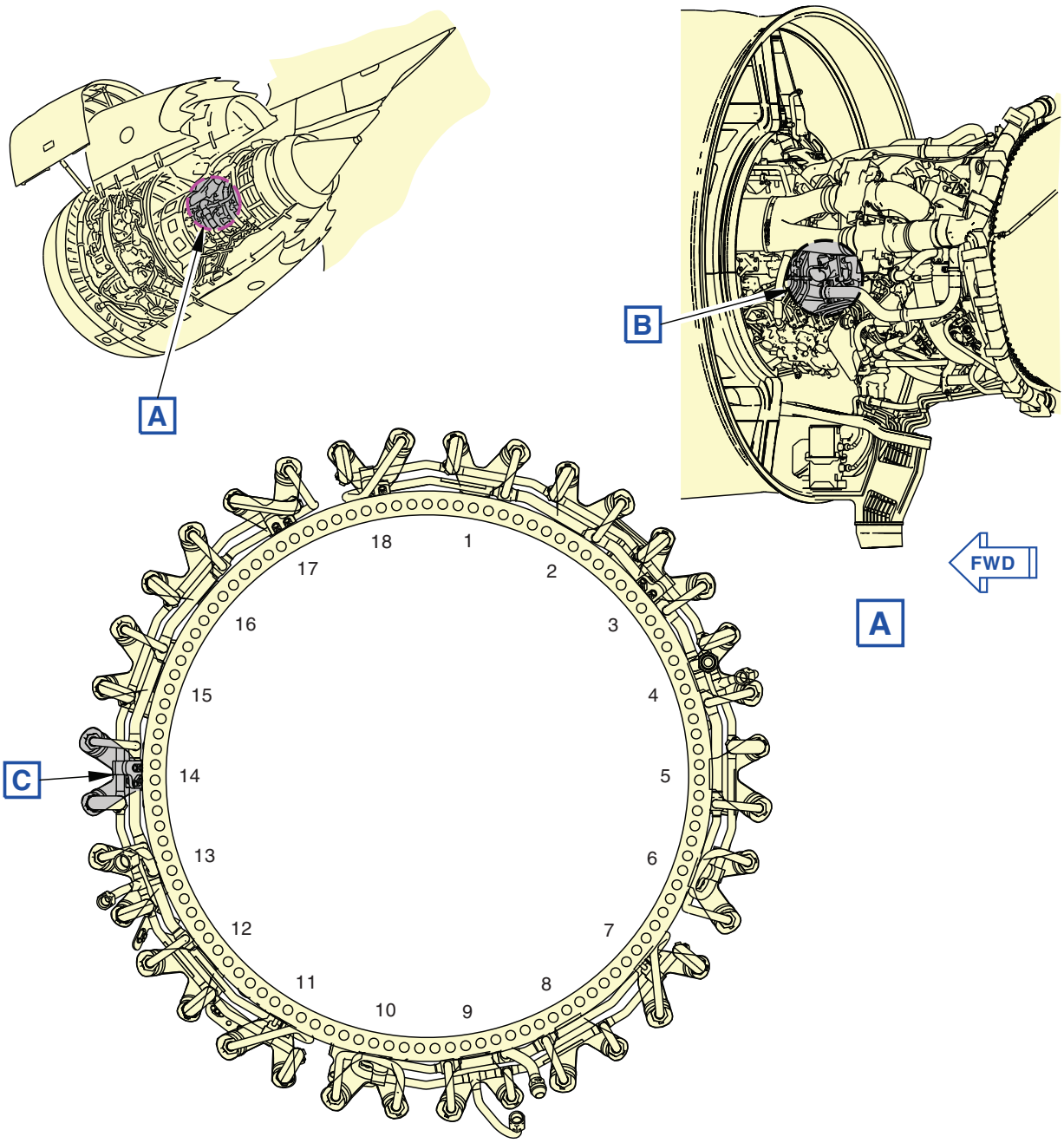


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Fuel Nozzle (Position 13) Installation
Figure 414/73-11-05-990-900-G00 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE FORWARD DIRECTION)

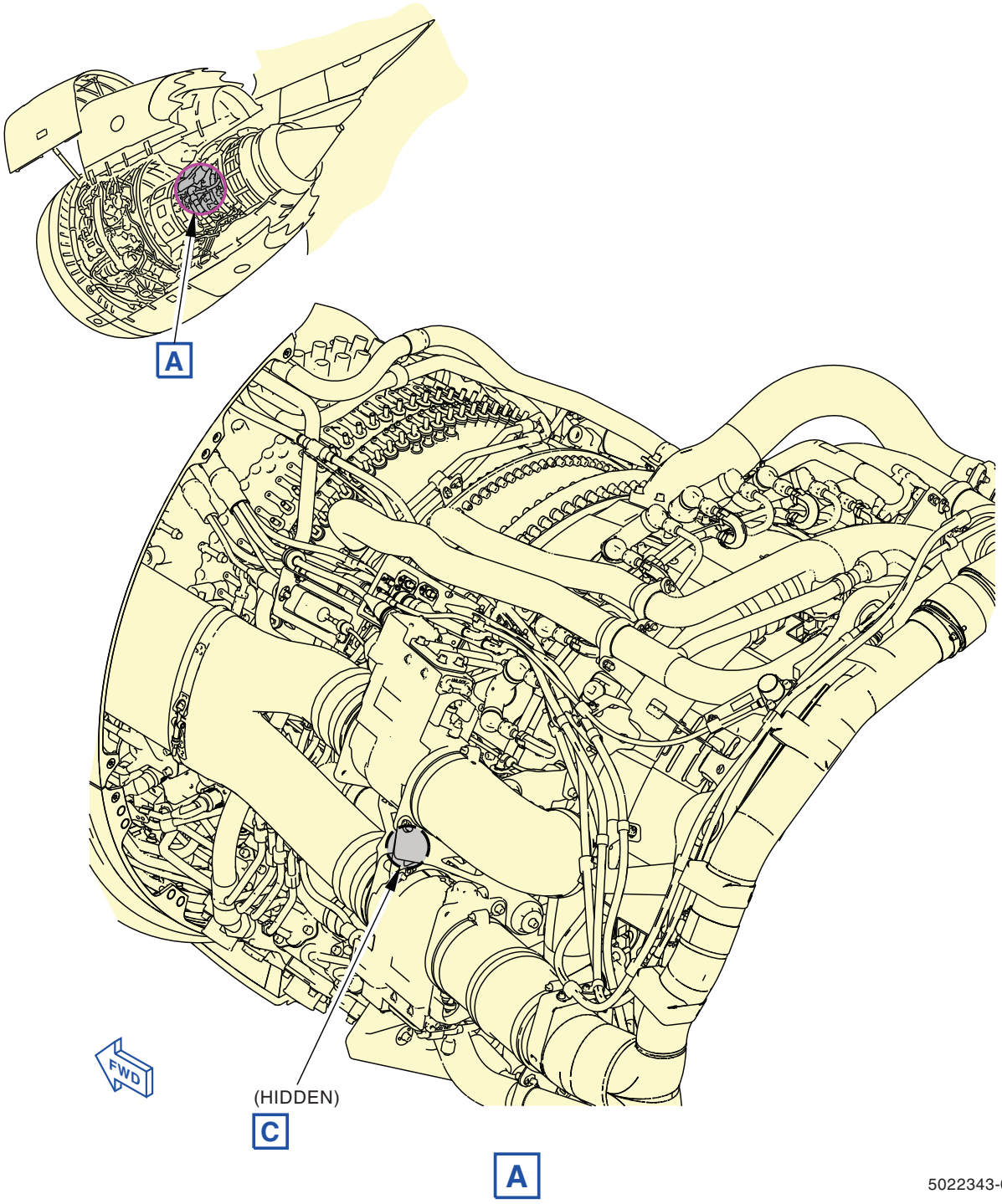
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5021979-00

2476630 S0000580896_V1

Fuel Nozzle (Position 14) Installation
Figure 415/73-11-05-990-901-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

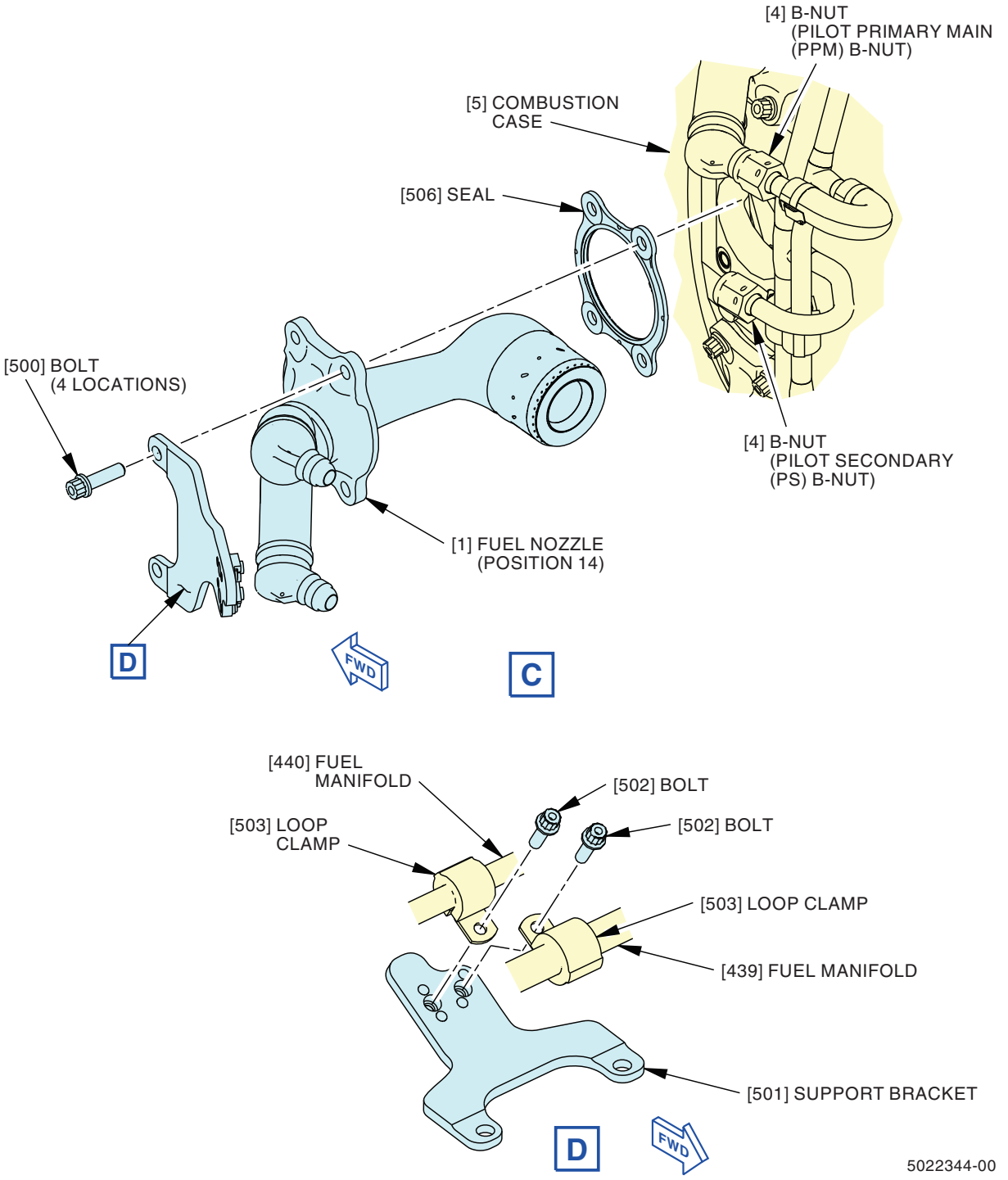


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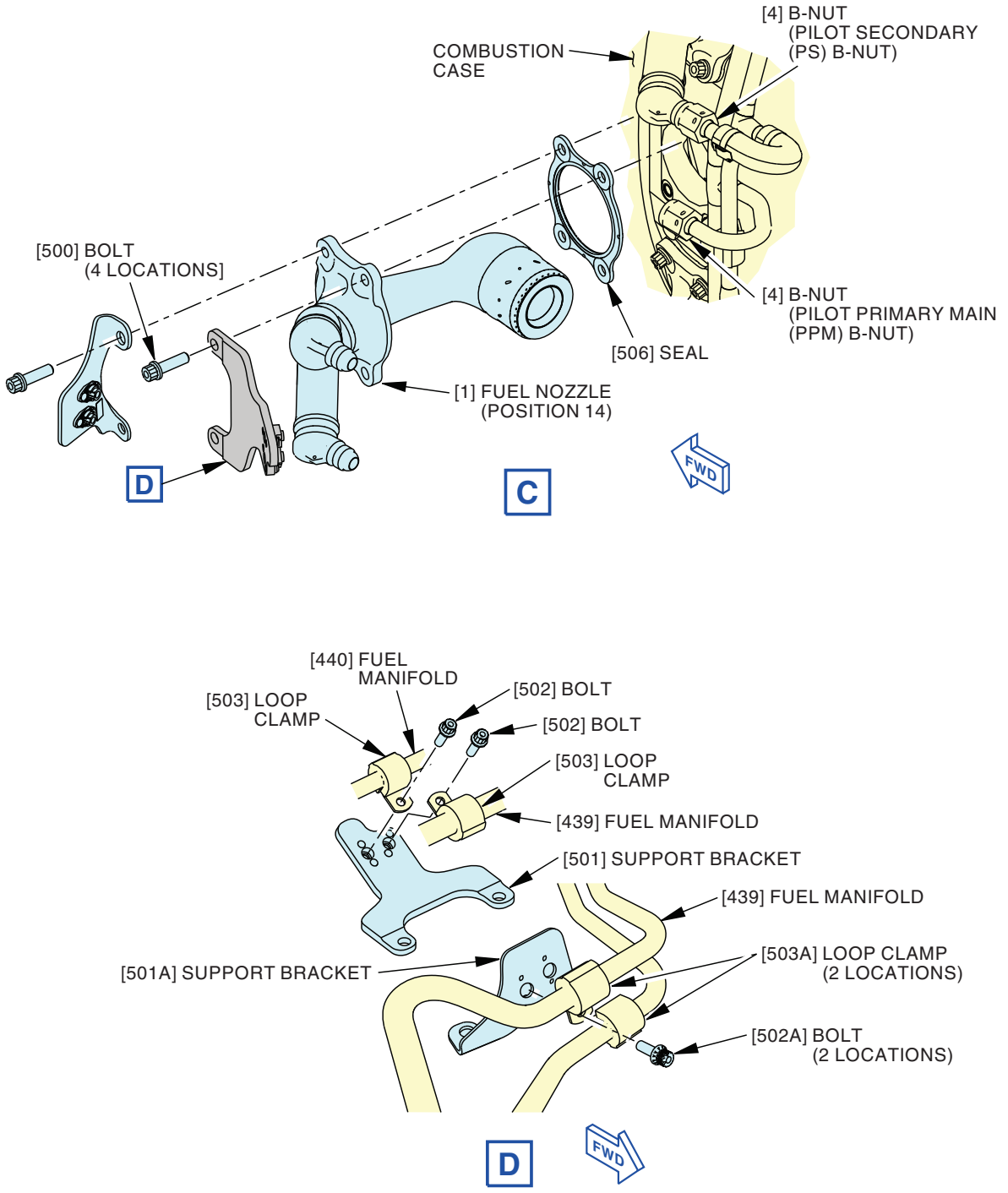
Fuel Nozzle (Position 14) Installation
Figure 415/73-11-05-990-901-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



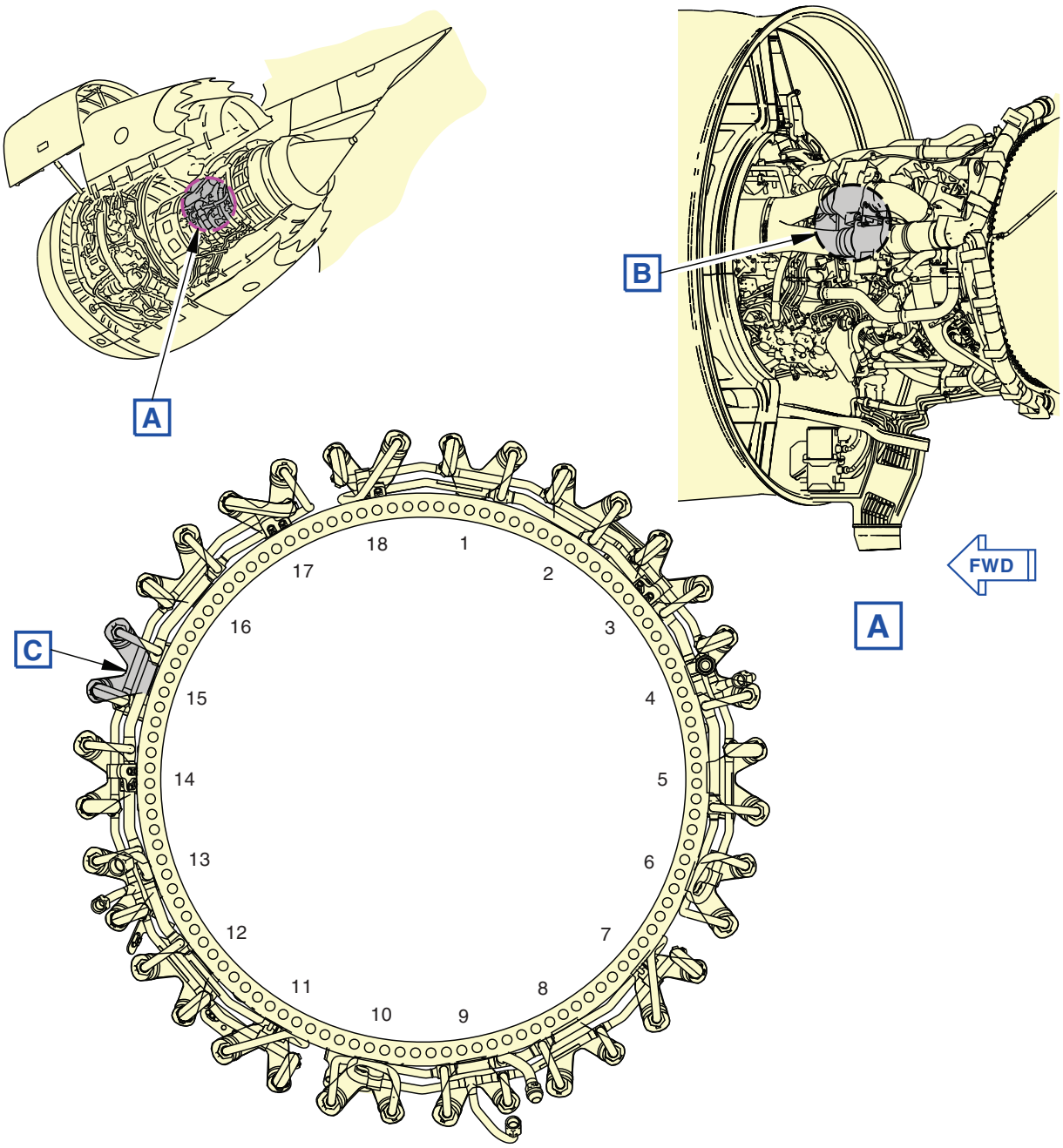
Fuel Nozzle (Position 14) Installation
Figure 415/73-11-05-990-901-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 14) Installation
Figure 415/73-11-05-990-901-G00 (Sheet 4 of 4)

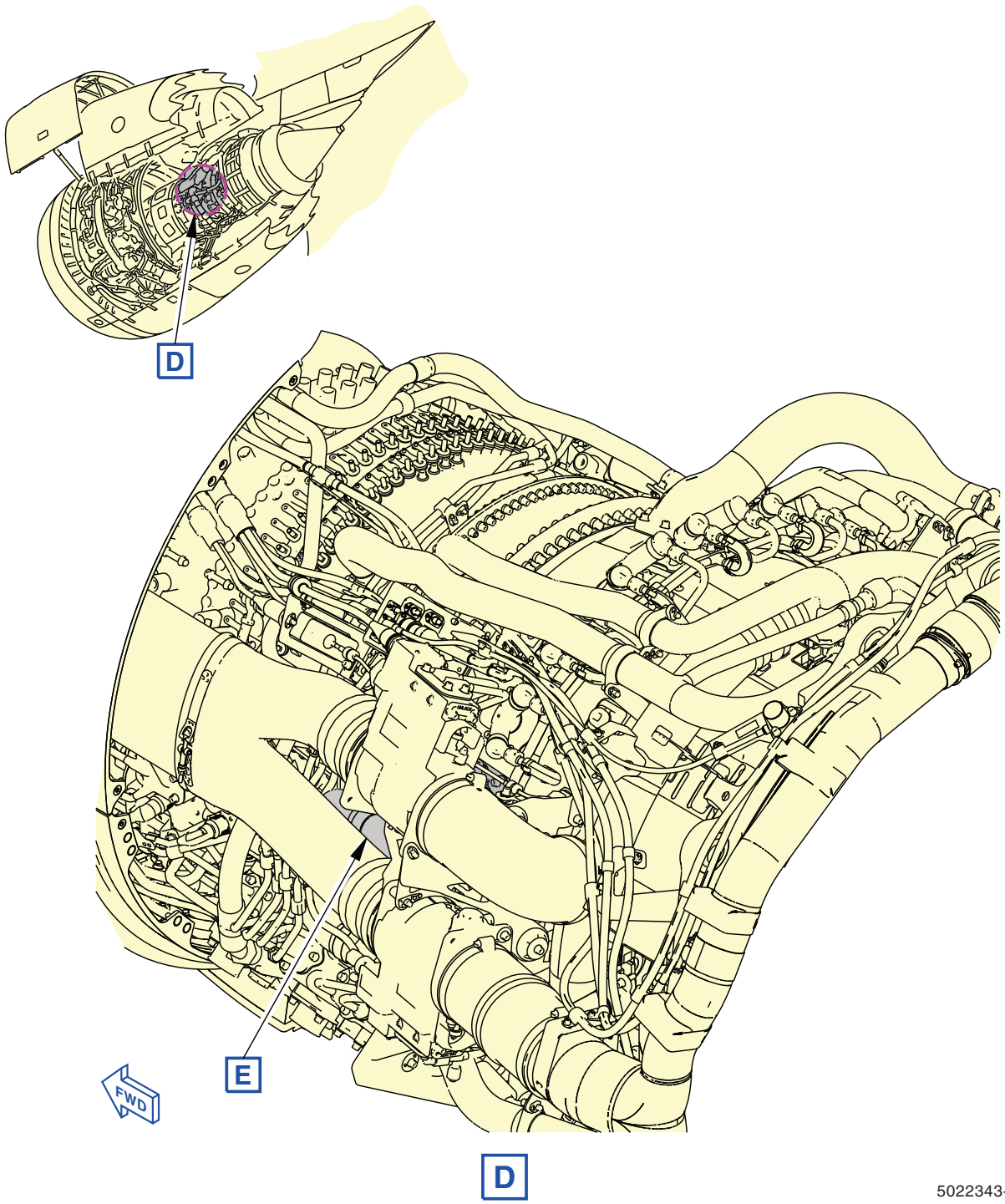
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Fuel Nozzle (Position 15) Installation
Figure 416/73-11-05-990-902-G00 (Sheet 1 of 4)

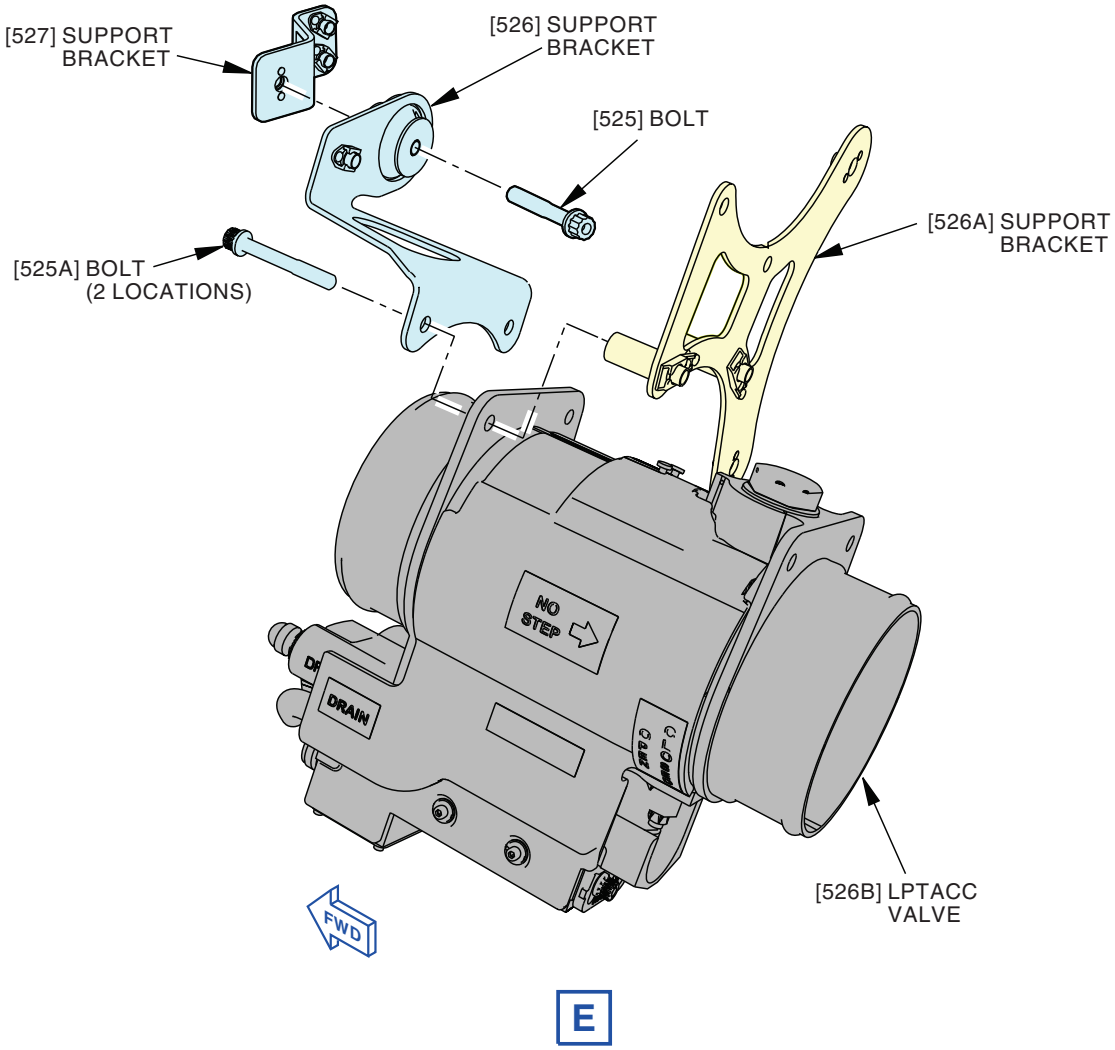
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 15) Installation
Figure 416/73-11-05-990-902-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

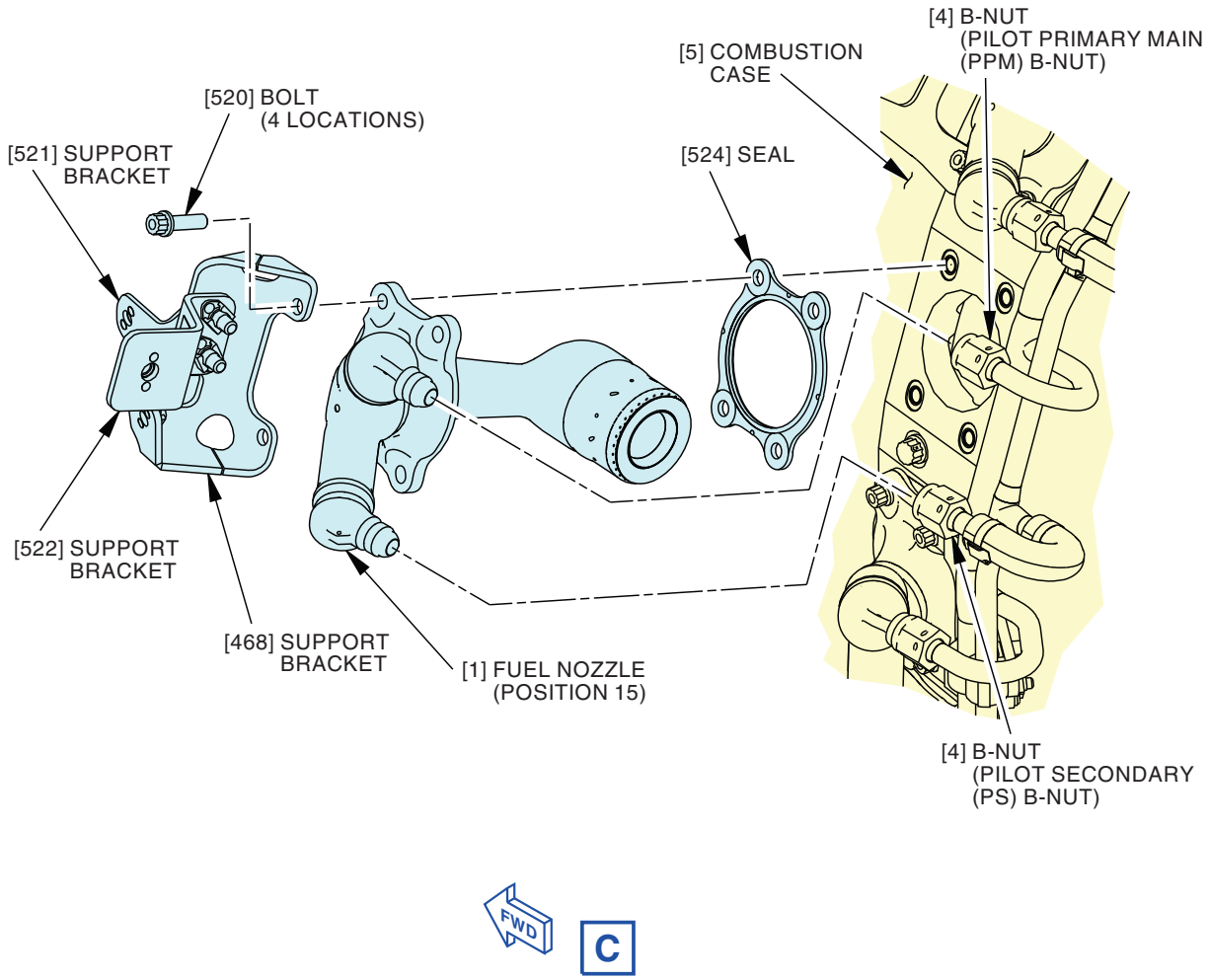


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Fuel Nozzle (Position 15) Installation
Figure 416/73-11-05-990-902-G00 (Sheet 3 of 4)

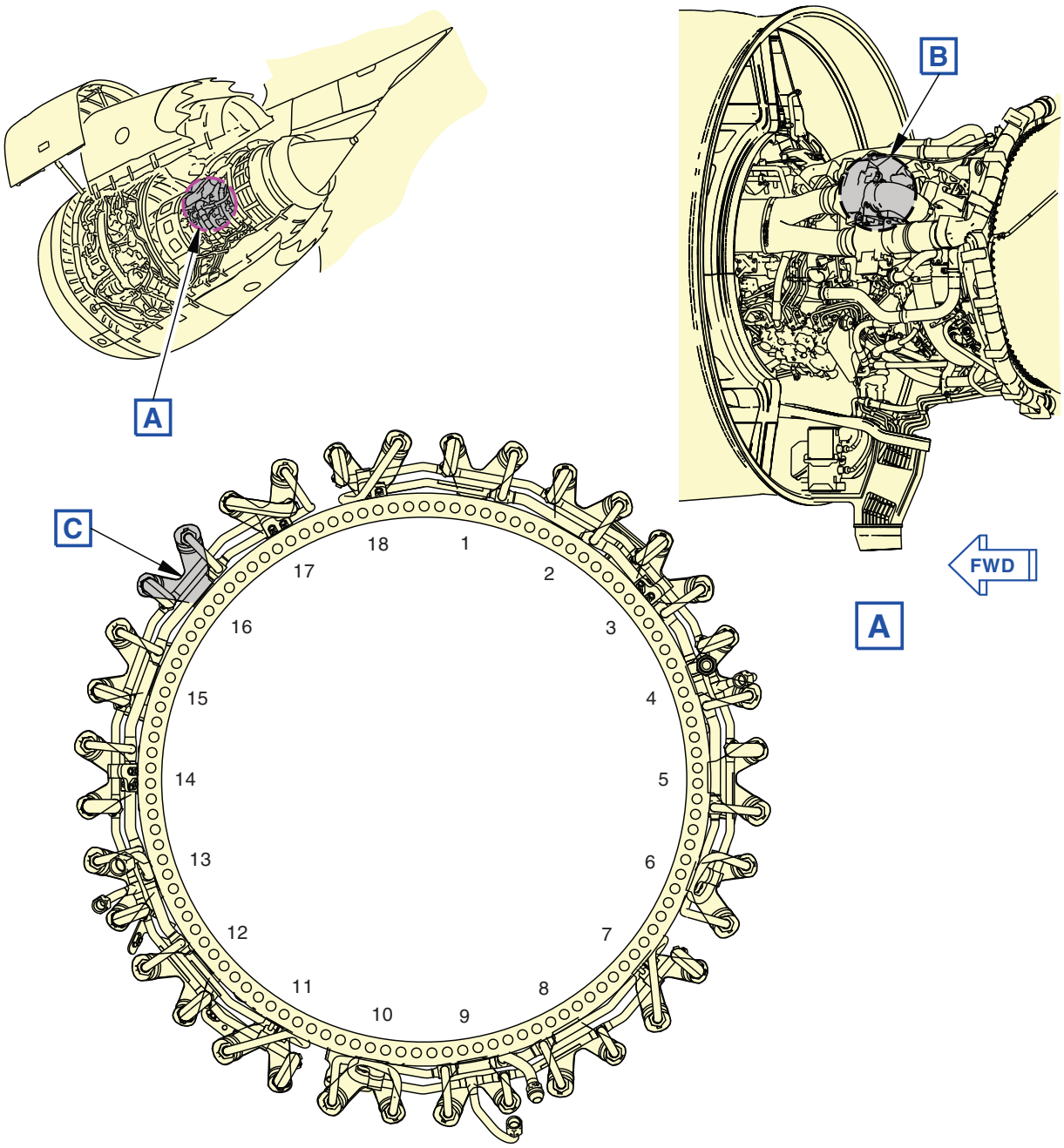
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Fuel Nozzle (Position 15) Installation
Figure 416/73-11-05-990-902-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE FORWARD DIRECTION)

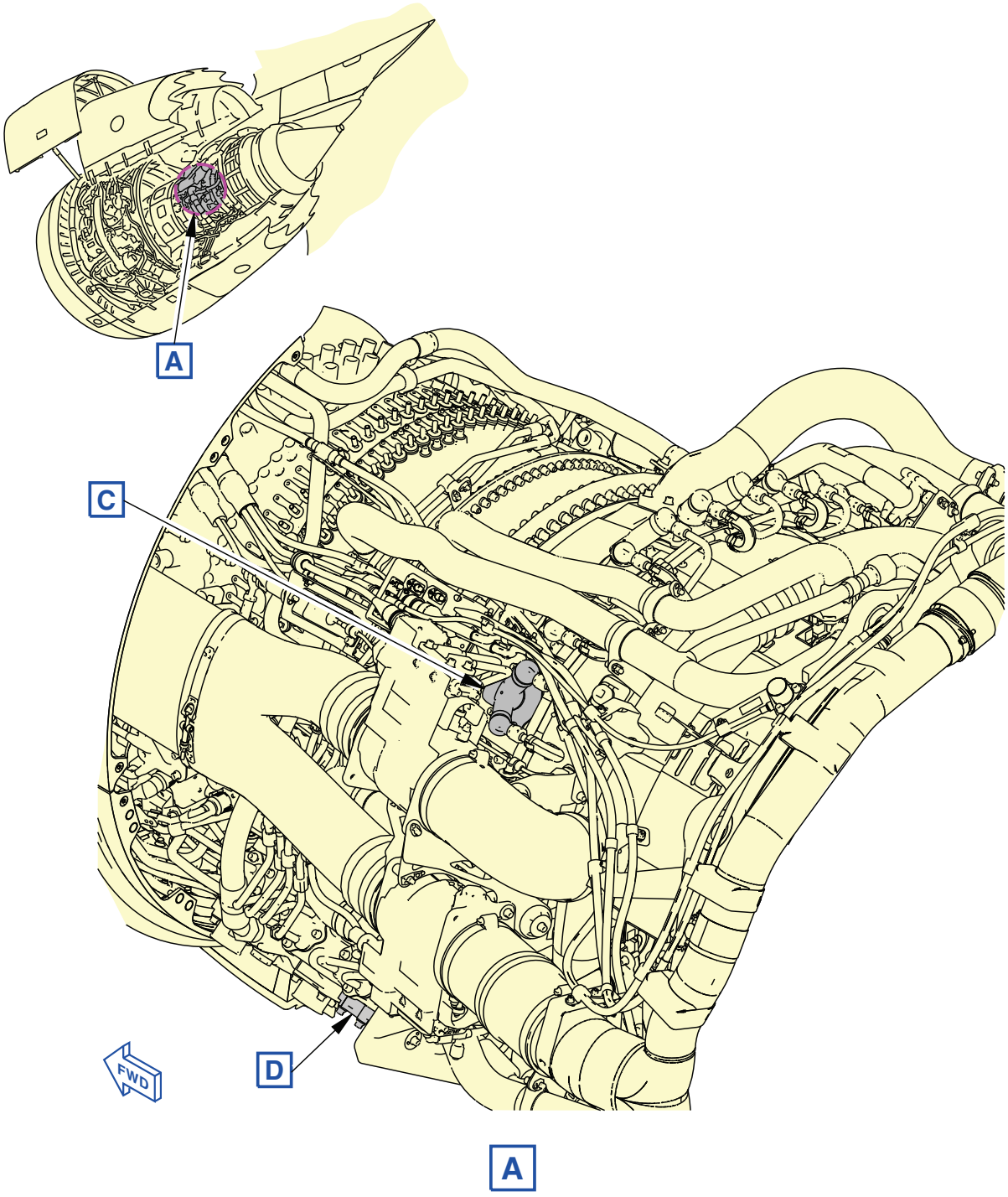
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Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 1 of 19)

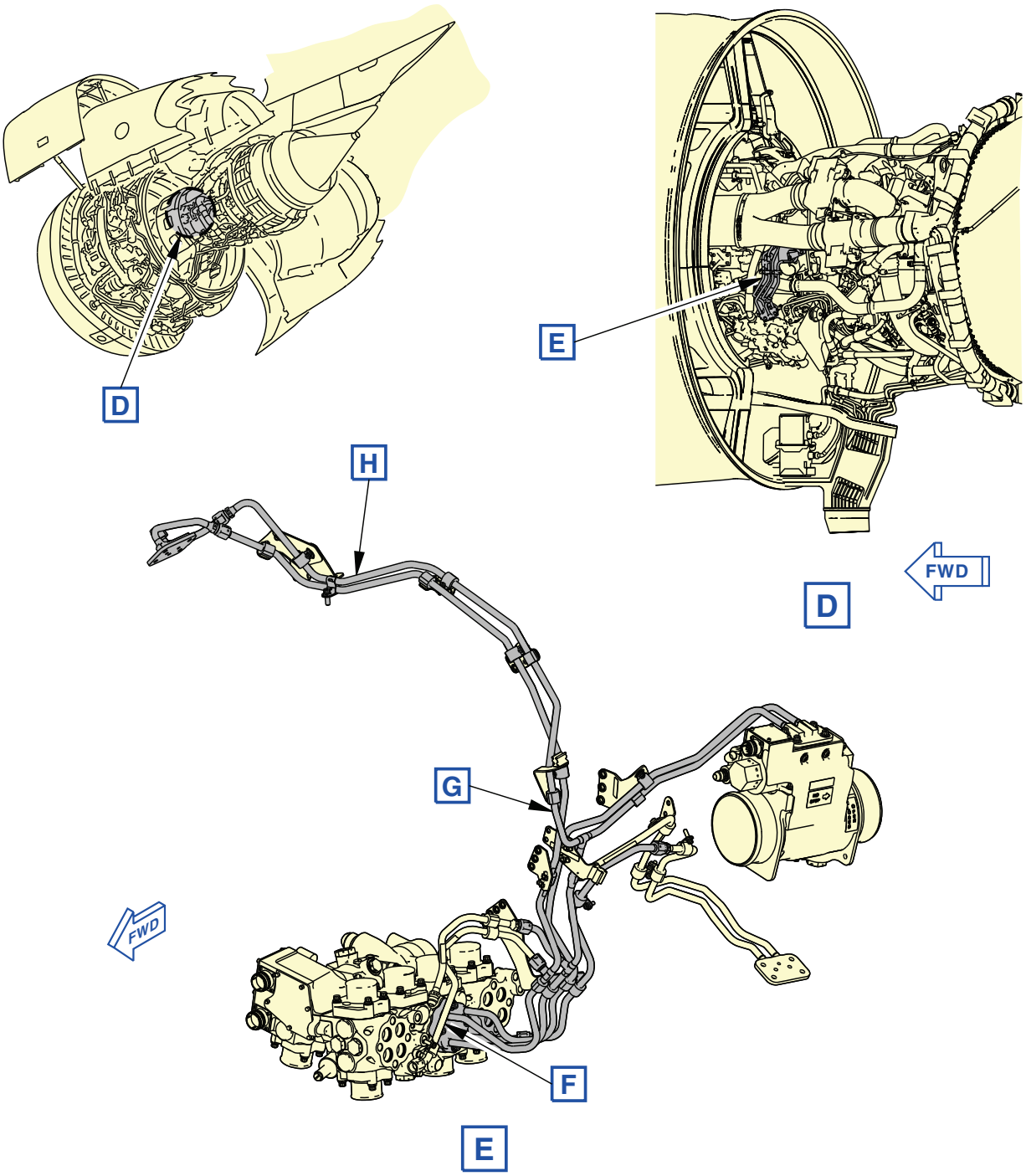
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Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 2 of 19)

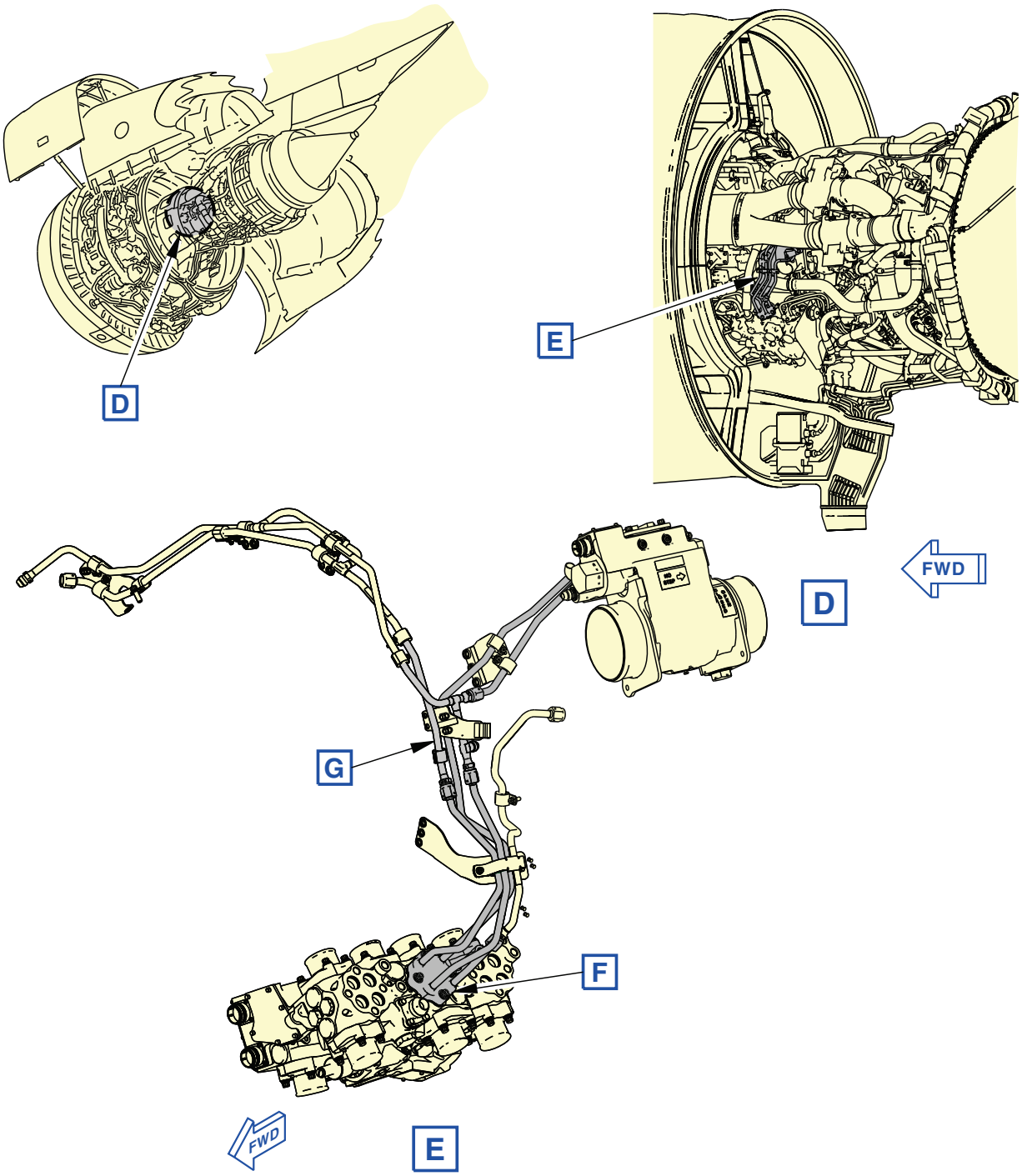
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Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 3 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

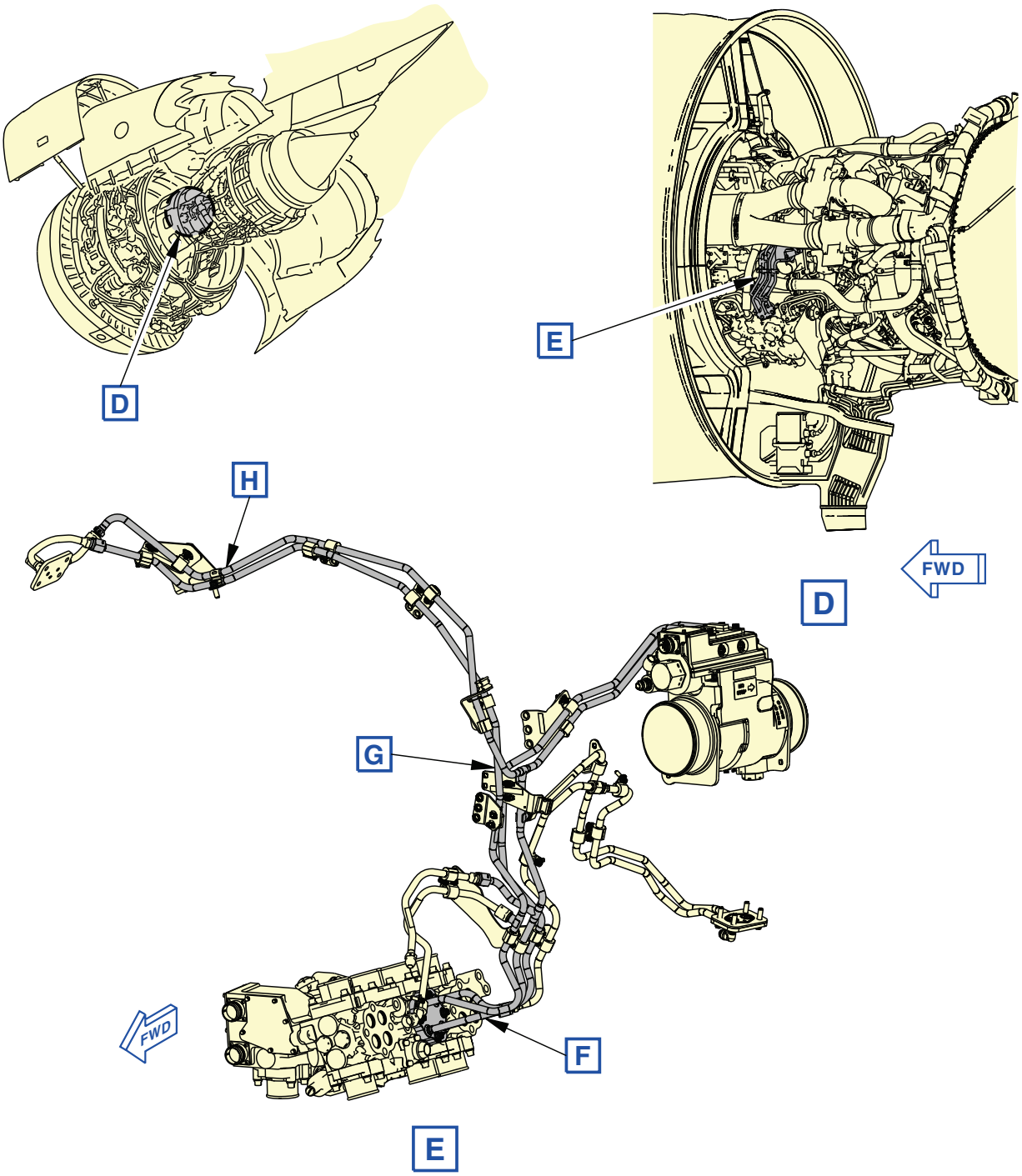


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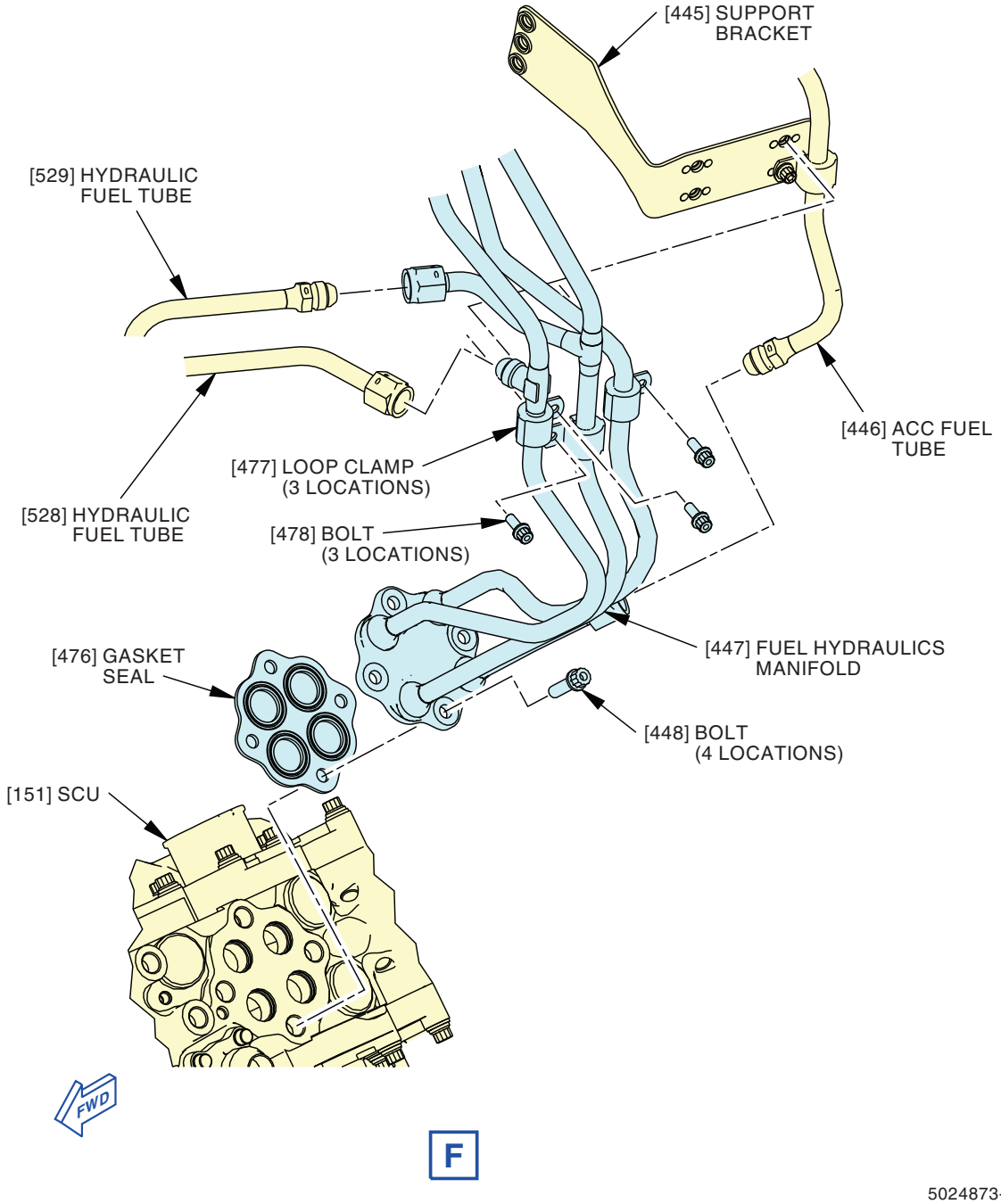
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 4 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



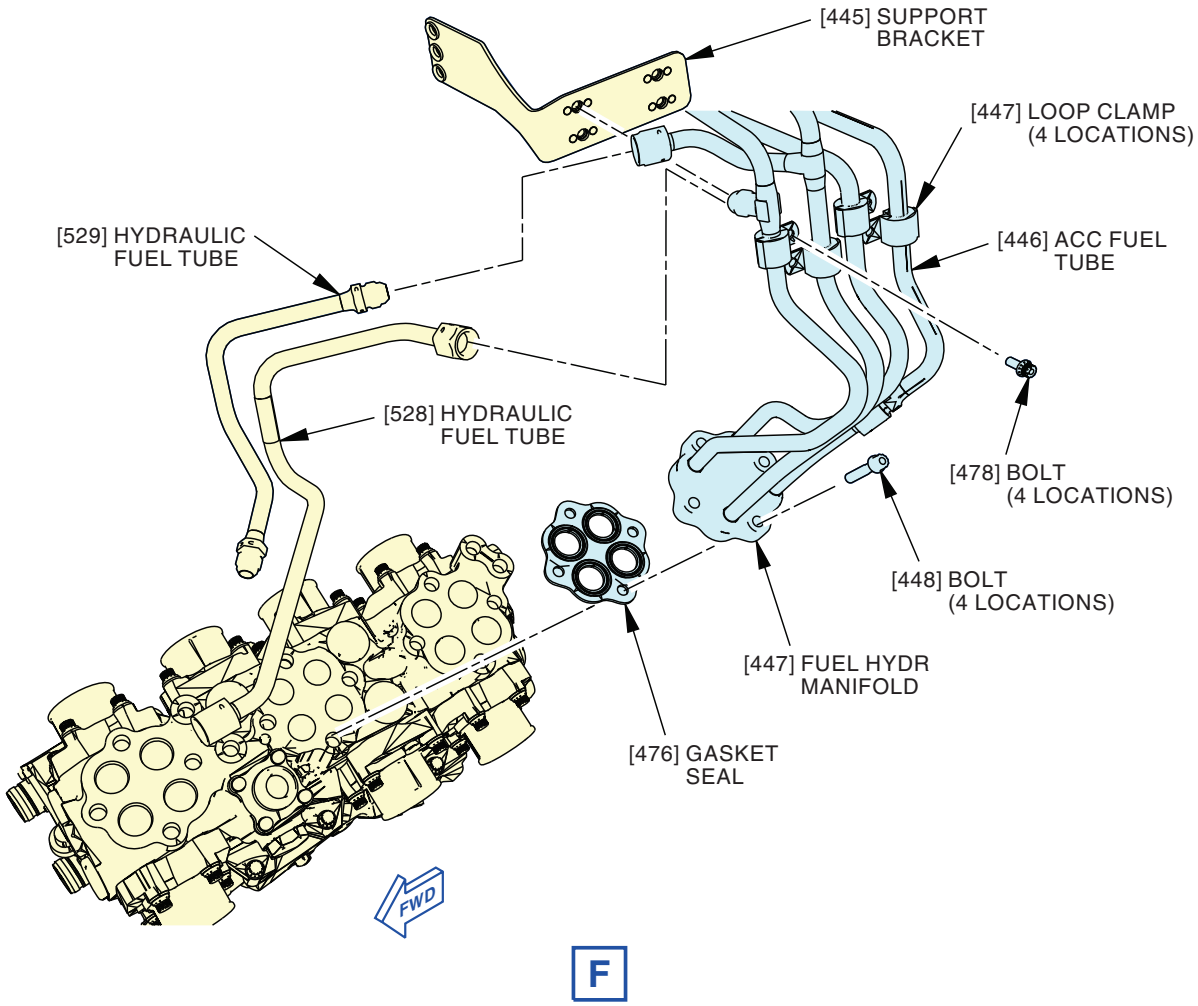
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 5 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 6 of 19)

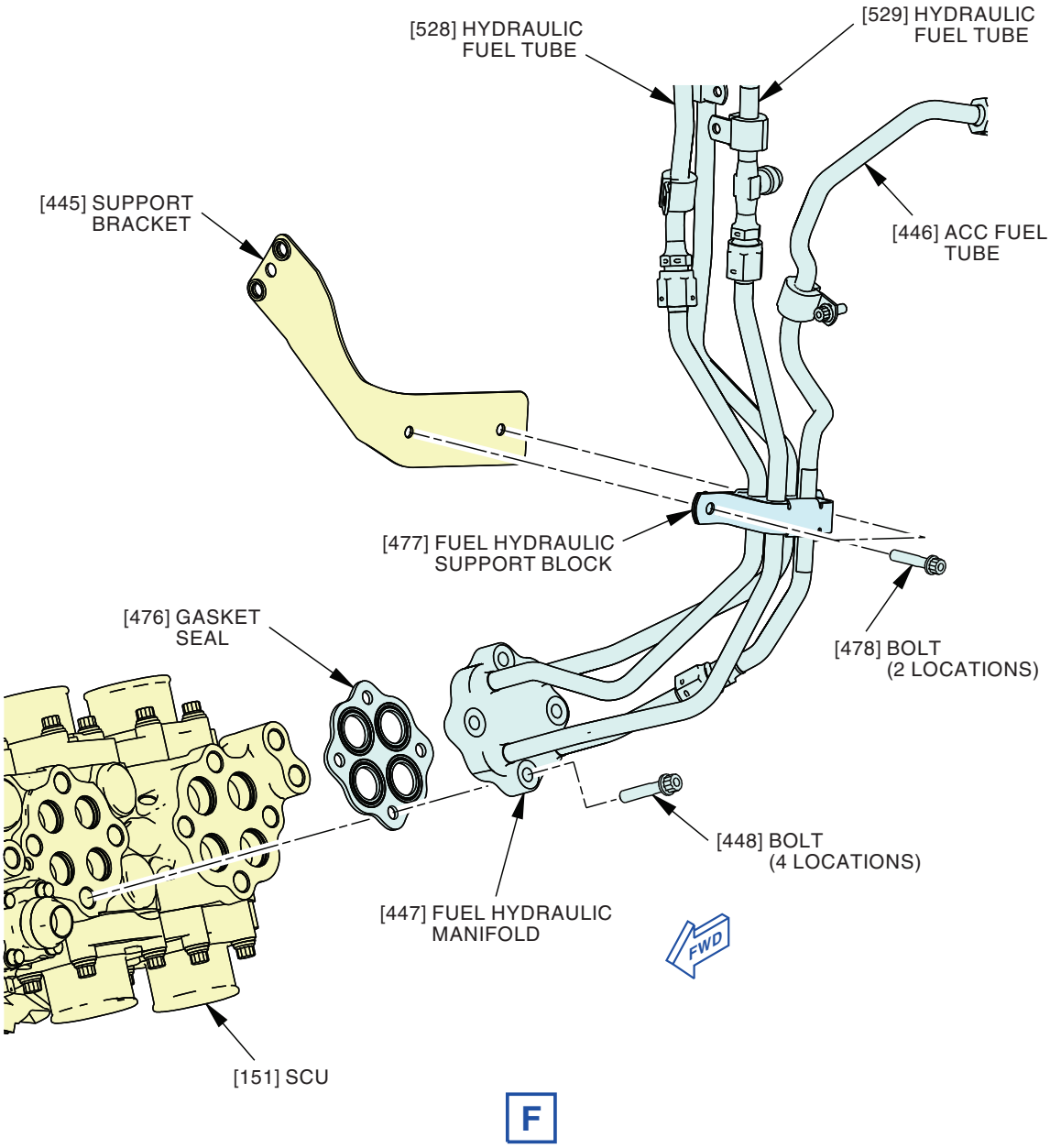
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Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 7 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

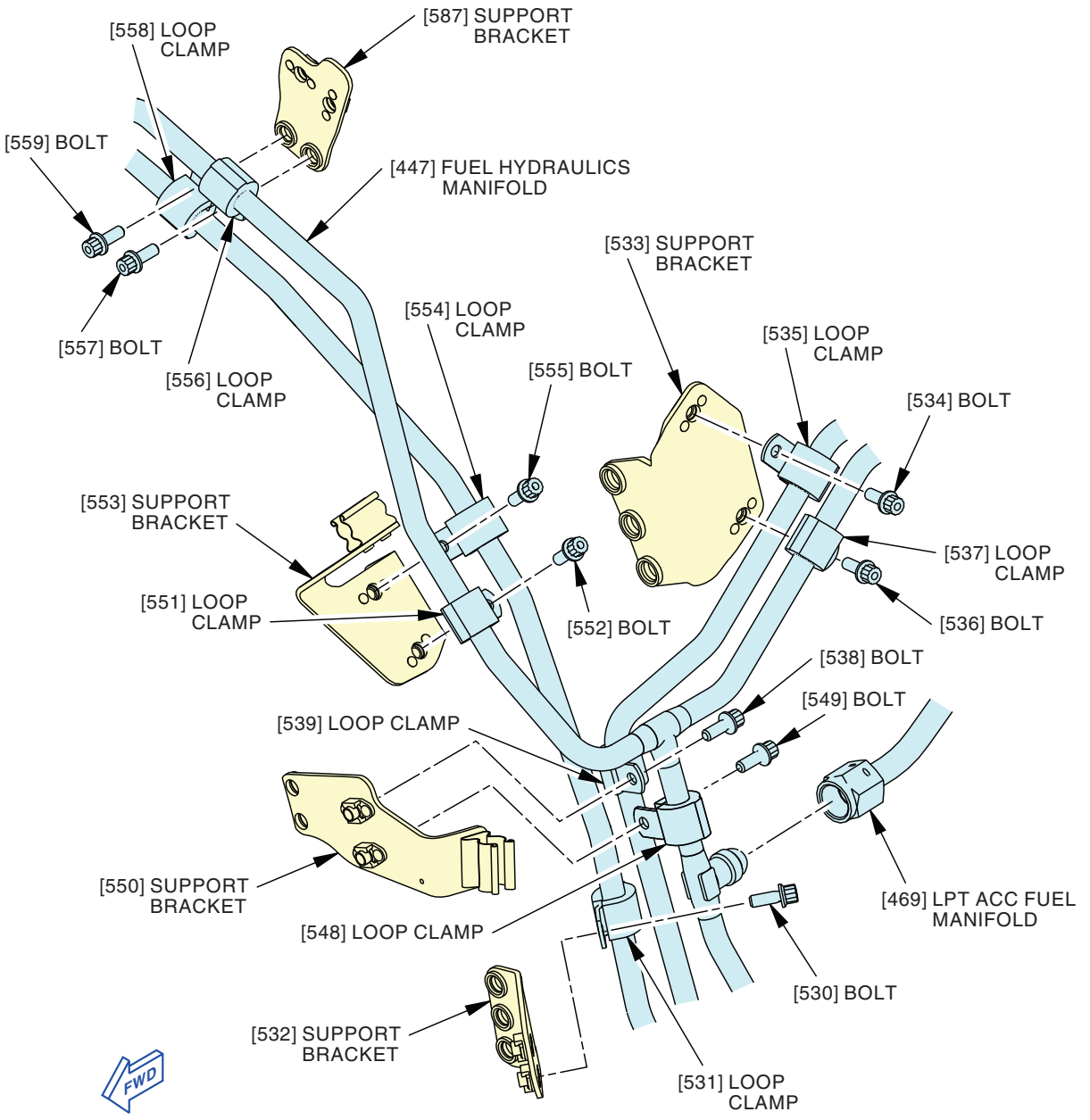


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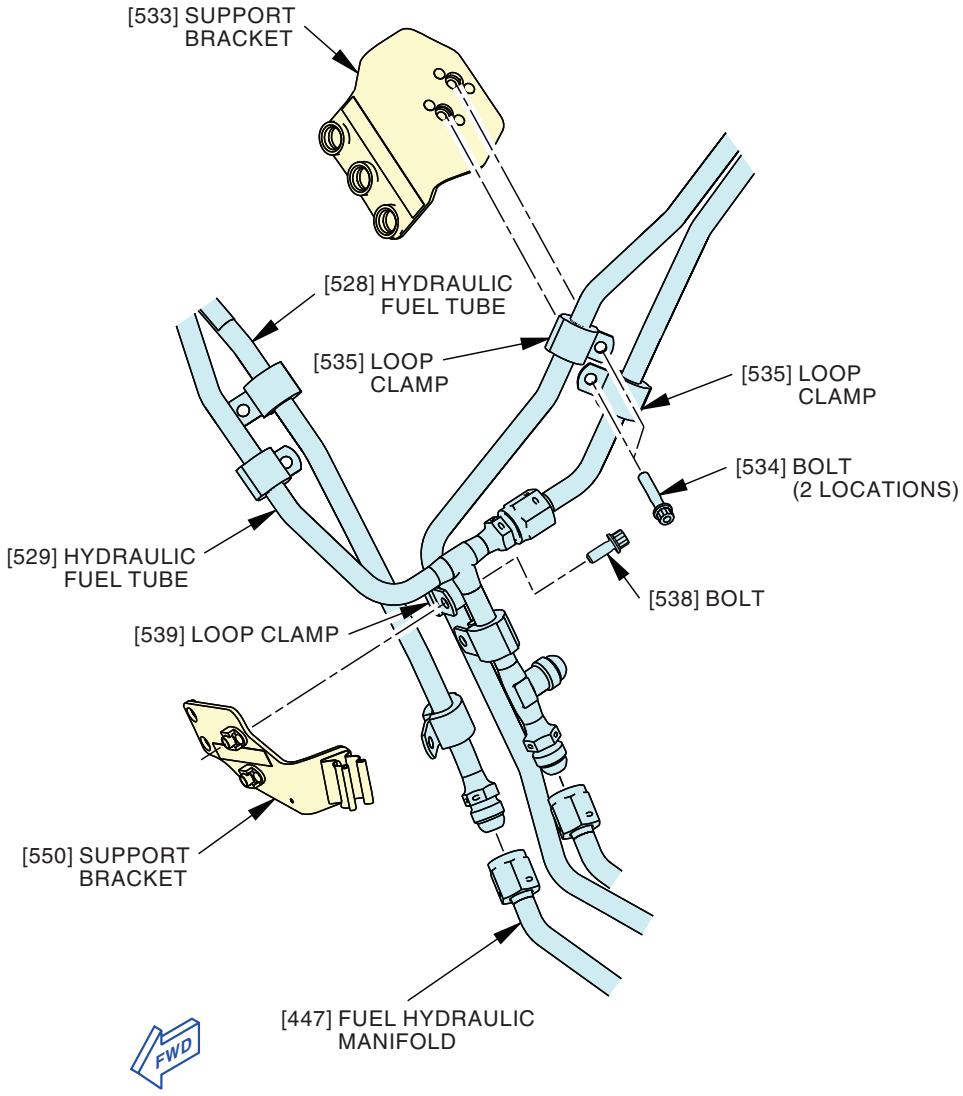
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 8 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



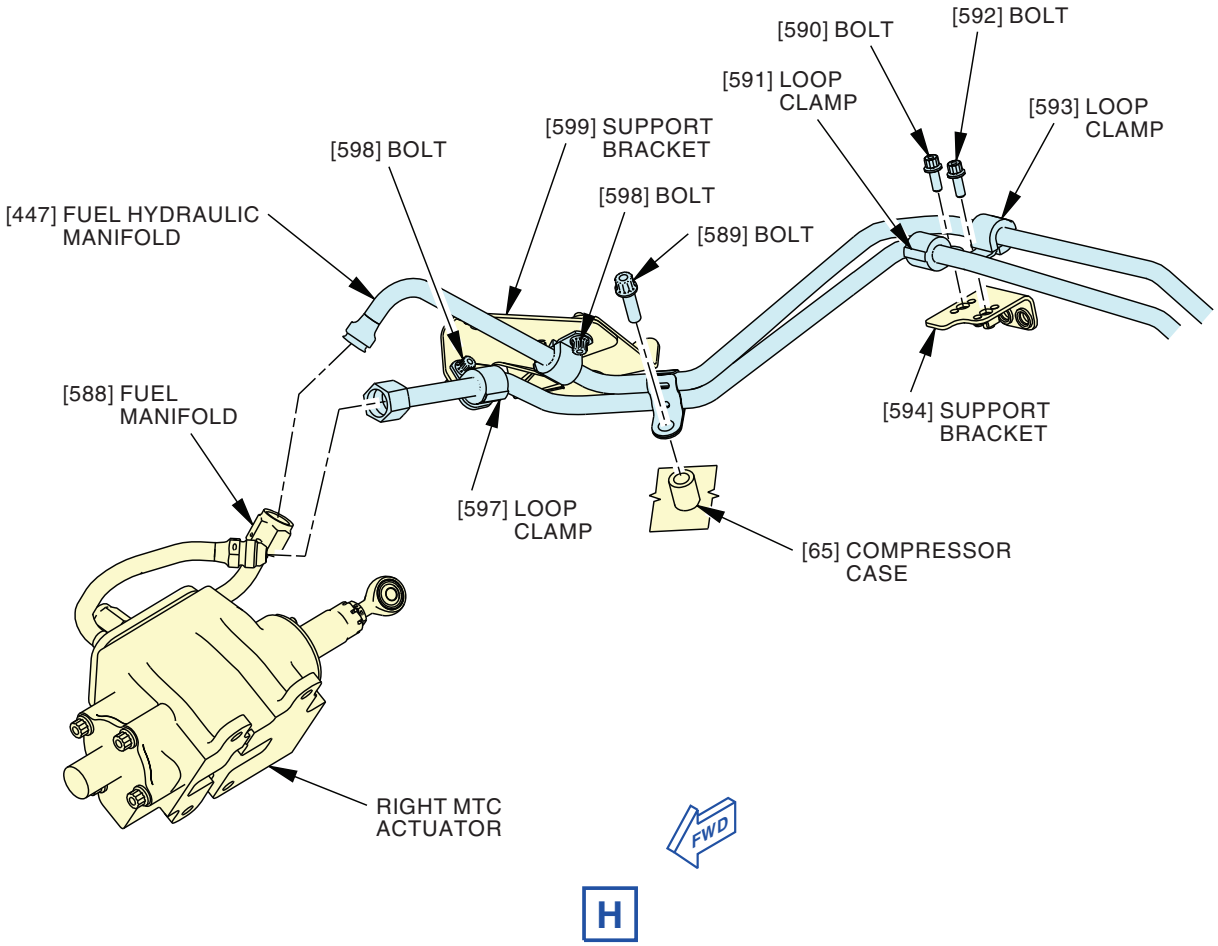
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 9 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 10 of 19)

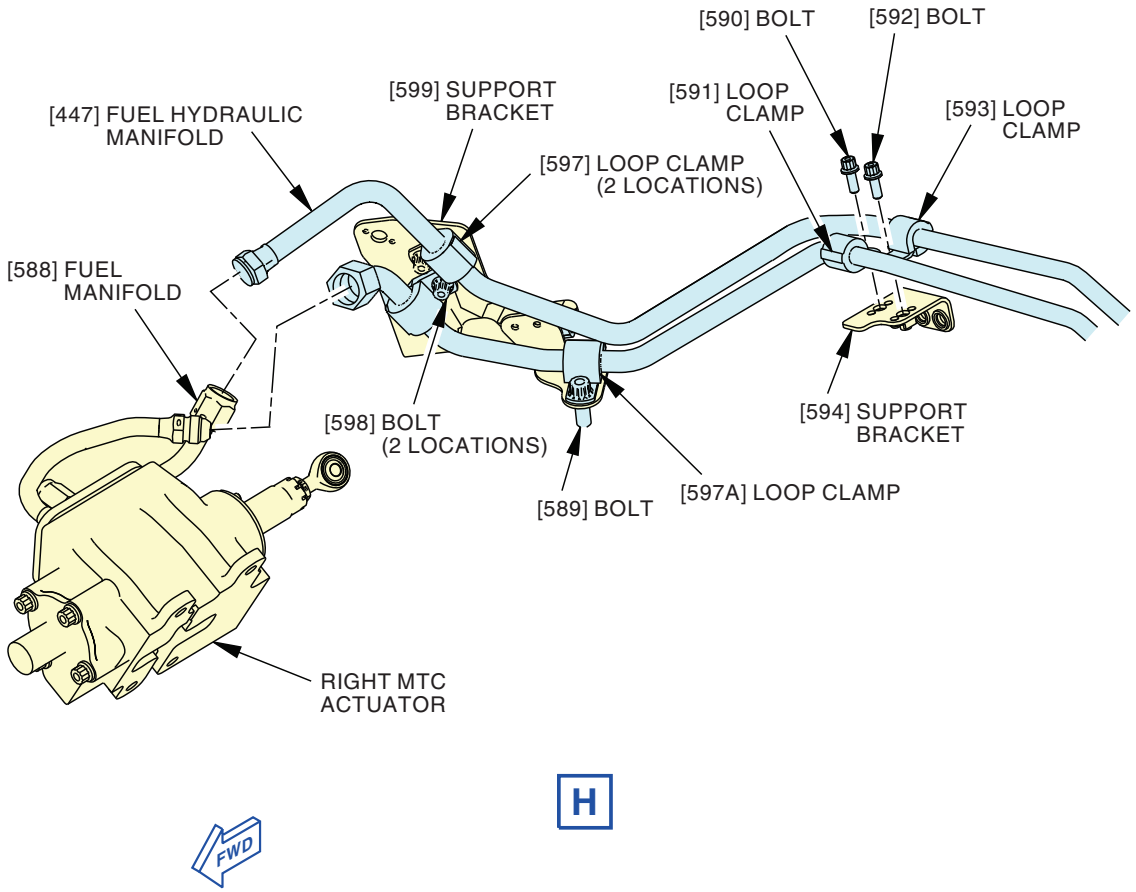
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 11 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

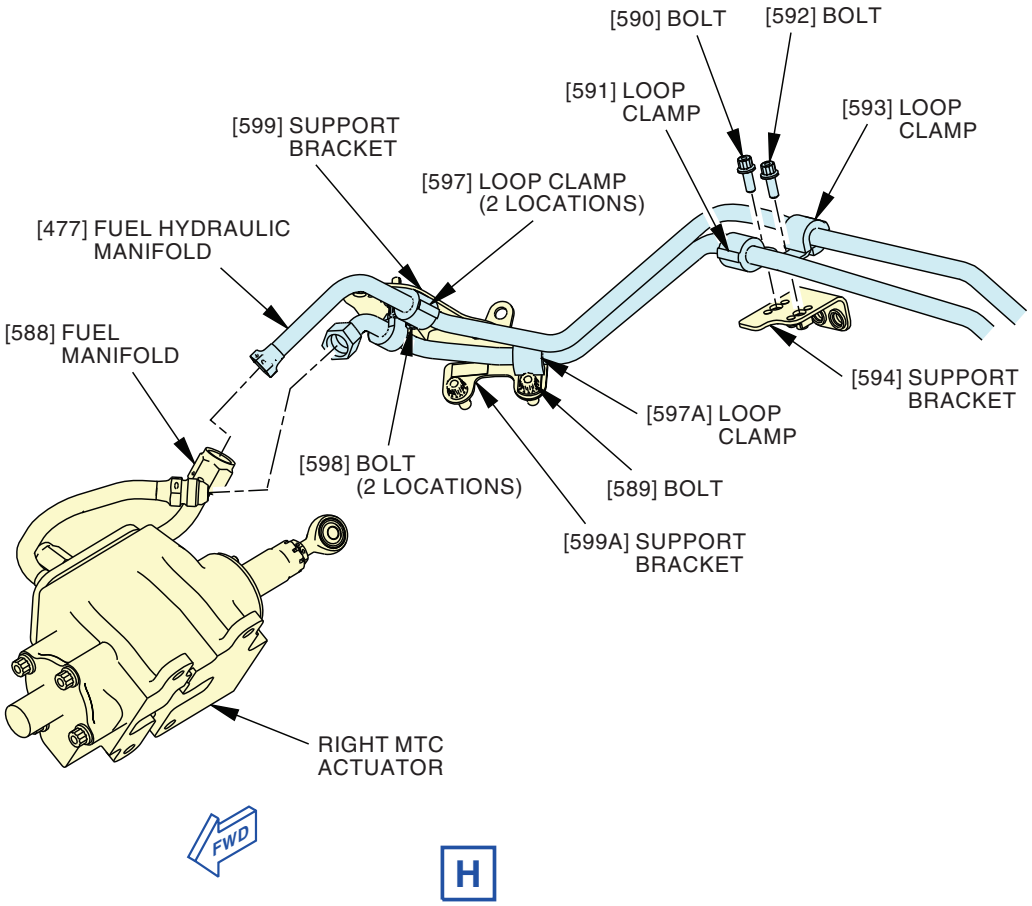


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Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 12 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

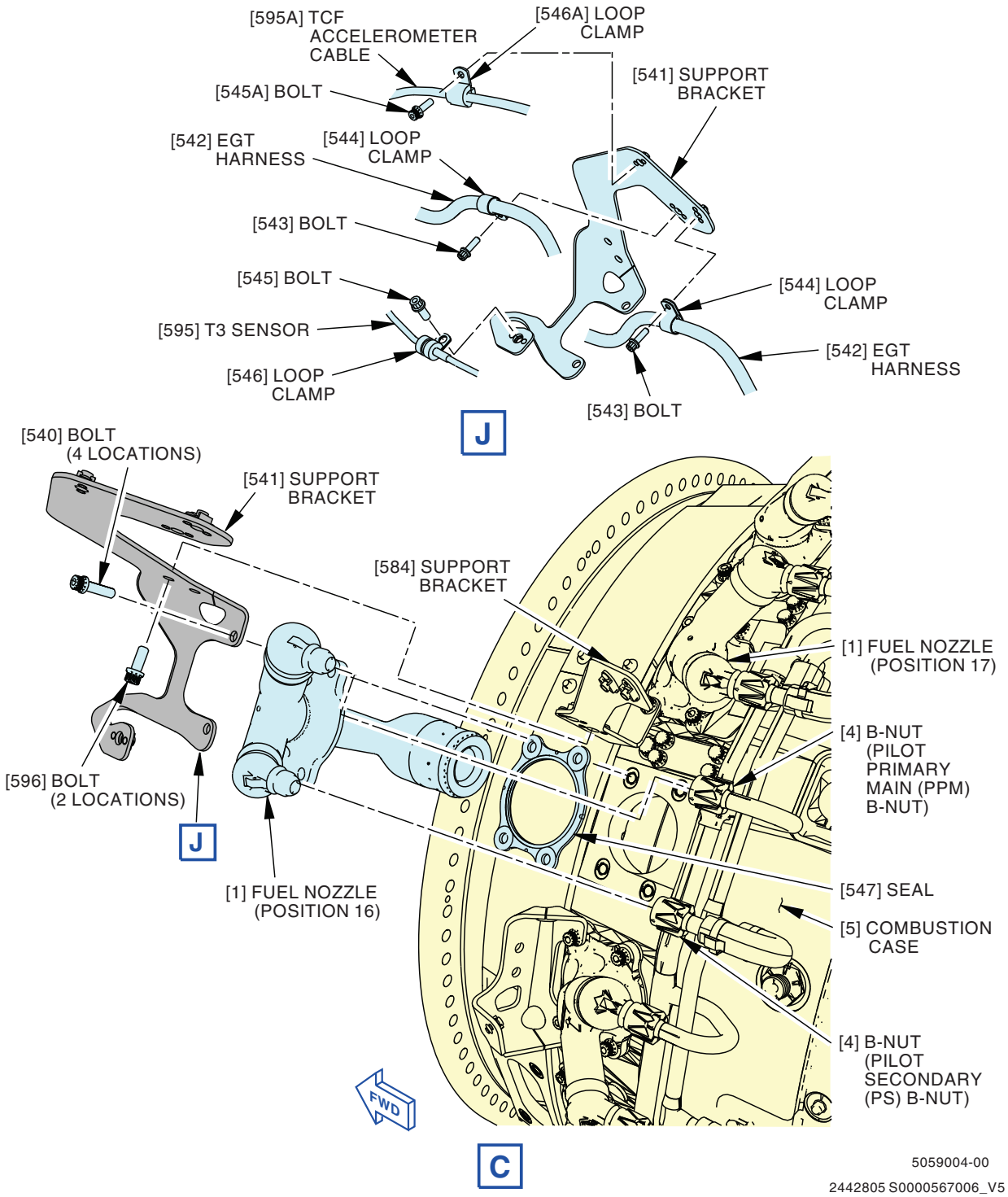


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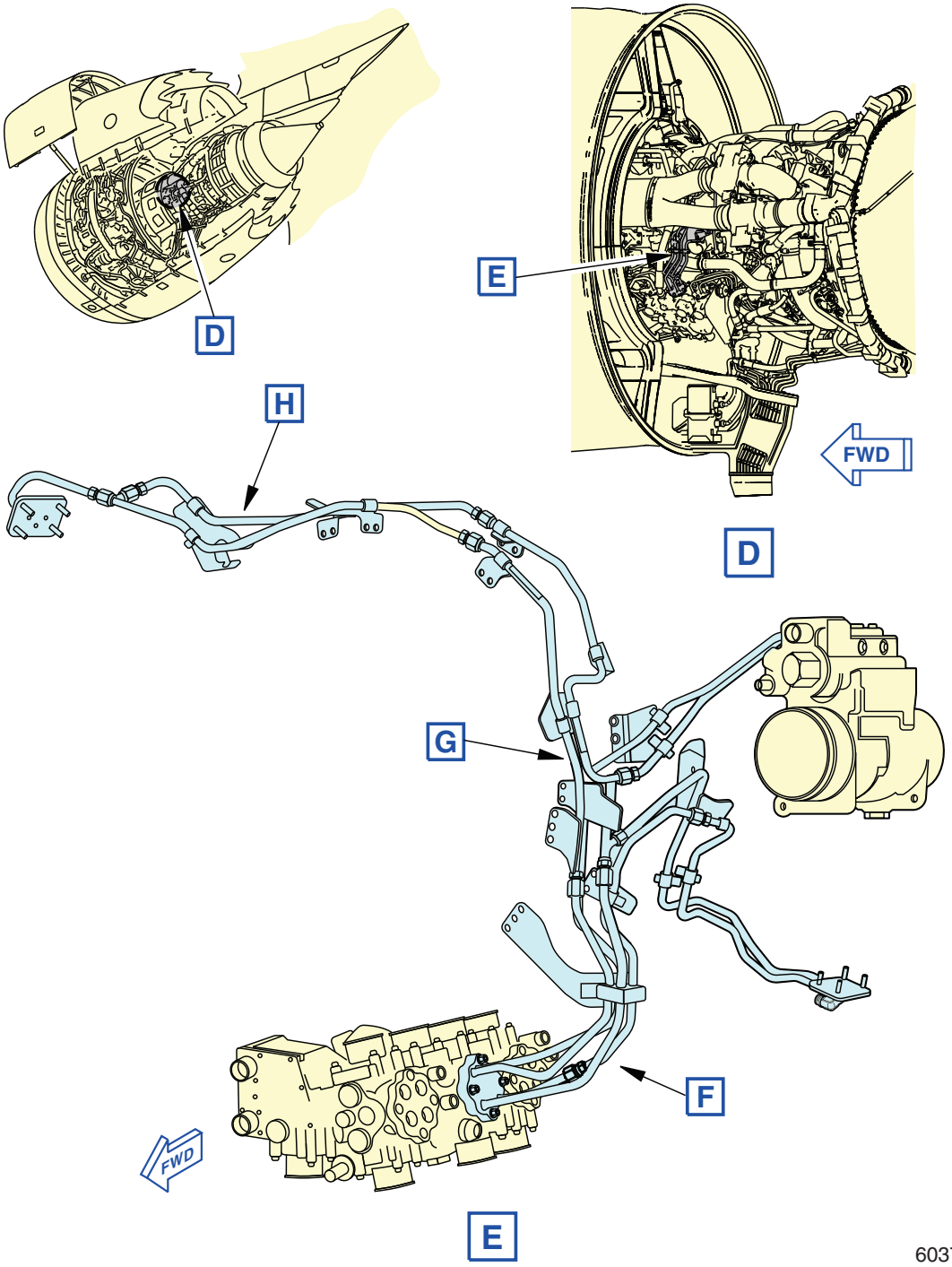
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 13 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



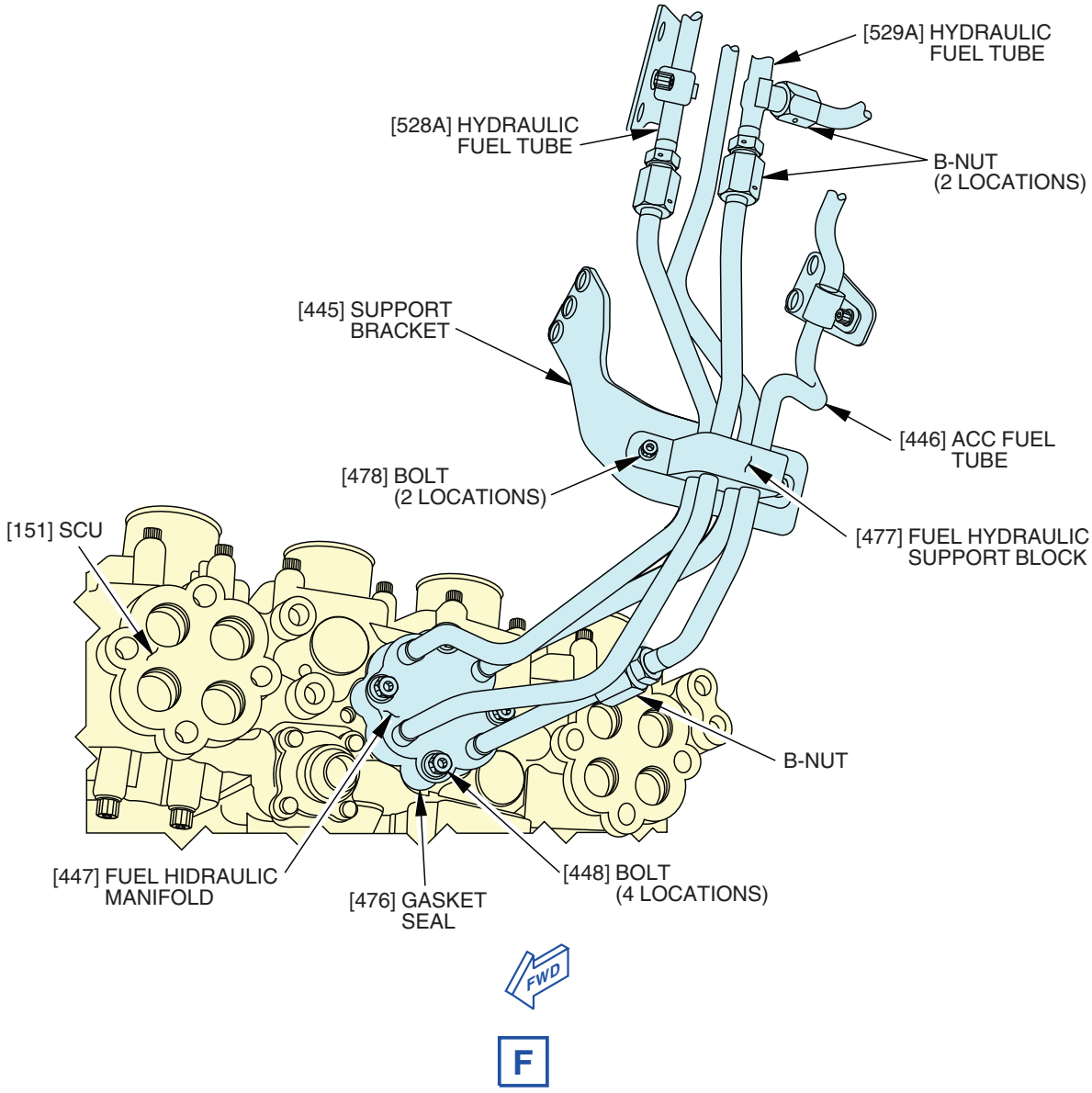
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 14 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



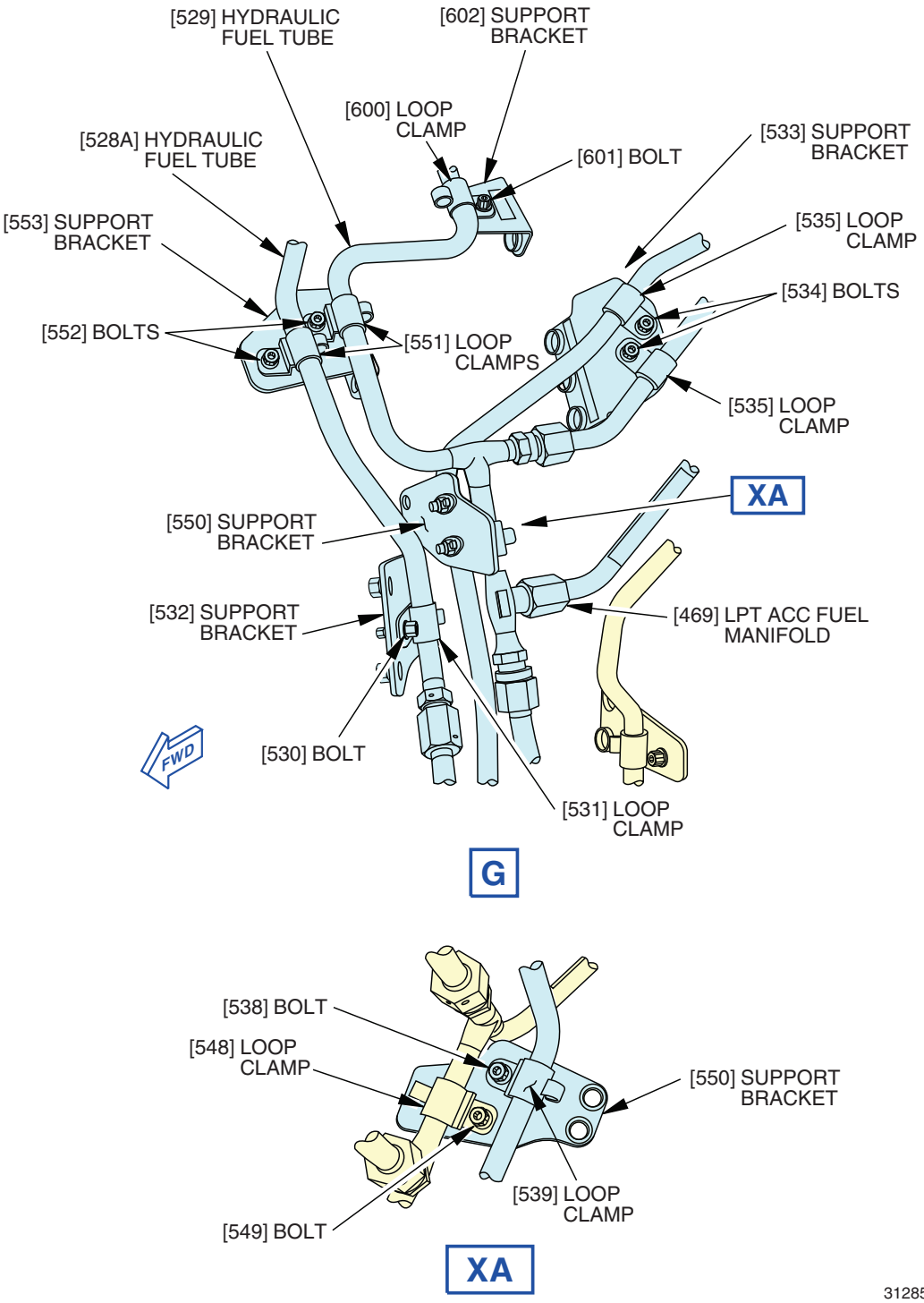
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 15 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 16 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

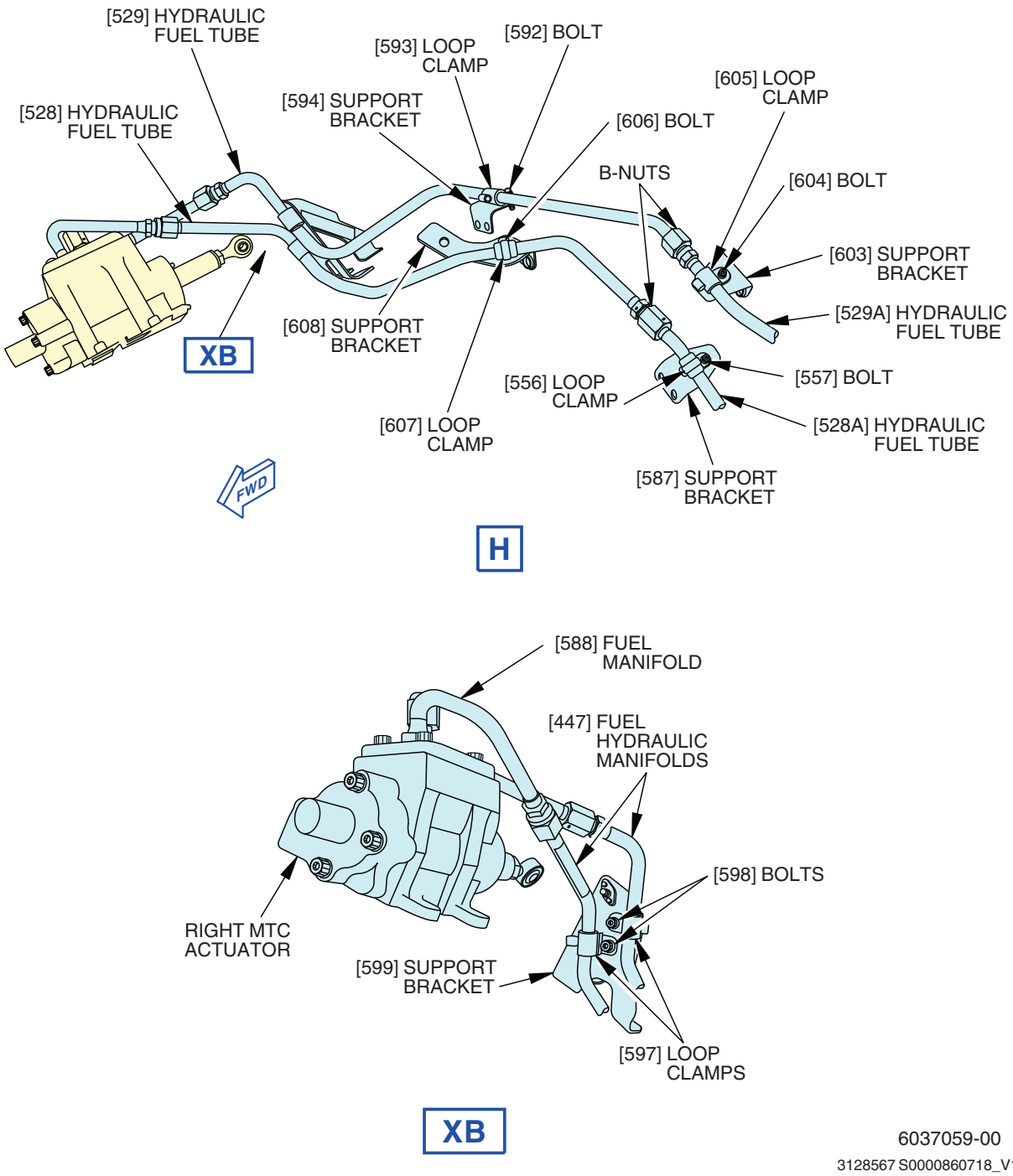


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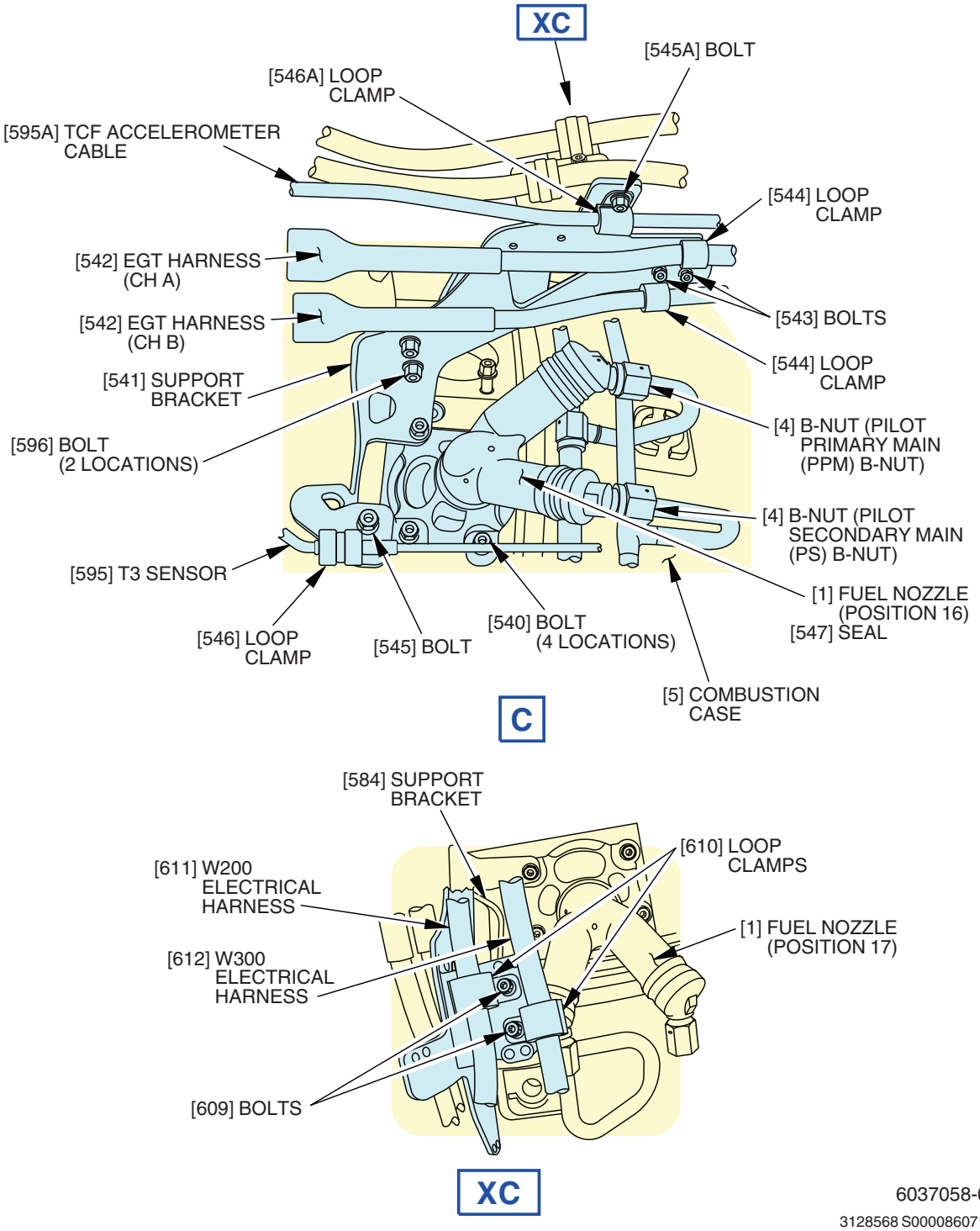
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 17 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 18 of 19)

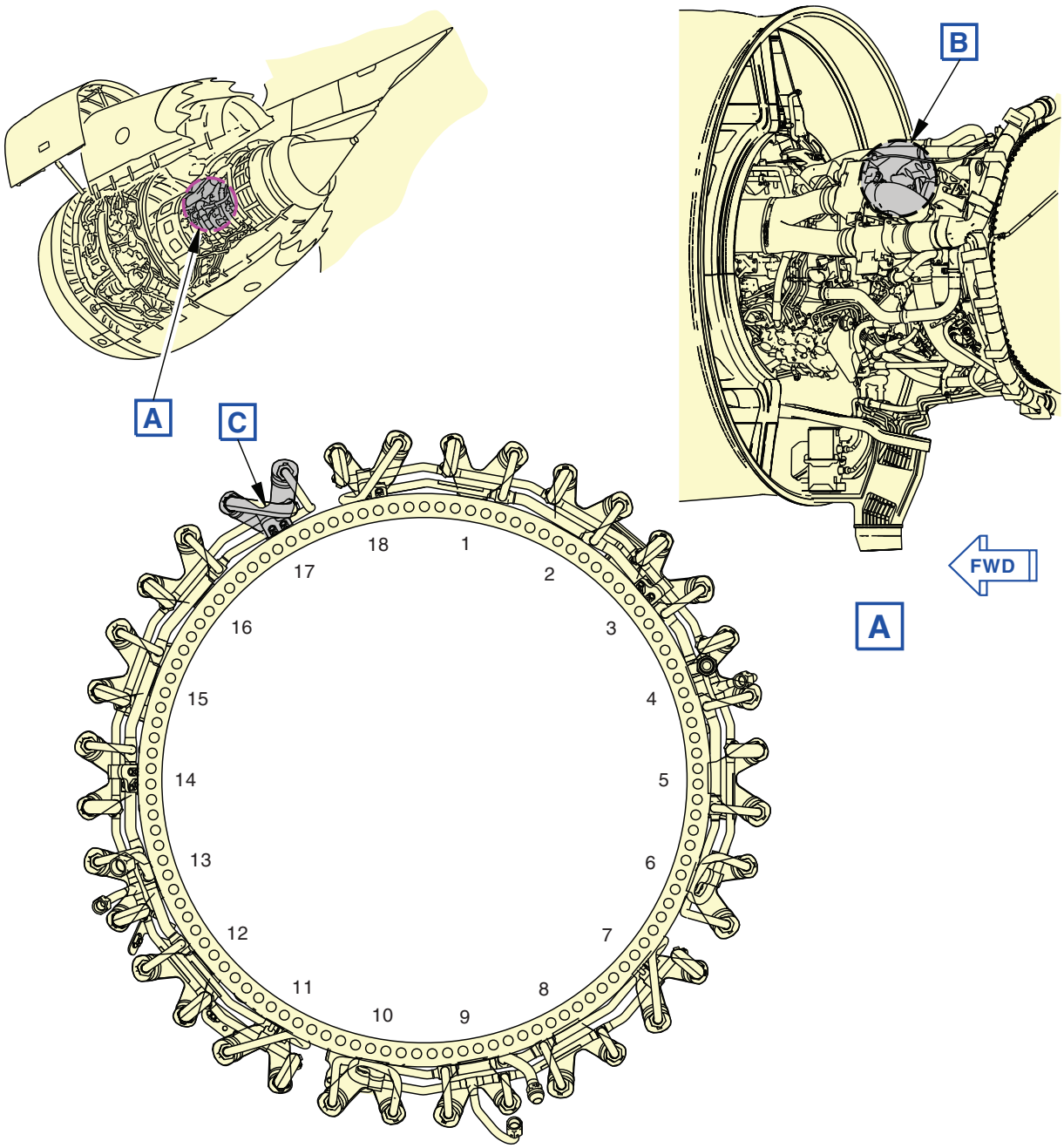
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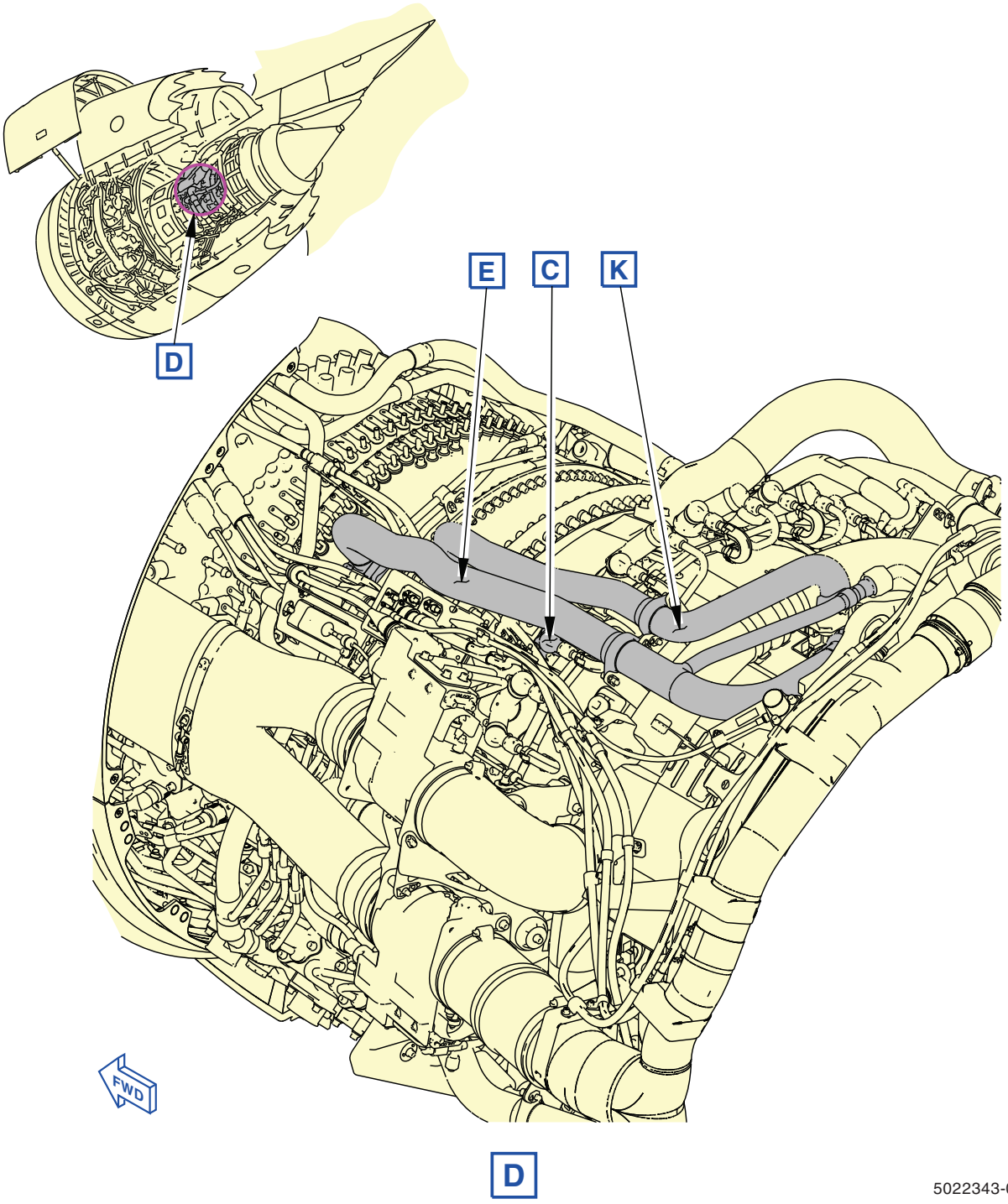
Fuel Nozzle (Position 16) Installation
Figure 417/73-11-05-990-903-G00 (Sheet 19 of 19)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 1 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

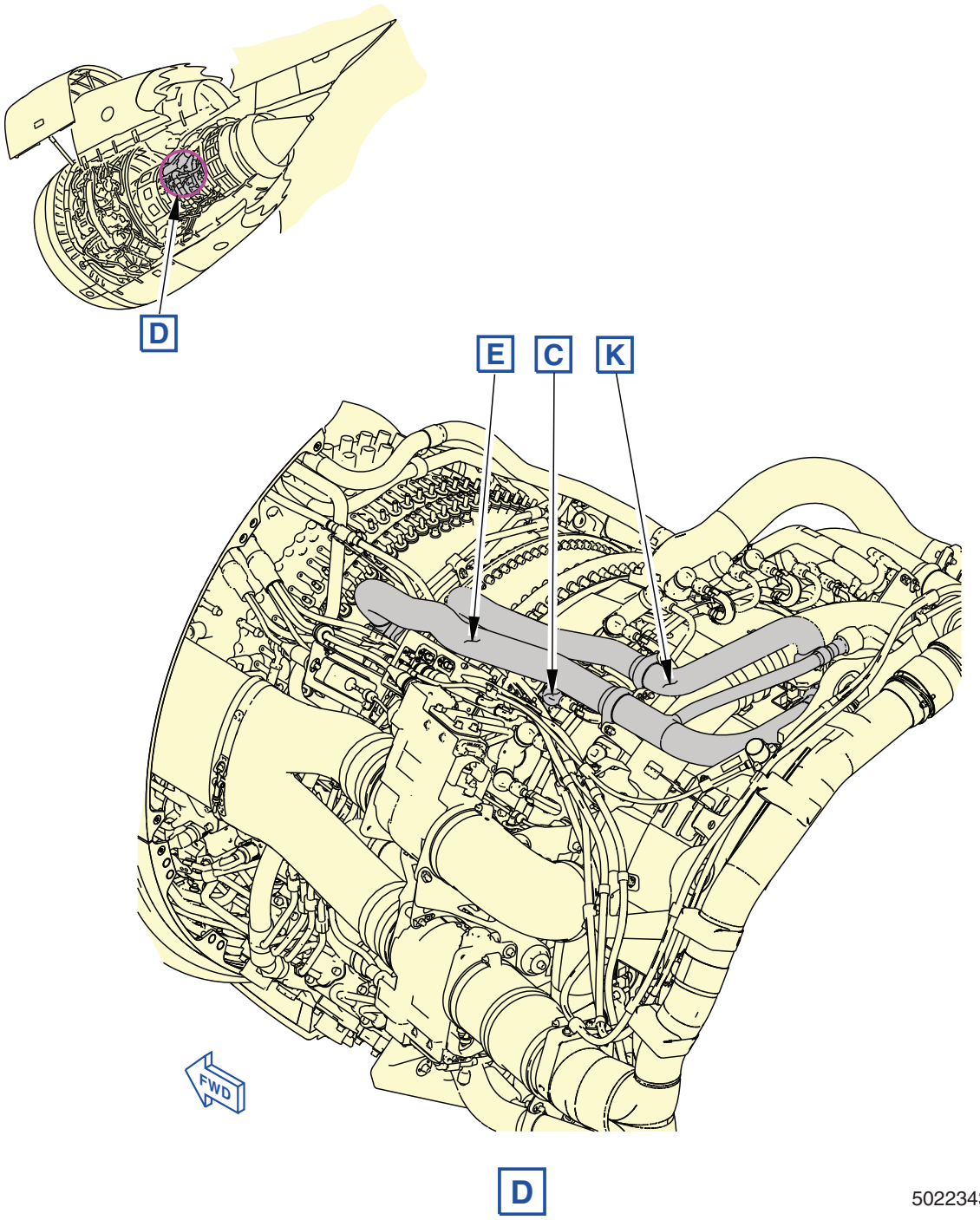


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Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 2 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

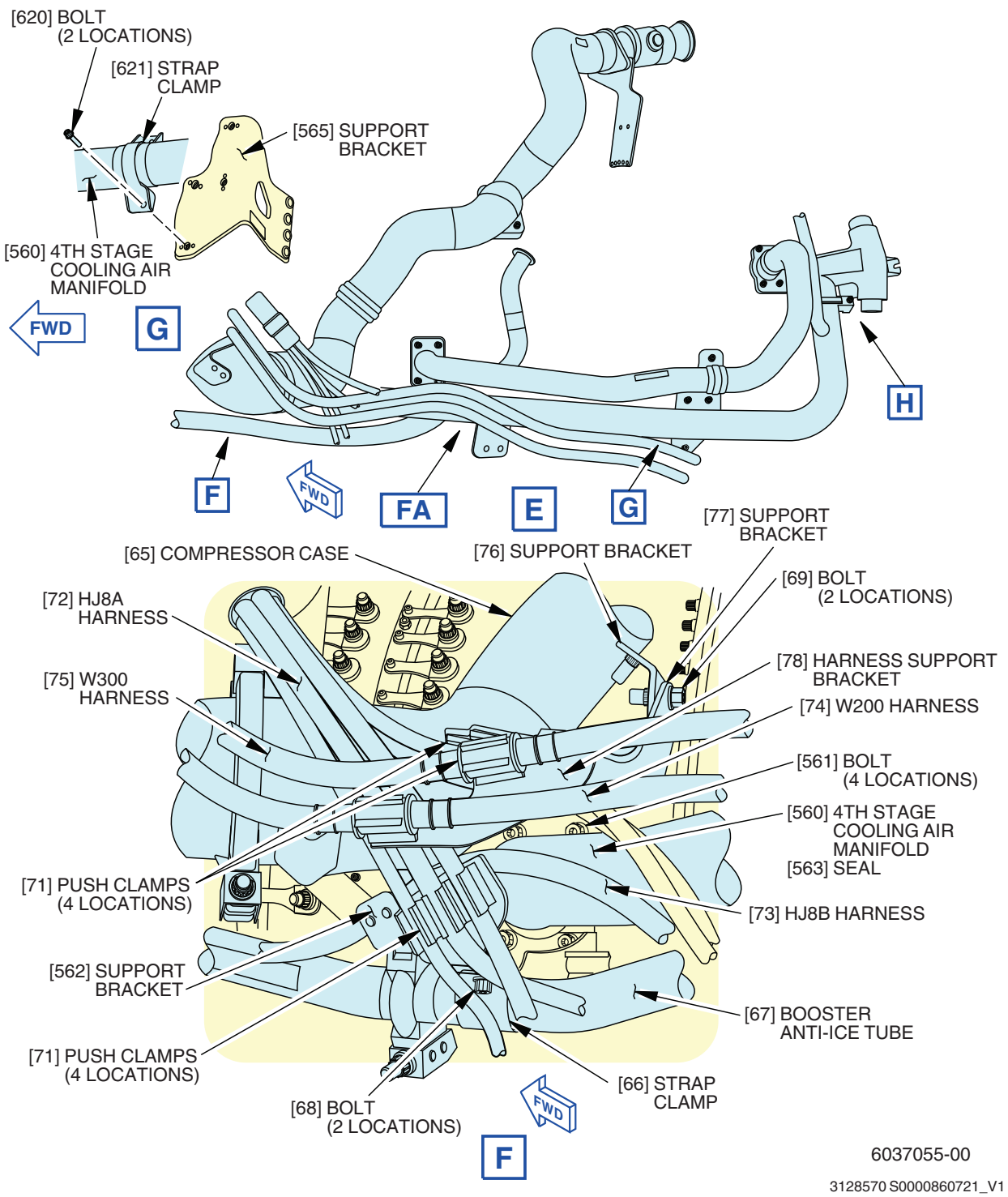


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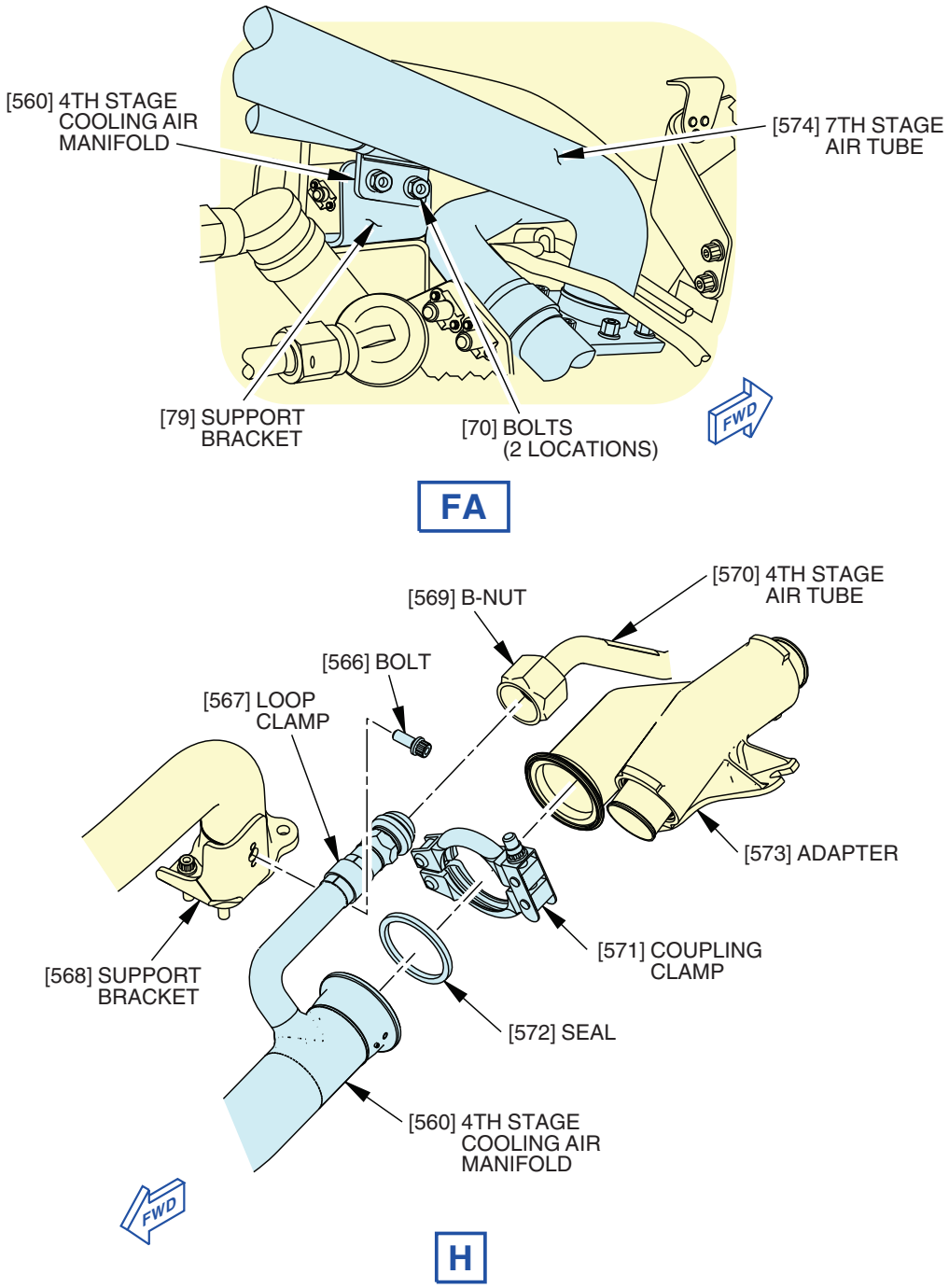
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 3 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 4 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

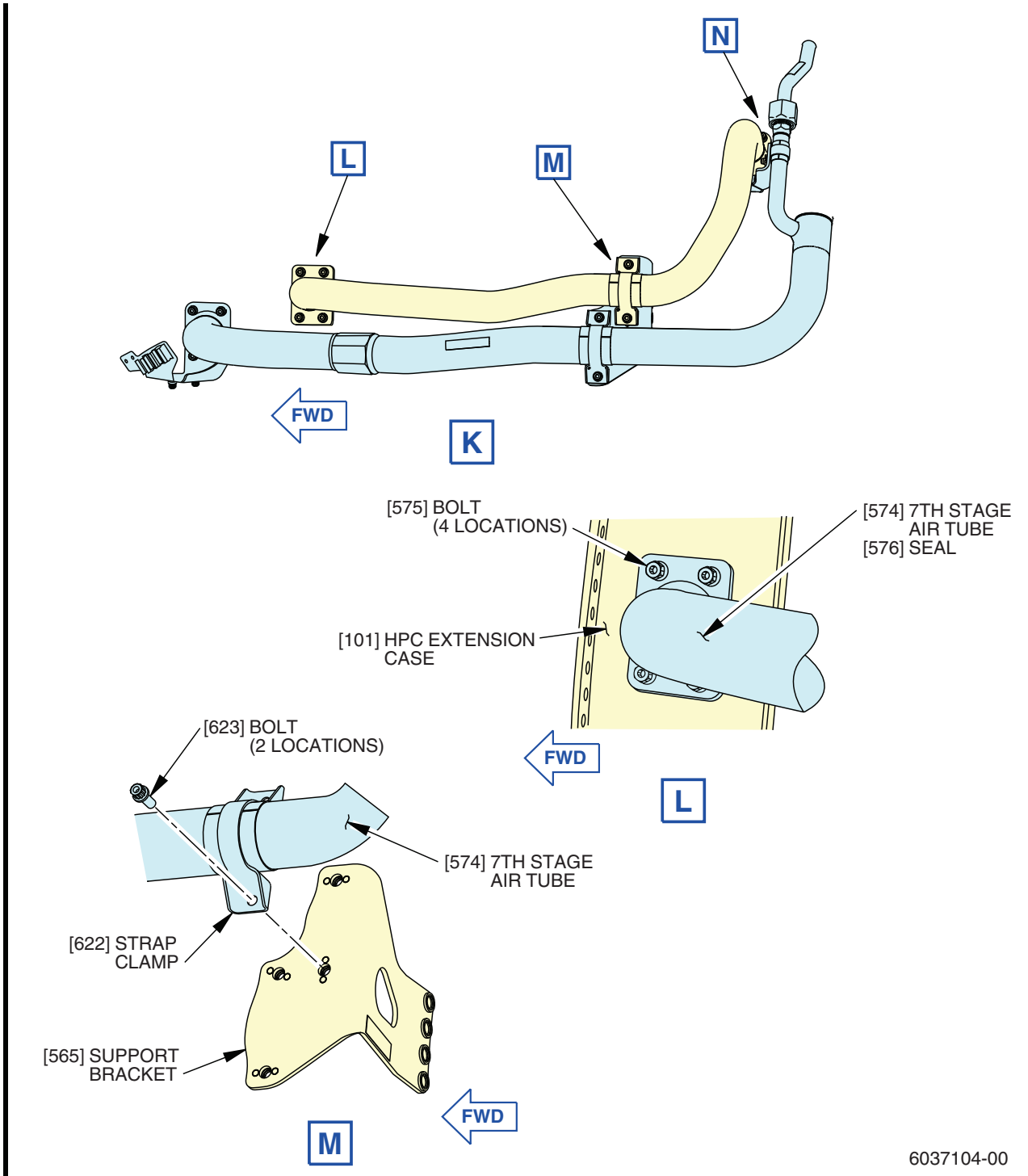


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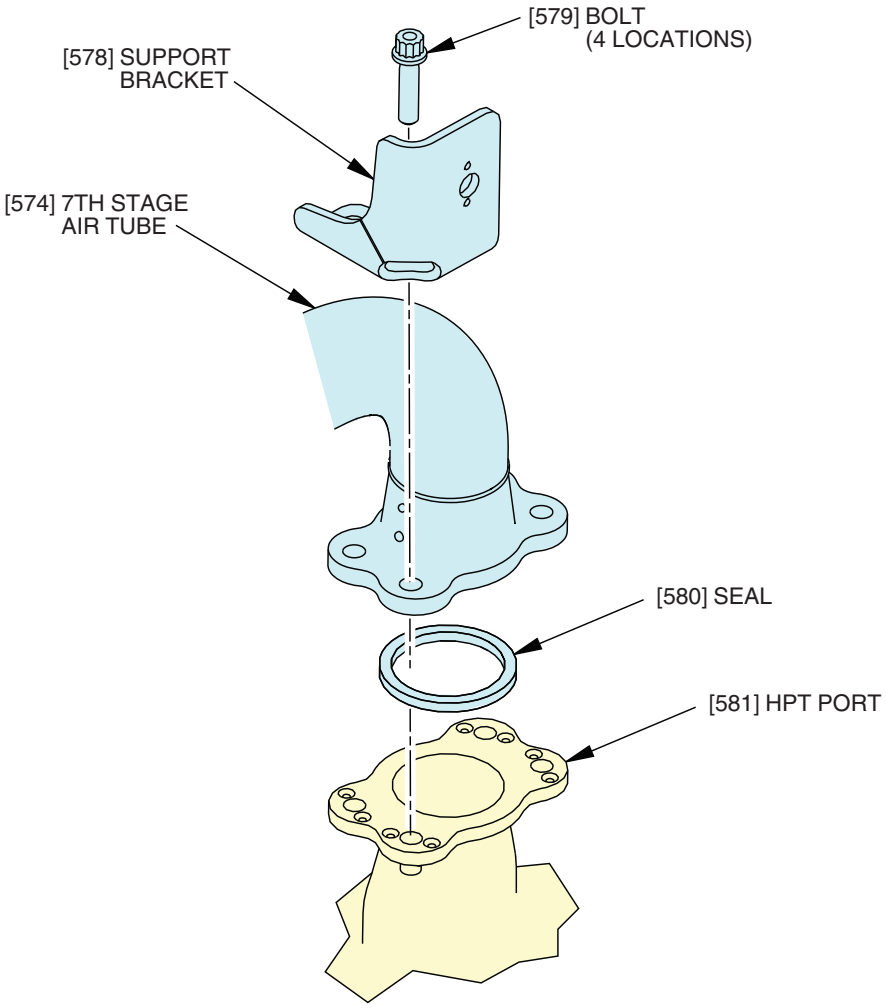
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 5 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 6 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



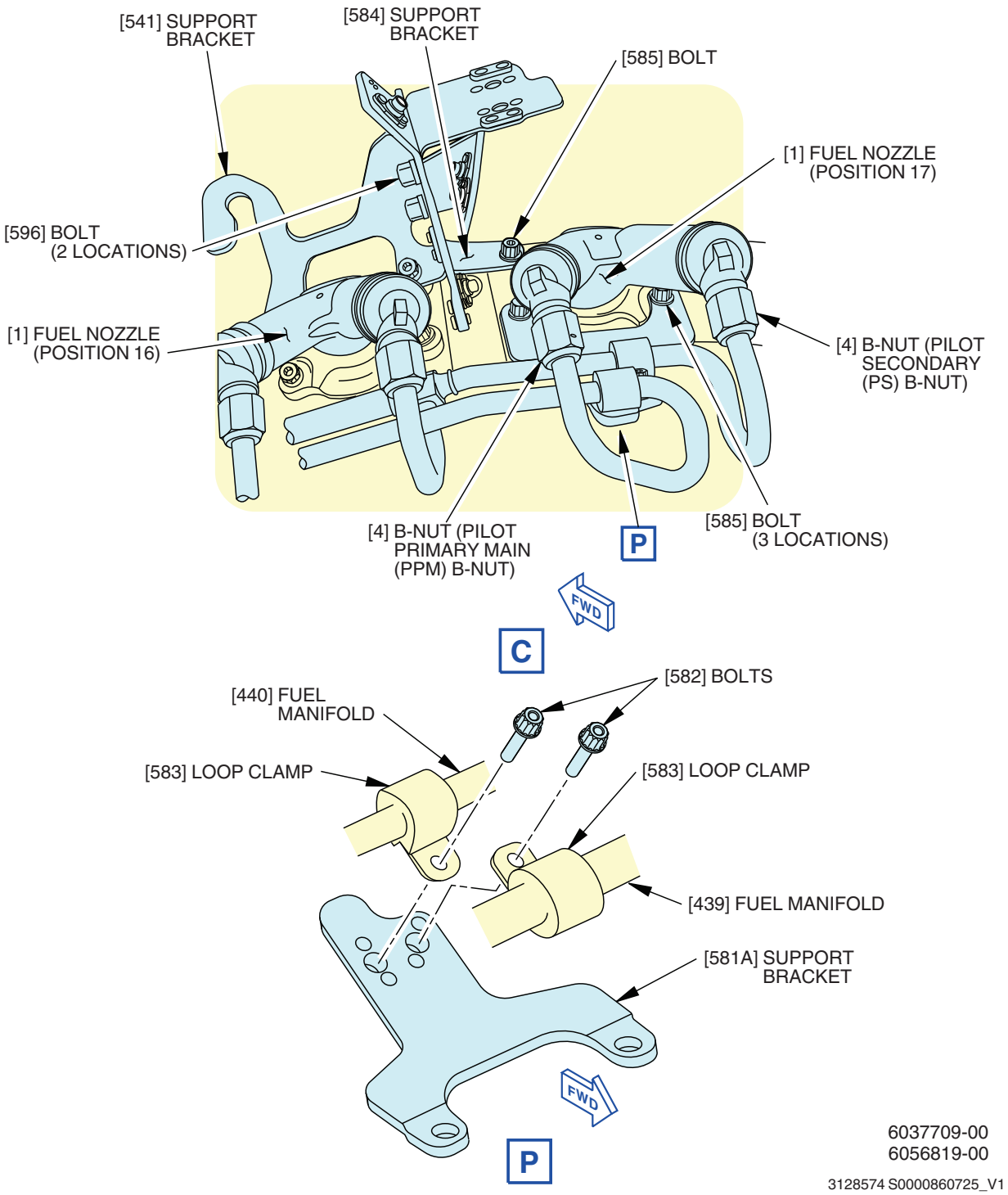
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Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 7 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

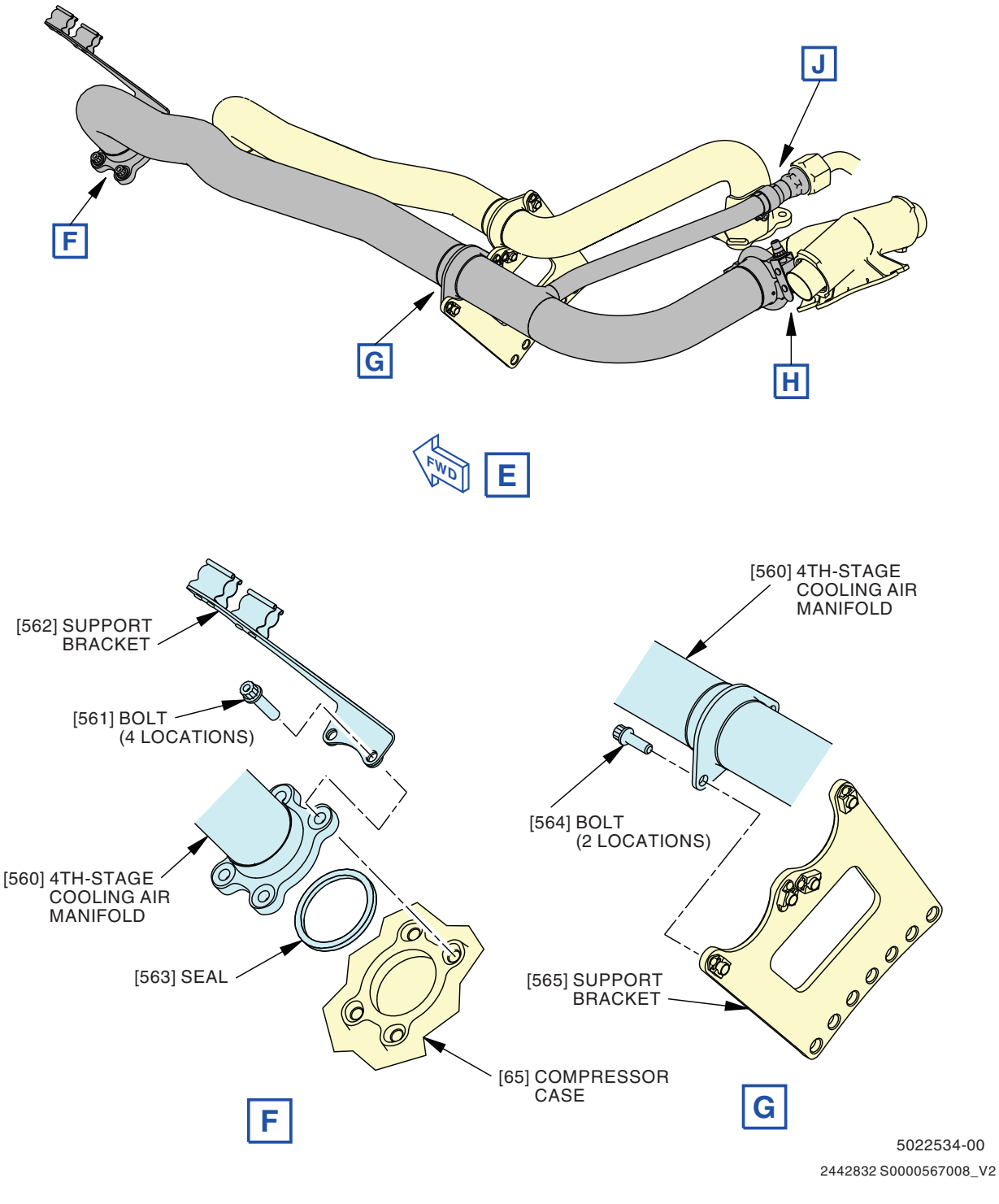


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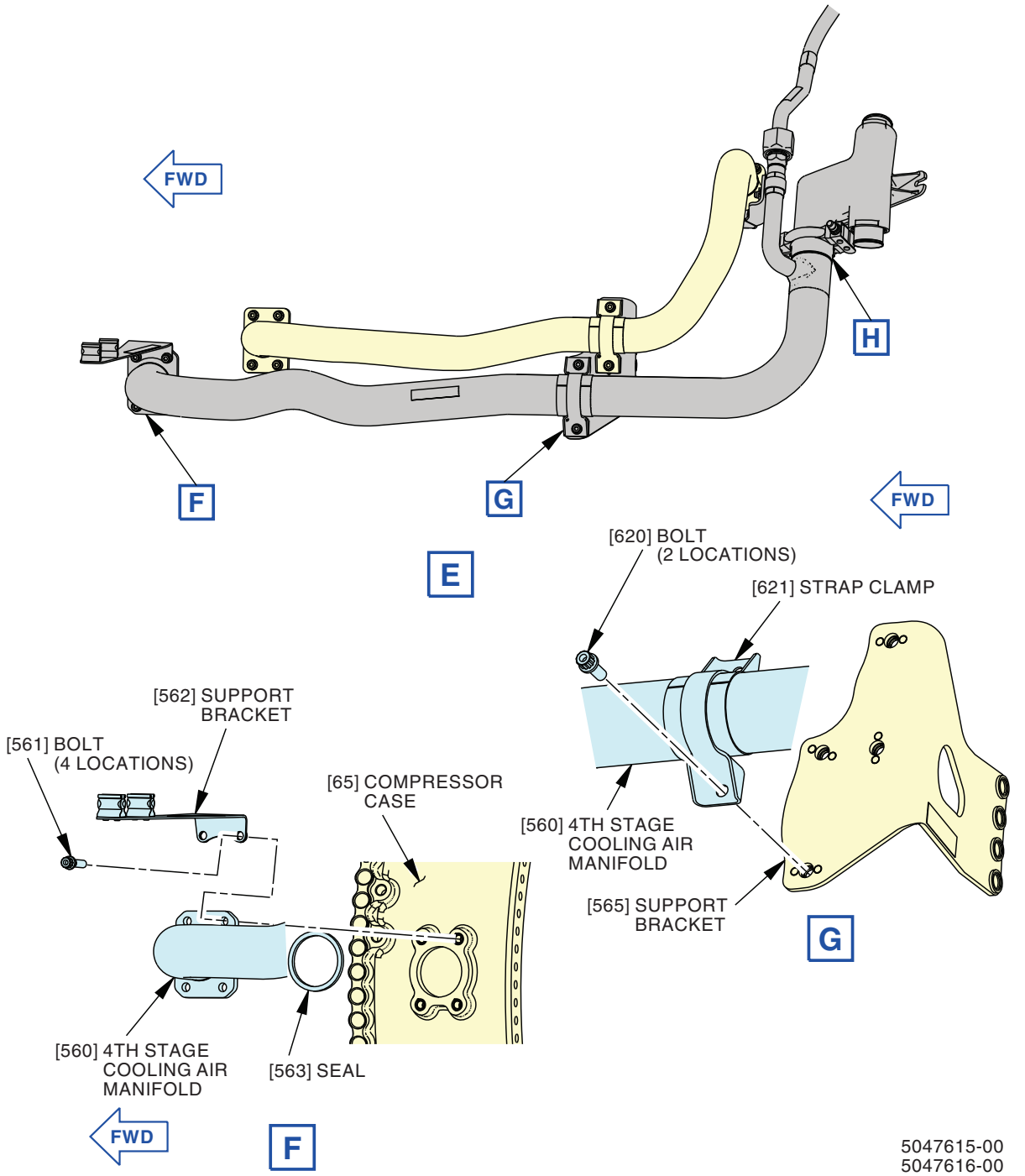
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 8 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



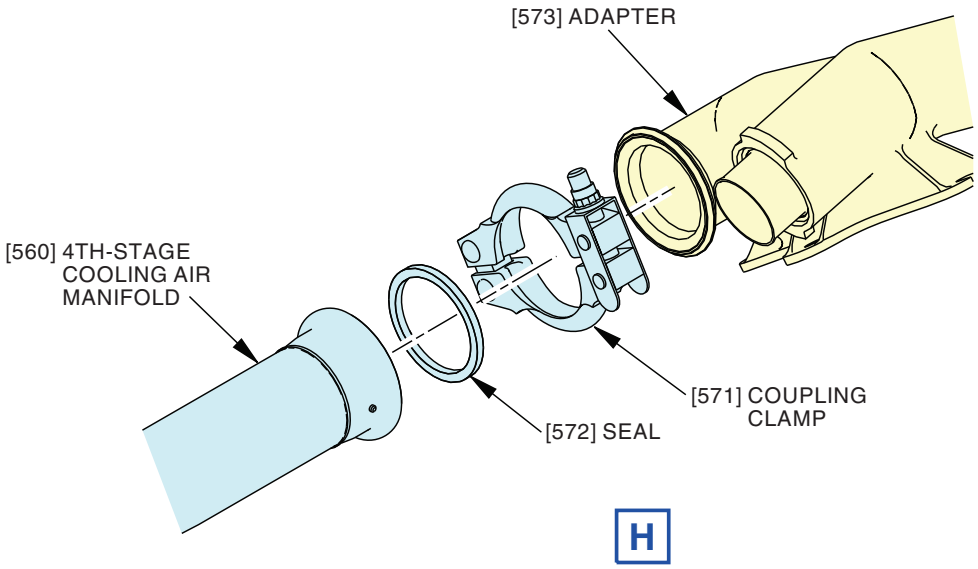
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 9 of 16)

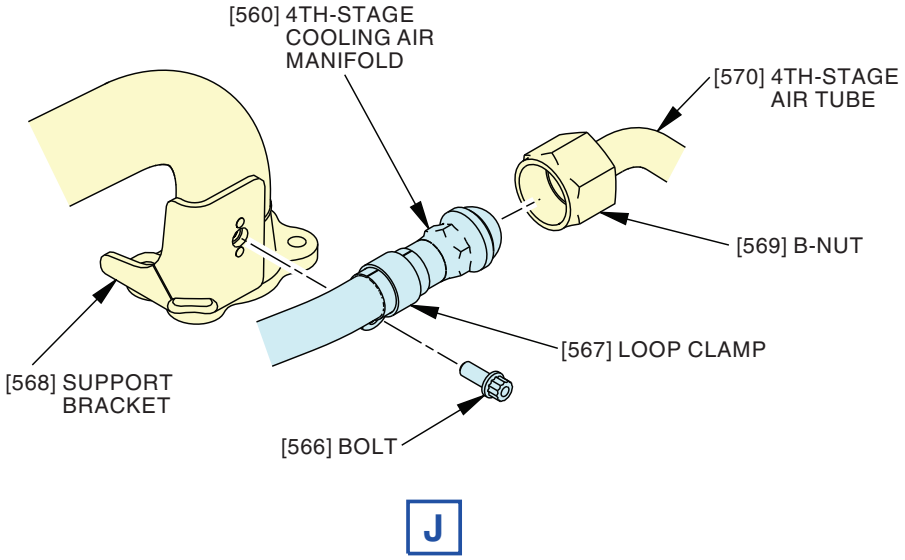
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 10 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



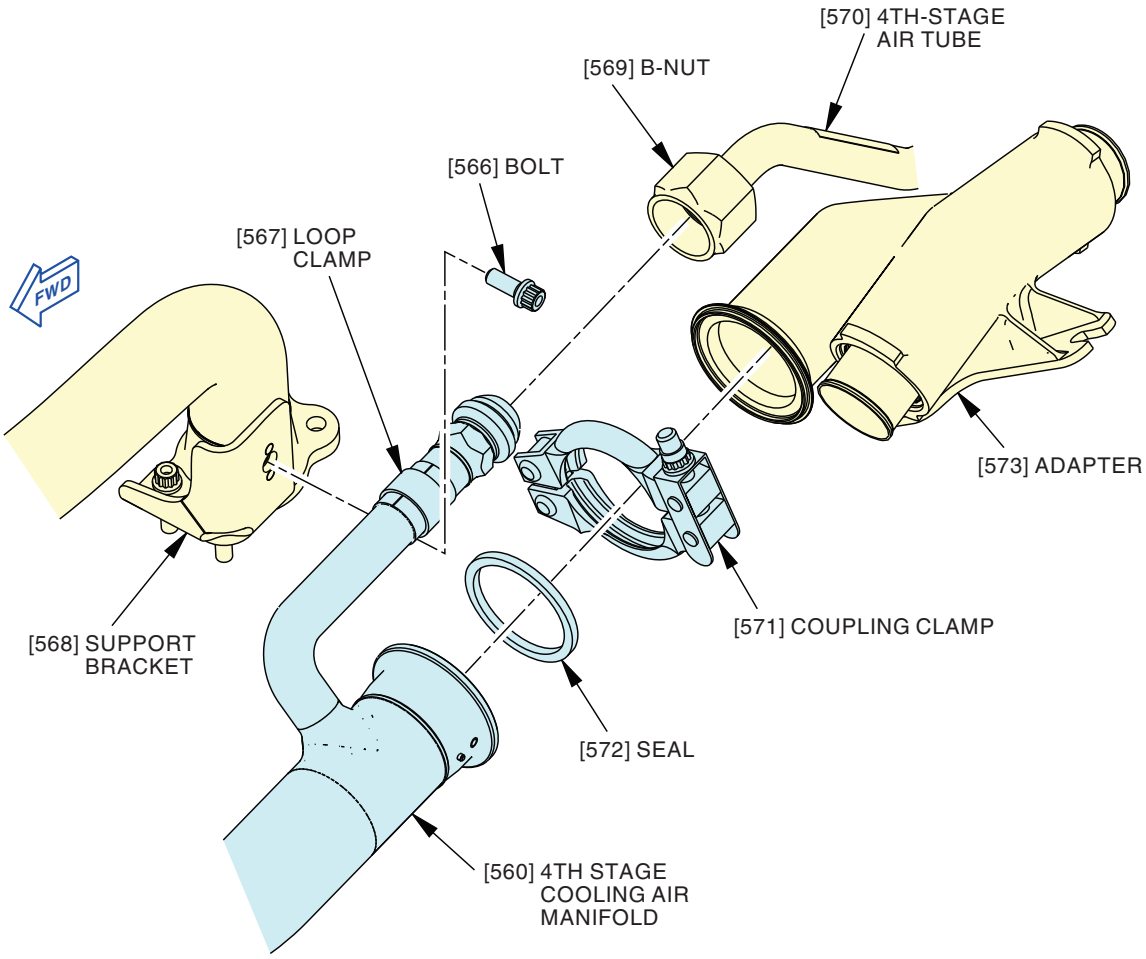


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Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 11 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

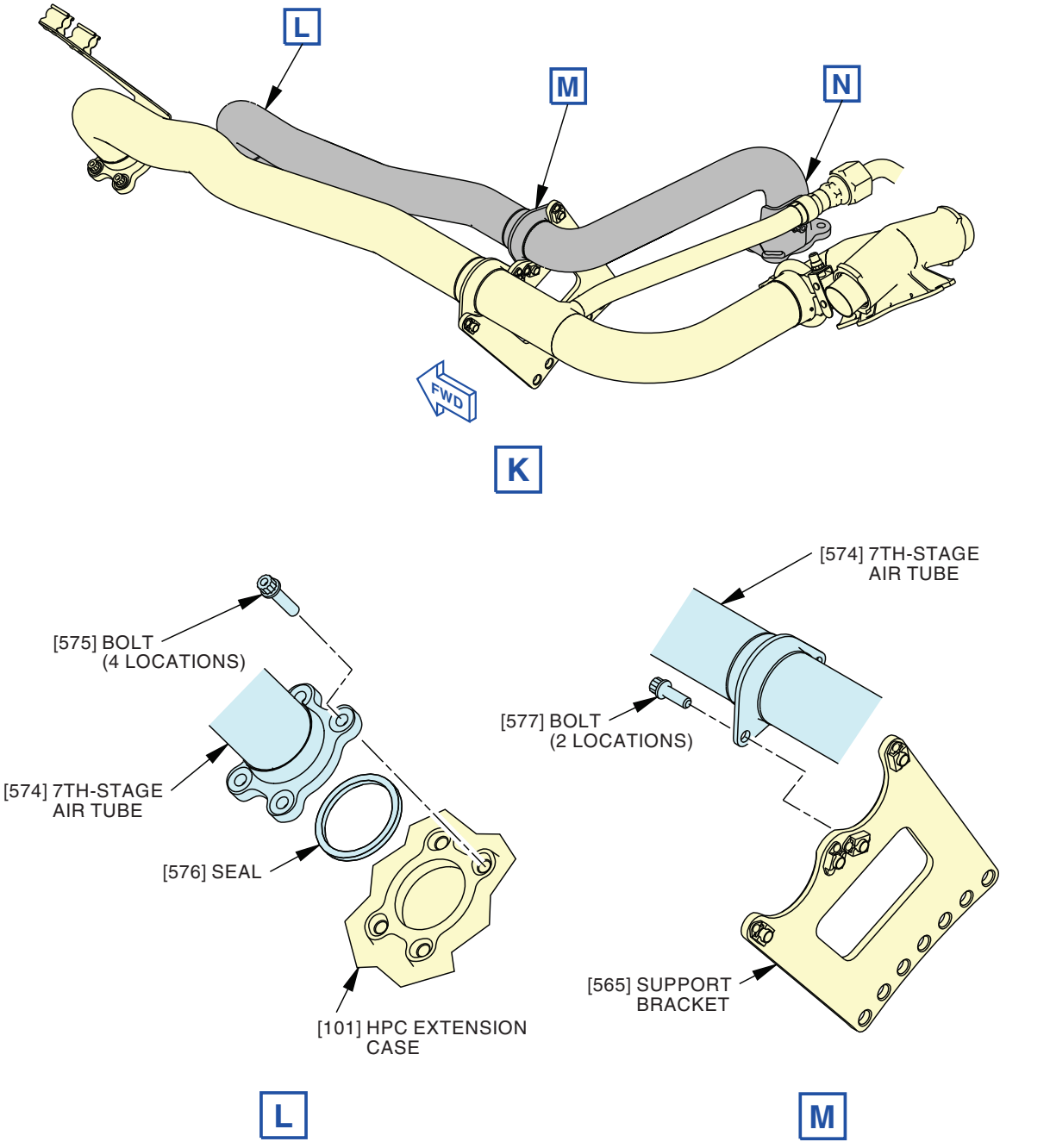


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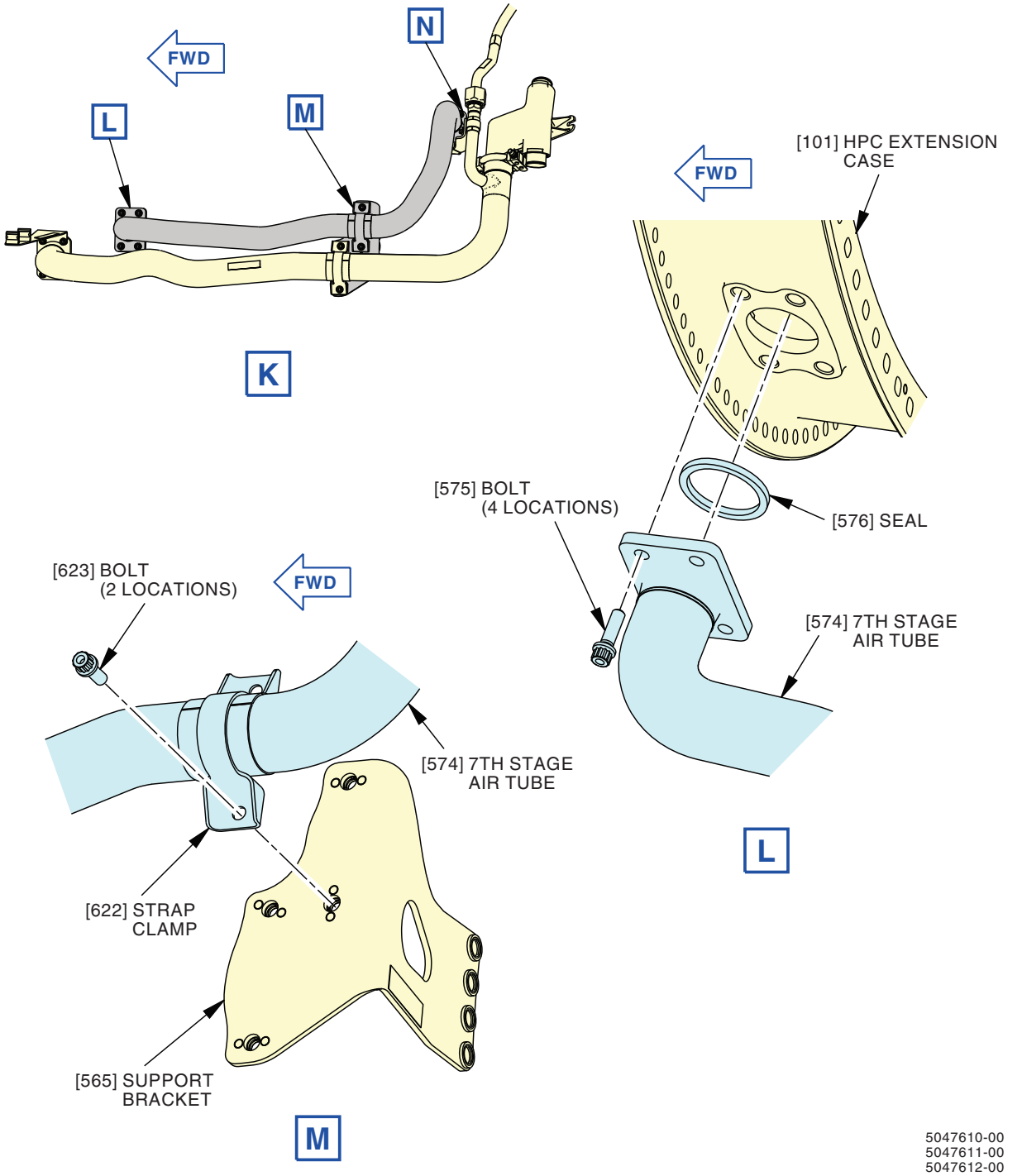
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 12 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



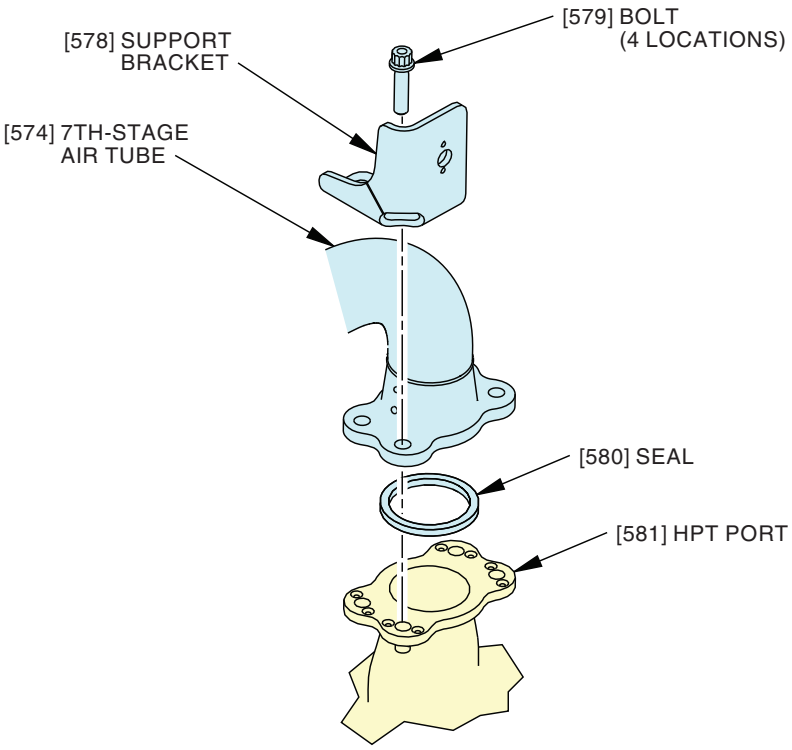
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 13 of 16)

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Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 14 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

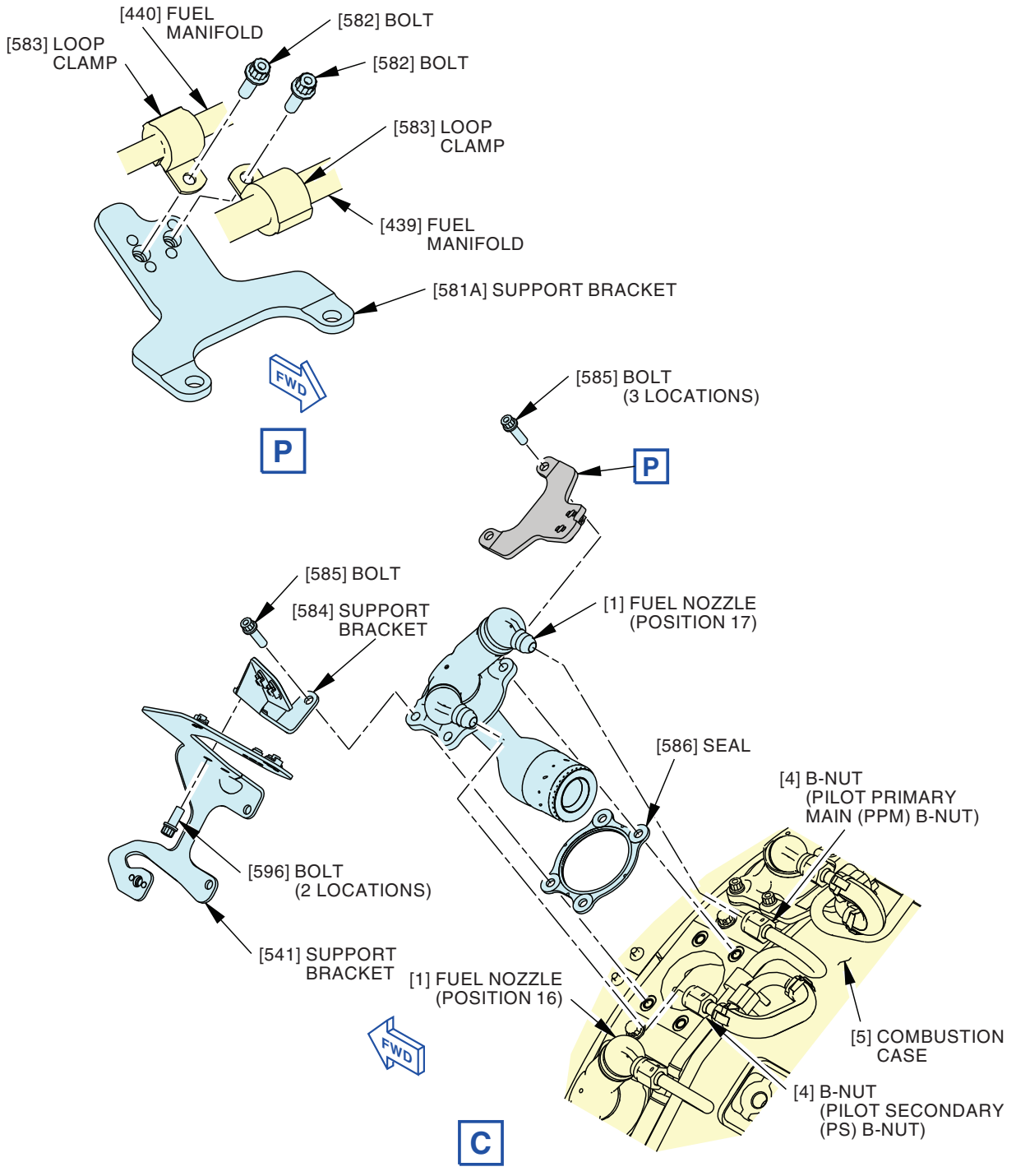


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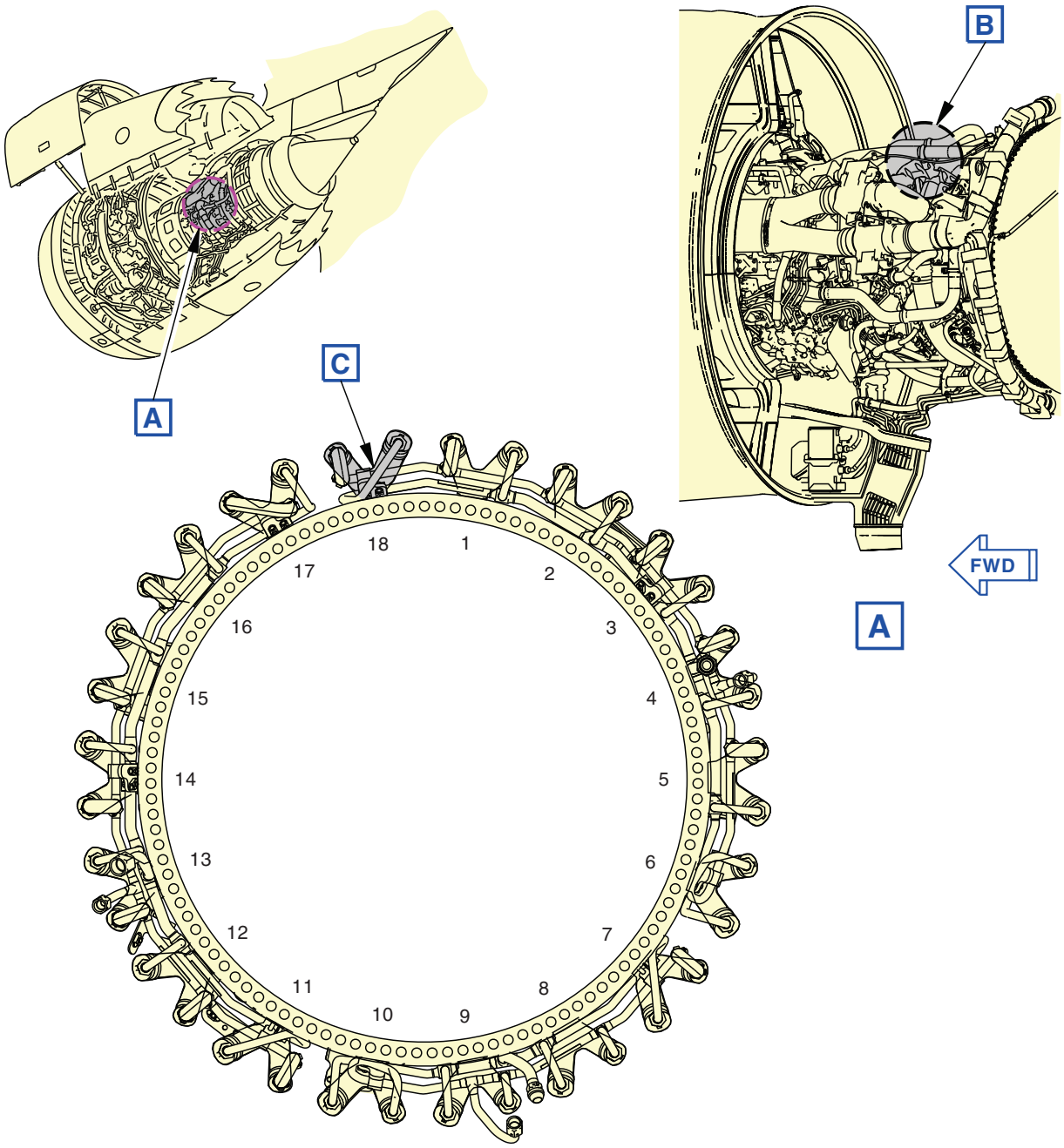
Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 15 of 16)

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Fuel Nozzle (Position 17) Installation
Figure 418/73-11-05-990-904-G00 (Sheet 16 of 16)

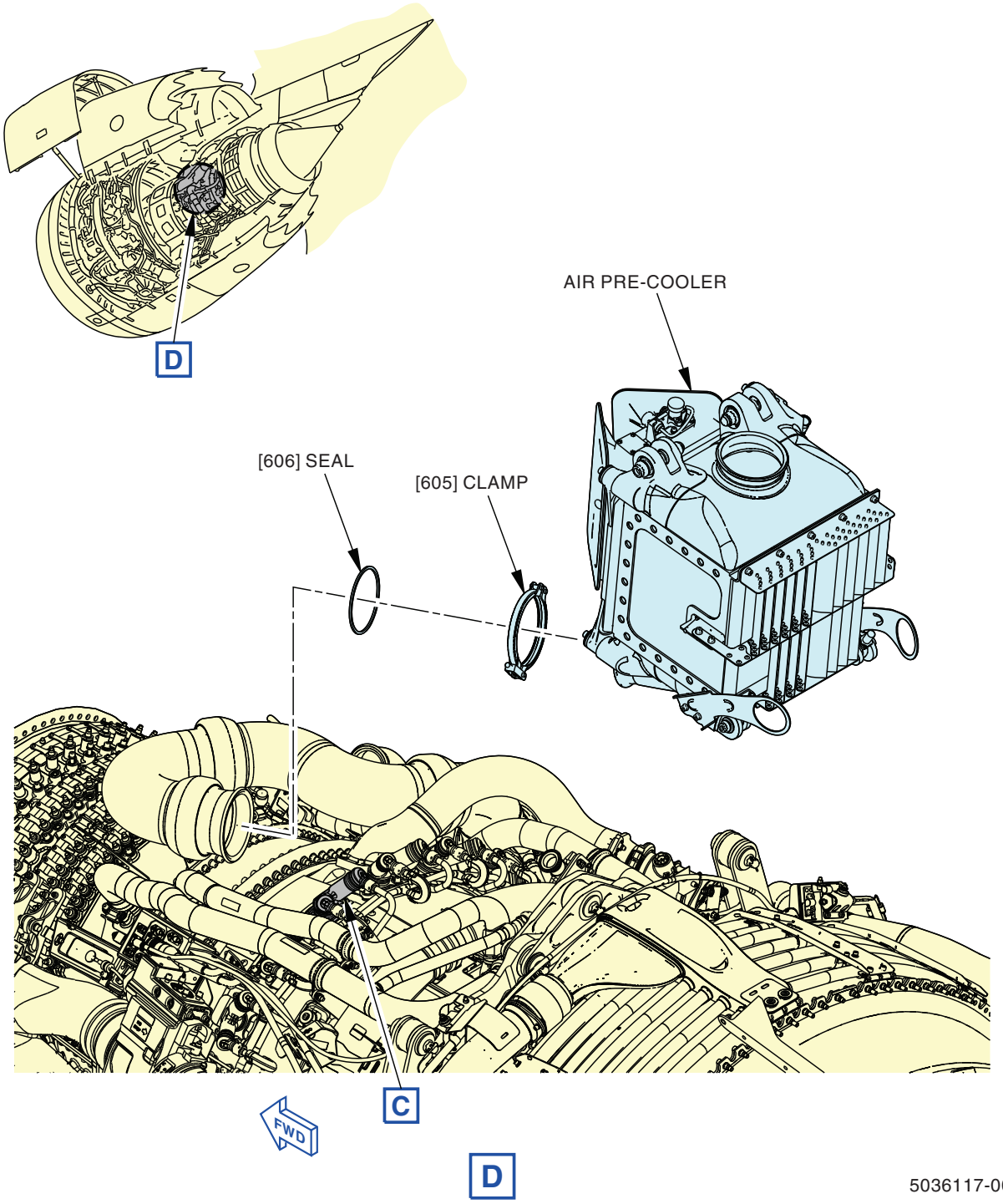
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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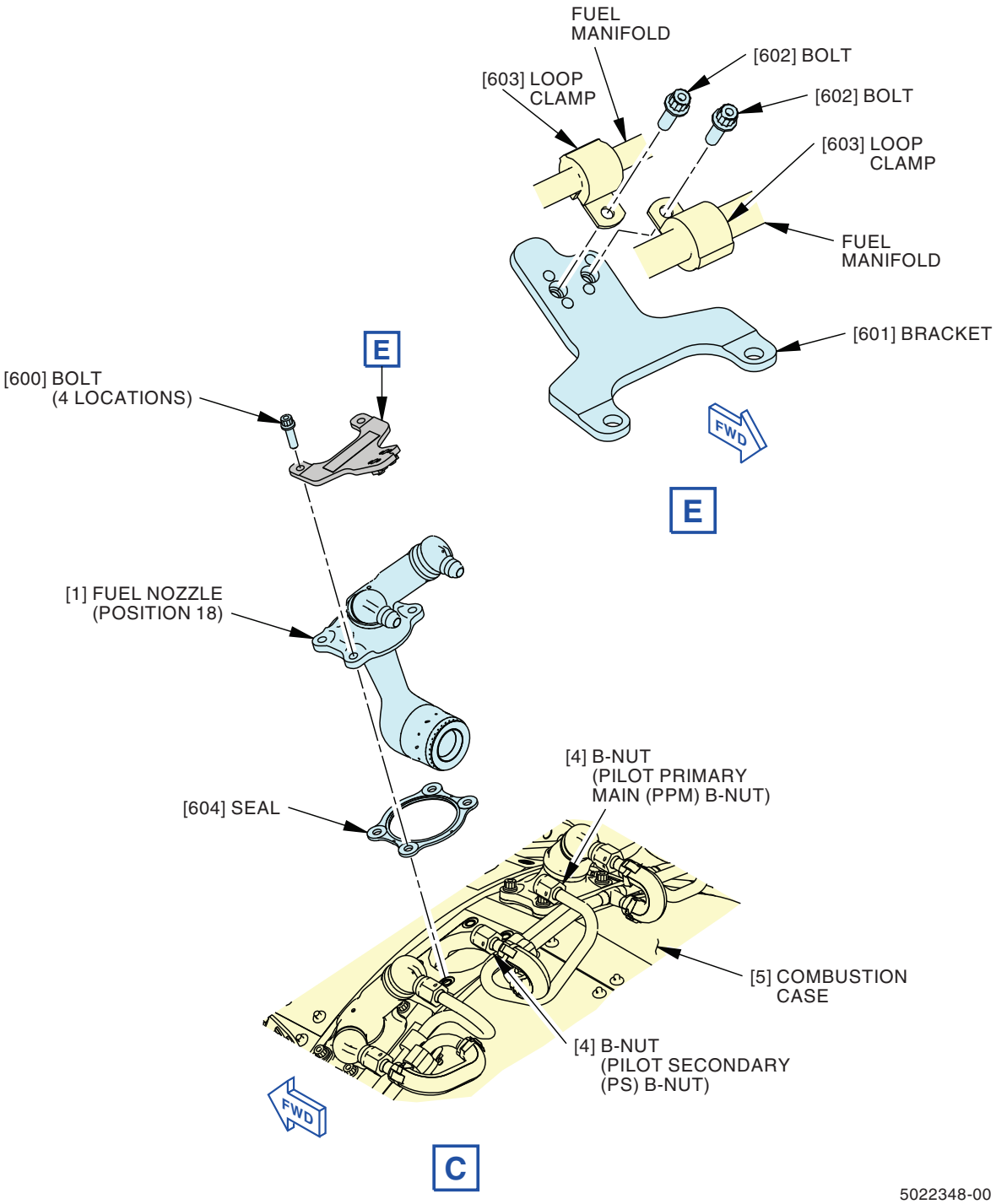
Fuel Nozzle (Position 18) Installation
Figure 419/73-11-05-990-905-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



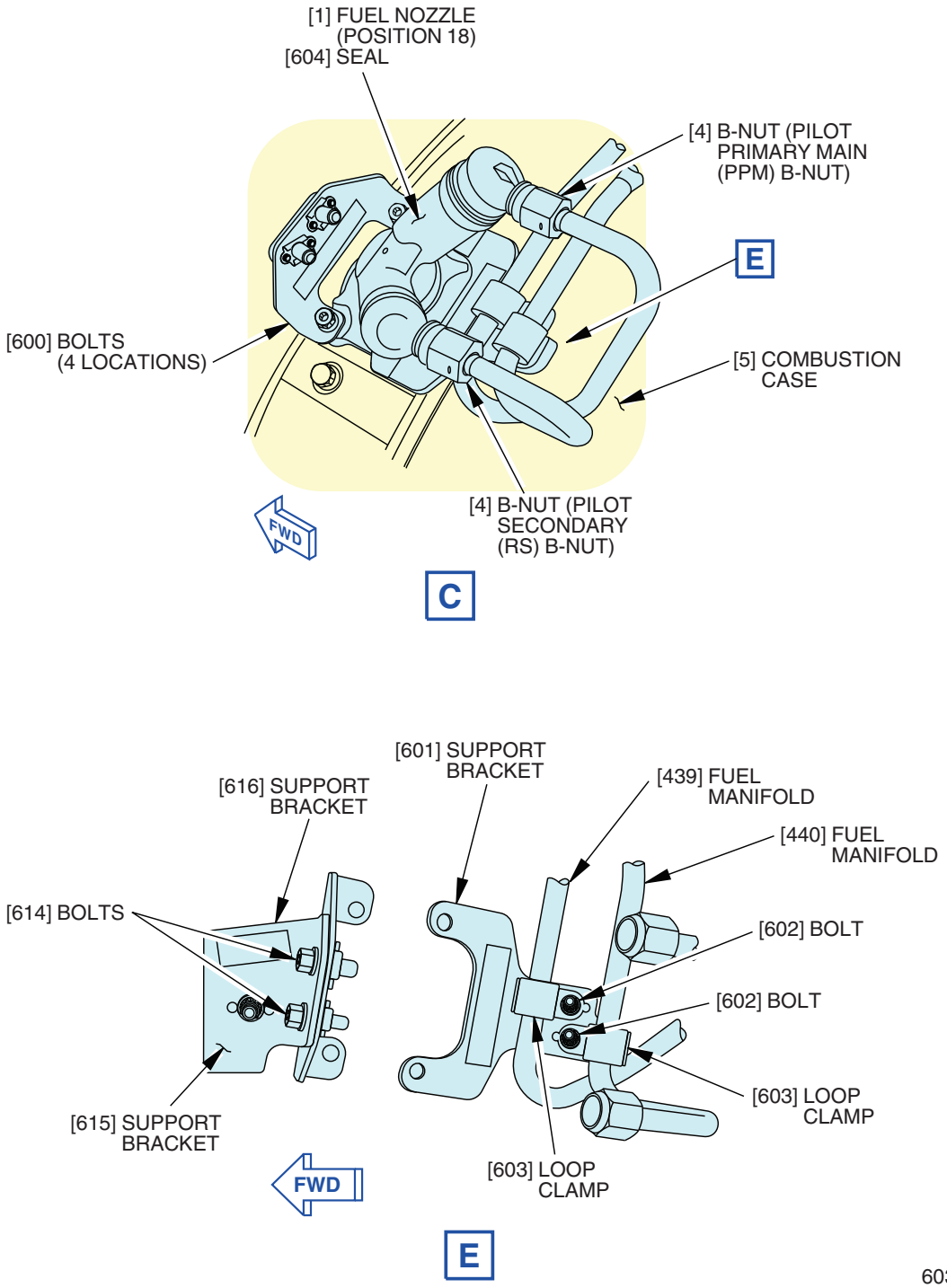
Fuel Nozzle (Position 18) Installation
Figure 419/73-11-05-990-905-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



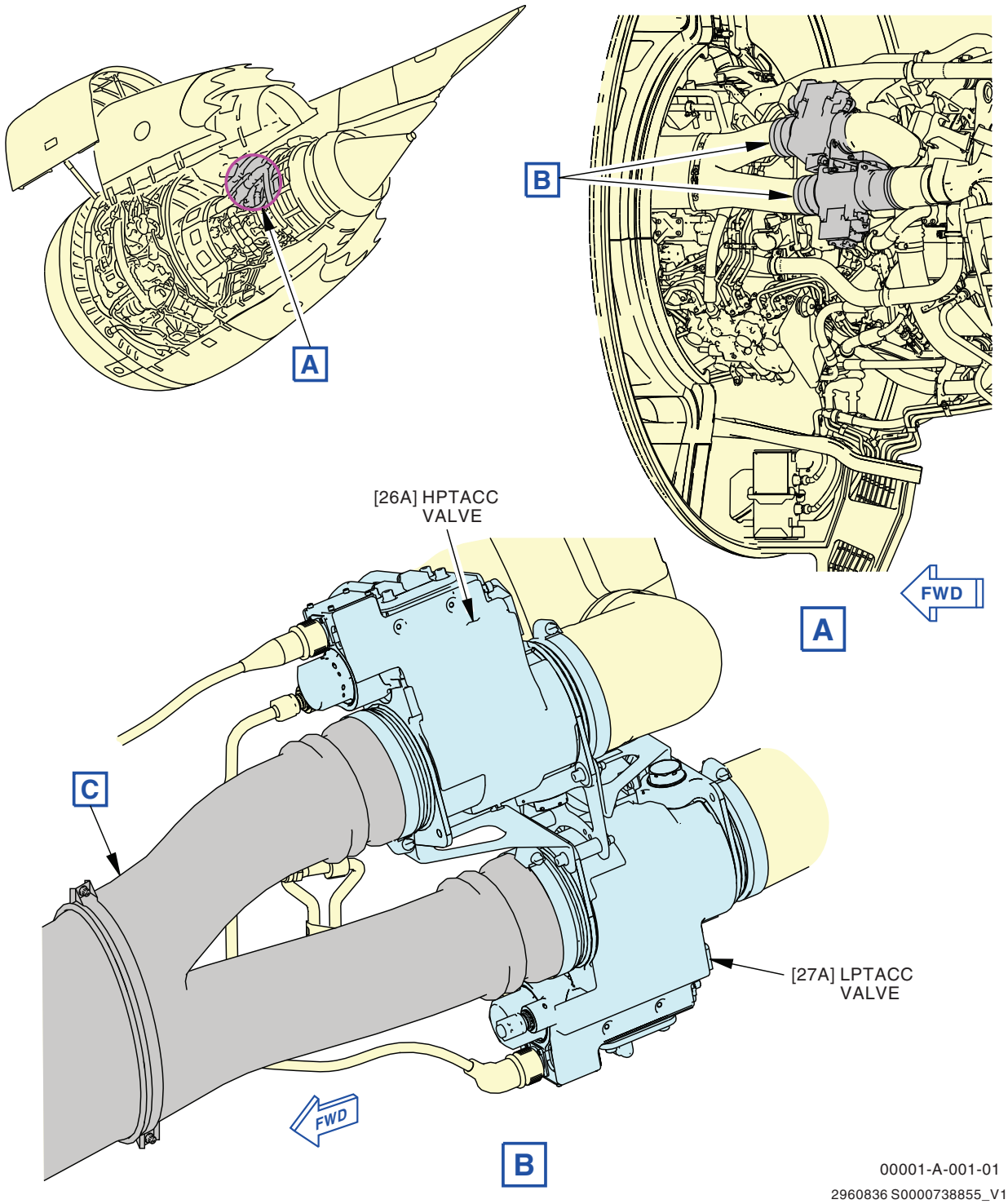
Fuel Nozzle (Position 18) Installation
Figure 419/73-11-05-990-905-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



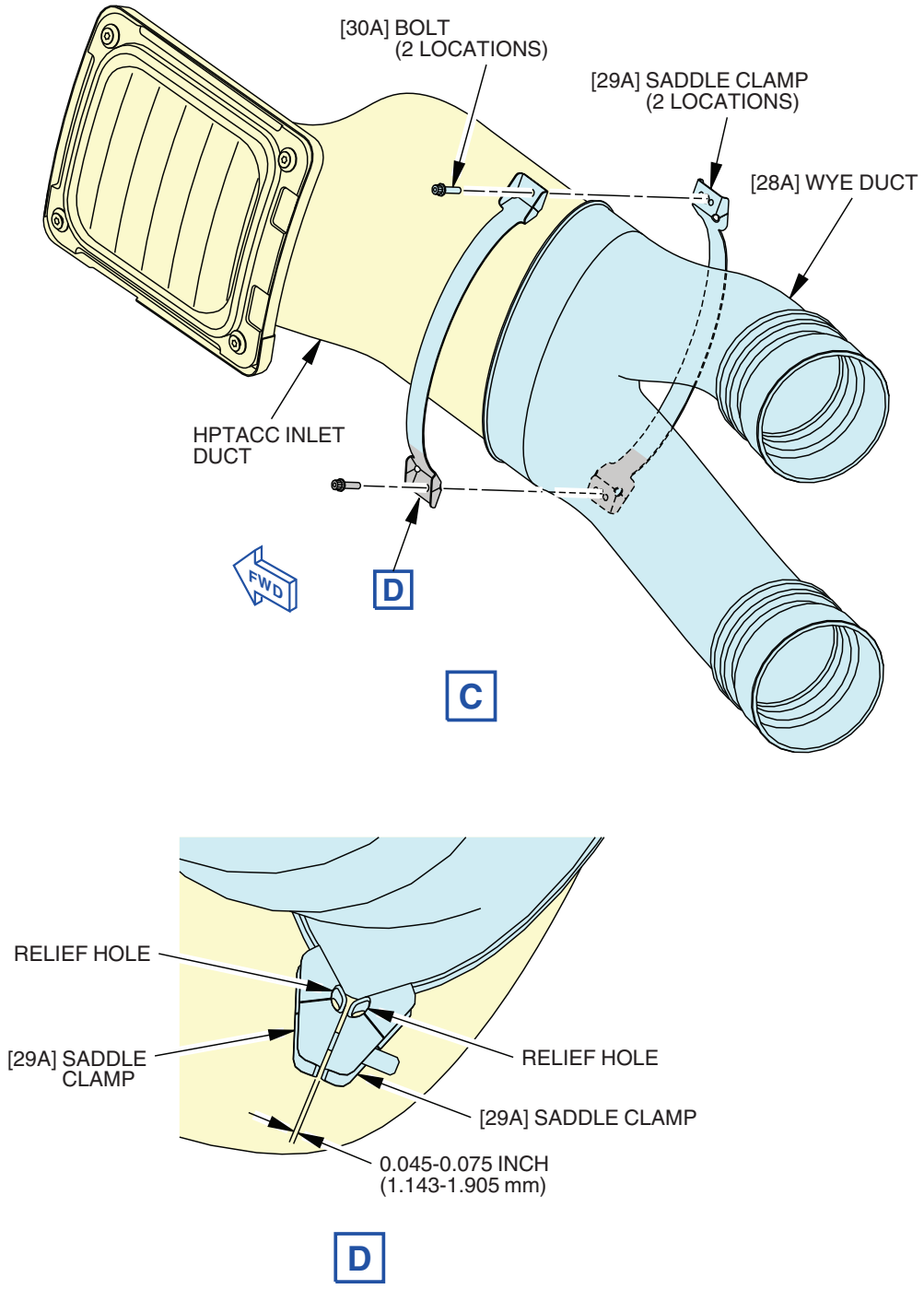
Fuel Nozzle (Position 18) Installation
Figure 419/73-11-05-990-905-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HPT and LPT Valve Installation
Figure 420/73-11-05-990-906-G00 (Sheet 1 of 2)

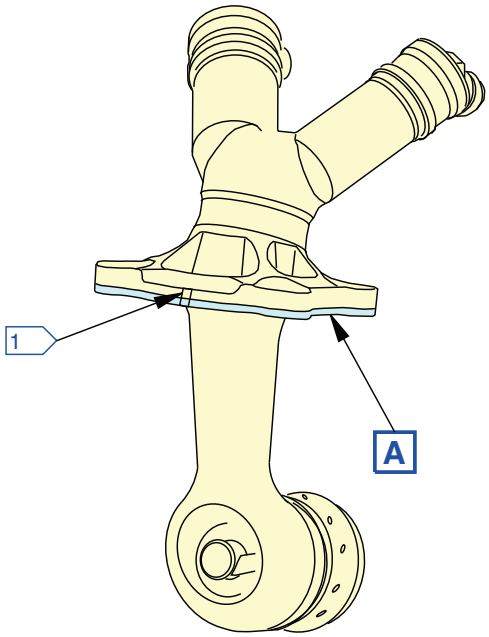
ECCN 9E991 BOEING PROPRIETARY - See title page for details

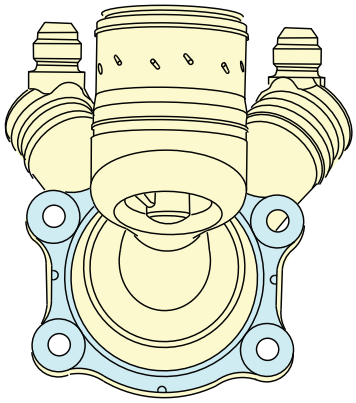


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HPT and LPT Valve Installation
Figure 420/73-11-05-990-906-G00 (Sheet 2 of 2)

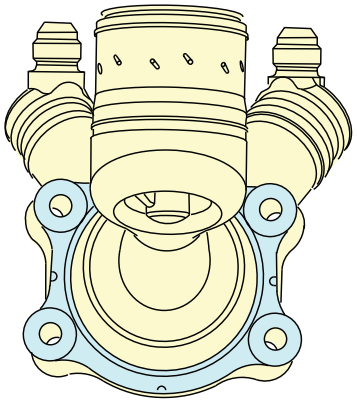
ECCN 9E991 BOEING PROPRIETARY - See title page for details





SEAL ALIGNED
(CORRECT POSITION)

A



SEAL NOT ALIGNED
(INCORRECT POSITION)

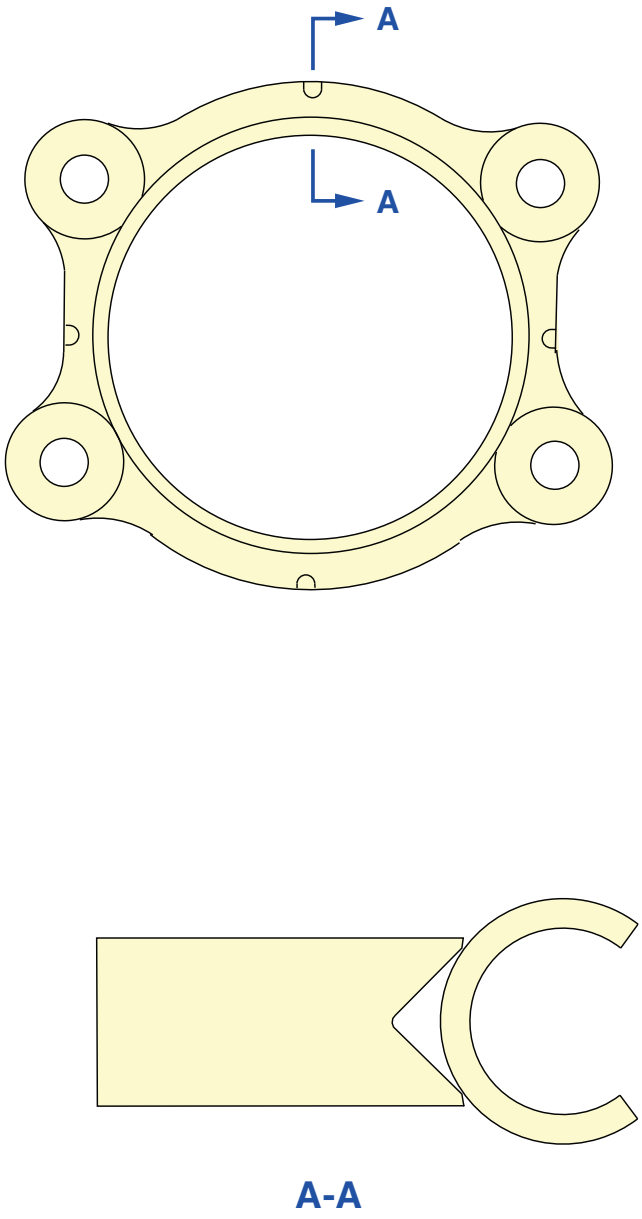
A

1 THE MARK ON THE SEAL MUST BE ALIGNED WITH THE MARK ON THE FUEL NOZZLE.

2953777 S0000731186_V1

Fuel Nozzle Seal Inspection
Figure 421/73-11-05-990-907-G00 (Sheet 1 of 2)

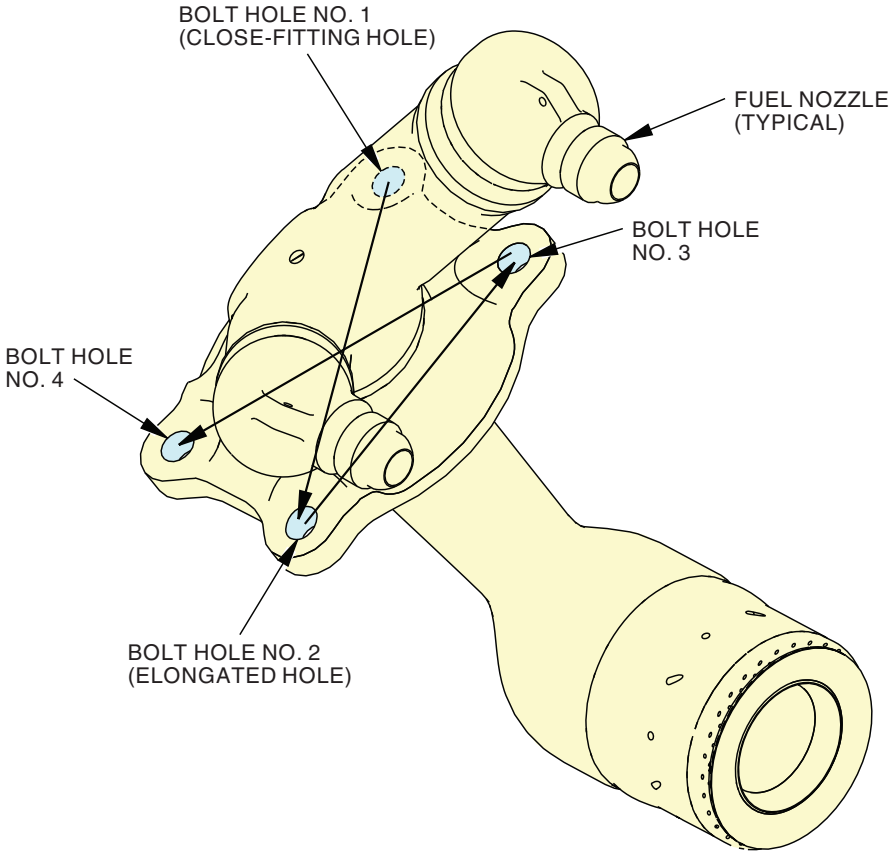
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2953822 S0000731208_V1

Fuel Nozzle Seal Inspection
Figure 421/73-11-05-990-907-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

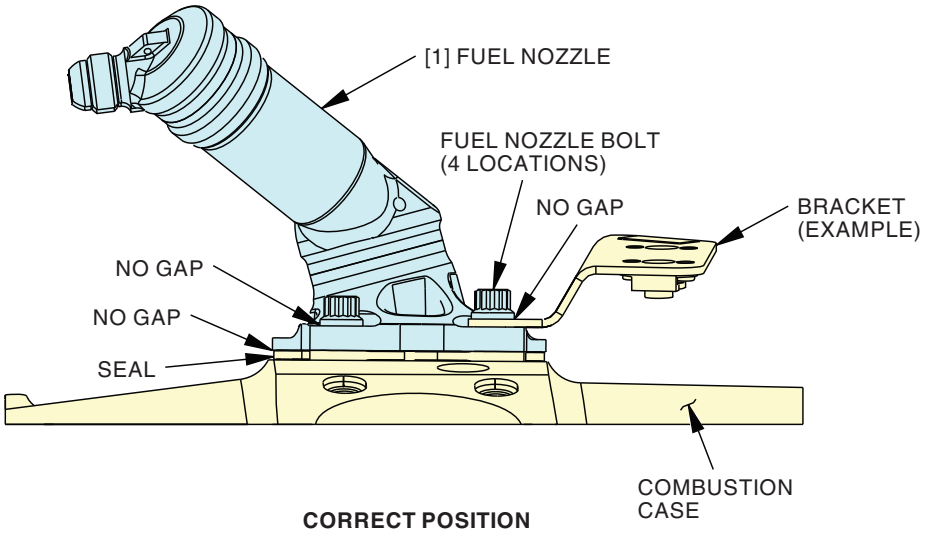


FUEL NOZZLE TORQUE SEQUENCE

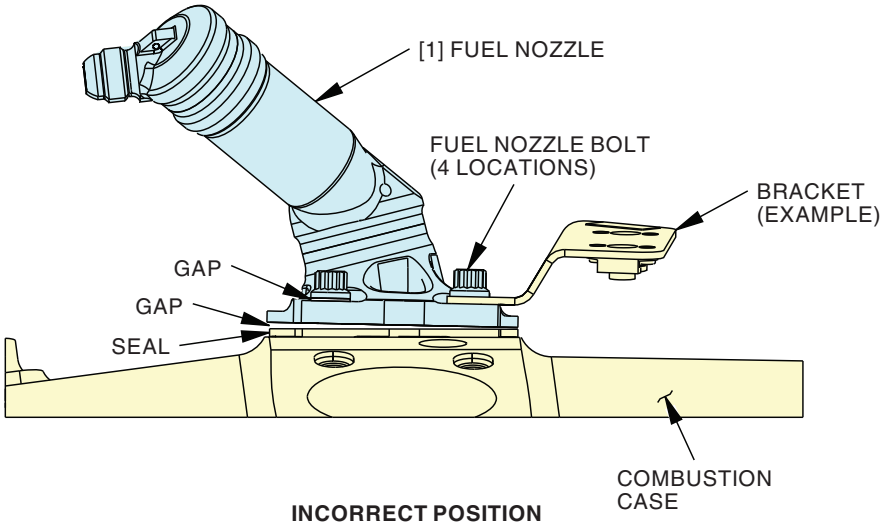
2947512 S0000724407_V2

Fuel Nozzle Torque Sequence
Figure 422/73-11-05-990-908-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CORRECT POSITION

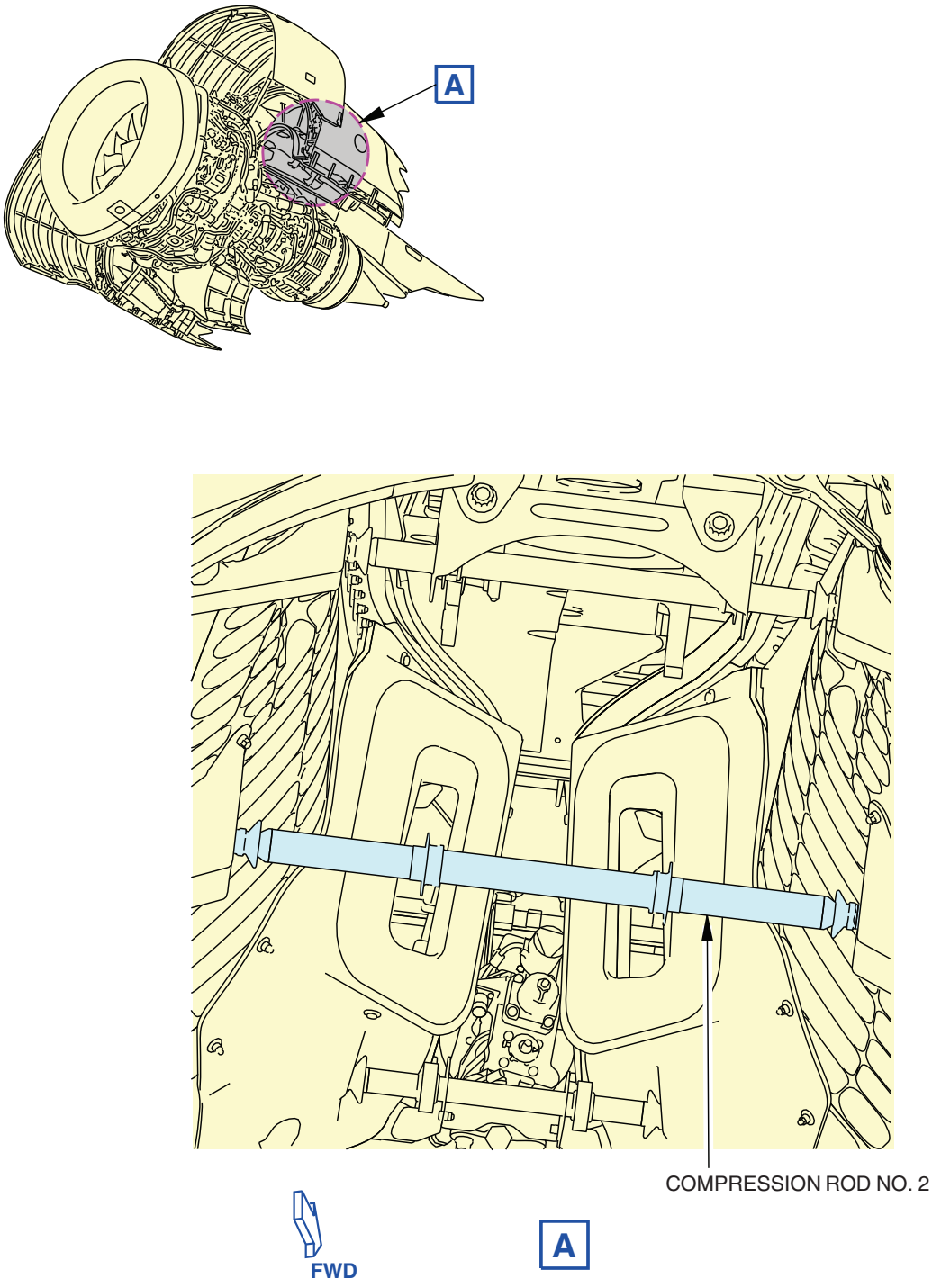


INCORRECT POSITION

2978976 S0000757000_V1

Fuel Nozzle Bolt Seated Position Inspection
Figure 423/73-11-05-990-909-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Compression Rod Installation
Figure 424/73-11-05-990-913-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-05-400-801-G00

3. Fuel Nozzle Installation

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408, Figure 409, Figure 410, Figure 411, Figure 412, Figure 413, Figure 414, Figure 415, Figure 416, Figure 417, Figure 418, Figure 419, Figure 420, Figure 421, Figure 422, Figure 423, and Figure 424)

A. General

(1) This task gives the instructions to install one or more fuel nozzles.

(2) There are eighteen fuel nozzles on the combustion case.

(3) The fuel nozzle position numbers are in a Clockwise (CW) sequence when viewed from the rear of the engine facing forward or Aft Looking Forward (ALF).

NOTE: All bolts that attach fuel nozzle to the case have the same part number.

NOTE: The offset hole on the combustion case aft flange is the Top Vertical Centerline (TVCL). The first fuel nozzle clockwise, aft looking forward, from the TVCL is No. 1.

(4) The task is divided into eighteen installation topics.

(5) You can install the fuel nozzles individually per instructions in the individual topic.

(6) If you will install multiple or a full set of fuel nozzles at the same time, it is preferable that you install the fuel nozzles in sets.

NOTE: It may be necessary to loosen and partially remove an adjacent fuel nozzle in order to get access to rotate the fuel nozzle into the combustion case during fuel nozzle installation. Replace the fuel nozzle seal assembly if three or more fuel nozzle bolts are loosened.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

NOTE: The left side Modulated Turbine Cooling (MTC) valve and actuator (aft looking forward) are removed.

PTF ALL

NOTE: Right side MTC valve is installed between Fuel nozzles 3 & 4 and Left side MTC valve is installed between Fuel nozzles 12 & 13. Nozzle installation in sets may avoid MTC valve installation (if removed during Fuel Nozzle removal).

(7) To install the fuel nozzles [1] (positions 1 through 18), turn the fuel nozzles [1] as follows:

NOTE: The installation sequence is only for reference and can be changed for best achievement, such as down-top installation sequence.

(a) Fuel Nozzle (FN) at positions 1, 2, 3, 18, and 17 (installed at the 1:30 o'clock position).

1) The FN 1 turn to the LEFT and into the case.

2) The FN 2 turn to the RIGHT and into the case.

3) The FN 3 turn to the RIGHT and into the case.

4) The FN 18 turn to the LEFT and into the case.

5) The FN 17 turn to the LEFT and into the case.

6) Move the fuel nozzles [1] aft to engage the swirl cup.

(b) FN 4, 5, 6, 7, 8, and 9 (installed at the 3 to 5 o'clock position).

1) The FN 4 turn UP and into the case.

2) The FN 5 turn DOWN and into the case.

3) The FN 6 turn DOWN and into the case.

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4) The FN 7 turn DOWN and into the case.
5) The FN 8 turn to the LEFT and into the case.
6) The FN 9 turn to the LEFT and into the case.
7) Move the fuel nozzles [1] aft to engage the swirl cup.

(c) FN 11, 10, and 12 (installed at the 7 to 8 o'clock position).
1) The FN 11 turn DOWN and into the case.
2) The FN 10 turn RIGHT and into the case.
3) The FN 12 turn UP and into the case.
4) Move the fuel nozzles [1] aft to engage the swirl cup.

(d) FN 16, 15, 14, and 13 (installed at the 9 o'clock position).
1) The FN 16 turn UP and into the case.
2) The FN 15 turn DOWN and into the case.
3) The FN 13 turn DOWN and into the case.
4) Move the fuel nozzles [1] aft to engage the swirl cup.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
36-11-01-400-801	Engine Pneumatic Duct Installation (P/B 401)
36-11-04-400-801	PRSOV Installation (P/B 401)
36-11-07-400-801	HPSOV Installation (P/B 401)
36-12-01-400-803	Bleed Air Precooler Installation with Engine Installed (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00 P/B 201	LOCKING METHODS - MAINTENANCE PRACTICES
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
70-51-00-800-801-G00	Instructions for Torque (Miscellaneous) (P/B 201)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-700-811-G00	Test No. 35 - Erase All Channel A Faults Special Functions Test (P/B 501)
71-00-00-700-812-G00	Test No. 36 - Erase All Channel B Faults Special Functions Test (P/B 501)
71-00-00-790-801-G00	Test No. 1 - Pneumatic Leak Test (P/B 501)
71-00-00-790-803-G00	Test No. 3 - Wet Motor Leak Test (P/B 501)
71-00-00-790-804-G00	Test No. 4 - Idle Leak Test (P/B 501)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-64-400-801-G00	T3 Sensor Installation (P/B 401)
74-21-02-400-801-G00	Left Igniter Plug Installation (P/B 401)
74-21-02-400-802-G00	Right Igniter Plug Installation (P/B 401)
75-23-02-400-801-G00	Left Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-23-02-400-802-G00	Right Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)

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(Continued)

Reference	Title
75-24-01-400-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation (P/B 401)
75-24-04-400-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation (P/B 401)
75-34-01-400-801-G00	Reverse Bleed Check Valve Assembly Installation (P/B 401)
77-21-04-400-801-G00	Exhaust Gas Temperature Sensor Harness 1 Installation (P/B 401)
77-21-05-400-801-G00	Exhaust Gas Temperature Sensor Harness 2 Installation (P/B 401)
77-31-06-400-801-G00	Turbine Center Frame Accelerometer Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-600	Mirror - Inspection
STD-858	Tag - DO NOT OPERATE
STD-6087	Brush - Acid Resistant

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50034 [CP2619]	Compound - Antiseize - Pure Nickel Special, Never - Seez	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G50530 [CP5061]	Marker - Temporary	
G51403 [CP2166]	Tape - Polyester-Backed, Adhesive	
G51435 [CP2635]	Wax - Stick	
G51669 [CP2975]	Tape - Silicone Rubber, Glass Cloth Fabric	MIL-I-22444

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel nozzle	73-11-05-01-020	PTF ALL
3	Seal	73-11-05-01-030	PTF ALL
21A	Seal	75-23-00-02-310	PTF ALL
23	Metal seal ring	75-23-00-02-350	PTF ALL
41	Seal	73-11-05-01-030	PTF ALL
41A	Seal	72-00-00-25-160	PTF ALL
45A	Seal	72-00-00-25-070	PTF ALL
52A	Seal	75-11-00-03-230	PTF ALL
60	Seal	75-23-00-02-380	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
67	Seal ring	75-23-00-02-440	PTF ALL
70	Seal	73-11-05-01-030	PTF ALL
92	Seal ring	75-23-00-02-270	PTF ALL
100	Seal ring	75-23-00-02-280	PTF ALL
103	Seal	73-11-05-01-030	PTF ALL
150	Seal gasket	73-11-00-20-140	PTF ALL
175	Seal	73-11-05-01-030	PTF ALL
195	Seal	73-11-05-01-030	PTF ALL
221	Seal	73-11-05-01-030	PTF ALL
222	Seal	75-23-00-02-100	PTF ALL
226	Seal	75-23-00-02-110	PTF ALL
312	Seal	75-23-00-01-070	PTF ALL
317	Seal	75-23-00-01-060	PTF ALL
378	Seal	73-11-05-01-030	PTF ALL
392	Seal	73-11-05-01-030	PTF ALL
415	Seal	75-23-00-01-160	PTF ALL
423	Seal	75-23-00-01-120	PTF ALL
429	Seal	73-11-05-01-030	PTF ALL
454	Seal	75-23-00-01-210	PTF ALL
458	Seal	75-23-00-01-200	PTF ALL
471	Flange seal gasket	73-11-00-20-270	PTF ALL
475	Seal	73-11-05-01-030	PTF ALL
506	Seal	73-11-05-01-030	PTF ALL
524	Seal	73-11-05-01-030	PTF ALL
547	Seal	73-11-05-01-030	PTF ALL
563	Seal	75-23-00-01-300	PTF ALL
572	Seal	75-23-00-01-260	PTF ALL
576	Seal	75-23-00-01-370	PTF ALL
580	Seal	75-23-00-01-360	PTF ALL
586	Seal	73-11-05-01-030	PTF ALL
604	Seal	73-11-05-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Fuel Nozzle Installation

SUBTASK 73-11-05-420-198-G00

(1) Do the following steps:

NOTE: The steps below apply to the fuel nozzle pad bolts and any bolts that can be attached to the combustion case during the installation of the fuel nozzle.

(a) At the required fuel nozzle [1] position, new fuel nozzle bolts must be used.

(b) Prepare all the fuel nozzles bolts and case bolts with Tiodize T8E-H grease, D50355 [CP2961], with the acid resistant brush, STD-6087.

NOTE: You can put a label on the serviceable (new or overhauled) fuel nozzle to make installation easy.

(c) Place all clean fuel nozzles [1] on a bench and mark the fuel nozzles [1] (No. 1 through No.18) as applicable per number of nozzles being installed.

(d) Make a record of the fuel nozzle [1] position with the related fuel nozzle serial number, and part number.

(e) Remove the DO NOT OPERATE tags from the fuel nozzle [1].

(f) Remove 3M 851 Tape, G51403 [CP2166], from the fuel nozzle [1].

(g) Do a general visual inspection of the fuel nozzle [1].

(h) Do a general visual inspection of the new seal that connects to the fuel nozzle [1] to see if there is an inner C-seal (Figure 421).

1) Do an inspection for deformation or damage across mating joints.

CAUTION

TURN THE FUEL NOZZLE SEAL UNTIL BOLT HOLES ALIGN. THE ORIENTATION OF THE SEAL TURNS IN ONE DIRECTION. THE BOLT HOLES WILL NOT LINE UP IN THE OTHER DIRECTION. DO NOT USE USED FUEL NOZZLE SEALS. IF YOU DO NOT OBEY, DAMAGE TO THE FUEL NOZZLE CAN OCCUR.

CAUTION

YOU MUST INSTALL A NEW FUEL NOZZLE SEAL EACH TIME. THIS WILL PREVENT A HOT AIR LEAK AND HIGH TEMPERATURE BELOW THE COWLING. IF YOU DO NOT OBEY, DAMAGE TO THE FUEL NOZZLE CAN OCCUR.

(i) Install the new seal on the fuel nozzle [1].

NOTE: It is recommended that the new set of fuel nozzles and their related C-seals are marked with a match mark as a visual aid to make the alignment between the fuel nozzles and gasket seals easy during the assembly process (Figure 421).

(j) Remove the protective covers from the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(k) Remove 3M 851 Tape, G51403 [CP2166], from the combustion case [5].

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(l) Do an inspection of the new or overhauled fuel nozzle [1] inlet strainers for any foreign objects.

I. Fuel Nozzle Installation (Position 1)

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

SUBTASK 73-11-05-420-199-G00

(1) Install the fuel nozzle [1] (position 1) as follows (View C, Figure 402):

(a) Install the fuel nozzle [1] (position 1) with the new seal [3] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 1) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [3] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 1) aft to engage the fuel nozzle [1] (position 1) tip in the combustor radial mixer (cup).

(d) Align the seal [3] and fuel nozzle [1] (position 1) bolt holes to the combustion case [5].

(e) Do not tighten the fuel nozzle bolts at this time.

(f) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(g) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 1) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(h) Tighten the bolts [2] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the fuel nozzle bolts [2].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(i) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

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2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(j) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

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a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the pilot secondary (PS) B-nut first followed by the pilot primary main (PPM) B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with the safety cable, safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-010-016-G00

(2) If disconnected, install the bleed air precooler (TASK 36-12-01-400-803).

J. Fuel Nozzle Installation (Position 2)

WARNING	DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION	DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

SUBTASK 73-11-05-420-200-G00

(1) Install the fuel nozzle [1] (position 2) as follows (View L, Figure 403):

(a) Install the fuel nozzle [1] (position 2) with the new seal [41] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 2) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [41] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 2) aft to engage the fuel nozzle [1] (position 2) tip in the combustor radial mixer (cup).

(d) Align the seal [41] and fuel nozzle [1] (position 2) bolt holes to the combustion case [5].

(e) Install the booster anti-ice tube bracket [45] as follows (View L, Figure 403):

NOTE: The booster anti-ice bracket is two brackets assembled together that span across fuel nozzle (position 2) and fuel nozzle (position 3).

1) Install the two bolts [40] that attach the bracket [45] to the fuel nozzle [1] (position 2).

2) Install the bolt [46] that attaches the bracket [45] to the fuel nozzle [1] (position 3).

(f) Install the remaining two bolts [40] that attach the fuel nozzle [1] (position 2) to the combustion case [5].

(g) Do not tighten the fuel nozzle bolts at this time.

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(h) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(i) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 2) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(j) Tighten the bolts [40] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the fuel nozzle bolts [40].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m)..

NOTE: One of the four fuel nozzle bolt holes is elongated.

(k) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION

MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION

MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(l) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with the safety cable, safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-201-G00

(2) Install the booster anti-ice tube [37A] as follows (View E, Figure 403):

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(a) Support and secure the booster anti-ice tube [38A] and booster anti-ice tube [39A] from damage.

(b) Support and secure the booster anti-ice tube [37A] from damage.

(c) Install the new seal [41A] and coupling clamp [40A] that attach the booster anti-ice tube [37A] to the booster anti-ice tube [44A].

1) Do not tighten the coupling clamp [40A] at this moment.

(d) Install the two coupling clamps [46A] and two new seals [45A] that attach the booster anti-ice tube [37A] to the booster anti-ice tube [38A] and booster anti-ice tube [39A].

1) Do not tighten the two coupling clamps [46A] at this moment.

(e) Apply oil, D00599 [CP2442], on the threads of the bolts [42A] and bolts [47A].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(f) Install the four bolts [42A] and two strap clamps [43A] that attach the booster anti-ice tube [37A] and booster anti-ice tube [38A] to the support brackets.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(g) Install the two bolts [42A] and strap clamp [43A] that attach the booster anti-ice tube [37A] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 OR PRE LEAP-1B SERVICE BULLETIN 72-0144

(h) Install the bolts [47A] and loop clamp [48A] that attach the booster anti-ice tube [37A] and booster anti-ice tube [39A] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0144

(i) Install the two bolts [47A] and strap clamp [43A] that attach the booster anti-ice tube [37A] and booster anti-ice tube [39A] to the support bracket.

PTF ALL

(j) Tighten the coupling clamp [40A] and coupling clamps [46A] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(k) Tighten the bolts [42A] and bolts [47A] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-202-G00

(3) Install the booster anti-ice tube [44A] as follows (View G, Figure 403):

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [50A] and bolt [53A].

(b) Install the new seal [52A] and coupling clamp [51A] that attaches the booster anti-ice tube [44A] to the Start Bleed Valve (SBV) at the 4:00 o'clock position (View J, Figure 403).

1) Do not tighten the coupling clamp [51A] at this time.

(c) Support and tightly attach the booster anti-ice tube [44A].

(d) Install the bolt [50A], bolt [53A], and clamps [49A] that attach the booster anti-ice tube [44A] to the support bracket.

1) Do not tighten the bolt [50A] and bolt [53A] at this time.

(e) Tighten the coupling clamp [51A], bolt [50A], and bolt [53A] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

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SUBTASK 73-11-05-420-203-G00

(4) Install the transient bleed air tube [19] as follows (View C and D, Figure 403):

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [22].

(b) Apply a layer of stick wax, G51435 [CP2635], to the new metal seal ring [23] to hold it during installation.

(c) Attach the transient bleed air tube [19] and new metal seal ring [23] to the combustion case [5] with the four bolts [22].

1) Do not tighten the bolts [22] at this time.

(d) Apply oil, D00599 [CP2442], on the threads of the bolts [18].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0229

(e) Install the three bolts [18] that attach the transient bleed air tube [19] to the support bracket [20].

1) Do not tighten the bolts [18] at this time.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0229

(f) Install the two bolts [18] that attach the transient bleed air tube [19] to the support bracket [20].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

NOTE: The transient bleed air tube [19] is referenced as the reverse bleed check valve assembly in Post LEAP-1B Service Bulletin 72-0378.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0229

1) Do not tighten the bolts [18] at this time.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(g) Install the reverse bleed valve assembly (TASK 75-34-01-400-801-G00).

PTF ALL

(h) Install the new seal [21A] on the transient bleed air tube [19].

(i) Install the coupling clamp [21] on the transient bleed valve [47].

1) Tighten the coupling clamp [21] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Tighten the bolts [22] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(k) Tighten the bolts [18] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-204-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Install the engine crossover duct assembly (TASK 36-11-01-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

K. Fuel Nozzle Installation (Position 3)

SUBTASK 73-11-05-420-205-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 3) as follows (View G, Figure 404):

(a) Install the fuel nozzle [1] (position 3) with the new seal [70] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 3) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [70] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 3) aft to engage the fuel nozzle [1] (position 3) tip in the combustor radial mixer (cup).

(d) Align the seal [70] and fuel nozzle [1] (position 3) bolt holes to the combustion case [5].

(e) Install the booster anti-ice tube bracket [74] as follows:

NOTE: The booster anti-ice bracket is two brackets assembled together that span across fuel nozzle (position 2) and fuel nozzle (position 3).

1) Install the two bolts [68] that attach the bracket [74] to the fuel nozzle [1] (position 2).

2) Install the bolt [69] that attaches the bracket [74] to the fuel nozzle [1] (position 3).

(f) Install the bracket [75] as follows (Figure 404):

1) Install the bolts [69] that attach the bracket [75] to the fuel nozzle [1] (position 3).

(g) Install the remaining bolt [69] that attach the fuel nozzle [1] (position 3) to the combustion case [5].

(h) Do not tighten the fuel nozzle bolts at this time.

(i) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(j) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 3) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(k) Tighten the bolts [69] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the bolts [69].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

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3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(l) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(m) Install the bolts [76] that attach the loop clamps [77] to the bracket [75].

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(n) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-206-G00

(2) If removed, install the right MTC valve (TASK 75-23-02-400-802-G00).

SUBTASK 73-11-05-420-207-G00

(3) Install the 4th-stage cooling air manifold [58] as follows (View C, D, and E, Figure 404):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0004

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [66] and two bolts [64].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0004

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [66] and two bolts [634].

PTF ALL

(c) Install the seal ring [67] on the compressor case [65].

(d) Install the two bolts [66] that attach the harness support bracket [78] and 4th-stage cooling air manifold [58] to the compressor case [65].

1) Do not tighten the bolts [66] at this time.

(e) Install the bolt [66] that attaches the support bracket [79] and 4th-stage cooling air manifold [58] to the compressor case [65].

1) Do not tighten the bolt [66] at this time.

(f) Install the bolt [66] that attaches the 4th-stage cooling air manifold [58] to the compressor case [65].

1) Do not tighten the bolt [66] at this time.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0004

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [80].

(h) Install the bolt [80] that attaches the harness support bracket [78] to the compressor case [65].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0004 (Continued)

1) Tighten the bolt [80] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

PTF ALL

(i) Install the seal [60] and coupling clamp [61] that attaches the 4th-stage cooling air manifold [58] to the air supply adapter tube [62].

1) Do not tighten the coupling clamp [61] at this time.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0004

(j) Install the two bolts [64] that attach the 4th-stage cooling air manifold [58] at the air tube bracket [63].

1) Tighten the bolts [64] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0004

(k) Install the two bolts [634] and strap clamp [635] that attach the 4th-stage cooling air manifold [58] at the air tube bracket [63].

1) Tighten the bolts [634] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

(l) Attach the LPT air supply tube [59] to the 4th-stage cooling air manifold [58].

1) Tighten the LPT air supply tube [59] B-nut to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

(m) Tighten the bolts [66] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(n) Tighten the coupling clamp [61] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

SUBTASK 73-11-05-420-208-G00

(4) Install the booster anti-ice tube [37A] as follows (View E, Figure 403):

(a) Support and secure the booster anti-ice tube [38A] and booster anti-ice tube [39A] from damage.

(b) Support and secure the booster anti-ice tube [37A] from damage.

(c) Install the new seal [41A] and coupling clamp [40A] that attach the booster anti-ice tube [37A] to the booster anti-ice tube [44A].

1) Do not tighten the coupling clamp [40A] at this moment.

(d) Install the two coupling clamps [46A] and two new seals [45A] that attach the booster anti-ice tube [37A] to the booster anti-ice tube [38A] and booster anti-ice tube [39A].

1) Do not tighten the two coupling clamps [46A] at this moment.

(e) Apply oil, D00599 [CP2442], on the threads of the bolts [42A] and bolts [47A].

(f) Install the four bolts [42A] and two strap clamps [43A] that attach the booster anti-ice tube [37A] and booster anti-ice tube [38A] to the support brackets.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(g) Install the bolts [42A] and strap clamp [43A] that attach the booster anti-ice tube [37A] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 OR PRE LEAP-1B SERVICE BULLETIN 72-0144

(h) Install the bolts [47A] and loop clamp [48A] that attach the booster anti-ice tube [37A] and booster anti-ice tube [39A] to the support bracket.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0144

(i) Install the two bolts [47A] and strap clamp [43A] that attach the booster anti-ice tube [37A] and booster anti-ice tube [39A] to the support bracket.

PTF ALL

(j) Tighten the coupling clamp [40A] and coupling clamps [46A] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(k) Tighten the bolts [42A] and bolts [47A] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-400-035-G00

(5) Install the booster anti-ice tube [44A] as follows (View G, Figure 403):

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [50A] and bolt [53A].

(b) Install the new seal [52A] and coupling clamp [51A] that attaches the booster anti-ice tube [44A] to the SBV at the 4:00 o'clock position (View J, Figure 403).

1) Do not tighten the coupling clamp [51A] at this time.

(c) Support and tightly attach the booster anti-ice tube [44A].

(d) Install the bolt [50A], bolt [53A], and clamps [49A] that attach the booster anti-ice tube [44A] to the support bracket.

1) Do not tighten the bolt [50A] and bolt [53A] at this time.

(e) Tighten the coupling clamp [51A], bolt [50A], and bolt [53A] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-210-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Install the engine crossover duct assembly (TASK 36-11-01-400-801).

L. Fuel Nozzle Installation (Position 4)

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

SUBTASK 73-11-05-420-211-G00

(1) Install the fuel nozzle [1] (position 4) as follows (View G, Figure 405):

(a) Install the fuel nozzle [1] (position 4) with the new seal [103] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 4) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [103] and combustion case [5] pad.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Move the fuel nozzle [1] (position 4) aft to engage the fuel nozzle [1] (position 4) tip in the combustor radial mixer (cup).

(d) Align the seal [103] and fuel nozzle [1] (position 4) bolt holes to the combustion case [5].

(e) Do not tighten the fuel nozzle bolts at this time.

(f) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(g) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 4) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(h) Install the four bolts [102] that attach the fuel nozzle [1] (position 4) to the combustion case [5].

1) Tighten the bolts [102] as follows (Figure 422):

a) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the fuel nozzle bolts [102].

b) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

c) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(i) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION

MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION

MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(j) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-212-G00

(2) If removed, install the right MTC valve (TASK 75-23-02-400-802-G00).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-213-G00

(3) Install the bracket [98] as follows (Figure 406):

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [152] and two bolts [126].

(b) Install the bolt [152] that attaches the CCC air manifold [120] to the bracket [98] (View S, Figure 406).

(c) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98] (View J, Figure 406).

(d) Tighten the bolt [152] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(e) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-214-G00

(4) Install the bracket [98] as follows (Figure 406):

(a) Apply oil, D00599 [CP2442], on the threads of the bolts [191], bolt [197], and bolts [206].

(b) Install two bolts [206] that attach the bracket [98] to the support bracket [207].

(c) Install the bolt [197] that attaches the loop clamp [205] and drain tube [208] to the bracket [98].

(d) Install the two bolts [191] that attach the strap clamp [191A] and BAI air tube [27] to the bracket [98].

(e) Install the bolt [152] that attaches the CCC air manifold [120] to the bracket [98] (View S, Figure 406).

(f) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98] (View J, Figure 406).

(g) Tighten the bolt [152] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(h) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Tighten the bolts [206] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(j) Tighten the bolt [197] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(k) Tighten the bolts [191] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

SUBTASK 73-11-05-420-215-G00

(5) Install the 7th-stage air tube [90] as follows (View C, D, and E, Figure 405):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [95], two bolts [97], and four bolts [99].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [95], and four bolts [99].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(c) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [632], and four bolts [99].

PTF ALL

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(d) Apply a layer of stick wax, G51435 [CP2635], to the seal ring [92] and seal ring [100] to hold them during installation.

(e) Install the new seal ring [92] at the aft end of the 7th-stage air tube [90] on the HPT case [94].

(f) Install the new seal ring [100] at the forward end of the 7th-stage air tube [90] on the HPC extension case [101].

(g) Put the 7th-stage air tube [90] in position between the 7th-stage air tube bracket [96] and HPT case [94].

1) Install the bolts [91] that attach the bracket [93] and 7th-stage air tube [90] to the HPT case [94].

(h) Install the four bolts [99] that attach the forward end of the 7th-stage air tube [90] to the HPC extension case [101].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(i) Install the two bolts [95] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

(j) Install the two bolts [97] that attach the 7th-stage air tube [90] to the bracket [98].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(k) Install the two bolts [95] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(l) Install the two bolts [632] and strap clamp [633] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL

(m) Tighten the bolts [91] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(n) Tighten the bolts [99] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(o) Tighten the bolts [95] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(p) Tighten the bolts [97] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(q) Tighten the bolts [95] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(r) Tighten the bolts [632] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(s) Apply a layer of oil, D00599 [CP2442], on the threads of the bolt [87].

(t) Install the bolt [87] that attaches the air tube [89] to the bracket [93].

(u) Tighten the bolt [87] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-05-420-216-G00

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Install the engine crossover duct assembly (Engine Pneumatic Duct Installation, TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-217-G00

(7) Install the Pressure Regulating and Shutoff Valve (PRSOV) (TASK 36-11-04-400-801).

M. Fuel Nozzle Installation (Position 5)

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

SUBTASK 73-11-05-420-218-G00

(1) Install the fuel nozzle [1] (position 5) as follows (View C, Figure 406):

(a) Install the fuel nozzle [1] (position 5) with the new seal [175] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 5) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [175] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 5) aft to engage the fuel nozzle [1] (position 5) tip in the combustor radial mixer (cup).

(d) Align the seal [175] and fuel nozzle [1] (position 5) bolt holes to the combustion case [5].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(e) Install the two bolts [173] that attach the bracket [177] and fuel nozzle [1] (position 5) to the combustion case [5].

PTF ALL

(f) Install the remaining two bolts [173] that attach the fuel nozzle [1] (position 5) to the combustion case [5].

(g) Install the remaining two bolts [173] to attach the bracket [177] to the combustion case [5].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0050 AND PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(h) Install the remaining two bolts [173] to attach the bracket [181] to the combustion case [5].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(i) Install the two bolts [173] to attach bracket [202A] to fuel nozzle [1] (position 5).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378 (Continued)

(j) Install the two bolts [173] that attach the support bracket [181] to combustion case [5].

PTF ALL

(k) Do not tighten the fuel nozzle bolts at this time.

(l) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(m) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1](position 5) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(n) Tighten the bolts [173] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the fuel nozzle bolts [173].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m)).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(o) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION

MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION

MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(p) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-219-G00

(2) Install the PS3 tube [164] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(a) Apply a layer of Tiodize T8E-H grease, D50355 [CP2961], on the threads of the fitting [165].

NOTE: The pure nickel special compound, D50034 [CP2619], can be used as an alternative.

(b) Connect the PS3 tube [164] to the PS3 port fitting [165] on the combustion case [5].

(c) Connect the PS3 tube [164] to the PS3 sense line [172].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [167].

1) Install the bolt [167] that attaches the loop clamp [168] and PS3 tube [164] to the support bracket [166].

2) Tighten the bolt [167] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolt [169].

1) Install the bolt [169] that attaches the loop clamp [170] and PS3 tube [164] to the tube bracket [171].

2) Tighten the bolt [169] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(f) Tighten the PS3 tube [164] B-nuts at the sense line [172] and combustion case [5] as follows:

1) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the two B-nuts until you can turn them with your fingers.

3) Tighten the two B-nuts again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the two B-nuts for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Make sure that the B-nuts are fully seated.

6) If the B-nuts are not fully seated, do the above steps again to tighten the PS3 tube [164] B-nut(s).

7) Secure the PS3 tube [164] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-221-G00

(3) Install the PS3 tube [164] as follows (View Y, Z, AA, and BB, Figure 406):

(a) Apply a layer of Tiodize T8E-H grease, D50355 [CP2961], on the threads of the fitting [165].

NOTE: The pure nickel special compound, D50034 [CP2619], can be used as an alternative.

(b) Connect the PS3 tube [164] to the PS3 port fitting [165] on the combustion case [5].

(c) Connect the PS3 tube [164] to the PS3 sense line [172].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [167].

1) Install the bolt [167] that attaches the loop clamp [168] and PS3 tube [164] to the support bracket [166].

2) Tighten the bolt [167] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolt [185].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

1) Install the bolt [185] that attaches the loop clamp [186] and PS3 tube [164] to the support bracket [181].
2) Tighten the bolt [185] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(f) Tighten the PS3 tube [164] B-nuts at the sense line [172] and combustion case [5] as follows:

1) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
2) Loosen the two B-nuts until you can turn them with your fingers.
3) Tighten the two B-nuts again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
4) Tighten the two B-nuts for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
5) Make sure that the B-nuts are fully seated.
6) If the B-nuts are not fully seated, do the above steps again to tighten the PS3 tube [164] B-nut(s).
7) Secure the PS3 tube [164] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0135 AND POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-220-G00

(4) Install the PS3 tube [164] as follows (View Y, Z, AA, and BB, Figure 406):

(a) Apply a layer of Tiodize T8E-H grease, D50355 [CP2961], on the threads of the fitting [165].

NOTE: The pure nickel special compound, D50034 [CP2619], can be used as an alternative.

(b) Connect the PS3 tube [164] to the PS3 port fitting [165] on the combustion case [5].

(c) Connect the PS3 tube [164] to the PS3 sense line [172].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [639].

1) Install the bolt [639] that attaches the loop clamp [640] and PS3 tube [164] to the support bracket [641].
2) Tighten the bolt [639] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolt [636].

1) Install the bolt [636] that attaches the loop clamp [637] and PS3 tube [164] to the support bracket [638].
2) Tighten the bolt [636] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(f) Apply oil, D00599 [CP2442], on the threads of the bolt [642].

1) Install the bolt [642] that attaches the loop clamp [643] and PS3 tube [164] to the tube bracket [171].
2) Tighten the bolt [642] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

(g) Tighten the PS3 tube [164] B-nuts at the sense line [172] and combustion case [5] as follows:

1) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
2) Loosen the two B-nuts until you can turn them with your fingers.

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0135 AND POST SB-CFM-LEAP-1B-72-0378 (Continued)

3) Tighten the two B-nuts again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the two B-nuts for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Make sure that the B-nuts are fully seated.

6) If the B-nuts are not fully seated, do the above steps again to tighten the PS3 tube [164] B-nut(s).

7) Secure the PS3 tube [164] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

PTF ALL

SUBTASK 73-11-05-420-222-G00

(5) Install the 7th-stage air tube [90] as follows (View C, D, and E, Figure 405):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [95], two bolts [97], and four bolts [99].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [95], and four bolts [99].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(c) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [632], and four bolts [99].

PTF ALL

(d) Apply a layer of stick wax, G51435 [CP2635], to the seal ring [92] and seal ring [100] to hold them during installation.

(e) Install the new seal ring [92] at the aft end of the 7th-stage air tube [90] on the HPT case [94].

(f) Install the new seal ring [100] at the forward end of the 7th-stage air tube [90] on the HPC extension case [101].

(g) Put the 7th-stage air tube [90] in position between the 7th-stage air tube bracket [96] and HPT case [94].

1) Install the bolts [91] that attach the bracket [93] and 7th-stage air tube [90] to the HPT case [94].

(h) Install the four bolts [99] that attach the forward end of the 7th-stage air tube [90] to the HPC extension case [101].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(i) Install the two bolts [95] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

(j) Install the two bolts [97] that attach the 7th-stage air tube [90] to the bracket [98].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(k) Install the two bolts [95] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(l) Install the two bolts [632] and strap clamp [633] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL

(m) Tighten the bolts [91] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(n) Tighten the bolts [99] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(o) Tighten the bolts [95] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(p) Tighten the bolts [97] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(q) Tighten the bolts [95] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(r) Tighten the bolts [632] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(s) Apply a layer of oil, D00599 [CP2442], on the threads of the bolt [87].

(t) Install the bolt [87] that attaches the air tube [89] to the bracket [93].

(u) Tighten the bolt [87] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-223-G00

(6) Install the CCC air manifold [120] as follows (View R, S, T, U, V, W, and X, Figure 406):

(a) Install and align the CCC air manifold [120] in the rubber bellows [161] and flexible duct [162].

(b) Tighten the hose clamps [163] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(c) Tighten the hose clamp [180] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [159].

(e) Install the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(f) Tighten the bolt [159] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [156].

(h) Install the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(i) Tighten the bolt [156] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [153].

(k) Install the two bolts [153] that attach the strap clamp [154] to the bracket [155].

(l) Tighten the bolts [153] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(m) Apply oil, D00599 [CP2442], on the threads of the bolt [152].

(n) Install the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(o) Tighten the bolt [152] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-224-G00

(7) Install the CCC air manifold [120] as follows (View R, S, T, U, V, W, DD, and X, Figure 406):

(a) Install and align the CCC air manifold [120] in the rubber bellows [161] and flexible duct [162].

(b) Install the bolt [189] that attaches the bracket [155] to the fuel nozzle [1] (position 6) (View C, Figure 407).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(c) Apply oil, D00599 [CP2442], on the threads of the bolts [209].

(d) Install the two bolts [209] that attach the CCC air manifold [120] to the CCC duct support bracket [210].

(e) Tighten the bolts [209] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(f) Tighten the hose clamp [163] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

(g) Tighten the hose clamp [180] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(h) Apply oil, D00599 [CP2442], on the threads of the bolt [159].

(i) Install the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(j) Tighten the bolt [159] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the bolt [156].

(l) Install the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(m) Tighten the bolt [156] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the bolts [153].

(o) Install the bolts [153] that attaches the strap clamp [154] to the bracket [155].

(p) Tighten the bolts [153] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the bolt [152].

(r) Install the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(s) Tighten the bolt [152] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-225-G00

(8) Do these steps to install the bracket [202] (View C, Figure 406):

(a) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the one bolt [189].

NOTE: You can use oil, D00599 [CP2442], as an alternative.

(b) Install the bolt [189] to attach the bracket [202] and fuel nozzle (position 6) to the combustion case [5].

(c) Tighten the bolt [189] to 87 in-lb (9.83 N·m) - 103 in-lb (11.64 N·m).

(d) If Tiodize T8E-H grease, D50355 [CP2961], is used as a lubricant, final tighten the bolt [189] to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

(e) If oil, D00599 [CP2442], is used as a lubricant, final tighten the bolt [189] to 189 in-lb (21.35 N·m) - 221 in-lb (24.97 N·m).

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378 (Continued)

(f) Apply a layer of oil, D00599 [CP2442], on the threads of the bolts [131A].

(g) Attach the bracket [202] to bracket [202A] with the two bolts [131A].

(h) Tighten the bolts [131A] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-226-G00

(9) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(f) Install the four bolts [145] that attach the hinged loop clamp [146] to the tube bracket [147] and tube bracket [148].

(g) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(l) Install the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(o) Install the two bolts [137] that attach the strap clamp [136] and SBV TBV manifold [125] to the support bracket [139].

(p) Tighten the bolts [137] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the two bolts [134].

(r) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Tighten the bolts [134] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(t) Connect the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(u) Connect the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(w) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(x) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(z) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(aa) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(ac) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(ad) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-227-G00

(10) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two Transient Bleed Valve (TBV) fittings.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(g) Install the four bolts [145] that attach the hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [140] and bolt [140A].

(k) Install the two bolts [140] that attach the hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(l) Install the bolt [140A] that attaches the hinged loop clamps [141] to the fuel manifold bracket [142].

(m) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(n) Install the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(o) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(p) Install the two bolts [134] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(q) Install the two bolts [134A] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(s) Install the two bolts [131] that attach the loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(t) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(u) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(w) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(x) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(y) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(z) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(aa) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ac) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ad) Tighten the bolt [140A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ae) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ag) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ah) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(aj) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-420-228-G00

(11) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the two bolts [145].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(g) Install the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the bolt [179] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(k) Install the two bolts [140] that attach the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(l) Install the bolt [140] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Apply oil, D00599 [CP2442], on the threads of the three bolts [137] and bolt [137A].

(n) Install the three bolts [137] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(o) Install the bolt [137A] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(p) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(q) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Install the two bolts [134A] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(t) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(u) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(w) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(x) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(z) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(aa) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ab) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ac) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(ad) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ae) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ag) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ah) Tighten the bolts [137A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(aj) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ak) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(al) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(am) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL

SUBTASK 73-11-05-420-229-G00

(12) Connect the HJ8A harness [118] as follows (View E, F, and G, Figure 406):

(a) Install the HJ8A harness [118] in the push clamp [178] on the CCC air manifold [120].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [121].

(c) Install the bolt [121] that attaches the loop clamp [122] to the CCC air manifold [120].

(d) Tighten the bolt [121] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolt [119].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(f) Install the bolt [119] that attaches the HJ8A harness [118] wye to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(g) Install the bolt [119] that attaches the HJ8A harness [118] wye to the bracket [202].

PTF ALL

(h) Tighten the bolt [119] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Connect the electrical connector [116] and electrical connector [117] of the HJ8A harness [118] to the TBV valve [123] and SBV valve [124] (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-05-420-230-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(13) Install the engine mix manifold duct assembly (TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-231-G00

(14) Install the PRSOV (TASK 36-11-04-400-801).

N. Fuel Nozzle Installation (Position 6)

SUBTASK 73-11-05-420-232-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 6) as follows (View C and D, Figure 407):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the fuel nozzle [1] (position 6) with the new seal [195] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 6) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [195] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 6) aft to engage the fuel nozzle [1] (position 6) tip in the combustor radial mixer (cup).

(d) Align the seal [195] and fuel nozzle [1] (position 6) bolt holes to the combustion case [5].

(e) Install the two bolts [193] that attach the bracket [194] and fuel nozzle [1] (position 6) to the combustion case [5].

(f) Install the bolt [189] that attaches the bracket [155] or bracket [202] and fuel nozzle [1] (position 6) to the combustion case [5].

(g) Install the remaining bolt [193] that attaches the fuel nozzle [1] (position 6) to the combustion case [5].

(h) Install the bolt [153] that attaches the strap clamp [154] and bracket [155] to the bracket [192].

(i) Do not tighten the fuel nozzle bolts at this time.

(j) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] fittings.

(k) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 6) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(l) Tighten the bolts [193] and bolt [189] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the fuel nozzle bolts [193] and bolt [189].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(m) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(n) Tighten the bolt [153] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(o) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

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b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with the safety cable, safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

(p) Apply oil, D00599 [CP2442], on the threads of the bolts [201].

(q) Install the two bolts [201] that attach the two loop clamps [199], fuel manifold [198], and fuel manifold [196] to the bracket [194].

(r) Tighten the bolts [201] to 32 in-lb (3.62 N·m) - 36 in-lb (4.07 N·m).

SUBTASK 73-11-05-420-233-G00

(2) Install the 7th-stage air tube [90] as follows (View C, D, and E, Figure 405):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [95], two bolts [97], and four bolts [99].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [95], and four bolts [99].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(c) Apply oil, D00599 [CP2442], on the threads of the four bolts [91], two bolts [632], and four bolts [99].

PTF ALL

(d) Apply a layer of stick wax, G51435 [CP2635], to the seal ring [92] and seal ring [100] to hold them during installation.

(e) Install the new seal ring [92] at the aft end of the 7th-stage air tube [90] on the HPT case [94].

(f) Install the new seal ring [100] at the forward end of the 7th-stage air tube [90] on the HPC extension case [101].

(g) Put the 7th-stage air tube [90] in position between the 7th-stage air tube bracket [96] and HPT case [94].

1) Install the bolts [91] that attach the bracket [93] and 7th-stage air tube [90] to the HPT case [94].

(h) Install the four bolts [99] that attach the forward end of the 7th-stage air tube [90] to the HPC extension case [101].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(i) Install the two bolts [95] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

(j) Install the two bolts [97] that attach the 7th-stage air tube [90] to the bracket [98].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(k) Install the two bolts [95] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(l) Install the two bolts [632] and strap clamp [633] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL

(m) Tighten the bolts [91] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(n) Tighten the bolts [99] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(o) Tighten the bolts [95] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(p) Tighten the bolts [97] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(q) Tighten the bolts [95] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(r) Tighten the bolts [632] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(s) Apply a layer of oil, D00599 [CP2442], on the threads of the bolt [87].

(t) Install the bolt [87] that attaches the air tube [89] to the bracket [93].

(u) Tighten the bolt [87] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-234-G00

(3) Install the CCC air manifold [120] as follows (View R, S, T, U, V, W, and X, Figure 406):

(a) Install and align the CCC air manifold [120] in the rubber bellows [161] and flexible duct [162].

(b) Tighten the hose clamps [163] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(c) Tighten the hose clamp [180] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [159].

(e) Install the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(f) Tighten the bolt [159] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [156].

(h) Install the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(i) Tighten the bolt [156] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [153].

(k) Install the two bolts [153] that attach the strap clamp [154] to the bracket [155].

(l) Tighten the bolts [153] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(m) Apply oil, D00599 [CP2442], on the threads of the bolt [152].

(n) Install the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(o) Tighten the bolt [152] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-235-G00

(4) Install the CCC air manifold [120] as follows (View R, S, T, U, V, W, DD, and X, Figure 406):

(a) Install and align the CCC air manifold [120] in the rubber bellows [161] and flexible duct [162].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [209].

(c) Install the two bolts [209] that attach the CCC air manifold [120] to the CCC duct support bracket [210].

(d) Tighten the bolts [209] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(e) Tighten the hose clamp [163] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

(f) Tighten the hose clamp [180] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [159].

(h) Install the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(i) Tighten the bolt [159] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the bolt [156].

(k) Install the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(l) Tighten the bolt [156] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(m) Apply oil, D00599 [CP2442], on the threads of the bolts [153].

(n) Install the bolts [153] that attaches the strap clamp [154] to the bracket [155].

(o) Tighten the bolts [153] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(p) Apply oil, D00599 [CP2442], on the threads of the bolt [152].

(q) Install the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(r) Tighten the bolt [152] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

SUBTASK 73-11-05-420-236-G00

(5) Do these steps to install the bracket [202A] (View C, Figure 407):

I (a) Apply oil, D00599 [CP2442], on the threads of the two bolts [134A].

(b) Attach the bracket [202] to the bracket [202A] with two bolts [131A].

(c) Tighten the two bolts [131A] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-237-G00

(6) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(e) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(f) Install the four bolts [145] that attach the hinged loop clamp [146] to the tube bracket [147] and tube bracket [148].

(g) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(l) Install the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(o) Install the two bolts [137] that attach the strap clamp [136] and SBV TBV manifold [125] to the support bracket [139].

(p) Tighten the bolts [137] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the two bolts [134].

(r) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Tighten the bolts [134] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(t) Connect the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(u) Connect the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

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2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(w) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(x) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(z) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(aa) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(ac) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(ad) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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SUBTASK 73-11-05-420-238-G00

(7) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):
   (a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].
   (b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].
   (c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].
   (d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.
   (e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.
   (f) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].
   (g) Install the four bolts [145] that attach the hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].
   (h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].
   (i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].
   (j) Apply oil, D00599 [CP2442], on the threads of the two bolts [140] and bolt [140A].
   (k) Install the two bolts [140] that attach the hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].
   (l) Install the bolt [140A] that attaches the hinged loop clamps [141] to the fuel manifold bracket [142].
   (m) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(n) Install the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(o) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(p) Install the two bolts [134] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(q) Install the two bolts [134A] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(s) Install the two bolts [131] that attach the loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(t) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(u) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(w) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(x) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(y) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.

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c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(z) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(aa) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ac) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ad) Tighten the bolt [140A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ae) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ag) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ah) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(aj) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

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SUBTASK 73-11-05-420-239-G00

(8) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the two bolts [145].

(g) Install the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the bolt [179] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(k) Install the two bolts [140] that attach the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(l) Install the bolt [140] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Apply oil, D00599 [CP2442], on the threads of the three bolts [137] and bolt [137A].

(n) Install the three bolts [137] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(o) Install the bolt [137A] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(p) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(q) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Install the two bolts [134A] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

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(t) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

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(u) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the bracket [202].

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(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

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(w) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

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(x) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the bracket [202].

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(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(z) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(aa) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

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(ab) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ac) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(ad) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ae) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ag) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ah) Tighten the bolts [137A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(aj) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ak) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(al) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(am) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL

SUBTASK 73-11-05-420-240-G00

(9) Connect the HJ8A harness [118] as follows (View E, F, and G, Figure 406):

(a) Install the HJ8A harness [118] in the push clamp [178] on the CCC air manifold [120].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [121].

(c) Install the bolt [121] that attaches the loop clamp [122] to the CCC air manifold [120].

(d) Tighten the bolt [121] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolt [119].

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(f) Install the bolt [119] that attaches the HJ8A harness [118] wye to the CCC air manifold [120].

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(g) Install thebolt [119] that attaches the HJ8A harness [118] wye to the bracket [202].

PTF ALL

(h) Tighten the bolt [119] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Connect the electrical connector [116] and electrical connector [117] of the HJ8A harness [118] to the TBV valve [123] and SBV valve [124] (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-05-420-241-G00

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(10) Install the engine mix manifold duct assembly (TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-242-G00

(11) Install the High Pressure Shutoff Valve (HPSOV) (TASK 36-11-07-400-801).

O. Fuel Nozzle Installation (Position 7)

SUBTASK 73-11-05-420-243-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 7) as follows (View C, Figure 408):

(a) Install the fuel nozzle [1] (position 7) with the new seal [221] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 7) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [221] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 7) aft to engage the fuel nozzle [1] (position 7) tip in the combustor radial mixer (cup).

(d) Align the seal [221] and fuel nozzle [1] (position 7) bolt holes to the combustion case [5].

(e) Install the support bracket [214] on the fuel nozzle [1] (position 7).

(f) Install the two bolts [383] that attach the support bracket [214] and fuel nozzle [1] (position 7) to the combustion case [5].

(g) Install the two bolts [213] that attach the support bracket [215] and fuel nozzle [1] (position 7) to the combustion case [5].

(h) Do not tighten the fuel nozzle bolts at this time.

(i) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(j) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 7) fittings, before you tighten the fuel nozzle bolts.

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1) Do not tighten the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(k) Tighten the bolts [213] and bolts [383] as follows (Figure 408):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the threads of the fuel nozzle the bolts [213] and bolt [383].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(l) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(m) Apply oil, D00599 [CP2442], on the threads of the bolt [218].

(n) Install the bolt [218] that attaches the support bracket [215] to the bracket [203].

(o) Tighten the bolt [218] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(p) Apply oil, D00599 [CP2442], on the threads of the two bolts [220].

(q) Install the two bolts [220] that attach the start bleed bracket [219] to the combustion case [5].

(r) Install the two bolts [384] that attach the bracket [385] to the support bracket [214].

(s) Tighten the bolts [384] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(t) Tighten the bolts [220] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(u) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

(v) Apply oil, D00599 [CP2442], on the threads of the bolts [217].

(w) Install the two bolts [217] that attach the two loop clamp [216] to the support bracket [215].

(x) Tighten the bolts [217] to 32 in-lb (3.62 N·m) - 36 in-lb (4.07 N·m).

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SUBTASK 73-11-05-420-244-G00

(2) Install the start bleed tube [224] as follows (View F, Figure 408):

(a) Apply oil, D00599 [CP2442], on the threads of the bolts [223] and bolts [225].

(b) Apply a layer of stick wax, G51435 [CP2635], to the new seal [226] to hold it during installation.

(c) Install the new seal [226] on the start bleed tube [224] or on the HPC extension case [101].

(d) Install the new seal [222] and coupling clamp [227] that attach the start bleed tube [224] to the SBV valve [124].

(e) Tighten the coupling clamp [227] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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(f) Install the two bolts [223] that attach the start bleed tube [224] to the start bleed bracket [219].

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(g) Install the two bolts [223] and start bleed clamp [610] that attach the start bleed tube [224] to the start bleed bracket [219].

PTF ALL

(h) Tighten the bolts [223] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(i) Install the four bolts [225] that attach the start bleed tube [224] to the HPC extension case [101].

(j) Tighten the bolts [225] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-245-G00

(3) Connect the HJ8A harness [118] as follows (View E, F, and G, Figure 406):

(a) Install the HJ8A harness [118] in the push clamp [178] on the CCC air manifold [120].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [121].

(c) Install the bolt [121] that attaches the loop clamp [122] to the CCC air manifold [120].

(d) Tighten the bolt [121] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the bolt [119].

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(f) Install the bolt [119] that attaches the HJ8A harness [118] wye to the CCC air manifold [120].

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(g) Install thebolt [119] that attaches the HJ8A harness [118] wye to the bracket [202].

PTF ALL

(h) Tighten the bolt [119] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Connect the electrical connector [116] and electrical connector [117] of the HJ8A harness [118] to the TBV valve [123] and SBV valve [124] (TASK 70-00-01-910-803-G00).

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SUBTASK 73-11-05-420-246-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Install the engine lower CDP duct assembly (TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-247-G00

(5) Install the HPSOV (TASK 36-11-07-400-801).

P. Fuel Nozzle Installation (Position 8)

SUBTASK 73-11-05-420-248-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 8) as follows (View C, Figure 409):

(a) Install the fuel nozzle [1] (position 8) with the new seal [3] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 8) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [3] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 8) aft to engage the fuel nozzle [1] (position 8) tip in the combustor radial mixer (cup).

(d) Align the seal [3] and fuel nozzle [1] (position 8) bolt holes to the combustion case [5].

(e) Do not tighten the fuel nozzle bolts at this time.

(f) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(g) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 8) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(h) Tighten the bolts [2] as follows (Figure 409):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [2].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

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(i) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(j) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

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f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-249-G00

(2) Install the ECS middle flange [262] as follows (View E, Figure 409):

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [260].

(b) Put the metal ring seal [261] in position on the ECS middle flange [262].

(c) Put the ECS middle flange [262] in position on the combustion case [5].

(d) Install the four bolts [260] that attach the ECS middle flange [262] to the combustion case [5].

(e) Tighten the bolts [260] to 225 in-lb (25.42 N·m) - 265 in-lb (29.94 N·m).

(f) Connect the sense lines manifold [277] B-nut to the ECS middle flange [262].

(g) Triple tighten the sense lines manifold [277] B-nut as follows:

1) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the B-nut until it turns freely.

3) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Secure the sense lines manifold [277] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

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SUBTASK 73-11-05-420-250-G00

(3) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(f) Install the four bolts [145] that attach the hinged loop clamp [146] to the tube bracket [147] and tube bracket [148].

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(g) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(l) Install the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(o) Install the two bolts [137] that attach the strap clamp [136] and SBV TBV manifold [125] to the support bracket [139].

(p) Tighten the bolts [137] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the two bolts [134].

(r) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Tighten the bolts [134] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(t) Connect the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(u) Connect the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

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c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(w) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(x) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(z) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(aa) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(ac) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(ad) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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SUBTASK 73-11-05-420-251-G00

(4) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(g) Install the four bolts [145] that attach the hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [140] and bolt [140A].

(k) Install the two bolts [140] that attach the hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(l) Install the bolt [140A] that attaches the hinged loop clamps [141] to the fuel manifold bracket [142].

(m) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(n) Install the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(o) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(p) Install the two bolts [134] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(q) Install the two bolts [134A] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(s) Install the two bolts [131] that attach the loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(t) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(u) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(w) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(x) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(y) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(z) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(aa) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ac) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ad) Tighten the bolt [140A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ae) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ag) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ah) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(aj) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-420-252-G00

(5) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the two bolts [145].

(g) Install the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the bolt [179] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(k) Install the two bolts [140] that attach the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(l) Install the bolt [140] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Apply oil, D00599 [CP2442], on the threads of the three bolts [137] and bolt [137A].

(n) Install the three bolts [137] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(o) Install the bolt [137A] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(p) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(q) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Install the two bolts [134A] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(t) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(u) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(w) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(x) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(z) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(aa) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ab) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ac) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(ad) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ae) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ag) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ah) Tighten the bolts [137A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(aj) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ak) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(al) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(am) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL

SUBTASK 73-11-05-420-253-G00

(6) Install the start bleed bracket [219] as follows (View C, Figure 408):
   (a) Apply oil, D00599 [CP2442], on the threads of the two bolts [220].
   (b) Install the two bolts [220] that attach the start bleed bracket [219] to the combustion case [5].
   (c) Tighten the two bolts [220] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-254-G00

(7) Install the start bleed tube [224] as follows (View F, Figure 408):
   (a) Apply oil, D00599 [CP2442], on the threads of the bolts [223] and bolts [225].
   (b) Apply a layer of stick wax, G51435 [CP2635], to the new seal [226] to hold it during installation.
   (c) Install the new seal [226] on the start bleed tube [224] or on the HPC extension case [101].
   (d) Install the seal [222] and coupling clamp [227] that attach the start bleed tube [224] to the SBV valve [124].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Tighten the coupling clamp [227] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0134

(f) Install the two bolts [223] that attach the start bleed tube [224] to the start bleed bracket [219].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0134

(g) Install the two bolts [223] and start bleed clamp [610] that attach the start bleed tube [224] to the start bleed bracket [219].

PTF ALL

(h) Tighten the bolts [223] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(i) Install the four bolts [225] that attach the start bleed tube [224] to the HPC extension case [101].

(j) Tighten the bolts [225] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-255-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Install the engine lower CDP duct assembly (TASK 36-11-01-400-801).

Q. Fuel Nozzle Installation (Position 9)

SUBTASK 73-11-05-420-256-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 9) as follows (View C, Figure 410):

(a) Install the fuel nozzle [1] (position 9) with the new seal [3] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 9) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [3] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 9) aft to engage the fuel nozzle [1] (position 9) tip in the combustor radial mixer (cup).

(d) Align the seal [3] and fuel nozzle [1] (position 9) bolt holes to the combustion case [5].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(e) Install the four bolts [2] that attach the fuel nozzle [1] (position 9) to the combustion case [5].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(f) Put the bracket [650] in position on the fuel nozzle [1] (position 9) and align with the bolt holes.

(g) Install the four bolts [2] that attach the fuel nozzle [1] (position 9) and bracket [650] to the combustion case [5].

PTF ALL

(h) Do not tighten the fuel nozzle bolts at this time.

(i) Install the fuel manifold [376] and fuel manifold [377] to the fuel nozzle [1] (positions 7, 8, and 10) (View G, Figure 410).

1) Apply oil, D00599 [CP2442], on the threads of the four bolts [373].

2) Install the two bolts [373] to the two loop clamps [374] that attach the fuel manifold [377] to the bracket [396].

3) Install the two bolts [373] to the two loop clamps [374] that attach the fuel manifold [376] to the bracket [396].

4) Hand tighten the fuel manifold [376] B-nuts to the fuel nozzle [1] (positions 7, 8, and 10).

a) Do not torque the B-nuts at this time.

5) Hand tighten the fuel manifold [377] B-nuts to the fuel nozzle [1] (positions 7, 8, and 10).

a) Do not torque the B-nuts at this time.

6) Hand tighten the B-nuts [395] that connect the fuel hose [376A] to fuel manifold [376] and fuel manifold [377] to fuel manifold [377A].

7) Tighten the bolts [373] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(j) Hand tighten the fuel manifold [376] and fuel manifold [377] B-nuts to the fuel nozzle [1] (position 9) pilot-primary and pilot-secondary fittings.

1) Do not torque the B-nuts at this time.

(k) Tighten the bolts [2] as follows (Figure 410):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [2].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(l) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

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2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(m) Tighten the fuel manifold B-nuts [4] at all four fuel nozzle [1] positions 7, 8, 9, and 10 to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

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a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

(n) For the B-nuts [395] (fuel hose [376A] to fuel manifold [376] and fuel manifold [377] to fuel manifold [377A]) triple tighten and install safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002].

SUBTASK 73-11-05-420-257-G00

(2) Install the 4th stage air manifold [419] as follows (View E, X, Y, and Z, Figure 413):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(a) Apply oil, D00599 [CP2442], on the threads of the six bolts [414].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [414] and two bolts [615A].

PTF ALL

(c) Install the new seal [423] between the 4th stage air manifold [419] and air adapter [426].

(d) Install the new seal [415] between the 4th stage air manifold [419] and compressor case [65].

(e) Install the four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

(f) Tighten the bolts [414] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(g) Install the two bolts [420] that attach the 4th stage air manifold [419] to the fuel tube bracket [421].

1) Tighten the bolts [420] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(h) Install the two bolts [615A] and strap clamp [616A] that attach the 4th stage air manifold [419] to the bracket [617A].

1) Tighten the bolts [615A] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

(i) Install the coupling clamp [422] that attaches the 4th stage air manifold [419] to the air adapter [426].

1) Tighten the coupling clamp [422] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(j) Connect the LPT air supply tube [424] and B-nut [425] to the 4th stage air manifold [419].

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1) Tighten the B-nut [425] to 79 ft-lb (107.1 N·m) - 91 ft-lb (123.4 N·m).

SUBTASK 73-11-05-420-258-G00

(3) Install the bifurcation drain manifold [265] as follows (View D, E, F, and R, Figure 410):

WARNING THE OIL SUPPLY MANIFOLD IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL OR INSTALLATION TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Apply oil, D00599 [CP2442], on the threads of all loop clamp and strap clamp bolts.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(b) Connect the bifurcation drain manifold [265] at four locations in the TCF/HPT area [304] and B-nuts at the FWD end of the manifold.

(c) Connect the B-nut [303] on the ACC valve drain tube [291] to the bifurcation drain manifold [265].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(d) Connect the drain tube [652], drain tube [653], drain tube [654], drain tube [655], drain tube [656], drain tube [657], drain tube [658], and drain tube [659] to the bifurcation drain manifold [265].

(e) Connect the drain tube [652], drain tube [653], drain tube [655] to the auxiliary drain oil tube [305], C drain oli tube [306], and B drain oil tube [307].

PTF ALL

(f) Do not tighten at this time.

(g) Support and secure the bifurcation drain manifold [265] in place.

(h) Install the two bolts [297] that attach the two loop clamps [298] to the sense line bracket [397].

(i) Tighten the bolts [297] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(j) Install the two bolts [266] that attach the strap clamp [267] and bifurcation drain manifold [265] to the support bracket [268].

(k) Tighten the bolts [266] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(l) Apply a layer of oil, D00599 [CP2442], on the threads of the bolt [309] and bolt [481].

(m) Install the bolt [309] that attaches the drain manifold to the bracket [308].

(n) Install the bolt [481] that attaches the clamp [482] to the bifurcation drain manifold [265].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(o) Install the two bolts [266] that attach the block clamp [651], drain tube [652], drain tube [653], drain tube [654], and drain tube [655] to the support bracket [268].

(p) Tighten the bolts [266] to 50 in-lb (5.65 N·m) - 60 in-lb (6.78 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(q) Tighten the drain manifold to the B drain oil tube [307] to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(r) Tighten the drain manifold to the C drain oli tube [306] to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135 (Continued)

(s) Tighten the drain manifold to the auxiliary drain oil tube [305] to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(t) Tighten the bifurcation drain manifold [265] B-nut that connects to the VBV RHS drain manifold [449] to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

(u) Tighten the No.3 bearing drain tube [479] B-nut that connects to the bifurcation drain manifold [265] to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

(v) Tighten the VBV LHS drain tube [480] B-nut that connects to the bifurcation drain manifold [265] to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

(w) Tighten the ACC valve drain tube [291] and B-nut [303] that connects to the bifurcation drain manifold [265] to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

(x) Tighten the bolt [309] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(y) Tighten the bolt [481] to 32 in-lb (3.6 N·m) - 38 in-lb (4.3 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(z) Tighten the drain tube [653] B-nut that connects to the auxiliary drain oil tube [305] to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(aa) Tighten the C drain oli tube [306] B-nut that connects drain tube [652] to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(ab) Tighten the B drain oil tube [307] B-nut that connects to drain tube [655] to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

PTF ALL

(ac) Tighten the B-nut at the anti-ice drain tube [269] to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

SUBTASK 73-11-05-420-259-G00

(4) Install the oil tube [661] as follows (View A, S, T, U, V, W, and X, Figure 410):

(a) Put the oil tube [661] onto the core module between the Turbine Center Frame (TCF) and High Pressure Compressor (HPC) forward case.

(b) Connect the oil tube [661] to the oil tube [666].

1) Tighten the coupling nut with your hands.

(c) Connect the oil tube [661] to the oil tube [678].

1) Tighten the coupling nut with your hands.

(d) Apply oil, D00599 [CP2442], on the threads and pressure faces of the bolts [668].

NOTE: The bolts must not be tighten at this time. You must tighten them with your hands.

(e) Install the bolt [672] and clamp [673] that attaches the oil tube [661] to the support bracket [671].

(f) Install the bolts [672] and clamps [673] that attach the oil tube [661] to the bracket [674].

(g) Install the bolt [672] and clamp [673] that attaches the oil tube [661] to the bracket [675].

(h) Install the bolt [672] and clamp [673] that attaches the oil tube [661] to the bracket [676].

(i) Triple-tighten the oil tube [661] coupling nut to the oil tube [666] as follows:

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 (Continued)

1) Tighten the coupling nut to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).
2) Loosen each coupling nut until it turns freely.
3) Tighten the coupling nuts to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).
4) Tighten the coupling nuts again to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(j) Install safety cable, G50065 [CP8006], or safety wire, G02345 [CP8001], to the oil tube [661] coupling nut.

(k) Triple-tighten the oil tube [678] coupling nuts to the oil tube [661] as follows:
1) Tighten the coupling nut to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).
2) Loosen each coupling nut until it turns freely.
3) Tighten the coupling nuts to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).
4) Tighten the coupling nuts again to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(l) Install safety cable, G50065 [CP8006], or safety wire, G02345 [CP8001], to the oil tube [678] coupling nuts.

(m) Tighten the bolts [672] to 35 in-lb (4 N·m).

SUBTASK 73-11-05-420-260-G00

(5) Install the oil tube [662] as follows (View A, S, T, U, V, W, X, and Y, Figure 410):
(a) Put the oil tube [662] onto the core module between the TCF and HPC forward case.
(b) Connect the oil tube [662] at its FWD end to the oil tube [664] and oil tube [663].
1) Tighten the coupling nut with your hands.
(c) Connect the oil tube [662] at the TCF area to the air adapter [426].
1) Tighten the coupling nut with your hands.
(d) Make sure that the oil tube [662] is not under strain.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(e) Apply oil, D00599 [CP2442], to the threads and pressure faces of the bolts [668].
NOTE: The bolts must not be tightened at this time. You must tighten them with your hands.
(f) Install the bolt [668] and clamp [669] that attaches the oil tube [662] to the support bracket [670].
(g) Install the bolt [668] and clamp [669] that attaches the oil tube [662] to the support bracket [671].
(h) Install the bolts [668] and clamps [669] that attach the oil tube [662] to the bracket [674].
(i) Install the bolt [668] and clamp [669] that attaches the oil tube [662] to the bracket [675].
(j) Install the bolt [668] and clamp [669] that attaches the oil tube [662] to the bracket [676].
(k) Install the bolt [668] and clamp [669] that attaches the oil tube [662] to the bracket [677].
(l) Triple-tighten the oil tube [662] coupling nut to the air adapter [426] as follows:
1) Tighten the coupling nut to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 (Continued)

2) Loosen each coupling nut until it turns freely.

3) Tighten the coupling nut to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

4) Tighten the coupling nut again to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

(m) Triple-tighten the oil tube [662] coupling nut that connects to the oil tube [664] as follows:

1) Tighten the coupling nut to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

2) Loosen each coupling nut until it turns freely.

3) Tighten the coupling nut to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

4) Tighten the coupling nut again to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

(n) Triple-tighten the B-nut on the oil tube [663] that connects to the oil tube [662] as follows:

1) Tighten the coupling nut to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

2) Loosen each coupling nut until it turns freely.

3) Tighten the coupling nut to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

4) Tighten the coupling nut again to 262 in-lb (29.6 N·m) - 308 in-lb (34.8 N·m).

(o) Install safety cable, G50065 [CP8006], or safety wire, G02345 [CP8001], to the oil tube [662] and oil tube [663] coupling nuts.

(p) Tighten the bolts [668] to 35 in-lb (4 N·m).

PTF ALL

SUBTASK 73-11-05-420-261-G00

(6) Connect the ignition leads [296] as follows (View F, Figure 410):

(a) Apply oil, D00599 [CP2442], on the threads of the bolts [297].

(b) Install the two bolts [297] that attaches the two loop clamps [298] to the sense lines manifold [277].

(c) Tighten the bolts [297] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(d) Install the igniter cooling shroud halves and left ignition lead on the igniter (TASK 74-21-02-400-801-G00).

(e) Install the igniter cooling shroud halves and right ignition lead on the igniter (TASK 74-21-02-400-802-G00).

SUBTASK 73-11-05-420-262-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Install the engine lower CDP duct assembly (TASK 36-11-01-400-801).

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R. Fuel Nozzle Installation (Position 10)

SUBTASK 73-11-05-420-263-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 10) as follows (View C, Figure 411):

(a) Install the fuel nozzle [1] (position 10) with the new seal [378] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 10) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [378] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 10) aft to engage the fuel nozzle [1] (position 10) tip in the combustor radial mixer (cup).

(d) Align the seal [378] and fuel nozzle [1] (position 10) bolt holes to the combustion case [5].

(e) Install the support bracket [367], support bracket [368], and support bracket [372] in position on the fuel nozzle [1] (position 10).

(f) Install the four bolts [366] that attach the fuel nozzle [1] (position 10), support bracket [367], support bracket [368], and support bracket [372] to the combustion case [5].

(g) Do not tighten the fuel nozzle bolts at this time.

(h) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(i) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 10) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(j) Tighten the bolts [366] as follows (Figure 411):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [366].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(k) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

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1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(l) Attach the loop clamp [370] to the support bracket [367] and support bracket [368] with the three bolts [369].

(m) Tighten the bolts [369] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Attach the two loop clamps [374] to the support bracket [372] with the two bolt [373].

(o) Tighten the bolt [373] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(p) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-264-G00

(2) Install the fuel manifold bracket [142] as follows (View N, Figure 406):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(a) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(b) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(c) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(d) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(e) Install the three bolts [140] that attach the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143] with the hinged loop clamps [141].

(f) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(g) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(h) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(i) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [140] and bolt [140A].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(k) Install the bolts [140] that attach theSBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144] with the two hinged loop clamps [141].

1) Tighten the bolts [140] with your hands.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(l) Install the bolt [140A] that attaches the SBV TBV manifold [125] to the fuel manifold bracket [142] with a hinged loop clamp [141].

1) Tighten the bolt [140A] with your hands.

(m) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Tighten the bolt [140A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(o) Apply oil, D00599 [CP2442], on the threads of the bolts [179].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(p) Install the bolt [140] that attaches the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143] with a hinged loop clamps [141].

1) Tighten the bolt [140] with your hands.

(q) Install the two bolts [140] that attach the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144] with the two hinged loop clamps [141].

1) Tighten the bolts [140] with your hands.

(r) Tighten the bolt [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(s) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-265-G00

(3) Install the HPT ACC valve drain tube [352] as follows (View R, S, T, U, V, and W, Figure 411):

(a) Connect the LPT ACC valve drain tube [364] and B-nut [365] to the HPT ACC valve drain tube [352].

(b) Connect the HPT ACC valve drain tube [352] and B-nut [361] to the drain tube [436].

(c) Connect the HPT ACC valve drain tube [352] and B-nut [380] to the ACC valve [379].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [359].

(e) Install the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142], fuel manifold bracket [143], and fuel manifold bracket [144].

(f) Tighten the bolt [359] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [362].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(h) Install the bolt [362] that attaches the loop clamp [363] to the tube bracket [147] and tube bracket [148].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(i) Install the bolt [362] that attaches loop clamp [363] to the tube bracket [147].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(j) Tighten the bolt [362] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the bolt [356].

(l) Install the bolt [356] that attach the loop clamp [357] and HPT ACC valve drain tube [352] to the ACC drain bracket [358].

(m) Tighten the bolt [356] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the bolt [353].

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PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378 (Continued)

(o) Install the bolt [353] that attaches the loop clamp [354] to the support bracket [355].

(p) Tighten the bolt [353] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(q) Tighten the B-nut [361], B-nut [380], and B-nut [365] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

SUBTASK 73-11-05-420-326-G00

(4) Install the fuel hydraulic drain tube [383] as follows (View R, U, V, and W, Figure 411):

(a) Connect the HPT ACC valve drain tube [352] and B-nut [384] to the fuel hydraulic drain tube [383].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [356], bolt [362], and bolt [359].

(c) Install the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

(d) Tighten the bolt [356] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Install the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

(f) Tighten the bolt [362] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Install the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(h) Tighten the bolt [359] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Connect the fuel hydraulic drain tube [383] and B-nut [361] to the drain tube [436].

(j) Tighten the B-nut [384] and B-nut [361] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

SUBTASK 73-11-05-420-266-G00

(5) Install the 7-th stage air tube [310] as follows (View E, F, G, and H, Figure 411):

(a) Apply a layer of stick wax, G51435 [CP2635], to the seal [312] and seal [317] to hold them during installation.

(b) Install the new seal [312] on the 7-th stage air tube [310] or on the compressor case [65].

(c) Install the new seal [317] on the 7-th stage air tube [310] or on the HPT port [318].

(d) Apply oil, D00599 [CP2442], on the threads of the four bolts [311].

(e) Install the four bolts [311] that attach the 7-th stage air tube [310] to the compressor case [65].

(f) Tighten the bolts [311] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the four bolts [316].

(h) Install the four bolts [316] that attach the 7-th stage air tube [310] to the HPT port [318].

(i) Tighten the bolts [316] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the four bolts [315].

(k) Install the four bolts [315] that attach the 7-th stage air tube [310] to the support support bracket [313] and support bracket [314].

(l) Tighten the bolts [315] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

SUBTASK 73-11-05-420-267-G00

(6) Install the 7-th stage air tube [310] as follows (View E, F, G, and H, Figure 411):

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 (Continued)

(a) Apply a layer of stick wax, G51435 [CP2635], to the new seal [312], and new seal [317] to hold them during installation.

(b) Install the new seal [312] on the 7-th stage air tube [310] or on the compressor case [65].

(c) Install the new seal [317] on the 7-th stage air tube [310] or on the HPT port [318].

(d) Apply oil, D00599 [CP2442], on the threads of the four bolts [311].

(e) Install the four bolts [311] that attach the 7-th stage air tube [310] to the compressor case [65].

(f) Tighten the bolts [311] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the four bolts [316].

(h) Install the four bolts [316] that attach the 7-th stage air tube [310] to the HPT port [318].

(i) Tighten the bolts [316] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [614] and two bolts [615].

(k) Install the two bolts [614] that attach the bracket [613] to the support bracket [313].

(l) Install the two bolts [615] and strap clamp [616] that attach the 7-th stage air tube [310] to the bracket [617].

(m) Install the two bolts [611] that attach the block clamp [612] to the bracket [613].

(n) Tighten the bolts [614] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(o) Tighten the bolts [615] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(p) Tighten the bolts [611] to 50.6 in-lb (5.72 N·m) - 59.4 in-lb (6.71 N·m).

PTF ALL

SUBTASK 73-11-05-420-268-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Install the engine upper CDP duct assembly (TASK 36-11-01-400-801).

S. Fuel Nozzle Installation (Position 11)

SUBTASK 73-11-05-420-269-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 11) as follows (View C, Figure 412):

(a) Install the fuel nozzle [1] (position 11) with the new seal [392] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

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(b) Turn the fuel nozzle [1] (position 11) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [392] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 11) aft to engage the fuel nozzle [1] (position 11) tip in the combustor radial mixer (cup).

(d) Align the seal [392] and fuel nozzle [1] (position 11) bolt holes to the combustion case [5].

(e) Install the fuel supply bracket [391] in position on the fuel nozzle [1] (position 11).

(f) Install the tube bracket [147] in position on the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 12).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [390A] and bolt [390].

(h) Install the tube bracket [147A] as follows:

1) Install the bolt [390] that attaches the tube bracket [147A] to the fuel nozzle [1] (position 12).

2) Install the bolt [390A] that attaches the tube bracket [147A] to the tube bracket [147].

PTF ALL

(i) Apply oil, D00599 [CP2442], on the threads of the bolt [393].

(j) Align the fuel manifold bracket [394] to the fuel supply bracket [391] and install the bolt [393].

(k) Install the five bolts [390] that attach the fuel nozzle [1] (position 11), fuel nozzle [1] (position 12), fuel supply bracket [391] and tube bracket [147] to the combustion case [5].

(l) Do not tighten the fuel nozzle bolts at this time.

(m) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(n) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 11) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(o) Tighten the bolts [390] as follows (Figure 412):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [390A].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(p) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

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2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(q) Tighten the bolt [393] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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(r) Tighten the bolt [390A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(s) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

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3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzleB-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

(t) Apply oil, D00599 [CP2442], on the threads of the bolt [443] and bolt [444].

(u) Install the bolt [443] that attaches the loop clamp [442] and fuel manifold [440] to the fuel supply bracket [391].

(v) Tighten the bolt [443] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(w) Install the bolt [444] that attaches the loop clamp [441] and fuel manifold [439] to the fuel supply bracket [391].

(x) Tighten the bolt [444] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

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SUBTASK 73-11-05-420-270-G00

(2) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(f) Install the four bolts [145] that attach the hinged loop clamp [146] to the tube bracket [147] and tube bracket [148].

(g) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(l) Install the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

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(o) Install the two bolts [137] that attach the strap clamp [136] and SBV TBV manifold [125] to the support bracket [139].

(p) Tighten the bolts [137] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the two bolts [134].

(r) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Tighten the bolts [134] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(t) Connect the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(u) Connect the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(w) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(x) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

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(z) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(aa) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(ac) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(ad) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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SUBTASK 73-11-05-420-271-G00

(3) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(g) Install the four bolts [145] that attach the hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach thefuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [140] and bolt [140A].

(k) Install the two bolts [140] that attach the hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(l) Install the bolt [140A] that attaches the hinged loop clamps [141] to the fuel manifold bracket [142].

(m) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(n) Install the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(o) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(p) Install the two bolts [134] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(q) Install the two bolts [134A] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

(s) Install the two bolts [131] that attach the loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(t) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(u) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(w) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(x) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(y) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(z) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(aa) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ac) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ad) Tighten the bolt [140A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ae) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ag) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(ah) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(aj) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-420-272-G00

(4) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the two bolts [145].

(g) Install the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the bolt [179] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(k) Install the two bolts [140] that attach the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(l) Install the bolt [140] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Apply oil, D00599 [CP2442], on the threads of the three bolts [137] and bolt [137A].

(n) Install the three bolts [137] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(o) Install the bolt [137A] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(p) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(q) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Install the two bolts [134A] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(t) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(u) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the bracket [202].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(w) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(x) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(z) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(aa) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ab) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ac) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(ad) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ae) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ag) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ah) Tighten the bolts [137A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(aj) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ak) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(al) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(am) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-273-G00

(5) Install the HPT ACC valve drain tube [352] as follows (View R, S, T, U, V, and W, Figure 411):

(a) Connect the LPT ACC valve drain tube [364] and B-nut [365] to the HPT ACC valve drain tube [352].

(b) Connect the HPT ACC valve drain tube [352] and B-nut [361] to the drain tube [436].

(c) Connect the HPT ACC valve drain tube [352] and B-nut [380] to the ACC valve [379].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [359].

(e) Install the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142], fuel manifold bracket [143], and fuel manifold bracket [144].

(f) Tighten the bolt [359] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [362].

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(h) Install the bolt [362] that attaches the loop clamp [363] to the tube bracket [147] and tube bracket [148].

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(i) Install the bolt [362] that attaches loop clamp [363] to the tube bracket [147].

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(j) Tighten the bolt [362] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the bolt [356].

(l) Install the bolt [356] that attach the loop clamp [357] and HPT ACC valve drain tube [352] to the ACC drain bracket [358].

(m) Tighten the bolt [356] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the bolt [353].

(o) Install the bolt [353] that attaches the loop clamp [354] to the support bracket [355].

(p) Tighten the bolt [353] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(q) Tighten the B-nut [361], B-nut [380], and B-nut [365] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-327-G00

(6) Install the fuel hydraulic drain tube [383] as follows (View R, U, V, and W, Figure 411):

(a) Connect the HPT ACC valve drain tube [352] and B-nut [384] to the fuel hydraulic drain tube [383].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [356], bolt [362], and bolt [359].

(c) Install the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

(d) Tighten the bolt [356] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Install the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

(f) Tighten the bolt [362] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Install the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(h) Tighten the bolt [359] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Connect the fuel hydraulic drain tube [383] and B-nut [361] to the drain tube [436].

(j) Tighten the B-nut [384] and B-nut [361] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

PTF ALL

SUBTASK 73-11-05-420-274-G00

(7) Install the fuel hose manifold [323] as follows (View J, K, M, and N, Figure 411):

NOTE: The fuel hose manifold can also be identified as Pilot Primary and Main non-enriched (PPMne) fuel manifold.

(a) Connect the left primary fuel manifold [344] and B-nut [375] to the fuel hose manifold [323].

(b) Connect the fuel hose manifold [323] and B-nut [346] at the fuel tube [347].

(c) Connect the fuel hose manifold [323] and B-nut [327] at the SCU [151].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [324].

(e) Install the bolt [324] that attaches the loop clamp [325] to the support bracket [326].

(f) Apply oil, D00599 [CP2442], on the threads of the bolt [338].

(g) Install the bolt [338] that attaches the loop clamp [339] to the support bracket [340].

(h) Apply oil, D00599 [CP2442], on the threads of the bolt [341].

(i) Install the bolt [341] that attaches the loop clamp [342] to the support bracket [343].

(j) Tighten the B-nut [327] at the SCU [151] as follows:

1) Tighten the B-nut [327] to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

2) Loosen the B-nut [327] until you can turn them with your fingers.

3) Tighten the B-nut [327] again to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

4) Tighten the B-nut [327] for a third time to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

5) Make sure that the B-nut [327] is fully seated.

6) If the B-nut [327] is not fully seated, do the above steps again to tighten the B-nut [327].

(k) Tighten the B-nut [375] as follows:

1) Tighten the B-nut [375] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

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2) Loosen the B-nut [375] until you can turn them with your fingers.

3) Tighten the B-nut [375] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

4) Tighten the B-nut [375] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

5) Make sure that the B-nut [375] is fully seated.

6) If the B-nut [375] is not fully seated, do the above steps again to tighten the B-nut [375].

7) Secure the B-nut [375] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(l) Tighten the B-nut [346] as follows:

1) Tighten the B-nut [346] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Loosen the B-nut [346] until you can turn them with your fingers.

3) Tighten the B-nut [346] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

4) Tighten the B-nut [346] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

5) Make sure that the B-nut [346] is fully seated.

6) If the B-nut [346] is not fully seated, do the above steps again to tighten the B-nut [346].

7) Secure the B-nut [346] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

SUBTASK 73-11-05-420-275-G00

(8) Connect the fuel tube hose [319] to the SCU [151] as follows (View K, Figure 411):

NOTE: The fuel tube hose can also be identified as PS fuel manifold.

(a) Apply oil, D00599 [CP2442], on the threads of a bolt [320].

(b) Install the bolt [320] that attaches the loop clamp [321] and fuel tube hose [319] to the support bracket [326].

1) Tighten the bolt [320] by hands.

(c) Connect the fuel tube hose [319] and B-nut [322] to the SCU [151].

(d) Tighten the B-nut [322] as follows:

1) Tighten the B-nut [322] to 55.2 ft-lb (74.8 N·m) - 64.8 ft-lb (87.9 N·m).

2) Loosen the B-nut [322] until you can turn them with your fingers.

3) Tighten the B-nut [322] again to 55.2 ft-lb (74.8 N·m) - 64.8 ft-lb (87.9 N·m).

4) Tighten the B-nut [322] for a third time to 55.2 ft-lb (74.8 N·m) - 64.8 ft-lb (87.9 N·m).

5) Make sure that the B-nut [322] is fully seated.

6) If the B-nut [322] is not fully seated, do the above steps again to tighten the B-nut [322].

7) Install safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002], on the B-nut [322](TASK 70-41-00-400-802-G00).

(e) Tighten the bolt [320] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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SUBTASK 73-11-05-420-276-G00

(9) Connect the fuel manifold hose [329] on the SCU [151] as follows (View K, Figure 411):

NOTE: The fuel manifold hose can also be identified as Pilot Primary and Main enriched (PPMe) fuel manifold.

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [328].

(b) Install the bolt [328] that attaches loop clamp [330] and fuel manifold hose [329] to the support bracket [326].

1) Tighten the bolt [328] by hands.

(c) Connect the fuel manifold hose [329] and B-nut [331] on the SCU [151].

(d) Tighten the B-nut [331] as follows:

1) Tighten the B-nut [331] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Loosen the B-nut [331] until you can turn them with your fingers.

3) Tighten the B-nut [331] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

4) Tighten the B-nut [331] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

5) Make sure that the B-nut [331] is fully seated.

6) If the B-nut [331] is not fully seated, do the above steps again to tighten the B-nut [331].

7) Install safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002], on the B-nut [331] (TASK 70-41-00-400-802-G00).

(e) Tighten the bolt [328] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

SUBTASK 73-11-05-420-277-G00

(10) Connect the electrical connector [438] to the fuel manifold pressure sensor [437] (TASK 70-00-01-910-803-G00) (View K, Figure 411).

SUBTASK 73-11-05-420-278-G00

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Install the engine upper CDP duct assembly (TASK 36-11-01-400-801).

T. Fuel Nozzle Installation (Position 12)

SUBTASK 73-11-05-420-279-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 12) as follows (View C, Figure 413):

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(a) Install the fuel nozzle [1] (position 12) with the new seal [429] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 12) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [429] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 12) aft to engage the fuel nozzle [1] (position 12) tip in the combustor radial mixer (cup).

(d) Align the seal [429] and fuel nozzle [1] (position 12) bolt holes to the combustion case [5].

(e) Install the drain tube bracket [428] in position on the fuel nozzle [1] (position 12) and on the fuel nozzle [1] (position 13).

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(f) Install the tube bracket [147] and tube bracket [148] in position on the fuel nozzle [1] (position 11) and in position on the fuel nozzle [1] (position 12).

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(g) Install the tube bracket [147] in position on the fuel nozzle [1] (position 11) and in position on the fuel nozzle [1] (position 12).

(h) Apply oil, D00599 [CP2442], on the threads of the bolt [390A].

(i) Install the tube bracket [147A] as follows:

1) Install the tube bracket [147A] in position on the fuel nozzle [1] (position 12).

2) Install the bolt [390A] that attaches the tube bracket [147A] to the tube bracket [147].

PTF ALL

(j) Install the six bolts [427] that attach the fuel nozzle [1] (position 11), fuel nozzle [1] (position 12), fuel nozzle [1] (position 13), tube bracket [147], tube bracket [148], and drain tube bracket [428] to the combustion case [5].

(k) Do not tighten the fuel nozzle bolts at this time.

(l) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(m) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 12) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(n) Tighten the bolts [427] as follows (Figure 413):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [427].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(o) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

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1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

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(p) Tighten the bolt [390A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(q) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-280-G00

(2) If removed, install the left MTC valve (TASK 75-23-02-400-801-G00).

SUBTASK 73-11-05-420-281-G00

(3) Install the HPT ACC valve drain tube [352] as follows (View R, S, T, U, V, and W, Figure 411):

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Connect the LPT ACC valve drain tube [364] and B-nut [365] to the HPT ACC valve drain tube [352].

(b) Connect the HPT ACC valve drain tube [352] and B-nut [361] to the drain tube [436].

(c) Connect the HPT ACC valve drain tube [352] and B-nut [380] to the ACC valve [379].

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [359].

(e) Install the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142], fuel manifold bracket [143], and fuel manifold bracket [144].

(f) Tighten the bolt [359] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the bolt [362].

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(h) Install the bolt [362] that attaches the loop clamp [363] to the tube bracket [147] and tube bracket [148].

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(i) Install the bolt [362] that attaches loop clamp [363] to the tube bracket [147].

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(j) Tighten the bolt [362] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the bolt [356].

(l) Install the bolt [356] that attach the loop clamp [357] and HPT ACC valve drain tube [352] to the ACC drain bracket [358].

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PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378 (Continued)

(m) Tighten the bolt [356] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the bolt [353].

(o) Install the bolt [353] that attaches the loop clamp [354] to the support bracket [355].

(p) Tighten the bolt [353] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(q) Tighten the B-nut [361], B-nut [380], and B-nut [365] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-328-G00

(4) Install the fuel hydraulic drain tube [383] as follows (View R, U, V, and W, Figure 411):

(a) Connect the HPT ACC valve drain tube [352] and B-nut [384] to the fuel hydraulic drain tube [383].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [356], bolt [362], and bolt [359].

(c) Install the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

(d) Tighten the bolt [356] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Install the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

(f) Tighten the bolt [362] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Install the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(h) Tighten the bolt [359] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Connect the fuel hydraulic drain tube [383] and B-nut [361] to the drain tube [436].

(j) Tighten the B-nut [384] and B-nut [361] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-282-G00

(5) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(e) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(f) Install the four bolts [145] that attach the hinged loop clamp [146] to the tube bracket [147] and tube bracket [148].

(g) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(k) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(l) Install the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(o) Install the two bolts [137] that attach the strap clamp [136] and SBV TBV manifold [125] to the support bracket [139].

(p) Tighten the bolts [137] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the two bolts [134].

(r) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Tighten the bolts [134] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(t) Connect the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(u) Connect the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings.

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(w) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(x) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(z) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(aa) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(ac) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(ad) Tighten the bolts [126] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-420-283-G00

(6) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the four bolts [145].

(g) Install the four bolts [145] that attach the hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the two bolts [179] that attach thefuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [140] and bolt [140A].

(k) Install the two bolts [140] that attach the hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(l) Install the bolt [140A] that attaches the hinged loop clamps [141] to the fuel manifold bracket [142].

(m) Apply oil, D00599 [CP2442], on the threads of the two bolts [137].

(n) Install the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(o) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(p) Install the two bolts [134] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(q) Install the two bolts [134A] that attach the loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(s) Install the two bolts [131] that attach the loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

(t) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

(u) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(w) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(x) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(y) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(z) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(aa) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ab) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(ac) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ad) Tighten the bolt [140A] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ae) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ag) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ah) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(aj) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-420-284-G00

(7) Install the SBV TBV manifold [125] as follows (View H, J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Install the new seal gasket [150] on the SBV TBV manifold [125] or on the SCU [151].

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [149].

(c) Install the four bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(d) Hand tighten the two SBV TBV manifold [125] B-nut to the two TBV fittings.

(e) Hand tighten the two SBV TBV manifold [125] B-nut to the two SBV fittings.

(f) Apply oil, D00599 [CP2442], on the threads of the two bolts [145].

(g) Install the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [179].

(i) Install the bolt [179] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142].

(j) Apply oil, D00599 [CP2442], on the threads of the three bolts [140].

(k) Install the two bolts [140] that attach the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(l) Install the bolt [140] that attaches the hinged loop clamp [141] and SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(m) Apply oil, D00599 [CP2442], on the threads of the three bolts [137] and bolt [137A].

(n) Install the three bolts [137] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(o) Install the bolt [137A] that attach the loop clamp [136B] and SBV TBV manifold [125] to the support bracket [139].

(p) Apply oil, D00599 [CP2442], on the threads of the two bolts [134] and bolts [134A].

(q) Install the two bolts [134] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(r) Install the two bolts [134A] that attach the two loop clamps [135] and SBV TBV manifold [125] to the manifold bracket [133].

(s) Apply oil, D00599 [CP2442], on the threads of the two bolts [131].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(t) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(u) Install the two bolts [131] that attach the two loop clamps [132] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [129].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(w) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(x) Install the two bolts [129] that attach the two loop clamps [130] and SBV TBV manifold [125] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(y) Apply oil, D00599 [CP2442], on the threads of the two bolts [126].

(z) Install the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98].

(aa) Tighten the two SBV TBV manifold [125] B-nuts to the two TBV valve [123] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
   d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ab) Tighten the two SBV TBV manifold [125] B-nuts to the two SBV valve [124] fittings as follows:

1) Tighten the larger SBV TBV manifold [125] B-nut as follows:
   a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   b) Loosen the B-nut until you can turn it with your fingers.
   c) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
   d) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).
2) Tighten the smaller SBV TBV manifold [125] B-nut as follows:

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

a) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

3) Secure the SBV TBV manifold [125] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(ac) Tighten the bolts [149] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(ad) Tighten the bolts [145] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ae) Tighten the bolts [179] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(af) Tighten the bolts [140] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ag) Tighten the bolts [137] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(ah) Tighten the bolts [137A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ai) Tighten the bolts [134] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

(aj) Tighten the bolts [134A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(ak) Tighten the bolts [131] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(al) Tighten the bolts [129] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(am) Tighten the bolts [126] to 51 in-lb (5.76 N·m) - 59 in-lb (6.67 N·m).

PTF ALL

SUBTASK 73-11-05-420-285-G00

(8) Install the LPT ACC valve drain tube [364] as follows (View D, Figure 413):

(a) Connect the LPT ACC valve drain tube [364] and B-nut [410] to the LPT ACC fuel manifold [469].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [411].

(c) Install the bolt [411] that attaches the loop clamp [412] to the drain tube bracket [413].

(d) Tighten the bolt [411] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) Tighten the B-nut [410] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

SUBTASK 73-11-05-420-286-G00

(9) Install the fuel hose manifold [323] as follows (View K, N, and M, Figure 411):

NOTE: The fuel hose manifold can also be identified as Pilot Primary and Main non-enriched (PPMne) fuel manifold.

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [324], bolt [341], and bolt [338].

(b) Install the bolt [324] that attaches the loop clamp [325] and fuel hose manifold [323] to the support bracket [326].

1) Tighten the bolt [324] by hands.

(c) Install the bolt [341] that attaches the loop clamp [342] and fuel hose manifold [323] to the support bracket [343].

1) Tighten the bolt [341] by hands.

(d) Install the bolt [338] that attaches the loop clamp [339] and fuel hose manifold [323] to the support bracket [340].

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1) Tighten the bolt [338] by hands.

(e) Connect the fuel hose manifold [323] and B-nut [346] to the fuel tube [347] (View M, Figure 411).

1) Tighten the B-nut [346] as follows:

a) Tighten the B-nut [346] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut [346] until you can turn them with your fingers.

c) Tighten the B-nut [346] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut [346] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

e) Make sure that the B-nut [346] is fully seated.

f) If the B-nut [346] is not fully seated, do the above steps again to tighten the B-nut [346].

g) Secure the B-nut [346] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(f) Connect the left primary fuel manifold [344] and B-nut [375] to the fuel hose manifold [323] (View N, Figure 411).

1) Tighten the B-nut [375] as follows:

a) Tighten the B-nut [375] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut [375] until you can turn them with your fingers.

c) Tighten the B-nut [375] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten theB-nut [375] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

e) Make sure that the B-nut [375] is fully seated.

f) If the B-nut [375] is not fully seated, do the above steps again to tighten the B-nut [375].

g) Secure the B-nut [375] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(g) Connect the fuel hose manifold [323] and B-nut [327] to the SCU [151] (View K, Figure 411).

1) Tighten the B-nut [327] at the split control unit as follows:

a) Tighten the B-nut [327] to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

b) Loosen the B-nut [327] until you can turn them with your fingers.

c) Tighten the B-nut [327] again to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

d) Tighten the B-nut [327] for a third time to 78.2 ft-lb (106.0 N·m) - 91 ft-lb (123.4 N·m).

e) Make sure that the B-nut [327] is fully seated.

f) If the B-nut [327] is not fully seated, do the above steps again to tighten the B-nut [327].

g) Tighten the bolt [324], bolt [341], and bolt [338] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

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SUBTASK 73-11-05-420-287-G00

(10) Connect the fuel manifold hose [329] on the SCU [151] as follows (View K, Figure 411):

NOTE: The fuel manifold hose can also be identified as Pilot Primary and Main enriched (PPMe) fuel manifold.

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [328].

(b) Install the bolt [328] that attaches loop clamp [330] and fuel manifold hose [329] to the support bracket [326].

1) Tighten the bolt [328] by hands.

(c) Connect thefuel manifold hose [329] and B-nut [331] on the SCU [151].

(d) Tighten the B-nut [331] as follows:

1) Tighten the B-nut [331] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

2) Loosen the B-nut [331] until you can turn them with your fingers.

3) Tighten the B-nut [331] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

4) Tighten theB-nut [331] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

5) Make sure that the B-nut [331] is fully seated.

6) If the B-nut [331] is not fully seated, do the above steps again to tighten the B-nut [331].

7) Install safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002], on the B-nut [331] (TASK 70-41-00-400-802-G00).

(e) Tighten the bolt [328] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

SUBTASK 73-11-05-420-288-G00

(11) Install the fuel tube hose [319] as follows (View K, N, and M, Figure 411):

NOTE: The fuel tube hose can also be identified as PS fuel manifold.

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [320] and bolt [341].

(b) Install the bolt [320] that attaches the loop clamp [321] and fuel tube hose [319] to the support bracket [326].

1) Tighten the bolt [320] by hands.

(c) Install the bolt [341] that attaches the loop clamp [342] and fuel tube hose [319] to the support bracket [343].

1) Tighten the bolt [341] by hands.

(d) Connect the fuel tube hose [319] and B-nut [322] at the SCU [151] (View K, Figure 411).

1) Tighten the B-nut [322] as follows:

a) Tighten the B-nut [322] to 55.2 ft-lb (74.84 N·m) - 64.8 ft-lb (87.86 N·m).

b) Loosen the B-nut [322] until you can turn them with your fingers.

c) Tighten the B-nut [322] again to 55.2 ft-lb (74.84 N·m) - 64.8 ft-lb (87.86 N·m).

d) Tighten the B-nut [322] for a third time to 55.2 ft-lb (74.84 N·m) - 64.8 ft-lb (87.86 N·m).

e) Make sure that the B-nut [322] is fully seated.

f) If the B-nut [322] is not fully seated, do the above steps again to tighten the B-nut [322].

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g) Secure the B-nut [322] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(e) Connect the fuel tube [382] and B-nut [381] at the fuel tube hose [319] (View N, Figure 411).

1) Tighten the B-nut [381] as follows:

a) Tighten the B-nut [381] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut [381] until you can turn them with your fingers.

c) Tighten the B-nut [381] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut [381] for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut [381] is fully seated.

f) If the B-nut [381] is not fully seated, do the above steps again to tighten the B-nut [381].

g) Secure the B-nut [381] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(f) Connect the fuel tube hose [319] and B-nut [348] to the fuel manifold [351] (View M, Figure 411).

1) Tighten the B-nut [348] to the fuel manifold [351] as follows:

a) Tighten the B-nut [348] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

b) Loosen the B-nut [348] until you can turn them with your fingers.

c) Tighten the B-nut [348] again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

d) Tighten the B-nut [348] for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

e) Make sure that the B-nut [348] is fully seated.

f) If the B-nut [348] is not fully seated, do the above steps again to tighten the B-nut [348].

g) Secure the B-nut [348] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

2) Tighten the bolt [320] and bolt [341] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

SUBTASK 73-11-05-420-289-G00

(12) Connect the electrical connector [438] to the fuel manifold pressure sensor [437] (TASK 70-00-01-910-803-G00) (View K, Figure 411).

SUBTASK 73-11-05-420-290-G00

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(13) Install the engine upper CDP duct assembly (TASK 36-11-01-400-801).

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U. Fuel Nozzle Installation (Position 13)

SUBTASK 73-11-05-420-291-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 13) as follows (View C, Figure 414):

(a) Install the fuel nozzle [1] (position 13) with the new seal [475] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 13) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [475] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 13) aft to engage the fuel nozzle [1] (position 13) tip in the combustor radial mixer (cup).

(d) Align the seal [475] and fuel nozzle [1] (position 13) bolt holes to the combustion case [5].

(e) Install the drain tube bracket [428] in position on the fuel nozzle [1] (position 13) and on the fuel nozzle [1] (position 12).

(f) Install the three bolts [474] that attach the fuel nozzle [1] (position 13) to the combustion case [5], bolt [474] that attaches the drain tube bracket [428] to the fuel nozzle [1] (position 13), and one bolt [474] that attaches the drain tube bracket [428] to the fuel nozzle [1] (position 12).

(g) Do not tighten the fuel nozzle bolts at this time.

(h) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(i) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 13) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(j) Tighten the bolts [474] as follows (Figure 414):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the bolts [474].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(k) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(l) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

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a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-292-G00

(2) If removed, install the left MTC valve (TASK 75-23-02-400-801-G00).

SUBTASK 73-11-05-420-293-G00

(3) Install the LPT ACC fuel manifold [469] as follows (View K, L, and M, Figure 414):

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [470].

(b) Install the new flange seal gasket [471] to the LPT ACC fuel manifold [469] or to the ACC valve [472].

(c) Install the four bolts [470] that attach the LPT ACC fuel manifold [469] to the ACC valve [472].

(d) Connect the LPT ACC fuel manifold [469] and B-nut [463] to the hydraulic fuel manifold [462].

(e) Connect the LPT ACC fuel tube [464] and B-nut [465] to the LPT ACC fuel manifold [469].

(f) Apply oil, D00599 [CP2442], on the threads of the two bolts [466].

(g) Install the two bolts [466] that attaches the two loop clamps [467] to the support bracket [468].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0071 AND POST SB-CFM-LEAP-1B-72-0378

(h) Apply oil, D00599 [CP2442], on the threads of the two bolts [502A].

(i) Install the two bolts [502A] that attach the two loop clamps [503A] to the support bracket [501A].

(j) Tighten the bolts [502A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

(k) Tighten the bolts [466] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(l) Tighten the bolts [470] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(m) Tighten the B-nut [463] to the hydraulic fuel manifold [462] as follows:

1) Tighten the B-nut [463] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the B-nut [463] until you can turn them with your fingers.

3) Tighten the B-nut [463] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the B-nut [463] for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Make sure that the B-nut [463] is fully seated.

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6) If the B-nut [463] is not fully seated, do the above steps again to tighten the B-nut [463].

7) Secure the B-nut [463] with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(n) Tighten the B-nut [465] to the LPT ACC fuel manifold [469] as follows:

1) Tighten the B-nut [465] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the B-nut [465] until you can turn them with your fingers.

3) Tighten the B-nut [465] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the B-nut [465] for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Make sure that the B-nut [465] is fully seated.

6) If the B-nut [465] is not fully seated, do the above steps again to tighten the B-nut [465].

SUBTASK 73-11-05-420-294-G00

(4) Install the LPT ACC valve drain tube [364] as follows (View D, Figure 413):

(a) Connect the LPT ACC valve drain tube [364] and B-nut [410] to the LPT ACC fuel manifold [469].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(b) Connect the LPT ACC valve drain tube [364] and B-nut [430] to the HPT ACC valve drain tube [352].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(c) Connect the LPT ACC valve drain tube [364] and B-nut [430] that attach the LPT ACC valve drain tube [364] to the fuel hydraulic drain tube [383].

PTF ALL

(d) Apply oil, D00599 [CP2442], on the threads of the bolt [411].

(e) Install the bolt [411] that attaches the loop clamp [412] to the drain tube bracket [413].

(f) Tighten the bolt [411] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(g) Tighten the B-nut [410] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

SUBTASK 73-11-05-420-295-G00

(5) Install the 7th-stage air manifold [452] as follows (View D, E, F, G, H, and J, Figure 414):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(a) Apply oil, D00599 [CP2442], on the threads of the two bolts [450], four bolts [453], two bolts [455], and four bolts [457].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [624], bolts [453], two bolts [626], and four bolts [457].

PTF ALL

(c) Apply a layer of stick wax, G51435 [CP2635], to the new seal [454] and seal [458] to hold them during installation.

(d) Put the 7th-stage air manifold [452] in position on the engine.

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(e) Install the new seal [454] on the 7th-stage air manifold [452] or on the compressor case [65].

(f) Install the new seal [458] on the 7th-stage air manifold [452] or the HPT port [459].

(g) Install the four bolts [453] that attach the 7th-stage air manifold [452] and seal [454] to the compressor case [65].

(h) Install the four bolts [457] that attach the 7th-stage air manifold [452] and seal [458] to the HPT port [459].

(i) Tighten the bolts [453] and bolts [457] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Connect the air tube [460] and B-nut [461] to the 7th-stage air manifold [452].

(k) Tighten the B-nut [461] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(l) Install the two bolts [450] that attach the 7th-stage air manifold [452] to the support bracket [451].

(m) Tighten the bolts [450] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(n) Install the two bolts [455] that attach the 7th-stage air manifold [452] to the support bracket [456].

(o) Tighten the bolts [455] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(p) Install the two bolts [624] and strap clamp [625] that attach the 7th-stage air manifold [452] to the support bracket [451].

(q) Tighten the bolts [624] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(r) Install the two bolts [626] and strap clamp [627] that attach the 7th-stage air manifold [452] to the support bracket [456].

(s) Tighten the bolts [626] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

SUBTASK 73-11-05-420-296-G00

(6) Install the 4th stage air manifold [419] as follows (View E, F, G, H, and J, Figure 413):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [414], two bolts [416] and two bolts [420].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0006

(b) Apply oil, D00599 [CP2442], on the threads of the four bolts [414], two bolts [416], and two bolts [628].

PTF ALL

(c) Install the seal [423] between the 4th stage air manifold [419] and air adapter [426].

(d) Install the seal [415] between the 4th stage air manifold [419] and compressor case [65].

(e) Install the four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

(f) Tighten the bolts [414] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(g) Install the two bolts [416] that attach the strap clamp [417] to the support bracket [418].

(h) Tighten the bolts [416] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006

(i) Install the two bolts [420] that attach the 4th stage air manifold [419] to the fuel tube bracket [421].

(j) Tighten the bolts [420] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0006

(k) Install the two bolts [628] and strap clamp [629] that attach the 4th stage air manifold [419] to the fuel tube bracket [421].

(l) Tighten the bolts [628] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

(m) Install the coupling clamp [422] to the 4th stage air manifold [419] to the air adapter [426].

(n) Tighten the coupling clamp [422] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(o) Connect the LPT air supply tube [424] and B-nut [425] to the 4th stage air manifold [419].

(p) Tighten the B-nut [425] to 79 ft-lb (107.1 N·m) - 91 ft-lb (123.4 N·m).

SUBTASK 73-11-05-420-297-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Install the engine upper CDP duct assembly (TASK 36-11-01-400-801).

V. Fuel Nozzle Installation (Position 14)

SUBTASK 73-11-05-420-298-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 14) as follows (View B, Figure 415):

(a) Install the fuel nozzle [1] (position 14) with the new seal [506] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn thefuel nozzle [1] (position 14) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [506] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 14) aft to engage the fuel nozzle [1] (position 14) tip in the combustor radial mixer (cup).

(d) Align the seal [506] and fuel nozzle [1] (position 14) bolt holes to the combustion case [5].

(e) Install the support bracket [501] in position on the fuel nozzle [1] (position 14).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(f) Install the four bolts [500] that attach the fuel nozzle [1] (position 14) and support bracket [501] to the combustion case [5].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(g) Install the two bolts [500] that attach the fuel nozzle [1] (position 14) and support bracket [501] to the combustion case [5].

(h) Install the two bolts [500] that attach the fuel nozzle [1] (position 14) and support bracket [501A] to the combustion case [5].

PTF ALL

(i) Do not tighten the fuel nozzle bolts at this time.

(j) Do not apply lubricant on the fuel nozzle [1] and B-nut [4] threads.

(k) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 14) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(l) Tighten the bolts [500] as follows (Figure 415):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [500].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(m) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [502].

(o) Install the two bolts [502] that attach the two loop clamps [503] to the support bracket [501].

(p) Tighten the bolts [502] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(q) Install the two bolts [502A] that attach the two loop clamps [503A] to the support bracket [501A].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(r) Tighten the bolts [502A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(s) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

   a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

   b) Loosen the two B-nuts [4] until you can turn them with your fingers.

   c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

   d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

   e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

   f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

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a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-299-G00

(2) Install the LPT ACC fuel manifold [469] as follows (View K, L, and M, Figure 414):

(a) Apply oil, D00599 [CP2442], on the threads of the four bolts [470].

(b) Install the new flange seal gasket [471] to the LPT ACC fuel manifold [469] or to the ACC valve [472].

(c) Install the four bolts [470] that attach the LPT ACC fuel manifold [469] to the ACC valve [472].

(d) Connect the LPT ACC fuel manifold [469] and B-nut [463] to the hydraulic fuel manifold [462].

(e) Connect the LPT ACC fuel tube [464] and B-nut [465] to the LPT ACC fuel manifold [469].

(f) Connect the LPT ACC valve drain tube [364] and B-nut [473] that attaches the LPT ACC valve drain tube [364] to the LPT ACC fuel manifold [469].

(g) Apply oil, D00599 [CP2442], on the threads of the two bolts [466].

(h) Install the two bolts [466] that attaches the two loop clamps [467] to the support bracket [468].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(i) Apply oil, D00599 [CP2442], on the threads of the two bolts [502A].

(j) Install the two bolts [502A] that attach the two loop clamps [503A] to the support bracket [501A].

(k) Tighten the bolts [502A] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

(l) Tighten the bolts [466] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(m) Tighten the bolts [470] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(n) Tighten the B-nut [473] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

(o) Tighten the B-nut [463] to the hydraulic fuel manifold [462] as follows:

1) Tighten the B-nut [463] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the B-nut [463] until you can turn them with your fingers.

3) Tighten the B-nut [463] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the B-nut [463] for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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